

SingleRAN

ALD Management Feature Parameter Description

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Contents

1 Change History.....	1
1.1 SRAN22.1 Draft A (2025-12-31).....	1
2 About This Document.....	3
2.1 General Statements.....	3
2.2 Applicable RAT.....	3
2.3 Features in This Document.....	4
2.4 Differences.....	5
3 General Principles.....	6
4 RET Antenna.....	23
4.1 Principles.....	23
4.1.1 Connections Between RET Antennas and RRUs/RFUs.....	24
4.1.2 Operations on RET Antennas.....	34
4.2 Network Analysis.....	37
4.2.1 Benefits.....	37
4.2.2 Impacts.....	37
4.3 Requirements.....	37
4.3.1 Licenses.....	37
4.3.2 Functions.....	37
4.3.2.1 Prerequisite Functions.....	38
4.3.2.2 Mutually Exclusive Functions.....	38
4.3.2.3 Function Impacts.....	38
4.3.3 Hardware.....	38
4.3.4 Networking.....	38
4.4 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB).....	38
4.4.1 When to Use.....	38
4.4.2 Precautions.....	39
4.4.3 Data Configuration.....	42
4.4.3.1 Data Preparation (Old Model).....	42
4.4.3.2 Data Preparation (New Model).....	81
4.4.3.3 Initial Configuration.....	129
4.4.3.4 Using MML Commands (Old Model).....	130
4.4.3.5 Using MML Commands (New Model).....	144

4.4.4 Commissioning.....	159
4.4.5 Activation Verification.....	160
4.4.6 Deactivation.....	162
4.4.7 Reconfiguration.....	163
4.5 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	165
4.5.1 When to Use.....	165
4.5.2 Precautions.....	166
4.5.3 Data Configuration.....	168
4.5.3.1 Data Preparation.....	169
4.5.3.2 Initial Configuration.....	169
4.5.3.3 Using MML Commands.....	170
4.5.4 Commissioning.....	171
4.5.5 Activation Verification.....	171
4.5.6 Deactivation.....	171
4.5.7 Reconfiguration.....	171
4.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	172
4.6.1 When to Use.....	172
4.6.2 Precautions.....	172
4.6.3 Data Configuration.....	174
4.6.3.1 Data Preparation.....	174
4.6.3.2 Initial Configuration.....	187
4.6.3.3 Using MML Commands.....	187
4.6.4 Commissioning.....	191
4.6.5 Activation Verification.....	191
4.6.6 Deactivation.....	192
4.6.7 Reconfiguration.....	192
4.7 Operation and Maintenance (Automatic ALD Deployment).....	193
4.7.1 When to Use.....	193
4.7.2 Data Configuration.....	193
4.7.2.1 Data Preparation.....	193
4.7.2.2 Creating an Automatic ALD Deployment Task.....	197
4.7.2.3 Initial Configuration.....	199
4.7.2.4 Using MML Commands.....	199
4.7.2.5 Using the MAE-Deployment.....	200
4.7.3 Commissioning.....	201
4.7.4 Activation Verification.....	202
4.7.5 Checking that Automatic ALD Deployment Is Complete.....	202
4.7.6 Deactivation.....	203
4.7.7 Reconfiguration.....	205
4.7.8 Appendix: Automatic ALD Configuration Process.....	205
5 TMA.....	222
5.1 Principles.....	222

5.1.1 Connections Between the TMA, RRU/RFU, and RET Antenna.....	223
5.1.2 Operations on the TMA.....	229
5.2 Network Analysis.....	230
5.2.1 Benefits.....	231
5.2.2 Impacts.....	231
5.3 Requirements.....	231
5.3.1 Licenses.....	231
5.3.2 Functions.....	231
5.3.2.1 Prerequisite Functions.....	231
5.3.2.2 Mutually Exclusive Functions.....	231
5.3.2.3 Function Impacts.....	231
5.3.3 Hardware.....	231
5.3.4 Others.....	232
5.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	232
5.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB).....	232
5.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	232
5.7 Operation and Maintenance (Automatic ALD Deployment).....	232
6 SASU.....	233
6.1 Principles.....	233
6.1.1 Connections Between the SASU, RRU/RFU, and RET Antenna.....	233
6.1.2 Operations on the SASU.....	235
6.2 Network Analysis.....	237
6.2.1 Benefits.....	237
6.2.2 Impacts.....	237
6.3 Requirements.....	237
6.3.1 Licenses.....	237
6.3.2 Functions.....	237
6.3.2.1 Prerequisite Functions.....	237
6.3.2.2 Mutually Exclusive Functions.....	237
6.3.2.3 Function Impacts.....	237
6.3.3 Hardware.....	237
6.3.4 Others.....	238
6.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	238
6.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB).....	238
6.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	238
6.7 Operation and Maintenance (Automatic ALD Deployment).....	238
7 RAE.....	239
7.1 Principles.....	239
7.1.1 Connections Between the RAE and RRU/RFU/AAU.....	239
7.1.2 Operations on the RAE.....	246
7.2 Network Analysis.....	248
7.2.1 Benefits.....	248

7.2.2 Impacts.....	248
7.3 Requirements.....	248
7.3.1 Licenses.....	248
7.3.2 Functions.....	248
7.3.2.1 Prerequisite Functions.....	248
7.3.2.2 Mutually Exclusive Functions.....	248
7.3.2.3 Function Impacts.....	248
7.3.3 Hardware.....	248
7.3.4 Others.....	249
7.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	249
7.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB).....	249
7.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	249
7.7 Operation and Maintenance (Automatic ALD Deployment).....	249
7.8 Operation and Maintenance.....	249
7.8.1 Data Configuration.....	249
7.8.1.1 Data Preparation.....	250
7.8.1.2 Using MML Commands.....	250
7.8.1.3 Using the MAE-Deployment.....	250
7.8.2 Activation Verification.....	250
7.8.3 Network Monitoring.....	251
8 AAS.....	252
8.1 Principles.....	252
8.1.1 Application of AAS Passive Antennas.....	255
8.1.2 Operations on AAS Modules.....	258
8.2 Network Analysis.....	262
8.2.1 Benefits.....	262
8.2.2 Impacts.....	262
8.3 Requirements.....	262
8.3.1 Licenses.....	262
8.3.2 Functions.....	262
8.3.2.1 Prerequisite Functions.....	262
8.3.2.2 Mutually Exclusive Functions.....	262
8.3.2.3 Function Impacts.....	262
8.3.3 Hardware.....	262
8.3.4 Others.....	263
8.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	263
8.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB).....	263
8.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	263
8.7 Operation and Maintenance (Automatic ALD Deployment).....	263
9 SALD/MALD.....	264
9.1 Principles.....	264
9.1.1 Connection Between the SALD/MALD and the RRU/AAU.....	264

9.1.2 SALD/MALD Operations.....	266
9.2 Network Analysis.....	269
9.2.1 Benefits.....	269
9.2.2 Impacts.....	269
9.3 Requirements.....	269
9.3.1 Licenses.....	269
9.3.2 Functions.....	270
9.3.2.1 Prerequisite Functions.....	270
9.3.2.2 Mutually Exclusive Functions.....	270
9.3.2.3 Function Impacts.....	270
9.3.3 Hardware.....	270
9.3.4 Others.....	270
9.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	270
9.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB).....	271
9.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	271
9.7 Operation and Maintenance (Automatic ALD Deployment).....	271
9.8 Operation and Maintenance (MALD Subunit Access Permission Setting).....	271
10 RAS.....	273
10.1 Principles.....	273
10.1.1 Connections Between RAS Units and RRUs/RFUs.....	273
10.1.2 Operations on RAS Units.....	277
10.2 Network Analysis.....	278
10.2.1 Benefits.....	279
10.2.2 Impacts.....	279
10.3 Requirements.....	279
10.3.1 Licenses.....	279
10.3.2 Functions.....	279
10.3.2.1 Prerequisite Functions.....	279
10.3.2.2 Mutually Exclusive Functions.....	279
10.3.2.3 Function Impacts.....	279
10.3.3 Hardware.....	279
10.3.4 Others.....	280
10.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	280
10.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB).....	280
10.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	280
10.7 Operation and Maintenance (Automatic ALD Deployment).....	280
11 RVD.....	281
11.1 Principles.....	281
11.1.1 Connections Between the RVD and RRU/RFU.....	281
11.1.2 Operations on the RVD.....	285
11.2 Network Analysis.....	286
11.2.1 Benefits.....	286

11.2.2 Impacts.....	286
11.3 Requirements.....	286
11.3.1 Licenses.....	286
11.3.2 Functions.....	287
11.3.2.1 Prerequisite Functions.....	287
11.3.2.2 Mutually Exclusive Functions.....	287
11.3.2.3 Function Impacts.....	287
11.3.3 Hardware.....	287
11.3.4 Others.....	287
11.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station).....	287
11.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)....	288
11.6 Operation and Maintenance (Manual ALD Deployment on the GBTS).....	288
11.7 Operation and Maintenance (Automatic ALD Deployment).....	288
12 Parameters.....	289
13 Counters.....	291
14 Glossary.....	292
15 Reference Documents.....	293

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 04 (2025-12-14).

Technical Changes

Change Description	Parameter Change
Added the RRU5522t. For details, see 3 General Principles .	None
Added the AAU5785t and HAAU5235w. For details, see 8.1 Principles .	None
Added support for querying the azimuth adjustable status of the RAS unit for digital antennas. For details, see 10.1.2 Operations on RAS Units .	None

Change Description	Parameter Change
Added compatibility of the DSP ALDSUBUNIT and DSP ALDDEVICEDATA commands with the old ALD model. For details, see: • 4.1.2 Operations on RET Antennas • 4.4.3.1 Data Preparation (Old Model) • 4.4.3.4 Using MML Commands (Old Model) • 8.1.2 Operations on AAS Modules • 10.1.1 Connections Between RAS Units and RRUs/RFUs • 10.1.2 Operations on RAS Units • 11.1.2 Operations on the RVD	None
Added the antenna attitude detection function for digital antennas. For details, see: • 7.1.1 Connections Between the RAE and RRU/RFU/AAU • 7.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)	Modified parameter: Added the ANT_ATT_DETECT_SW option to the Equipment.EqmOptFeatSw parameter.

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

Unless otherwise specified, in this document, LTE and eNodeB always include FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, and LTE NB-IoT are used. The same rules apply to eNodeB.

Unless otherwise specified, in this document, NR and gNodeB always include FDD and TDD. In scenarios where they need to be distinguished, NR FDD and NR TDD are used. The same rules apply to gNodeB.

The "G", "C", "U", "L", "T", "M", "N", "N (TDD)", and "N (FDD)" in RAT acronyms refer to GSM, CDMA, UMTS, LTE FDD, LTE TDD, LTE NB-IoT, NR, NR TDD, and NR FDD, respectively.

2.3 Features in This Document

This document describes antenna line device (ALD) management and provides engineering guidelines. ALD management includes the features listed in [Table 2-1](#).

Table 2-1 Features related to ALD management

RAT	Feature ID	Feature Name	Chapter/Section
GSM	MRFD-210601	Connection with TMA (Tower Mounted Amplifier)	5 TMA
	MRFD-210602	Remote Electrical Tilt	4 RET Antenna
UMTS	MRFD-210601	Connection with TMA (Tower Mounted Amplifier)	5 TMA
	MRFD-210602	Remote Electrical Tilt	4 RET Antenna
	WRFD-060003	Same Band Antenna Sharing Unit (900 MHz)	4 RET Antenna
LTE FDD	LBFD-001024	Remote Electrical Tilt Control	4 RET Antenna
LTE TDD	TDLBFD-001024	Remote Electrical Tilt Control	4 RET Antenna
LTE NB-IoT	MLBFD-12000421	Remote Electrical Tilt Control	4 RET Antenna

2.4 Differences

Differences Between NR FDD and NR TDD

Function Name	Difference
ALD management	Depends on the differences between RF modules.

Differences Between NSA and SA

Function Name	Difference
ALD management	None

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference
ALD management	Supported only in low frequency bands

3 General Principles

Antenna line device (ALD) is a generic term for antenna devices that may vary with AISG protocols.

AISG1.1 and AISG2.0 define the following ALDs by the logical functions:

- Remote electrical tilt (RET) antenna
- Tower-mounted amplifier (TMA)
- Same-band antenna sharing unit (SASU)
- Remote eAntenna Extension (RAE) unit
- Remote azimuth steering (RAS) unit
- Remote vendor-defined antenna line device (RVD)

AISG3.0 defines the following ALDs by the physical hardware devices:

- Single-primary antenna line device (SALD)
- Multi-primary antenna line device (MALD)

Subunits of the SALD and MALD perform different logical functions. AISG3.0 defines the following subunits:

- RET antenna
- TMA
- Antenna database (ADB)

A SALD/MALD has multiple subunits with different logical functions, including at least one ADB.

NOTE

This document describes ALDs by their logical functions. Unless otherwise specified, the ALDs mentioned in this document refer to the subunits with logical functions, such as the RET/TMA. If ALDs defined in AISG3.0 need to be specified, the SALD/MALD will be used.

New and old ALD models are used to differentiate ALDs complying with different AISG protocols. The old ALD model applies to ALDs complying with protocols earlier than AISG3.0. The new ALD model applies to ALDs complying with AISG3.0 and earlier protocols.

In Huawei's current RET control solution, a maximum of ten ALD serial numbers can be managed on a control port of an RF module (one ALD may have multiple

ALD serial numbers). That is, the total number of RET antennas, TMAs, SASUs, RAE units, RAS units, RVDs, SALDs, and MALDs is not greater than 10. For example, a maximum of 10 RET antennas can be cascaded.

Currently, antenna manufacturers generally use a multi-frequency and multi-array solution. One physical antenna includes multiple device serial numbers, and antenna devices corresponding to those device serial numbers are logically cascaded. To improve antenna scanning efficiency and avoid scanning timeout, it is recommended that the control ends of multi-band multi-array antennas be connected to the RF modules of their respective sectors. This prevents multiple such antennas from being managed by one RF module after being physically cascaded.

The AAS is an active antenna system (AAS). It integrates the RF unit and antenna.

Table 3-1 lists the capabilities of ALD configuration and management in GSM, UMTS, LTE FDD, LTE NB-IoT, LTE TDD, and NR. "Yes" indicates supported, and "No" indicates not supported.

Table 3-1 Capabilities to support different types of ALDs working in different RATs

RAT	RET	TMA	SASU	RAE	RAS	AAS	SALD	MALD
GSM	Yes	Yes	No	Yes	Yes	Yes	No	No
UMTS	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LTE FDD	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
LTE TDD	Yes	No	No	Yes	Yes	No	Yes	Yes
LTE NB-IoT	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
NR	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes

NOTE

The RET cannot be used when a 4T4R RRU serves as two 2T2R RRUs in LTE TDD mode. The RAE/AAS/RAS devices support only the eGBTS in GSM.

The following tables list the capabilities of RF modules to support different types of ALDs. "Yes" indicates supported, "No" indicates not supported, and "-" indicates N/A. If a port supports TMAs but does not support RET antennas, the port can provide 12 V output voltage but does not provide OOK signals.

Table 3-2 Capabilities of each port on the 2-RF-port RF modules to support RET antennas and TMAs

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by RET Port	TMA Supported by RET Port
CRFUd	Yes	Yes	No	Yes	-	-
CRFUe	Yes	Yes	No	Yes	-	-
DRFU	No	No	No	No	-	-
GRFU	Yes	Yes	No	Yes	-	-
LRFU	Yes	Yes	No	Yes	-	-
LRFUe	Yes	Yes	No	Yes	-	-
MRFU	Yes	Yes	No	Yes	-	-
MRFUc	Yes	Yes	No	Yes	-	-
MRFUd	Yes	Yes	No	Yes	-	-
MRFUd V6	Yes	Yes	No	Yes	-	-
MRFUe	Yes	Yes	No	Yes	-	-
WRFU	Yes	Yes	No	Yes	-	-
WRFUa	Yes	Yes	No	Yes	-	-
WRFUd	Yes	Yes	No	Yes	-	-
WRFUe	Yes	Yes	No	Yes	-	-
RRU3004	No	No	No	No	Yes	No
RRU3008	Yes	Yes	No	Yes	Yes	No
RRU3201	Yes	Yes	No	Yes	Yes	No
RRU3203	Yes	Yes	No	Yes	Yes	No
RRU3220	Yes	No	No	No	Yes	No
RRU3220E	No	No	No	No	-	-
RRU3230E	No	No	No	No	-	-
RRU3930E	No	No	No	No	-	-
RRU3221	Yes	Yes	No	Yes	Yes	No
RRU3222	Yes	Yes	No	Yes	Yes	No
RRU3229	Yes	Yes	No	Yes	Yes	No
RRU3249	Yes	Yes	No	Yes	Yes	No

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by RET Port	TMA Support ed by RET Port
RRU3250	Yes	No	No	No	Yes	No
RRU3251	Yes	No	No	No	Yes	No
RRU3268	Yes	Yes	No	Yes	Yes	No
RRU3606	Yes	Yes	No	Yes	Yes	No
RRU3628	Yes	Yes	No	Yes	Yes	No
RRU3772	Yes	No	No	No	Yes	No
RRU3775	Yes	No	No	No	Yes	No
RRU3801E	Yes	Yes	No	Yes	Yes	No
RRU3804	Yes	Yes	No	Yes	Yes	No
RRU3805	Yes	Yes	No	Yes	Yes	No
RRU3806	Yes	Yes	No	Yes	Yes	No
RRU3808	Yes	Yes	No	Yes	Yes	No
RRU3821E	No	No	No	No	-	-
RRU3824	Yes	Yes	No	Yes	Yes	No
RRU3826	Yes	Yes	No	Yes	Yes	No
RRU3828	Yes	Yes	No	Yes	Yes	No
RRU3829	Yes	Yes	No	Yes	Yes	No
RRU3838	Yes	Yes	No	Yes	Yes	No
RRU3839	Yes	Yes	No	Yes	Yes	No
RRU3908	Yes	Yes	No	Yes	Yes	No
RRU3926	Yes	Yes	No	Yes	Yes	No
RRU3928	Yes	Yes	No	Yes	Yes	No
RRU3929	Yes	Yes	No	Yes	Yes	No
RRU3936	Yes	Yes	No	Yes	Yes	No
RRU3938	Yes	Yes	No	Yes	Yes	No
RRU3939	Yes	Yes	No	Yes	Yes	No
RRU3959	Yes	Yes	No	Yes	Yes	No
RRU3959w	Yes	Yes	No	Yes	Yes	No

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by RET Port	TMA Support ed by RET Port
RRU3959a	Yes	Yes	No	Yes	Yes	No
RRU3961	Yes	Yes	No	Yes	Yes	No
RRU3965	Yes	Yes	No	Yes	Yes	No
RRU3236E	No	No	No	No	-	-
RRU5305	Yes	Yes	No	Yes	Yes	No
RRU5308	Yes	Yes	No	Yes	Yes	No
RRU5309	Yes	Yes	No	Yes	Yes	No
RRU5309w	Yes	Yes	No	Yes	Yes	No
RRU5505	Yes	Yes	No	Yes	Yes	No
RRU5505N	Yes	Yes	No	Yes	Yes	No
RRU5515	Yes	Yes	No	No	Yes	No
RRU5515t	Yes	Yes	No	Yes	Yes	No
RRU5525t	Yes	Yes	No	Yes	Yes	No
RRU5905	Yes	Yes	No	Yes	Yes	No
RRU5905w	Yes	Yes	No	Yes	Yes	No
RRU5909	Yes	Yes	No	Yes	Yes	No
RRU5909c	Yes	Yes	No	Yes	Yes	No
RRU5909N	Yes	Yes	No	Yes	Yes	No
RRU5909s	Yes	Yes	No	Yes	Yes	No
RRU5915	Yes	Yes	No	Yes	Yes	Yes
RRU5919	Yes	Yes	No	Yes	Yes	Yes

Table 3-3 Capabilities of each port on the 4-RF-port RF modules to support RET antennas and TMAs

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by ANT_C Port	TMA Supported by ANT_C Port	RET Antenna Supported by ANT_D Port	TMA Supported by ANT_D Port	RET Antenna Supported by RET Port	TMA Supported by RET Port
MRFUdw V6	No	No	No	No	Yes	Yes	No	Yes	-	-
RRU3232	Yes	No	No	No	No	No	No	No	Yes	No
RRU3235	Yes	No	No	No	No	No	No	No	Yes	No
RRU3240	Yes	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes
RRU3252	Yes	No	No	No	No	No	No	No	-	-
RRU3256	Yes	No	No	No	No	No	No	No	Yes	No
RRU3260	Yes	Yes	No	Yes	No	Yes	No	Yes	Yes	No
RRU3262 (700 MHz)	Yes	Yes	Yes	Yes	No	No	No	No	Yes	No
RRU3262 (2600 MHz)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No
RRU3276	No	No	No	No	No	No	No	No	Yes	No
RRU3281	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU3702	Yes	No	No	No	No	No	No	No	Yes	No
RRU3832	Yes	Yes	No	No	No	No	No	No	Yes	No
RRU3841	Yes	Yes	Yes	Yes	No	No	No	No	Yes	No
RRU3853	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No
RRU3942	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No
RRU3952	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No
RRU3952m	Yes	Yes	No	Yes	No	No	No	No	Yes	Yes
RRU3953	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU3953w	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No
RRU3962	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No
RRU3971	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by ANT_C Port	TMA Supported by ANT_C Port	RET Antenna Supported by ANT_D Port	TMA Supported by ANT_D Port	RET Antenna Supported by RET Port	TMA Supported by RET Port
RRU3971a	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5251	Yes	No	No	No	No	No	No	No	Yes	No
RRU5254 (2300 MHz +2600 MHz)	Yes	No	No	No	No	No	No	No	Yes	No
RRU5265E (3500 MHz)	No	No	No	No	No	No	No	No	-	-
RRU5265E (3600 MHz)	No	No	No	No	No	No	No	No	-	-
RRU5301	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5304	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5304w	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5310	Yes	Yes	Yes	Yes	No	No	No	No	Yes	No
RRU5336E	No	No	No	No	No	No	No	No	-	-
RRU5501	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5502	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5502N	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5502w	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5507	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5508 (700 MHz +900 MHz/ 700 MHz +850 MHz/ 800 MHz +900 MHz)	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by ANT_C Port	TMA Supported by ANT_C Port	RET Antenna Supported by ANT_D Port	TMA Supported by ANT_D Port	RET Antenna Supported by RET Port	TMA Supported by RET Port
RRU5508 (1800 MHz +2100 MHz)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5509t	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes
RRU5512	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5512f	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5512t	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5513	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5513t (1400 MHz +1800 MHz +2100 MHz)	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5513t (1800 MHz +2100 MHz +2600 MHz)	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5513w	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5516 (700 MHz +900 MHz)	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5516 (1800 MHz +2100 MHz)	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5516t	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5516w	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5517t	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5517et	Yes	Yes	No	No	No	No	No	No	Yes	Yes

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by ANT_C Port	TMA Supported by ANT_C Port	RET Antenna Supported by ANT_D Port	TMA Supported by ANT_D Port	RET Antenna Supported by RET Port	TMA Supported by RET Port
RRU5519et	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes
RRU5522	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5522t	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5522w	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5526 (600 MHz +850 MHz)	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5526(700M+850M)	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5526 (1800 MHz +2100 MHz)	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5526et	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5526t	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5526w	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5527et	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5527t	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5529et	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes
RRU5836E	No	No	No	No	No	No	No	No	-	-
RRU5901 (1800 MHz/ AWS)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5901N (1800 MHz)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5901 (2100 MHz)	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5901w	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes

RF Module	RET Antenna Supported by ANT_A Port	TMA Supported by ANT_A Port	RET Antenna Supported by ANT_B Port	TMA Supported by ANT_B Port	RET Antenna Supported by ANT_C Port	TMA Supported by ANT_C Port	RET Antenna Supported by ANT_D Port	TMA Supported by ANT_D Port	RET Antenna Supported by RET Port	TMA Supported by RET Port
RRU5903	Yes	Yes	No	Yes	No	No	No	No	Yes	Yes
RRU5903w	Yes	Yes	No	No	No	No	No	No	Yes	Yes
RRU5904	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5904N	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5904w	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5910 (1800 MHz)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5910 (800 MHz)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5910 (900 MHz)	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5913	Yes	Yes	No	Yes	No	No	No	No	Yes	Yes
RRU5916	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5916s	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5920 (2100 MHz)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
RRU5920w	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes
RRU5935E	No	No	No	No	No	No	No	No	-	-

Table 3-4 Capabilities of each port on the 8-RF-port RF modules to support RET antennas

RF Module	ANT_1 Port	ANT_2 Port	ANT_3 Port	ANT_4 Port	ANT_5 Port	ANT_6 Port	ANT_7 Port	ANT_8 Port	CAL Port	RET Port
RRU3168	No	No	No	No	No	No	No	No	Yes	No
RRU3253	No	No	No	No	No	No	No	No	Yes	No
RRU3259	No	No	No	No	No	No	No	No	Yes	Yes

RF Module	ANT_1 Port	ANT_2 Port	ANT_3 Port	ANT_4 Port	ANT_5 Port	ANT_6 Port	ANT_7 Port	ANT_8 Port	CAL Port	RET Port
RRU3278	No	No	No	No	No	No	No	No	Yes	Yes
RRU3278u	No	No	No	No	No	No	No	No	Yes	Yes
RRU3279	No	No	No	No	No	No	No	No	Yes	Yes
RRU5254 (2600 MHz)	Yes	No	No	No	No	No	No	No	-	Yes
RRU5258 (with a CAL port)	No	No	No	No	No	No	No	No	Yes	Yes
RRU5258 (without any CAL port)	Yes	No	No	No	No	No	No	No	-	Yes
RRU5810 (external calibration)	No	No	No	No	No	No	No	No	Yes	Yes
RRU5810 (internal calibration)	No	No	No	No	No	No	No	No	No	Yes
RRU5815 (external calibration)	No	No	No	No	No	No	No	No	Yes	Yes
RRU5815 (internal calibration)	No	No	No	No	No	No	No	No	No	Yes
RRU5818 (external calibration)	No	No	No	No	No	No	No	No	Yes	Yes
RRU5818 (internal calibration)	No	No	No	No	No	No	No	No	No	Yes
RRU5818w (external calibration)	No	No	No	No	No	No	No	No	Yes	Yes
RRU5818w (internal calibration)	No	No	No	No	No	No	No	No	No	Yes
RRU5827	No	No	No	No	No	No	No	No	Yes	Yes

RF Module	ANT_1 Port	ANT_2 Port	ANT_3 Port	ANT_4 Port	ANT_5 Port	ANT_6 Port	ANT_7 Port	ANT_8 Port	CAL Port	RET Port
RRU5829	No	No	No	No	No	No	No	No	Yes	Yes
RRU5862	No	No	No	No	No	No	No	No	Yes	Yes
RRU5865	No	No	No	No	No	No	No	No	Yes	Yes
RRU5866	No	No	No	No	No	No	No	No	Yes	Yes
RRU5867	No	No	No	No	No	No	No	No	Yes	Yes
RRU5872	No	No	No	No	No	No	No	No	Yes	Yes

Table 3-5 lists the capabilities of RF modules to support RET.

Table 3-5 Capabilities of RF modules to support RET

RF Module	AISG Protocol	RET Antenna Voltage	RET Antenna Current
CRFUd	1.1/2.0	12 V	2.3 A
CRFUe	1.1/2.0	12 V	2.3 A
GRFU	1.1/2.0	12 V	2.3 A
LRFU	1.1/2.0	12 V	2.3 A
LRFUe	1.1/2.0	12 V	2.3 A
MRFU	1.1/2.0	12 V	2.3 A
MRFUc	1.1/2.0	12 V	2.3 A
MRFUd	1.1/2.0	12 V	2.3 A
MRFUd V6	1.1/2.0	12 V	2.3 A
MRFUdw V6	1.1/2.0	12 V	2.3 A
MRFUe	1.1/2.0	12 V	2.3 A
WRFU	1.1/2.0	12 V	2.3 A
WRFUa	1.1/2.0	12 V	2.3 A
WRFUd	1.1/2.0	12 V	2.3 A
WRFUe	1.1/2.0	12 V	2.3 A
RRU3004	1.1	12 V	2.3 A
RRU3008	1.1/2.0	12 V	2.3 A

RF Module	AISG Protocol	RET Antenna Voltage	RET Antenna Current
RRU3168	1.1/2.0	24 V	2.3 A
RRU3201	1.1/2.0	12 V	2.3 A
RRU3203	1.1/2.0	12 V	2.3 A
RRU3220	1.1/2.0	12 V	2.3 A
RRU3221	1.1/2.0	12 V	2.3 A
RRU3222	1.1/2.0	12 V	2.3 A
RRU3229	1.1/2.0	12 V	2.3 A
RRU3232	1.1/2.0	24 V/12 V	2.3 A
RRU3235	1.1/2.0	12 V	2.3 A
RRU3240	1.1/2.0	12 V	2.3 A
RRU3249	1.1/2.0	12 V	2.3 A
RRU3250	1.1/2.0	12 V	2.3 A
RRU3251	1.1/2.0	12 V	2.3 A
RRU3252	1.1/2.0	24 V/12 V	2.3 A
RRU3253	1.1/2.0	24 V/12 V	2.3 A
RRU3256	1.1/2.0	24 V/12 V	2.3 A
RRU3259	1.1/2.0	12 V	2.3 A
RRU3260	1.1/2.0	12 V	2.3 A
RRU3262	1.1/2.0	12 V	2.3 A
RRU3268	1.1/2.0	12 V	2.3 A
RRU3276	1.1/2.0/2.1	12 V	2.3 A
RRU3278	1.1/2.0	12 V	2.3 A
RRU3278u	1.1/2.0	12 V	2.3 A
RRU3279	1.1/2.0	12 V	2.3 A
RRU3281	1.1/2.0	12 V	2.3 A
RRU3606	1.1/2.0	12 V	2.3 A
RRU3702	1.1/2.0	12 V	2.3 A
RRU3801E	1.1	12 V	2.3 A
RRU3804	1.1/2.0	12 V	2.3 A

RF Module	AISG Protocol	RET Antenna Voltage	RET Antenna Current
RRU3805	1.1/2.0	12 V	2.3 A
RRU3806	1.1/2.0	12 V	2.3 A
RRU3808	1.1/2.0	12 V	2.3 A
RRU3824	1.1/2.0	12 V	2.3 A
RRU3826	1.1/2.0	12 V	2.3 A
RRU3828	1.1/2.0	12 V	2.3 A
RRU3829	1.1/2.0	12 V	2.3 A
RRU3832	1.1/2.0	12 V	2.3 A
RRU3838	1.1/2.0	12 V	2.3 A
RRU3839	1.1/2.0	12 V	2.3 A
RRU3841	1.1/2.0	12 V	2.3 A
RRU3853	1.1/2.0	12 V	2.3 A
RRU3908	1.1/2.0	12 V	2.3 A
RRU3926	1.1/2.0	12 V	2.3 A
RRU3928	1.1/2.0	12 V	2.3 A
RRU3929	1.1/2.0	12 V	2.3 A
RRU3936	1.1/2.0	12 V	2.3 A
RRU3938	1.1/2.0	12 V	2.3 A
RRU3939	1.1/2.0	12 V	2.3 A
RRU3942	1.1/2.0	12 V	2.3 A
RRU3952	1.1/2.0	12 V	2.3 A
RRU3952m	1.1/2.0	12 V	2.3 A
RRU3953	1.1/2.0	12 V	2.3 A
RRU3953w	1.1/2.0	12 V	2.3 A
RRU3959	1.1/2.0	12 V	2.3 A
RRU3959w	1.1/2.0	12 V	2.3 A
RRU3959a	1.1/2.0	12 V	2.3 A
RRU3961	1.1/2.0	12 V	2.3 A
RRU3962	1.1/2.0	12 V	2.3 A

RF Module	AISG Protocol	RET Antenna Voltage	RET Antenna Current
RRU3964	1.1/2.0	12 V	2.3 A
RRU3971	1.1/2.0	12 V	2.3 A
RRU3971a	1.1/2.0	12 V	2.3 A
RRU5254	1.1/2.0/3.0	12 V	2.3 A
RRU5258	1.1/2.0/3.0	12 V	2.3 A
RRU5265E	1.1/2.0/3.0	12 V	2.3 A
RRU5301	1.1/2.0/3.0	12 V	2.3 A
RRU5304	1.1/2.0/3.0	12 V	2.3 A
RRU5304w	1.1/2.0/3.0	12 V	2.3 A
RRU5305	1.1/2.0/3.0	12 V	2.3 A
RRU5309	1.1/2.0/3.0	12 V	2.3 A
RRU5309w	1.1/2.0/3.0	12 V	2.3 A
RRU5310	1.1/2.0/3.0	12 V	2.3 A
RRU5336E	1.1/2.0/3.0	12 V	2.3 A
RRU5501	1.1/2.0/3.0	12 V	2.3 A
RRU5502	1.1/2.0/3.0	12 V	2.3 A
RRU5502N	1.1/2.0/3.0	12 V	2.3 A
RRU5502w	1.1/2.0/3.0	12 V	2.3 A
RRU5505	1.1/2.0/3.0	12 V	2.3 A
RRU5505N	1.1/2.0/3.0	12 V	2.3 A
RRU5507	1.1/2.0/3.0	12 V	2.3 A
RRU5508	1.1/2.0/3.0	12 V	2.3 A
RRU5509t	1.1/2.0/3.0	12 V	2.3 A
RRU5512	1.1/2.0/3.0	12 V	2.3 A
RRU5512f	1.1/2.0/3.0	12 V	2.3 A
RRU5512t	1.1/2.0/3.0	12 V	2.3 A
RRU5513	1.1/2.0/3.0	12 V	2.3 A
RRU5513t	1.1/2.0/3.0	12 V	2.3 A
RRU5513w	1.1/2.0/3.0	12 V	2.3 A

RF Module	AISG Protocol	RET Antenna Voltage	RET Antenna Current
RRU5515	1.1/2.0/3.0	12 V	2.3 A
RRU5515t	1.1/2.0/3.0	12 V	2.3 A
RRU5516	1.1/2.0/3.0	12 V	2.3 A
RRU5516t	1.1/2.0/3.0	12 V	2.3 A
RRU5516w	1.1/2.0/3.0	12 V	2.3 A
RRU5517t	1.1/2.0/3.0	12 V	2.3 A
RRU5517et	1.1/2.0/3.0	12 V	2.3 A
RRU5519et	1.1/2.0/3.0	12 V	2.3 A
RRU5522	1.1/2.0/3.0	12 V	2.3 A
RRU5522t	1.1/2.0/3.0	12 V	2.3 A
RRU5522w	1.1/2.0/3.0	12 V	2.3 A
RRU5525t	1.1/2.0/3.0	12 V	2.3 A
RRU5526	1.1/2.0/3.0	12 V	2.3 A
RRU5526et	1.1/2.0/3.0	12 V	2.3 A
RRU5526t	1.1/2.0/3.0	12 V	2.3 A
RRU5526w	1.1/2.0/3.0	12 V	2.3 A
RRU5527et	1.1/2.0/3.0	12 V	2.3 A
RRU5527t	1.1/2.0/3.0	12 V	2.3 A
RRU5529et	1.1/2.0/3.0	12 V	2.3 A
RRU5810	1.1/2.0/3.0	12 V	2.3 A
RRU5815	1.1/2.0/3.0	12 V	2.3 A
RRU5818	1.1/2.0/3.0	12 V	2.3 A
RRU5818w	1.1/2.0/3.0	12 V	2.3 A
RRU5827	1.1/2.0/3.0	12 V	2.3 A
RRU5829	1.1/2.0/3.0	12 V	2.3 A
RRU5862	1.1/2.0/3.0	12 V	2.3 A
RRU5865	1.1/2.0/3.0	12 V	2.3 A
RRU5866	1.1/2.0/3.0	12 V	2.3 A
RRU5867	1.1/2.0/3.0	12 V	2.3 A

RF Module	AISG Protocol	RET Antenna Voltage	RET Antenna Current
RRU5872	1.1/2.0/3.0	12 V	2.3 A
RRU5901	1.1/2.0/3.0	12 V	2.3 A
RRU5901N	1.1/2.0/3.0	12 V	2.3 A
RRU5901w	1.1/2.0/3.0	12 V	2.3 A
RRU5903w	1.1/2.0/3.0	12 V	2.3 A
RRU5904	1.1/2.0/3.0	12 V	2.3 A
RRU5904N	1.1/2.0/3.0	12 V	2.3 A
RRU5904w	1.1/2.0/3.0	12 V	2.3 A
RRU5905	1.1/2.0/3.0	12 V	2.3 A
RRU5905w	1.1/2.0/3.0	12 V	2.3 A
RRU5909	1.1/2.0/3.0	12 V	2.3 A
RRU5909c	1.1/2.0/3.0	12 V	2.3 A
RRU5909N	1.1/2.0/3.0	12 V	2.3 A
RRU5909s	1.1/2.0/3.0	12 V	2.3 A
RRU5910	1.1/2.0/3.0	12 V	2.3 A
RRU5913	1.1/2.0/3.0	12 V	2.3 A
RRU5915	1.1/2.0/3.0	12 V	2.3 A
RRU5916	1.1/2.0/3.0	12 V	2.3 A
RRU5916s	1.1/2.0/3.0	12 V	2.3 A
RRU5919	1.1/2.0/3.0	12 V	2.3 A
RRU5920	1.1/2.0/3.0	12 V	2.3 A
RRU5920w	1.1/2.0/3.0	12 V	2.3 A

4 RET Antenna

4.1 Principles

One RET antenna consists of one remote control unit (RCU) and one or more RET subunits.

- The RCU is the control unit of an RET antenna. It receives and executes the control commands from the base station and drives the stepper motor. The stepper motor drives the phase shifter inside the antenna device, and the phase shifter adjusts the antenna tilt. Interface RS485 functions as the control interface of the RCU.
- RET subunits are antenna devices that can be separately controlled.

An RET antenna may comprise more than one RET subunits combined in a single physical entity. The RET antenna is classified into the following types:

- RET antenna containing only one RET subunit (**SINGLE_RET**)
- RET antenna containing multiple RET subunits (**MULTI_RET**), each supporting the configuration file download and downtilt setting. An RET antenna containing multiple RET subunits can be regarded as a set of multiple RET antennas installed in a radome.

The signal coverage of an RET antenna can be changed by adjusting the tilt through the RCU.

RET antennas have the following benefits:

- Remote adjustment eliminates onsite operations and is not subject to site conditions such as weather and location.
- High adjustment efficiency reduces network optimization and maintenance costs.
- Prevention of coverage distortion when the tilt is large improves signal coverage and decreases interference to neighboring cells.

RET antennas have the following drawback:

- RET antennas are more expensive, and have more complex structures.

4.1.1 Connections Between RET Antennas and RRUs/RFUs

RET antennas and RRUs/RFUs can be connected in a regular or daisy chain scenario. When splitters are used, antennas and RF modules can be connected in a sector splitting scenario. A GBTS configured with double radio frequency units (DRFUs) requires a GSM antenna and TMA control module (GATM). In this case, the GATM connects to the RET antenna.

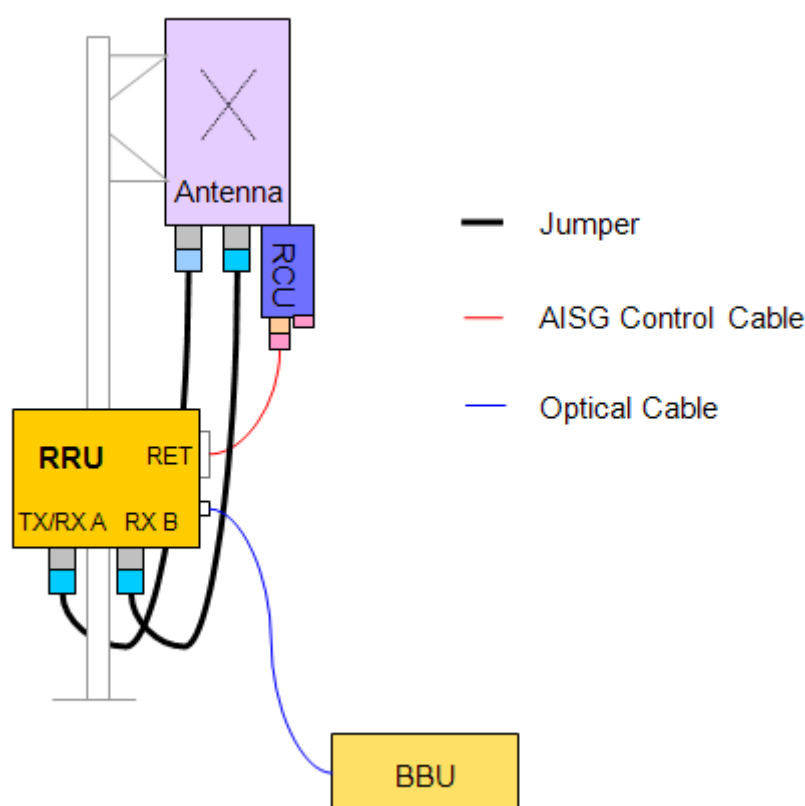
Regular Scenario

In a regular scenario, an RF module can be connected to one RET antenna by configuring the **RETPORT** or **ANTENNAPORT** MO. The RFU does not have any RET port and therefore can be connected to one RET antenna only through the antenna port.

- Connection to the RET antenna through the RET port

This scenario applies only to RRUs. An AISG multi-wire cable connects the RET port to the RCU, as shown in [Figure 4-1](#). With the integrated Bias Tee (BT), the RRU can send RS485 control signals and feed 12 V DC power to the RCU through the RET port. Therefore, no Smart Bias-Tee (SBT) is required. This connection scheme is recommended when the distance between the RRU and the RET antenna is shorter than 20 m.

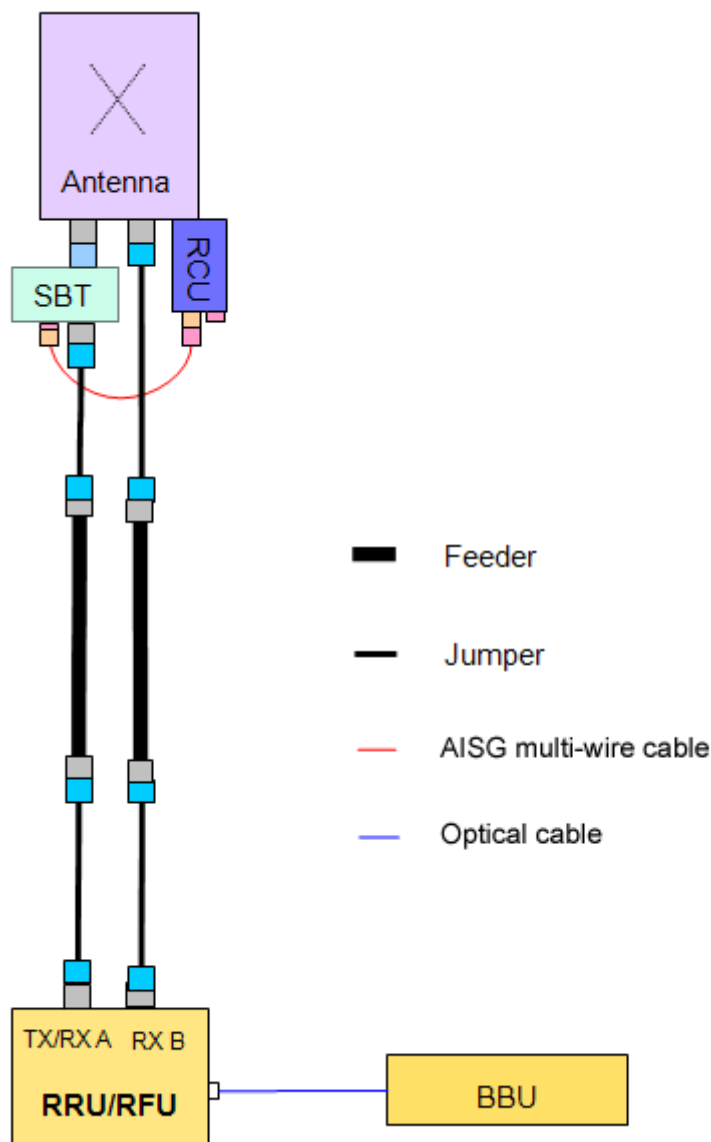
Figure 4-1 Connection to the RET antenna through the RET port



For the data configurations in this scheme, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 1 in [4.4.3.1 Data Preparation \(Old Model\)](#).
- GBTS: scenario 1 in [4.6.3.1 Data Preparation](#).
- Base station using automatic ALD deployment: scenario 1 or 4 in [4.7.2.1 Data Preparation](#).
- Connection to the RET antenna through the antenna port
An SBT is required in this connection scheme. Feeders and jumpers connect the RRU/RFU, SBT, and RET antenna, and an AISG multi-wire cable connects the SBT to the RCU, as shown in [Figure 4-2](#).
 - a. The RRU/RFU combines OOK signals, 12 V DC power, and RF signals. Then, it sends the combined signals to the SBT through feeders.
 - b. The SBT splits the combined signals into two paths. On one path, the RF signals are sent to the antenna. On the other path, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RCU through an AISG multi-wire cable.

Figure 4-2 Connection to the RET antenna through the antenna port



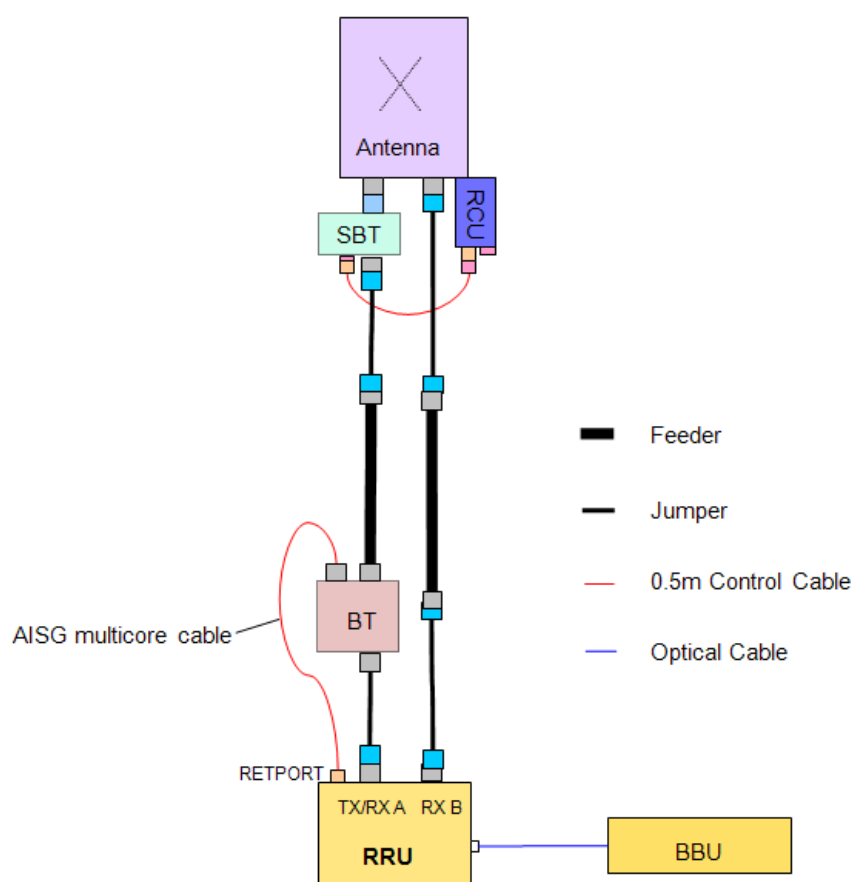
For the data configurations in this connection scheme, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 2 in [4.4.3.1 Data Preparation \(Old Model\)](#).
- GBTS: scenario 2 in [4.6.3.1 Data Preparation](#).
- Base station using automatic ALD deployment: scenario 1 or 4 in [4.7.2.1 Data Preparation](#).

The connections between the RET antenna and the RRU/RFU in multimode base stations are the same as those shown in [Figure 4-1](#) and [Figure 4-2](#). The RRU/RFU in the figures must support multiple modes. For details about data configuration, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

- Connection to the RET antenna through the RET port (with an external BT)
When an RRU without a BT such as the RRU3801C (20 W) for UMTS or RRU3220 for LTE is installed 20 m or further away from the RET antenna, RF ports on the RRU cannot provide 12 V DC power for the RET antenna. In this case, the RET port on the RRU must connect to an external BT, which connects to the RET antenna through an external SBT to provide power for the antenna. The connections between the RRU, external BT, external SBT, and RET antenna are shown in **Figure 4-3**. The BT is a passive component used on the base station side. It couples the RF signals and the OOK signals to the feeder.

Figure 4-3 Connection to the RET antenna through the RET port (with an external BT)



For the data configurations in this connection scheme, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 1 in **4.4.3.1 Data Preparation (Old Model)**.
- GBTS: scenario 1 in **4.6.3.1 Data Preparation**.
- Base station using automatic ALD deployment: scenario 1 or 4 in **4.5 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)**.

Daisy Chain Scenario

In this scenario, an upper-level RCU is connected to a lower-level RCU through an AISG multi-wire cable, and the upper-level RCU provides RS485 control signals and power for the lower-level RCU. Alternatively, the RRUs/RFUs are connected to a multi-band antenna through AISG multi-wire cables, and the multi-band antenna contains multiple RET antennas.

In [Figure 4-4](#), RET antennas are cascaded by connecting to the RRUs through the RET ports. This connection scheme applies to the scenario where antennas for multiple sectors are centrally installed, for example, on the same pole or tower.

In [Figure 4-5](#), RET antennas are cascaded by connecting to the RRUs/RFUs through the antenna ports. The connection through the antenna port requires SBTs, and cascading can reduce the number of required SBTs.

In [Figure 4-6](#), the RRUs/RFUs are connected to a multi-band antenna through the RET or antenna ports. This connection scheme applies to the scenario where the RET antennas for multiple frequency bands in a sector are integrated into one antenna.

For all connection schemes, the **SCENARIO** parameter for all RET antennas must be set to **DAISY_CHAIN**, and the RET antennas are identified by serial numbers.

- The scenario shown in [Figure 4-4](#) for an eGBTS/NodeB/eNodeB/gNodeB corresponds to scenario 1 in [4.4.3.1 Data Preparation \(Old Model\)](#). The connection for a GBTS corresponds to scenario 1 in [4.6.3.1 Data Preparation](#).
- The scenario shown in [Figure 4-5](#) for an eGBTS/NodeB/eNodeB/gNodeB corresponds to scenario 2 in [4.4.3.1 Data Preparation \(Old Model\)](#). The connection for a GBTS corresponds to scenario 2 in [4.6.3.1 Data Preparation](#).
- The scenario shown in [Figure 4-6](#) for an eGBTS/NodeB/eNodeB/gNodeB corresponds to scenario 2 in [4.4.3.1 Data Preparation \(Old Model\)](#). The connection for a GBTS corresponds to scenario 2 in [4.6.3.1 Data Preparation](#).
- If automatic deployment is applied, the connection corresponds to scenario 2 in [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

Figure 4-4 Connection to the RET antenna through the RET port

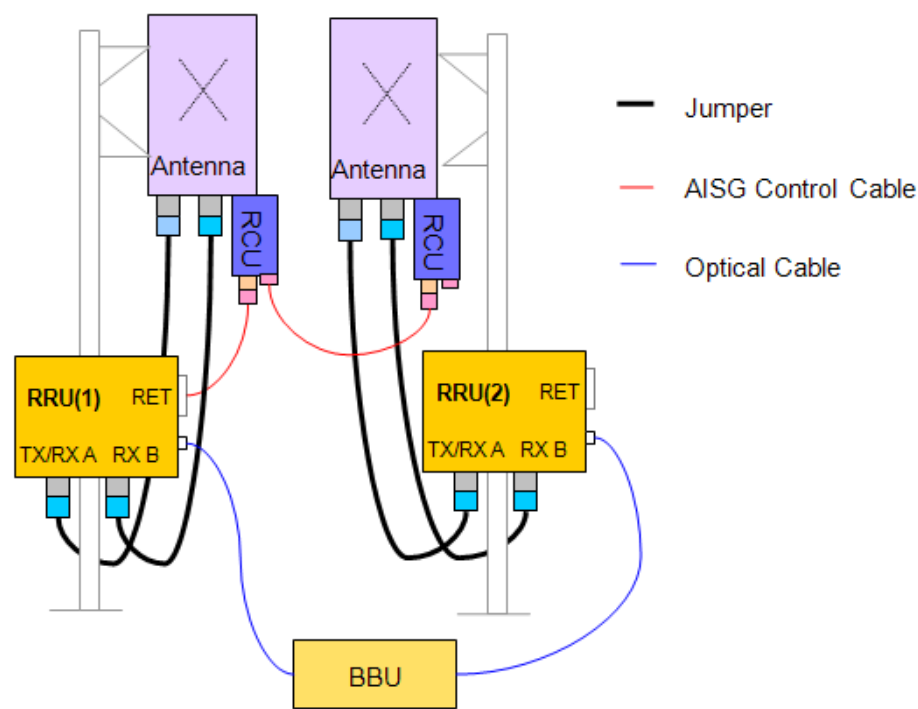


Figure 4-5 Connection to the RET antenna through the antenna port

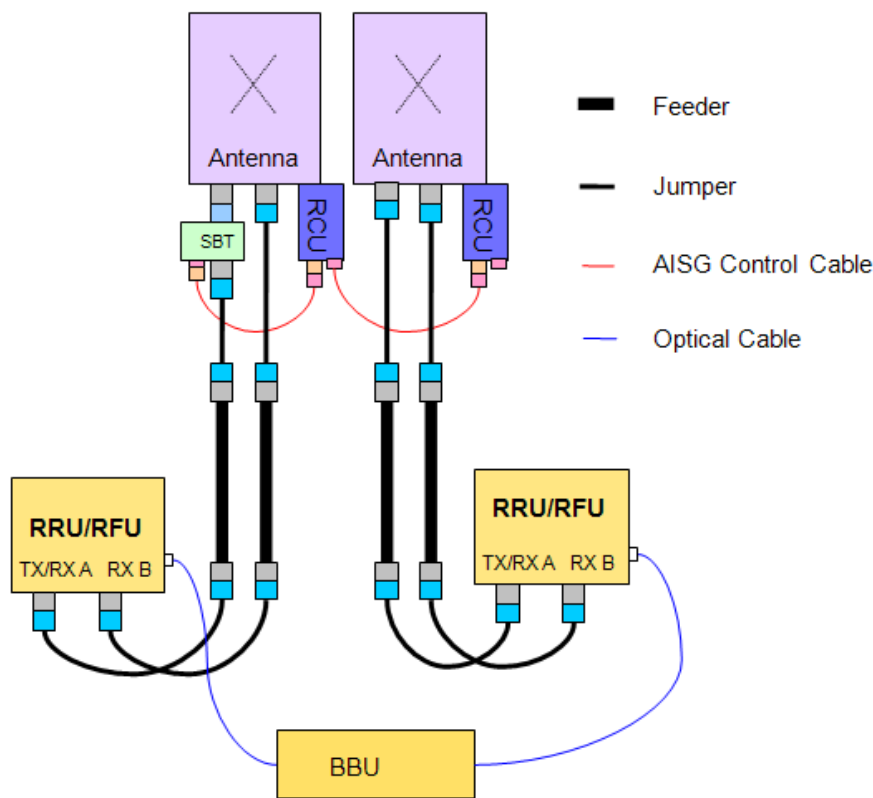
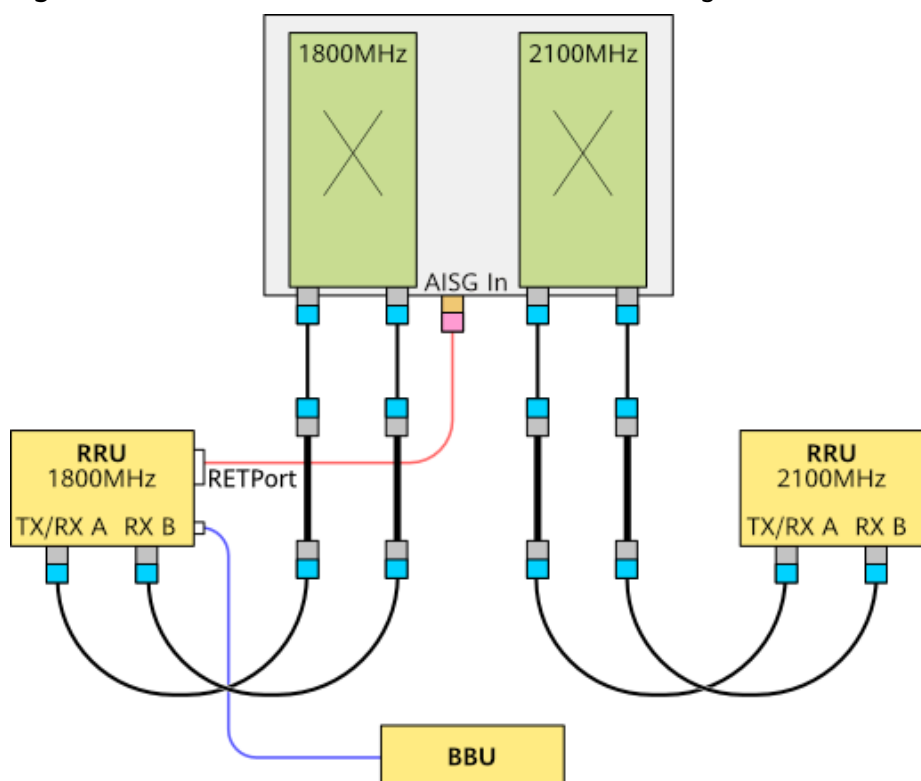


Figure 4-6 Connection to a multi-band antenna through the RET or antenna port

The connections between the RET antenna and the RRU/RFU in a multimode base station with separate antennas for each involved RAT are the same as those shown in [Figure 4-4](#) and [Figure 4-5](#). The RRU/RFU in the figures provides services for different NEs.

The RET antenna data of a multimode base station must be configured in a single-sided manner. For details about data configuration, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#). In [Figure 4-4](#),

- If RRU(1) serves a GBTS/eGBTS or eNodeB/gNodeB (the RET data is configured on the GBTS/eGBTS or eNodeB/gNodeB), the **SCENARIO** parameters must be set to **DAISY_CHAIN** for RET antennas connected to RRU(1) and RRU(2).
- If RRU(1) serves a NodeB (the RET data is configured on the NodeB), the **SCENARIO** parameter must be set to **DAISY_CHAIN** for RET antennas connected to RRU(1) and to **DAISY_CHAIN** or **2G_EXTENSION** for RET antennas connected to RRU(2).

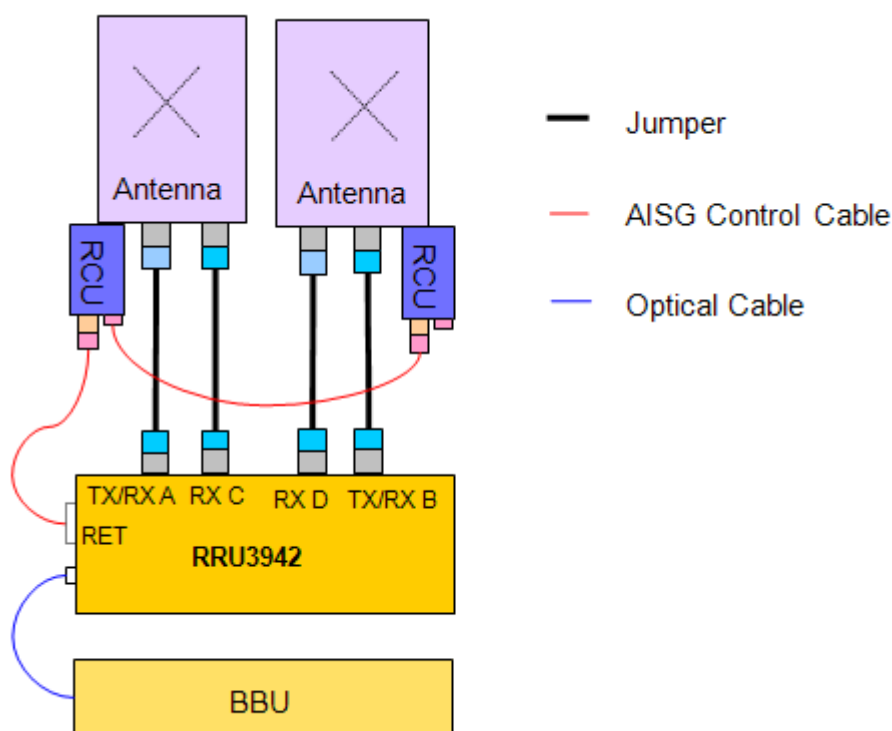
Some RF modules, such as the RRU3942 or RRU3841, have four RF ports. These RF modules have two or more RF ports that support RET functions besides the RET port. For details, see [Table 3-3](#). Two RF ports on these RF modules can be paired and connected to one antenna to provide various functions. For details about the RRU3942 and RRU3841, see *RRU3942 Hardware Description* and *RRU3841 Hardware Description* in *3900 & 5900 Series Base Station Product Documentation*. For details about other RRUs, see *RRU Hardware Description* of the corresponding RRU product in *3900 & 5900 Series Base Station Product Documentation*.

An RRU3278 has eight RF ports and can function as two 4-channel RRUs. After the split, the RRU can be connected to the RCU of the antenna only through the RET port and the RET antennas must be cascaded because the RRU3278 only has one RET port. For the connection to the RET antenna through the RET port, see [Figure 4-7](#). For details about the connections between the RF ports of the RRU3278 and the antennas, see *RRU3278 Hardware Description in 3900 & 5900 Series Base Station Product Documentation*.

The following description assumes that the RRU3942 uses a 2T4R configuration. "T" and "R" indicate transmission and reception, respectively. RF ports ANT_A and ANT_C on the RRU3942 are paired and connected to one antenna, and RF ports ANT_B and ANT_D are paired and connected to the other antenna.

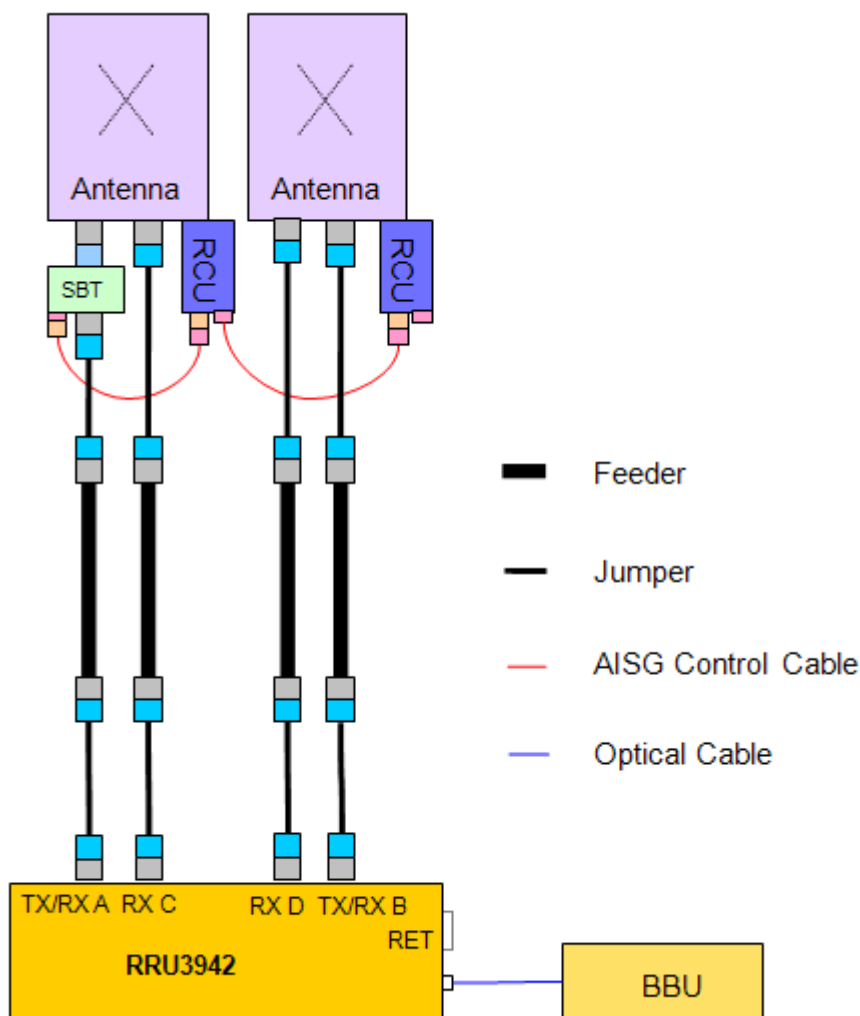
When the RRU3942 is installed close to RET antennas, the RRU3942 is connected to the RET antennas through the RET port. In this case, the RET antennas must be cascaded because the RRU3942 has only one RET port, as shown in [Figure 4-7](#).

Figure 4-7 RRU3942 connected to the RET antennas through the RET port



[Figure 4-8](#) shows how the RRU3942 and RET antennas are connected to achieve 2T4R when the RRU3942 modules are centrally installed far from the RET antennas.

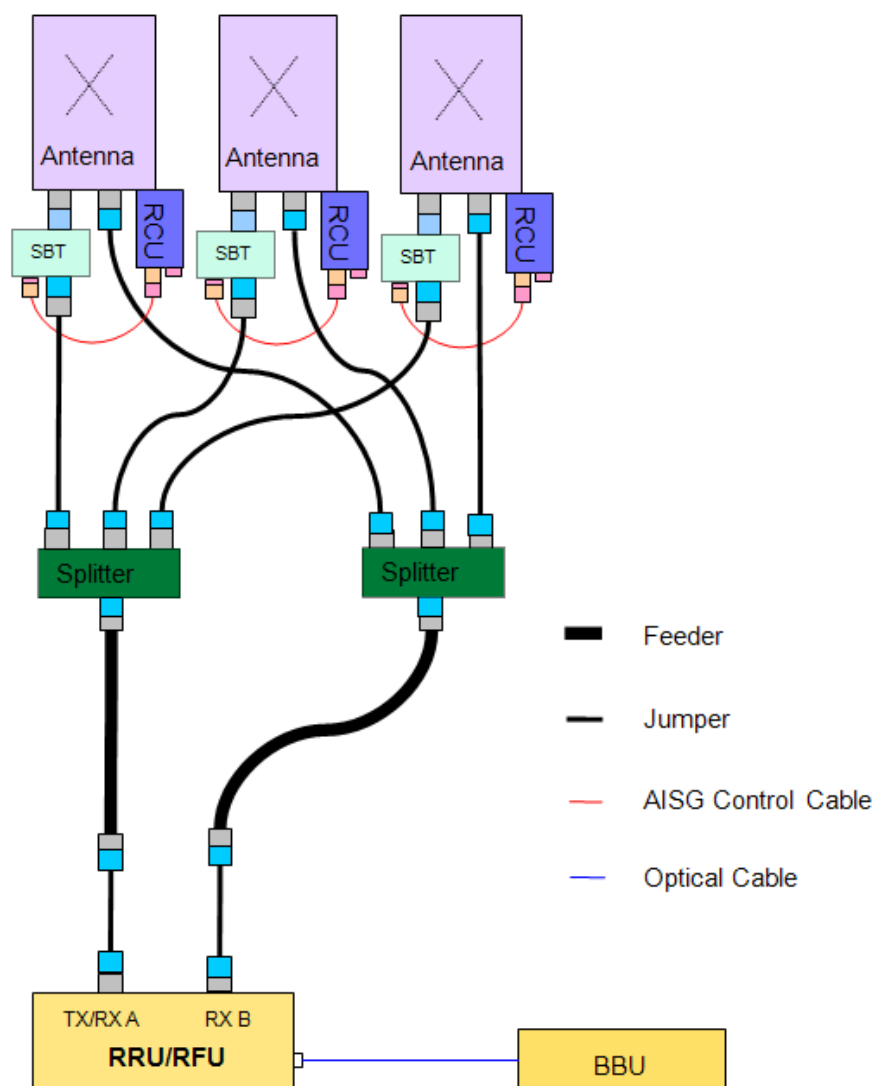
Figure 4-8 RRU3942 connected to the RET antennas through the antenna ports



In this connection scheme, RF port ANT_A is the control port for the two RET antennas connected to the RRU3942, as shown in [Figure 4-8](#). For the data configuration of this connection scheme for eGBTS/NodeB/eNodeB, refer to scenario 2 in [4.4.3.1 Data Preparation \(Old Model\)](#). For the data configuration of this connection scheme for GBTS, refer to scenario 2 in [4.6.3.1 Data Preparation](#).

Sector Splitting Scenario

In the sector splitting scenario, the RRU/RFU is connected to splitters and then RET antennas, as shown in [Figure 4-9](#).

Figure 4-9 Sector splitting scenario

In this scenario, the **SCENARIO** parameter for all RET antennas must be set to **SECTOR_SPLITTING**, and the RET antennas are identified by serial numbers. For the data configuration of this connection scheme for an eGBTS/NodeB/eNodeB/gNodeB, see scenario 2 in [4.4.3.1 Data Preparation \(Old Model\)](#). This connection scheme does not apply to the GBTS.

GATM Scenario

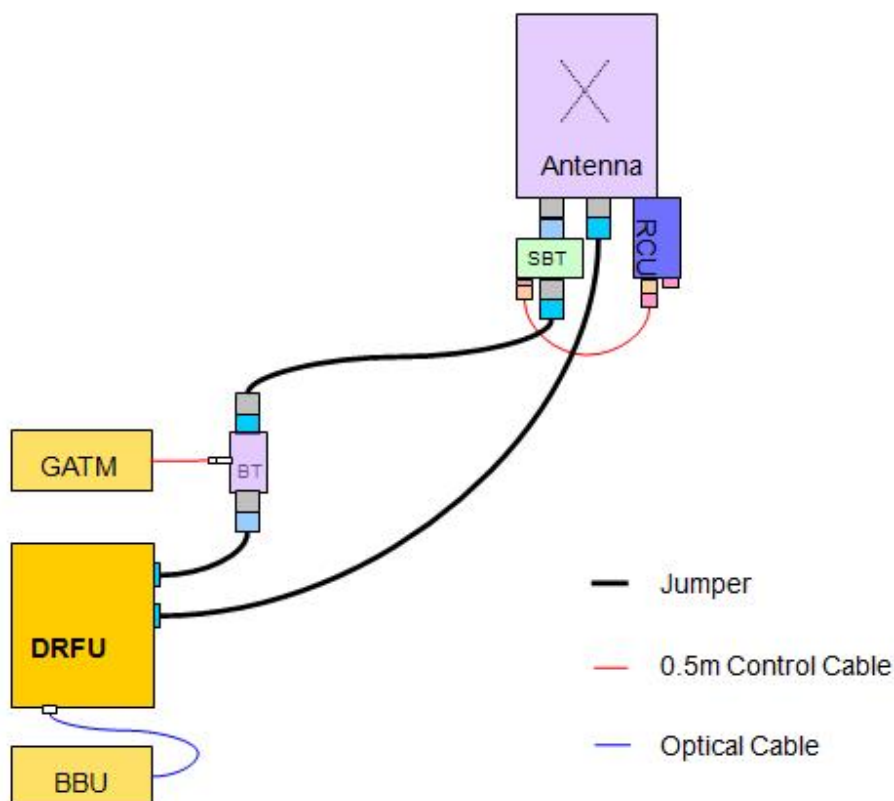
A GBTS configured with DRFUs requires a GATM. In this case, the GATM connects to the RET antenna. The DRFU, which cannot provide OOK signals or support BT function, can be connected to an RET antenna through a GATM, BT, and SBT, as shown in [Figure 4-10](#). In this scenario:

1. The GATM provides OOK signals and 12 V DC power for the BT.
2. The BT combines OOK signals, 12 V DC power, and RF signals sent by the DRFU and sends the combined signals to the SBT through feeders.

3. The SBT splits the combined signals into two paths. On one path, the RF signals are sent to the RET antenna. On the other path, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RCU through an AISG multi-wire cable.

Because of hardware limitations, the GATM supports only RET antennas compatible with AISG1.1 protocols and does not support RET antennas compatible with AISG2.0 protocols.

Figure 4-10 GATM scenario



For the data configuration for this connection scheme, see scenario 5 in [4.6.3.1 Data Preparation](#).

4.1.2 Operations on RET Antennas

The base station allows users to perform operations on each RET subunit separately, including configuration file loading, antenna calibration, and tilt setting.

Configuration File Loading

A configuration file, which is provided by the RET antenna manufacturer, describes the relationship between the RCU and the RET subunit tilt. Some RET antennas have been loaded with default configuration files before delivery. For RET antennas without default configuration files, run the following commands to load the configuration files:

- For a GBTS, run the **DLD BTSALDFILE** command to download the RET configuration files from the file server to the operation and maintenance unit (OMU) of the base station controller (BSC). Then run the **LOD BTSRETCFGDATA** command to load RET configuration files.
- For an eGBTS/NodeB/eNodeB/gNodeB, if the old ALD model is used, run the **DLD RETCFGDATA** command to load RET configuration files; if the new ALD model is used, run the **DLD ALDCFGDATA** command to load RET configuration files.

If an incorrect configuration file is loaded, the RET antenna may experience unexpected errors. To query the name of the latest loaded configuration file and the time when the configuration file was loaded, you can run the **DSP BTSRETSUBUNIT** (GBTS)/**DSP RETSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) (recommended)/**DSP ALDSUBUNIT** command if the old ALD model is used or **DSP ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command if the new ALD model is used. During the implementation, you are advised to select one or two base stations, and check whether the actual RET downtilts are the same as the configured tilts onsite to determine whether the loaded RET configuration file is correct. Ensure that a correct configuration file is loaded to the RET subunit before the calibration.

NOTE

- Run the **DSP BTSRETDEVICEDATA** (GBTS)/**DSP RETDEVICEDATA** (eGBTS/NodeB/eNodeB/gNodeB) or **DSP ALDDEVICEDATA** (eGBTS/NodeB/eNodeB/gNodeB) command to query dynamic information about RET additional data. If any information is incorrect in the command output, for example, if the values of **Max tilt** and **Min tilt** are **NULL**, no configuration file is loaded or the configuration file is lost.
- If ALM-26754 RET Antenna Data Loss is not reported, the RET device has been loaded with a configuration file. File reloading is not required unless the RCU is changed.
- The antennas with built-in RCUs have loaded configuration files before delivery, and therefore reloading is not required onsite. The antennas with external RCUs may need reloading configuration files according to the onsite conditions.

Antenna Calibration

After an RET antenna is installed, run the **CLB BTSRET** (GBTS)/**CLB RET** (eGBTS/NodeB/eNodeB/gNodeB) or **CLB ALD** (eGBTS/NodeB/eNodeB/gNodeB) command to calibrate the RET antenna. During the calibration, RCUs adjust the RET antennas within the tilt range until the RET antennas operate properly. If the calibration fails, ALM-26753 RET Antenna Not Calibrated is generated.

After the preceding calibration command is executed:

- If the RET antenna tilt has been configured on the base station, it will be restored to the configured value.
- If the RET antenna tilt has not been configured on the base station, it will be determined by the actual RET antenna implementation.

NOTE

- An RET antenna does not need to be calibrated again after it is reset or powered off.
- Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Tilt Setting

After the RET antenna is calibrated, run the **MOD BTSRETSUBUNIT** (GBTS)/**MOD RETSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) or **MOD ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command to set an antenna subunit tilt.

Before the configuration, run the **DSP BTSRETDEVICEDATA** (GBTS)/**DSP RETDEVICEDATA** (eGBTS/NodeB/eNodeB/gNodeB) or **DSP ALDDEVICEDATA** (eGBTS/NodeB/eNodeB/gNodeB) command to query the tilt range supported by the RET antenna. Setting the tilt of an RET subunit affects the coverage of the RET antenna. Therefore, set the tilt according to the engineering design.

NOTE

The tilt range of an RET antenna varies with the manufacturer and model.

The base station allows users to perform software downloading and RCU resetting on each RCU separately.

RCU Software Downloading

- Traditional RCU:

The RET manufacturers provide RCU software. For details, see the documents provided by the manufacturers.

For a GBTS, run the **DLD BTSALDFILE** command to download the RET software from the file server to the BSC OMU. Then, run the **LOD BTSALDSW** command to download the RCU software.

Perform the following operations for an eGBTS/NodeB/eNodeB/gNodeB to download the RCU software.

- a. Run the **DLD ALDSW** command and set **Device Type** to **RET**.
- b. Run the **DLD ALDFILE** command. Set **Device Type** to **RET** and **File Type** to **FIRMWARE**.

NOTE

After the software is loaded, the ALD automatically resets. The new software takes effect after the reset of the ALD. After the ALD is reset, run the **DSP ALDVER** command to query the software version of the target ALD to check whether the ALD has been successfully upgraded.

- Inner control RCU:

- a. Run the **DSP CHIPVER** command to check whether the base station is configured with the inner control RCU. If an AAU exists and **Chip Version** contains **RCU**, the base station is configured with the inner control RCU.
- b. Run the **ACT SOFTWARE: OT=BOARDTYPE, BT=RCU;** command to download and activate the software. After the command is executed successfully, the version of the RCU is consistent with that of the software package in use for the base station.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*.

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

RCU Resetting

Run the **RST BTSALD** (GBTS) or **RST ALD** (eGBTS/NodeB/eNodeB/gNodeB) command to reset the RCU. Resetting the RCU does not change the RET tilt.

4.2 Network Analysis

4.2.1 Benefits

None

4.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

No license is required for GSM, UMTS, LTE, and NR.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

For details, see [3 General Principles](#).

4.3.4 Networking

N/A

4.4 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

4.4.1 When to Use

It is recommended that this feature be enabled when ALDs complying with AISG protocols have been installed. The AISG protocol has three commonly used versions, AISG1.1, AISG2.0, and AISG3.0. AISG1.1 and AISG2.0 are supported in SRAN8.0 and later. AISG3.0 is supported as of SRAN17.0.

4.4.2 Precautions

This section describes precautions for configuring and operating ALDs, running a command for scanning ALDs, and setting the current alarm threshold type for ALD data.

NOTE

- The RET specifications vary according to the main control board type. A maximum of 512 RET objects can be configured. For details, see the MML command help.
- The RAE specifications vary according to the main control board type. A maximum of 72 RAE objects can be configured. For details, see the MML command help.
- The SALD/MALD specifications vary according to the main control board type. A maximum of 72 SALD/MALD objects can be configured. For details, see the MML command help.
- The RAS/RVD/TMA specifications vary according to the main control board type. A maximum of 128 RAS/RVD/TMA objects can be configured. For details, see the MML command help.
- Pay attention to the following restrictions when configuring ALD data:
 - The power switches in the **RETPORT** and **ANTENNAPORT** MOs for an RRU cannot be turned on simultaneously.
 - ALD scanning, calibration, tilt setting, software download, and configuration file download cannot be performed simultaneously on ALDs.
 - The common TMA does not support the AISG protocol. To configure a common TMA for an eGBTS/NodeB/eNodeB, run the **MOD ANTENNAPORT** command to turn on the power switch and set current alarm thresholds. (For data preparation details, see [Table 4-9](#).) Then, run the **MOD RXBRANCH** command to configure the RX channel attenuation based on the engineering design. (For data preparation details, see [Table 4-13](#).)
 - The AISG1.1-based dual TMA (DTMA) consists of two internal TMAs and performs the same functions as the two subunits of an AISG2.0-based TMA. The AISG1.1-based DTMA can be configured as two devices, each with one subunit or as one device comprising two subunits. For the configuration as two devices, batch loading of the TMA software may fail on one of the devices. For the configuration as one device comprising two subunits, the serial number cannot be configured for the TMA. Otherwise, only one subunit is operational. When an RRU with four ports is connected to multiple TMAs, the DTMA must be configured as two devices, each with one subunit, if the AISG1.1-based DTMA is used.
 - ALDs are automatically scanned when an RET antenna, TMA, RAS unit, or RAE is added.
 - ALDs are automatically scanned after an RRU/RFU is reset.
- Pay attention to the following restrictions when ALDs are scanned:
 - ALDs are scanned based on control link connections. The scanned result shows the ALDs physically connected to the base station regardless of whether the ALDs are configured or not.
 - ALDs can be detected only when the ALD control links are connected properly.

- After subunits are added to an AISG1.1-based TMA, all TMA subunits start to work only after you run the **SCN ALD** command.
 - If an AAS module with passive antennas is used and multiple RF modules share the AAS, there is a possibility that the RET can be found for only one RF module when you run the **SCN ALD** command to scan RET antennas for these RF modules. In this scenario, you can run this command separately to scan the RF module of which the RET antenna cannot be found.
- **Table 4-1** and **Table 4-2** list the current alarm threshold types for the control port and the reference current alarm thresholds.

Table 4-1 Reference current alarm thresholds for different current alarm threshold types (antenna port)

Reference Value	Description	Undercurrent Alarm Occur Threshold (mA)	Undercurrent Alarm Clear Threshold (mA)	Overcurrent Alarm Occur Threshold (mA)	Overcurrent Alarm Clear Threshold (mA)
TMA12DB_ONLY_NON_AISG	For 12 dB TMA only	30	40	170	150
TMA24DB_ONLY_NON_AISG	For 24 dB TMA only	40	60	310	280
RET_ONLY_COAXIAL	For RET antenna only (coaxial cable)	25	33	150	120
TMA12DB_AISG	For 12 dB TMA +RET antenna or 12 dB TMA only (AISG)	30	40	450	400
TMA24DB_AISG	For 24 dB TMA +RET antenna or 24 dB TMA only (AISG)	40	60	850	750
TMA12DB_AISU	For 12 dB TMA and AISU	30	40	450	500
TMA24DB_AISU	For 24 dB TMA and AISU	40	60	580	650
AISU_ONLY	For AISU only	40	15	300	350
UER_SELF_DEFINE	User-defined	For details, see the description below.			

Table 4-2 Reference current alarm thresholds for different current alarm threshold types (RET port)

Reference Value	Description	Undercurrent Alarm Occur Threshold (mA)	Undercurrent Alarm Clear Threshold (mA)	Overcurrent Alarm Occur Threshold (mA)	Overcurrent Alarm Clear Threshold (mA)
RET_ONLY_MULTICORE	For RET antenna only (multi-wire cable)	10	15	150	120
RET_AISU	For RET antenna and AISU	20	25	550	510
AISU_ONLY	For AISU only	10	15	350	300
RGPS_AISU	RGPS and AISU	10	15	350	200
RGPS_ONLY	RGPS only	0	10	300	200
UER_SELF_DEFINE	User-defined	For details, see the description below.			

For an eGBTS/NodeB/eNodeB/gNodeB, set the **THRESHOLDTYPE** parameter to **UER_SELF_DEFINE** in any of the following scenarios:

- For the AAU5240/AAU5243/AAU5246/AAU5940/AAU5942, set the **THRESHOLDTYPE** parameter to **UER_SELF_DEFINE**, and set the **Undercurrent Alarm Occur Threshold**, **Undercurrent Alarm Clear Threshold**, **Overcurrent Alarm Occur Threshold**, and **Overcurrent Alarm Clear Threshold** to 10 mA, 15 mA, 150 mA, and 120 mA, respectively.
- RET antennas are connected in a non-regular scenario.
- A smart TMA is used as a common TMA.
- The ALD model in use is not recommended by Huawei.
- SASU and RAS devices are used.

When the **THRESHOLDTYPE** parameter is set to **UER_SELF_DEFINE**, set current alarm thresholds for an eGBTS/NodeB/eNodeB/gNodeB based on the actual ALD type. Pay attention to the following restrictions:

- User-defined current alarm thresholds must meet the requirements:
Under Current Occur Threshold < Under Current Clear Threshold < Over Current Clear Threshold < Over Current Occur Threshold.
- Generally, set the Under Current Occur Threshold to 20% to 30% of the device rated operating current, and set the Under Current Clear Threshold to about 20 mA greater than the Under Current Occur Threshold. Set the Over Current Occur Threshold to 150% to 200% of the device rated operating current, and set the Over Current Clear Threshold to about 50 mA less than the Over Current Occur Threshold.
- If RET antennas are connected in a non-regular scenario, set the Under Current Occur Threshold to 20% to 30% of the total rated current of all ALDs controlled by the RRU. Set the Over Current Occur Threshold to

150% to 200% of the total rated current of all ALDs controlled by the RRU.

- If the ALD model in use is not recommended by Huawei, set the Under Current Occur Threshold to 20% to 30% of the total rated current of all ALDs controlled by the RRU. Set the Over Current Occur Threshold to 150% to 200% of the total rated current of all ALDs controlled by the RRU.

When an AAS module with passive antennas is used and the RET function is controlled by the RRU/RFU connecting to the AAS module, set these parameters to the values corresponding to **RET_ONLY_MULTICORE** if the control port is a RET port or to the values corresponding to **RET_ONLY_COAXIAL** if the control port is an antenna port.

4.4.3 Data Configuration

4.4.3.1 Data Preparation (Old Model)

When the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**, the old model is used for data preparation.

NOTE

The new and old ALD models in this document are selected based on the value of the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter. The old ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**. The new ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**.

Scenario 1: Connection to the RET Antenna Through the RET Port

The following operations also apply to the scenario in which the RET function on an AAS module is controlled by the RRU through the RET port. In this scenario, the AAS module functions as the conventional RET antennas. Therefore, the configuration can be performed in the same way as that for conventional RET antennas.

Table 4-3 describes the key parameters that must be set in the **RETPORT** MO.

Table 4-3 Key parameters that must be set in the **RETPORT** MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RETPORT.CN (LTE eNodeB, 5G gNodeB)	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU where the control port is located and the control port number. Set these parameters based on connections between the RET antenna and the RRU. Only one port on the RRU can be used as the control port for the RET antenna. In a daisy chain scenario, multiple RCUs share one control port.
Subrack No.	RETPORT.SRN (LTE eNodeB, 5G gNodeB)	
Slot No.	RETPORT.SN (LTE eNodeB, 5G gNodeB)	

Parameter Name	Parameter ID	Setting Notes
Port No.	RETPORT.PN (<i>LTE eNodeB, 5G gNodeB</i>)	
ALD Power Switch	RETPORT.PWRSWITCH (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to ON when an RET antenna is used. The default value is OFF .
Current Alarm Threshold Type	RETPORT.THRESHOLDTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	If the RCU is connected through the RET port in a regular scenario, set this parameter to RET_ONLY_MULTICORE . In other scenarios, set this parameter to UER_SELF_DEFINE .
Undercurrent Alarm Occur Threshold	RETPORT.UOTHD (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Undercurrent Alarm Clear Threshold	RETPORT.UCTHD (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Occur Threshold	RETPORT.OOTHD (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Clear Threshold	RETPORT.OCTHD (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

If the current alarm threshold included in the current alarm threshold type does not match the actual ALD power consumption, the **DEFCURRALMTHLD** MO can be used to modify the current alarm threshold included in this type. The following table describes the key parameters involved.

Table 4-4 Key parameters in the default current alarm threshold MO **DEFCURRALMTHLD**

Parameter Name	Parameter ID	Setting Notes
Port Type	DEFCURRALMTHLD.P ORTTYPE	Set this parameter to RETPORT .
Threshold Type	DEFCURRALMTHLD.T HRESHOLDTYPE	This parameter specifies the to-be-adjusted type associated with a set of default current alarm thresholds.

Parameter Name	Parameter ID	Setting Notes
Adjusted Undercurrent Alarm Occur Threshold	DEFCURRALMTHLD.U OTH	This parameter specifies the adjusted undercurrent alarm occur threshold. For the specified type, the default undercurrent alarm occur threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Undercurrent Alarm Clear Threshold	DEFCURRALMTHLD.U CTH	This parameter specifies the adjusted undercurrent alarm clear threshold. For the specified type, the default undercurrent alarm clear threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Occur Threshold	DEFCURRALMTHLD. OOTH	This parameter specifies the adjusted overcurrent alarm occur threshold. For the specified type, the default overcurrent alarm occur threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Clear Threshold	DEFCURRALMTHLD. OCTH	This parameter specifies the adjusted overcurrent alarm clear threshold. For the specified type, the default overcurrent alarm clear threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.

Table 4-5 describes the key parameters that must be set in the **RET** MO.

Table 4-5 Key parameters that must be set in the **RET** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RET.DEVICENO (LTE eNodeB, 5G gNodeB)	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.

Parameter Name	Parameter ID	Setting Notes
Device Name	RET.DEVICENAME (LTE eNodeB, 5G gNodeB)	This parameter identifies an RET antenna. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	RET.CTRLN (LTE eNodeB, 5G gNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located. Set these parameters based on the control relationship between the RET antenna and the RRU/RFU.
Control Port Subrack No.	RET.CTRLN (LTE eNodeB, 5G gNodeB)	
Control Port Slot No.	RET.CTRLN (LTE eNodeB, 5G gNodeB)	
RET Type	RET.RETTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to SINGLE_RET for the RET antenna with a single RET subunit. Set this parameter to MULTI_RET for the RET antenna with multiple RET subunits. Set this parameter to SINGLE_RET for the AAS passive antennas.
Number of RET Subunits	RET.SUBUNITNUM (LTE eNodeB, 5G gNodeB)	Set this parameter only when the RETTYPE parameter is set to MULTI_RET .
Polar Type	RET.POLARTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter based on the RET antenna specifications. Set this parameter based on the AAS specifications only when the AAS module with passive antennas is used.
Antenna Scenario	RET.SCENARIO (LTE eNodeB, 5G gNodeB)	Antenna scenario ¹
Vendor Code	RET.VENDORCODE (LTE eNodeB, 5G gNodeB)	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the manufacturer information, for example, KA for a Kathrein RET antenna, AN for an Andrew RET antenna, or HW for a Huawei Agissson RET antenna. Set this parameter to HW for an AAS module with passive antennas.

Parameter Name	Parameter ID	Setting Notes
Serial No.	RET.SERIALNO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number. If an AAS module with passive antennas is used, run the SCN ALD command to obtain the serial number of the AAS module. Then set this parameter according to the mapping between the serial number and antenna based on the hardware description specific to the AAS module.

 NOTE

- When the RET antenna is connected to an AAS module:
For an AAS module with passive antennas, this parameter must be set to **DAISY_CHAIN** even if you need to configure the RET function for only one antenna. In addition, the **VENDORCODE** and **SERIALNO** parameters must be specified to indicate the antenna to be configured for the RET function, as the antennas in the AAS module are working in daisy chain mode.
- When the RET antenna is connected to an RRU/RFU:
 - Set this parameter to **REGULAR** if the RET antenna is directly connected to the RRU/RFU. In this scenario, the **VENDORCODE** and **SERIALNO** parameters do not need to be specified.
 - Set this parameter to **DAISY_CHAIN** when two RET antennas are cascaded. In this scenario, the control port for RET antennas must be configured on the upper-level RRU/RFU of the daisy chain. The **VENDORCODE** and **SERIALNO** parameters must be specified.

Table 4-6 describes the key parameters that must be set in the **RETSUBUNIT** MO.

Table 4-6 Key parameters that must be set in the **RETSUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RETSUBUNIT.DEVICENO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the device number of the RET.
Subunit No.	RETSUBUNIT.SUBUNITNO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the RET subunit number, which starts from 1.
Tilt	RETSUBUNIT.TILT (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the engineering design.

Parameter Name	Parameter ID	Setting Notes
Connect Port 1 Cabinet No.	RETSUBUNIT.CONNCN1 <i>(LTE eNodeB, 5G gNodeB)</i>	If no AAS module with passive antennas is used, set these parameters based on connections between the RET subunits and the RF ports on the RRU/RFU. For a dual-polarized antenna, it is recommended that the port numbers of connect ports 1 and 2 be configured based on the polarization mapping relationship. For example, channels R0A and R0B of an RF module are connected to the positive and negative polarized antenna ports, respectively. In this case, the port numbers of connect port 1 and 2 are set to R0A and R0B, respectively. If an AAS module with passive antennas is used, set these parameters based on the connections between the AAS module and RF ports on the RRU/RFU.
Connect Port 1 Subrack No.	RETSUBUNIT.CONNSRN1 <i>(LTE eNodeB, 5G gNodeB)</i>	
Connect Port 1 Slot No.	RETSUBUNIT.CONNSN1 <i>(LTE eNodeB, 5G gNodeB)</i>	
Connect Port 1 Port No.	RETSUBUNIT.CONNPN1 <i>(LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Cabinet No.	RETSUBUNIT.CONNCN2 <i>(LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Subrack No.	RETSUBUNIT.CONNSRN2 <i>(LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Slot No.	RETSUBUNIT.CONNSN2 <i>(LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Port No.	RETSUBUNIT.CONNPN2 <i>(LTE eNodeB, 5G gNodeB)</i>	

[Table 4-7](#) describes the key parameter that must be set to configure the RET antenna tilt.

Table 4-7 Key parameter that must be set to configure the RET antenna tilt

Parameter Name	Parameter ID	Setting Notes
Tilt	RETSUBUNIT.TILT <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the engineering design.

[Table 4-8](#) describes the key parameters that must be set in the **RETDEVICEDATA** MO.

Table 4-8 Key parameters that must be set in the **RETDEVICEDATA** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RETDEVICEDATA.DEVICEN <i>O (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the device number of the RET.

Parameter Name	Parameter ID	Setting Notes
Subunit No.	RETDEVICEDATA.SUBUNIT NO (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the RET subunit.
Antenna Model Number	RETDEVICEDATA.MODEL NO (LTE eNodeB, 5G gNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Antenna Serial No.	RETDEVICEDATA.SERIAL NO (LTE eNodeB, 5G gNodeB)	
Band1	RETDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB)	These parameters specify the frequency bands supported by an antenna and the corresponding beamwidths. These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Beamwidth1	RETDEVICEDATA.BEAMWI DTH1 (LTE eNodeB, 5G gNodeB)	
Gain1	RETDEVICEDATA.GAIN1 (LTE eNodeB, 5G gNodeB)	
Band2	RETDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB)	
Beamwidth2	RETDEVICEDATA.BEAMWI DTH2 (LTE eNodeB, 5G gNodeB)	
Gain2	RETDEVICEDATA.GAIN2 (LTE eNodeB, 5G gNodeB)	
Band3	RETDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB)	
Beamwidth3	RETDEVICEDATA.BEAMWI DTH3 (LTE eNodeB, 5G gNodeB)	
Gain3	RETDEVICEDATA.GAIN3 (LTE eNodeB, 5G gNodeB)	
Band4	RETDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB)	
Beamwidth4	RETDEVICEDATA.BEAMWI DTH4 (LTE eNodeB, 5G gNodeB)	
Gain4	RETDEVICEDATA.GAIN4 (LTE eNodeB, 5G gNodeB)	

Parameter Name	Parameter ID	Setting Notes
Installation Date	RETDEVICEDATA.DATE <i>(LTE eNodeB, 5G gNodeB)</i>	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Installer's ID	RETDEVICEDATA.INSTALLERID <i>(LTE eNodeB, 5G gNodeB)</i>	
Base Station ID	RETDEVICEDATA.BSID <i>(LTE eNodeB, 5G gNodeB)</i>	
AISG Sector ID	RETDEVICEDATA.SECTORID <i>(LTE eNodeB, 5G gNodeB)</i>	
Antenna Bearing	RETDEVICEDATA.BEARING <i>(LTE eNodeB, 5G gNodeB)</i>	
Installed Mechanical Tilt	RETDEVICEDATA.TILT <i>(LTE eNodeB, 5G gNodeB)</i>	

Scenario 2: Connection to the RET Antenna Through the Antenna Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU through the antenna port. In this scenario, the AAS module functions as the conventional RET antennas. Therefore, the configuration can be performed in the same way as that for conventional RET antennas.

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and the key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

[Table 4-9](#) describes key parameters that must be set in the **ANTENNAPORT** MO.

Table 4-9 Key parameters that must be set in the **ANTENNAPORT MO**

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	ANTENNAPORT.CN (<i>LTE eNodeB, 5G gNodeB</i>)	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located and the control port number. Set these parameters based on connections between the RET antenna and the RRU/RFU. Only one port on the RRU/RFU can be used as the control port for the RET antenna. In a daisy chain scenario, multiple RCUs share one control port.
Subrack No.	ANTENNAPORT.SRN (<i>LTE eNodeB, 5G gNodeB</i>)	
Slot No.	ANTENNAPORT.SN (<i>LTE eNodeB, 5G gNodeB</i>)	
Port No.	ANTENNAPORT.PN (<i>LTE eNodeB, 5G gNodeB</i>)	
ALD Power Switch	ANTENNAPORT.PWRSWITCH (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to ON when an ALD is used. The default value is OFF .
Feeder Length	ANTENNAPORT.FEEDERLENGTH (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the length of the feeder connected to the RF port. Set this parameter to the actual feeder length.
Switch of Configuring Ultra Delay	ANTENNAPORT.ULTRADELAYSW (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the ultra-large antenna delay switch. This switch determines the configurable range of uplink delay and downlink delay. In normal sites, it is recommended that this parameter be set to OFF because the feeder delay is less than 4000 ns. In special sites such as optical fiber repeaters, it is recommended that this parameter be set to ON because the ultra-large delay is required.
DL Time Delay	ANTENNAPORT.DLDELAY (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the device specifications. Generally, the value is less than 30 ns.
UL Time Delay	ANTENNAPORT.ULDELAY (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the device specifications. Generally, the value is less than 30 ns.
Current Alarm Threshold Type	ANTENNAPORT.THRESHOLDTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter as required. For details, see Table 4-1 .
Undercurrent Alarm Occur Threshold	ANTENNAPORT.UOETHD (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

Parameter Name	Parameter ID	Setting Notes
Undercurrent Alarm Clear Threshold	ANTENNA PORT. <i>UCTHD</i> (LTE eNodeB, 5G gNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Occur Threshold	ANTENNA PORT. <i>OOTH</i> D (LTE eNodeB, 5G gNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Clear Threshold	ANTENNA PORT. <i>OCTHD</i> (LTE eNodeB, 5G gNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

If the current alarm threshold included in the current alarm threshold type does not match the actual ALD power consumption, the **DEFCURRALMTHLD** MO can be used to modify the current alarm threshold included in this type. The following table describes the key parameters involved.

Table 4-10 Key parameters in the default current alarm threshold MO **DEFCURRALMTHLD**

Parameter Name	Parameter ID	Setting Notes
Port Type	DEFCURRALMTHLD . <i>P</i> <i>ORTTYPE</i>	Set this parameter to ANTENNA PORT.
Threshold Type	DEFCURRALMTHLD . <i>T</i> <i>HRESHOLDTYPE</i>	This parameter specifies the to-be-adjusted type associated with a set of default current alarm thresholds.
Adjusted Undercurrent Alarm Occur Threshold	DEFCURRALMTHLD . <i>U</i> <i>OTH</i> D	This parameter specifies the adjusted undercurrent alarm occur threshold. For the specified type, the default undercurrent alarm occur threshold defined in the ANTENNA PORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Undercurrent Alarm Clear Threshold	DEFCURRALMTHLD . <i>U</i> <i>CTHD</i>	This parameter specifies the adjusted undercurrent alarm clear threshold. For the specified type, the default undercurrent alarm clear threshold defined in the ANTENNA PORT MO no longer takes effect, and the adjusted threshold is used instead.

Parameter Name	Parameter ID	Setting Notes
Adjusted Overcurrent Alarm Occur Threshold	DEFCURRALMTHLD. OOTH	This parameter specifies the adjusted overcurrent alarm occur threshold. For the specified type, the default overcurrent alarm occur threshold defined in the ANTENNAPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Clear Threshold	DEFCURRALMTHLD. OCTH	This parameter specifies the adjusted overcurrent alarm clear threshold. For the specified type, the default overcurrent alarm clear threshold defined in the ANTENNAPORT MO no longer takes effect, and the adjusted threshold is used instead.

Scenario 3: Connection to the RET Antenna Through the Antenna Port (with a TMA)

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and the key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-9](#).

[Table 4-11](#) describes the key parameters that must be set in the **TMA** MO.

Table 4-11 Key parameters that must be set in the **TMA** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	TMA.DEVICENO (<i>LTE eNodeB, 5G gNodeB</i>)	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.
Device Name	TMA.DEVICENAME (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the device name of the ALD, which identifies the ALD. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.

Parameter Name	Parameter ID	Setting Notes
Control Port Cabinet No.	<i>TMA.CTRLN (LTE eNodeB, 5G gNodeB)</i>	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located. Set these parameters based on connections between the TMA and the RRU/RFU.
Control Port Subrack No.	<i>TMA.CTRLN (LTE eNodeB, 5G gNodeB)</i>	
Control Port Slot No.	<i>TMA.CTRLN (LTE eNodeB, 5G gNodeB)</i>	
Number of TMA Subunits	<i>TMA.SUBUNITNUM (LTE eNodeB, 5G gNodeB)</i>	Set this parameter as required. Generally, there are two TMA subunits.
Vendor Code	<i>TMA.VENDORCODE (LTE eNodeB, 5G gNodeB)</i>	This parameter is required in a non-regular scenario. Set this parameter to the actual TMA manufacturer code.
Serial No.	<i>TMA.SERIALNO (LTE eNodeB, 5G gNodeB)</i>	This parameter is required in a non-regular scenario. Set this parameter to the actual TMA serial number.
TMA Type	<i>TMA.TMATYPE (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the TMA type. The value NORMAL_TMA indicates a normal TMA. The value VIRTUAL_TMA indicates a device other than a TMA which amplifies signals between an RF port and an antenna port.

Table 4-12 describes the key parameters that must be set in the **TMASUBUNIT** MO.

Table 4-12 Key parameters that must be set in the **TMASUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	<i>TMASUBUNIT.DEVICENO (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the device number of the TMA.
Subunit No.	<i>TMASUBUNIT.SUBUNITNO (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of a TMA subunit.
Connect Port Cabinet No.	<i>TMASUBUNIT.CONNEN (LTE eNodeB, 5G gNodeB)</i>	Set these parameters based on connections between the TMA and the RF port on the RRU/RFU.
Connect Port Subrack No.	<i>TMASUBUNIT.CONNSRN (LTE eNodeB, 5G gNodeB)</i>	
Connect Port Slot No.	<i>TMASUBUNIT.CONNSN (LTE eNodeB, 5G gNodeB)</i>	

Parameter Name	Parameter ID	Setting Notes
Connect Port No.	TMASUBUNIT.CONNPN <i>(LTE eNodeB, 5G gNodeB)</i>	
Work Mode	TMASUBUNIT.MODE <i>(LTE eNodeB, 5G gNodeB)</i>	The TMA subunit supports two working modes, normal mode and bypass mode: • In normal mode, the TMA subunit functions and the TMA amplifies uplink signals. • In bypass mode, the TMA subunit works as a straight-through feeder. It does not amplify any uplink signals. The default value is NORMAL .
Gain	TMASUBUNIT.GAIN <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the engineering design. The gain value range supported by the TMA varies with the manufacturer and model. Run the DSP TMADEVICEDATA command (eGBTS/NodeB/eNodeB) or the DSP BTSTMADEVICEDATA command (GBTS) to query the value range before setting the gain. If the gain is fixed, this parameter is optional, or you can set this parameter to its actual gain value.

[Table 4-13](#) describes the key parameters that must be set in the **RXBRANCH** MO.

Table 4-13 Key parameters that must be set in the **RXBRANCH** MO

Parameter Name	Parameter ID	Setting Notes
RX Channel No.	RXBRANCH.RXNO <i>(RFA LampSite, LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of the RX channel of the RRU/RFU.
Logical Switch of RX Channel	RXBRANCH.RXSW <i>(RFA LampSite, LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the logical switch for the RX channel of the RRU/RFU. The default value is ON .
Attenuation	RXBRANCH.ATTEN <i>(RFA LampSite, LTE eNodeB, 5G gNodeB)</i>	If no TMA is used, set this parameter to 0. If a 12 dB TMA is used, set this parameter to a value within the range from 4 dB to 11 dB. If a 24 dB TMA is used, set this parameter to a value within the range from 11 dB to 22 dB.

[Table 4-14](#) describes the parameters that must be set in the **TMADEVICEDATA** MO.

Table 4-14 Key parameters that must be set in the **TMADEVICEDATA** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	TMADEVICEDATA.DEVICE NO (LTE eNodeB, 5G gNodeB)	This parameter specifies the device number of the TMA.
Subunit No.	TMADEVICEDATA.SUBUN ITNO (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the TMA subunit.
Connect Antenna Model Number	TMADEVICEDATA.MODEL NO (LTE eNodeB, 5G gNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Antenna Serial No.	TMADEVICEDATA.SERIAL NO (LTE eNodeB, 5G gNodeB)	
Band1	TMADEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB)	These parameters specify the frequency bands supported by an antenna and the corresponding beamwidths. These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Beamwidth1	TMADEVICEDATA.BEAM WIDTH1 (LTE eNodeB, 5G gNodeB)	
Gain1	TMADEVICEDATA.GAIN1 (LTE eNodeB, 5G gNodeB)	
Band2	TMADEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB)	
Beamwidth2	TMADEVICEDATA.BEAM WIDTH2 (LTE eNodeB, 5G gNodeB)	
Gain2	TMADEVICEDATA.GAIN2 (LTE eNodeB, 5G gNodeB)	
Band3	TMADEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB)	
Beamwidth3	TMADEVICEDATA.BEAM WIDTH3 (LTE eNodeB, 5G gNodeB)	
Gain3	TMADEVICEDATA.GAIN3 (LTE eNodeB, 5G gNodeB)	

Parameter Name	Parameter ID	Setting Notes
Band4	TMADEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB)	
Beamwidth4	TMADEVICEDATA.BEAMWIDTH4 (LTE eNodeB, 5G gNodeB)	
Gain4	TMADEVICEDATA.GAIN4 (LTE eNodeB, 5G gNodeB)	
Installation Date	TMADEVICEDATA.DATE (LTE eNodeB, 5G gNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Installer's ID	TMADEVICEDATA.INSTALLERID (LTE eNodeB, 5G gNodeB)	
Base Station ID	TMADEVICEDATA.BSID (LTE eNodeB, 5G gNodeB)	
AISG Sector ID	TMADEVICEDATA.SECTORID (LTE eNodeB, 5G gNodeB)	
Antenna Bearing	TMADEVICEDATA.BEARING (LTE eNodeB, 5G gNodeB)	
Installed Mechanical Tilt	TMADEVICEDATA.TILT (LTE eNodeB, 5G gNodeB)	

Scenario 4: Connection to the RET Antenna Through the RET Port (with a TMA)

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and the key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **RETPORT** MO in this scenario, see [Table 4-3](#).

For key parameters that must be set in the **TMA**, **TMASUBUNIT**, **RXBRANCH**, and **TMADEVICEDATA** MOs in this scenario, see [Table 4-11](#), [Table 4-12](#), [Table 4-13](#), and [Table 4-14](#).

Scenario 5: Connection to the RET Antenna Through the Antenna Port (with an SASU, Supported only by NodeBs)

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and the key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-9](#).

[Table 4-15](#) describes the key parameters that must be set in the **SASU** MO.

Table 4-15 Key parameters that must be set in the **SASU** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	SASU.DEVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the SASU.
Device Name	SASU.DEVICENAME	This parameter specifies the device name of the ALD, which identifies the ALD. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	SASU.CTRLN	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located. Set these parameters based on connections between the SASU and the RRU/RFU.
Control Port Subrack No.	SASU.CTRLRN	
Control Port Slot No.	SASU.CTRLN	
DC Switch	SASU.DCSWITCH	If the SASU is directly connected to the RET antenna, set this parameter to OFF . If the SASU is connected to the RET antenna through a TMA, set this parameter to BS or UMTS .
Vendor Code	SASU.VENDORCODE	Set this parameter to the actual SASU manufacturer code.
Serial No.	SASU.SERIALNO	Set this parameter to the actual SASU serial number.

[Table 4-16](#) describes the key parameters that must be set in the **SASUSUBUNIT** MO.

Table 4-16 Key parameters that must be set in the **SASUSUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	SASUSUBUNIT.DEVICEN O	This parameter specifies the device number of the SASU.
Subunit No.	SASUSUBUNIT.SUBUNIT NO	This parameter specifies the number of an SASU subunit.
Connect Port Cabinet No.	SASUSUBUNIT.CONNCN	Set these parameters based on connections between the SASU subunits and the RF ports on the RRU/RFU.
Connect Port Subrack No.	SASUSUBUNIT.CONNSR N	
Connect Port Slot No.	SASUSUBUNIT.CONNSN	
Connect Port No.	SASUSUBUNIT.CONNPV	
Work Mode	SASUSUBUNIT.MODE	The SASU subunit supports two working modes, normal mode and bypass mode: • In normal mode, the SASU amplifies uplink signals. • In bypass mode, the SASU subunit works as a straight-through feeder. It does not amplify uplink signals. The default value is NORMAL .
GSM Gain	SASUSUBUNIT.BSGAIN	Set this parameter based on the engineering design. The gain value range of SASU varies with the manufacturer and model. Run the DSP SASUDEVICEDATA command to query the value range before setting the gain.
UMTS Gain	SASUSUBUNIT.UMTSGAI N	
DC Load	SASUSUBUNIT.DCLOAD	Set this parameter only if the DCSWITCH parameter is set to UMTS . If the SASU is connected to the RET antenna through a TMA, this parameter must be specified so that the TMA can be acknowledged by the BTS.

Scenario 6: Connection to the RET Antenna Through the Antenna Port (with an SASU and a TMA, Supported only by NodeBs)

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and the key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-9](#).

For key parameters that must be set in the **TMA**, **TMASUBUNIT**, **RXBRANCH**, and **TMADEVICEDATA** MOs in this scenario, see [Table 4-11](#), [Table 4-12](#), [Table 4-13](#), and [Table 4-14](#).

For key parameters that must be set in the **SASU** and **SASUSUBUNIT** MOs in this scenario, see [Table 4-15](#) and [Table 4-16](#).

Scenario 7: Using the AAS MU to Control the RET Function for AAS Passive Antennas

For key parameters that must be set in the **RETDEVICEDATA** MO on an AAS module with passive antennas, see [Table 4-8](#).

[Table 4-17](#) describes the key parameters that must be set in the **RET** MO on an AAS module with passive antennas.

Table 4-17 Key parameters that must be set in the **RET** MO on an AAS module with passive antennas (MU)

Parameter Name	Parameter ID	Setting Notes
Device No.	RET.DEVICENO (<i>LTE eNodeB, 5G gNodeB</i>)	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.
Device Name	RET.DEVICENAME (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter identifies an RET antenna. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	RET.CTRLN (<i>LTE eNodeB, 5G gNodeB</i>)	These parameters specify the numbers of the cabinet, subrack, and slot, respectively, where an AAS module is located.
Control Port Subrack No.	RET.CTRLSRN (<i>LTE eNodeB, 5G gNodeB</i>)	
Control Port Slot No.	RET.CTRLN (<i>LTE eNodeB, 5G gNodeB</i>)	
RET Type	RET.RETTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter to SINGLE_RET .
Number of RET Subunits	RET.SUBUNITNUM (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter only when the RETTYPE parameter is set to MULTI_RET . Set this parameter to 1 .
Polar Type	RET.POLARTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the AAS module specifications.

Parameter Name	Parameter ID	Setting Notes
Antenna Scenario	RET.SCENARIO (<i>LTE eNodeB, 5G gNodeB</i>)	For an AAS module with passive antennas, this parameter must be set to DAISY_CHAIN even if you need to configure the RET function for only one antenna. In addition, the VENDORCODE and SERIALNO parameters must be specified to indicate the antenna to be configured for the RET function, as the antennas in the AAS module are working in daisy chain mode.
Vendor Code	RET.VENDORCODE (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the manufacturer information, for example, HW for a Huawei Agissson RET antenna.
Serial No.	RET.SERIALNO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number.

Table 4-18 describes the key parameters that must be set in the **RETSUBUNIT** MO on an AAS module with passive antennas.

Table 4-18 Key parameters that must be set in the **RETSUBUNIT** MO on an AAS module with passive antennas

Parameter Name	Parameter ID	Setting Notes
Device No.	RETSUBUNIT.DEVICENO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the device number of the RET.
Subunit No.	RETSUBUNIT.SUBUNITNO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the RET subunit number, which starts from 1.
Subunit Name	RETSUBUNIT.SUBNAME (<i>LTE eNodeB, 5G gNodeB</i>)	The name of the RET subunit is optional, but the name configured for each RET subunit must be unique.
Connect Port 1 Cabinet No.	RETSUBUNIT.CONNCN1 (<i>LTE eNodeB, 5G gNodeB</i>)	Set these parameters based on connections between the AAS module and the RF ports on the RRU/RFU.
Connect Port 1 Subrack No.	RETSUBUNIT.CONNSRN1 (<i>LTE eNodeB, 5G gNodeB</i>)	
Connect Port 1 Slot No.	RETSUBUNIT.CONNSN1 (<i>LTE eNodeB, 5G gNodeB</i>)	

Parameter Name	Parameter ID	Setting Notes
Connect Port 1 Port No.	RETSUBUNIT.CONNPN1 (LTE eNodeB, 5G gNodeB)	
Connect Port 2 Cabinet No.	RETSUBUNIT.CONNCN2 (LTE eNodeB, 5G gNodeB)	
Connect Port 2 Subrack No.	RETSUBUNIT.CONNSRN2 (LTE eNodeB, 5G gNodeB)	
Connect Port 2 Slot No.	RETSUBUNIT.CONNSN2 (LTE eNodeB, 5G gNodeB)	
Connect Port 2 Port No.	RETSUBUNIT.CONNPN2 (LTE eNodeB, 5G gNodeB)	
Tilt	RETSUBUNIT.TILT (LTE eNodeB, 5G gNodeB)	Set this parameter based on the engineering design.

[Table 4-19](#) describes the key parameter that must be set in the **RETSUBUNIT** MO on an AAS module with passive antennas.

Table 4-19 Key parameter that must be set in the **RETSUBUNIT** MO on an AAS module with passive antennas

Parameter Name	Parameter ID	Setting Notes
Tilt	RETSUBUNIT.TILT (LTE eNodeB, 5G gNodeB)	Set this parameter based on the engineering design.

Scenario 8: Using the AAS RU to Control the RET function for AAS Passive Antennas

When the RU of the AAS module is used to control the RET function:

- For the AAU3910/AAU3911/AAU3961/AAU5246, the RET function on an AAS module is controlled by the RET port on the RU of the AAS module. [Table 4-3](#) describes the key parameters that must be set in the **RETPORT** MO.
- For the AAU3920/AAU3940/AAU5240/AAU5243/AAU5940/AAU5942, the RET function on an AAS module is controlled by the antenna port on the RU of the AAS module. [Table 4-9](#) of scenario 2 describes the key parameters that must be set in the **ANTENNAPORT** MO, in which the **PN** (LTE eNodeB, 5G gNodeB) parameter must be set to **ROA**.

Configuring the RET part for AAS passive antennas:

For the key parameters that must be set in the **RET** MO for AAS passive antennas, see [Table 4-5](#).

For the key parameters that must be set in the **RETSUBUNIT** MO for AAS passive antennas, see [Table 4-6](#).

For the key parameters that must be set to configure the RET antenna downtilt for AAS passive antennas, see [Table 4-7](#).

For key parameters that must be set in the **RETDEVICEDATA** MO for AAS passive antennas, see [Table 4-8](#).

Configuring the RVD part for AAS passive antennas:

[Table 4-20](#) describes the key parameters that must be set in the **RVD** MO.

Table 4-20 Key parameters that must be set in the **RVD** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RVD.DEVICENO (<i>LTE eNodeB, 5G gNodeB</i>)	The device number of the ALD must be unique.
Device Name	RVD.DEVICENAME (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the device name of the ALD, which identifies the ALD. The recommended format of the value is <i>sector_device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	RVD.CTRLN (<i>LTE eNodeB, 5G gNodeB</i>)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located.
Control Port Subrack No.	RVD.CTRLSRN (<i>LTE eNodeB, 5G gNodeB</i>)	
Control Port Slot No.	RVD.CTRLN (<i>LTE eNodeB, 5G gNodeB</i>)	
Number of RVD Subunits	RVD.SUBUNITNUM (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the number of configured RVD subunits.
Antenna Scenario	RVD.SCENARIO (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the actual hardware installation. The vendor code and serial number of the device must be correctly configured in a non-regular scenario. This parameter can be set to REGULAR or DAISY_CHAIN .
Vendor Code	RVD.VENDORCODE (<i>LTE eNodeB, 5G gNodeB</i>)	- If SCENARIO in an RVD MO is not set to REGULAR, VENDORCODE must be set. - When SCENARIO in an RVD MO is set to REGULAR, VENDORCODE must be set if SERIALNO is set.

Parameter Name	Parameter ID	Setting Notes
Serial No.	<i>RVD.SERIALNO (LTE eNodeB, 5G gNodeB)</i>	The vendor code and the serial number uniquely identify an ALD. • If SCENARIO in an RVD MO is not set to REGULAR, SERIALNO must be set. • When SCENARIO in an RVD MO is set to REGULAR, SERIALNO must be set if VENDORCODE is set.

Table 4-21 describes the key parameters that must be set in the **RVDSUBUNIT** MO.

Table 4-21 Key parameters that must be set in the **RVDSUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	<i>RVDSUBUNIT.DEVICENO (LTE eNodeB, 5G gNodeB)</i>	The device number of the ALD must be unique.
Subunit No.	<i>RVDSUBUNIT.SUBUNITNO (LTE eNodeB, 5G gNodeB)</i>	This parameter indicates the number of the RVD subunit, which starts from 1.
Subunit Name	<i>RVDSUBUNIT.SUBNAME (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the name of the RVD subunit. The name of the RVD subunit is optional, but the name configured for each RVD subunit must be unique.
Connect Port 1 Cabinet No.	<i>RVDSUBUNIT.CONNCN1 (LTE eNodeB, 5G gNodeB)</i>	These parameters specify the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 1, respectively.
Connect Port 1 Subrack No.	<i>RVDSUBUNIT.CONNSRN1 (LTE eNodeB, 5G gNodeB)</i>	
Connect Port 1 Slot No.	<i>RVDSUBUNIT.CONNSN1 (LTE eNodeB, 5G gNodeB)</i>	
Connect Port 1 Port No.	<i>RVDSUBUNIT.CONNPN1 (LTE eNodeB, 5G gNodeB)</i>	

Parameter Name	Parameter ID	Setting Notes
Connect Port 2 Cabinet No.	<i>RVDSUBUNIT.CONNCN2 (LTE eNodeB, 5G gNodeB)</i>	These parameters specify the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 2, respectively. They are valid when a dual-polarized antenna is used.
Connect Port 2 Subrack No.	<i>RVDSUBUNIT.CONNSRN 2 (LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Slot No.	<i>RVDSUBUNIT.CONNSN2 (LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Port No.	<i>RVDSUBUNIT.CONNPN2 (LTE eNodeB, 5G gNodeB)</i>	
Vertical BeamWidth	<i>RVDSUBUNIT.VBEAMWIDTH (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the vertical beamwidth of an RVD subunit.

[Table 4-22](#) describes the key parameter that must be set to configure the RVD information on an AAS module with passive antennas.

Table 4-22 Key parameter that must be set to configure the RVD information

Parameter Name	Parameter ID	Setting Notes
Vertical BeamWidth	<i>RVDSUBUNIT.VBEAMWIDTH (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the vertical beamwidth of an RVD subunit.

[Table 4-23](#) describes the key parameters that must be set in the RVDDEVICEDATA MO.

Table 4-23 Key parameters that must be set in the RVDDEVICEDATA MO

Parameter Name	Parameter ID	Setting Notes
Device No.	<i>RVDDEVICEDATA.DEVICENO (LTE eNodeB, 5G gNodeB)</i>	The device number of the ALD must be unique.

Parameter Name	Parameter ID	Setting Notes
Subunit No.	RVDDEVICEDATA.SUBUNITNO (LTE eNodeB, 5G gNodeB)	This parameter indicates the number of the RVD subunit, which starts from 1.
Antenna Model Number	RVDDEVICEDATA.MODELNO (LTE eNodeB, 5G gNodeB)	This parameter indicates the model of the RVD.
Antenna Serial No.	RVDDEVICEDATA.SERIALNO (LTE eNodeB, 5G gNodeB)	This parameter specifies the serial number of the RVD.
Band1	RVDDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB)	These parameters specify the bands supported by the antenna and the corresponding beamwidths. • In an RVDDEVICEDATA MO, when BAND4 is not set to UNUSED, BAND3 cannot be set to UNUSED. When BAND3 is not set to UNUSED, BAND2 cannot be set to UNUSED. When BAND2 is not set to UNUSED, BAND1 cannot be set to UNUSED. • In an RVDDEVICEDATA MO, when BAND4 is not set to UNUSED, BAND4 must be greater than BAND3. When BAND3 is not set to UNUSED, BAND3 must be greater than BAND2. When BAND2 is not set to UNUSED, BAND2 must be greater than BAND1.
Beamwidth1	RVDDEVICEDATA.BEAMWIDTH1 (LTE eNodeB, 5G gNodeB)	
Gain1	RVDDEVICEDATA.GAIN1 (LTE eNodeB, 5G gNodeB)	
Band2	RVDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB)	
Beamwidth2	RVDDEVICEDATA.BEAMWIDTH2 (LTE eNodeB, 5G gNodeB)	
Gain2	RVDDEVICEDATA.GAIN2 (LTE eNodeB, 5G gNodeB)	
Band3	RVDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB)	
Beamwidth3	RVDDEVICEDATA.BEAMWIDTH3 (LTE eNodeB, 5G gNodeB)	
Gain3	RVDDEVICEDATA.GAIN3 (LTE eNodeB, 5G gNodeB)	
Band4	RVDDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB)	
Beamwidth4	RVDDEVICEDATA.BEAMWIDTH4 (LTE eNodeB, 5G gNodeB)	
Gain4	RVDDEVICEDATA.GAIN4 (LTE eNodeB, 5G gNodeB)	
Installation Date	RVDDEVICEDATA.DATE (LTE eNodeB, 5G gNodeB)	This parameter specifies the date when the antenna is installed.

Parameter Name	Parameter ID	Setting Notes
Installer's ID	RVDDEVICEDATA.INSTALLERID (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the ID of the installation technician who installed the antenna.
Base Station ID	RVDDEVICEDATA.BSID (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the ID of a base station served by an antenna.
AISG Sector ID	RVDDEVICEDATA.SECTORID (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the AISG sector ID.
Antenna Bearing	RVDDEVICEDATA.BEARING (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the antenna azimuth.
Installed Mechanical Tilt	RVDDEVICEDATA.TILT (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the mechanical tilt of the installation.

Scenario 9: Connection to the RAE Through the RET Port

For key parameters that must be set in the **RETPORT** MO in this scenario, see [Table 4-3](#).

[Table 4-24](#) describes the key parameters that must be set in the **RAE** MO.

Table 4-24 Key parameters that must be set in the **RAE** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RAE.DEVICENO (<i>LTE eNodeB, 5G gNodeB</i>)	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RAE must differ from that of the RET antenna.
Device Name	RAE.DEVICENAME (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter identifies an RAE. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name–related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	RAE.CTRLN (<i>LTE eNodeB, 5G gNodeB</i>)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RF module where the control port is located. Set these parameters based on connections between the RAE and the RF module.
Control Port Subrack No.	RAE.CTRLSRN (<i>LTE eNodeB, 5G gNodeB</i>)	

Parameter Name	Parameter ID	Setting Notes
Control Port Slot No.	<i>RAE.CTRLN (LTE eNodeB, 5G gNodeB)</i>	
Number of RAE Subunits	<i>RAE.SUBUNITNUM (LTE eNodeB, 5G gNodeB)</i>	Set this parameter as required. Generally, there is one RAE subunit.
Antenna Scenario	<i>RAE.SCENARIO (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the scenario where an RAE is connected to the RF module. The antenna scenario must be set based on the hardware installation. Set this parameter to REGULAR if the RAE is directly connected to the RF module. In this scenario, the VENDORCODE and SERIALNO parameters do not need to be specified. Set this parameter to DAISY_CHAIN when two RAEs are cascaded. In this scenario, the control port for RAEs must be configured on the upper-level RF module of the daisy chain. The VENDORCODE and SERIALNO parameters must be specified.
Vendor Code	<i>RAE.VENDORCODE (LTE eNodeB, 5G gNodeB)</i>	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the manufacturer information.
Serial No.	<i>RAE.SERIALNO (LTE eNodeB, 5G gNodeB)</i>	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number.

[Table 4-25](#) describes the key parameters that must be set in the **RAESUBUNIT** MO.

Table 4-25 Key parameters that must be set in the **RAESUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	<i>RAESUBUNIT.DEVICENO (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the device number of the RAE.
Subunit No.	<i>RAESUBUNIT.SUBUNITNO (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of the RAE subunit.
Subunit Name	<i>RAESUBUNIT.SUBNAME (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the name of the RAE subunit. The name of the RAE subunit is optional, but the name configured for each RAE subunit must be unique.

Parameter Name	Parameter ID	Setting Notes
Connect Port Cabinet No.	RAESUBUNIT.CONNCN <i>(LTE eNodeB, 5G gNodeB)</i>	Set these parameters based on connections between the RAE subunits and the antenna ports on the RF module.
Connect Port Subrack No.	RAESUBUNIT.CONNSRN <i>(LTE eNodeB, 5G gNodeB)</i>	
Connect Port Slot No.	RAESUBUNIT.CONNSN <i>(LTE eNodeB, 5G gNodeB)</i>	

[Table 4-26](#) describes the key parameters that must be set in the **RAEDEVICEDATA** MO.

Table 4-26 Key parameters that must be set in the **RAEDEVICEDATA** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RAEDEVICEDATA.DEVICE NO <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the device number of the RAE.
Subunit No.	RAEDEVICEDATA.SUBUNIT NO <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of the RAE subunit.
Antenna Model Number	RAEDEVICEDATA.MODEL NO <i>(LTE eNodeB, 5G gNodeB)</i>	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Antenna Serial No.	RAEDEVICEDATA.SERIAL NO <i>(LTE eNodeB, 5G gNodeB)</i>	
Band1	RAEDEVICEDATA.BAND1 <i>(LTE eNodeB, 5G gNodeB)</i>	These parameters specify the frequency bands supported by an antenna and the corresponding beamwidths. These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Beamwidth1	RAEDEVICEDATA.BEAM WIDTH1 <i>(LTE eNodeB, 5G gNodeB)</i>	
Gain1	RAEDEVICEDATA.GAIN1 <i>(LTE eNodeB, 5G gNodeB)</i>	

Parameter Name	Parameter ID	Setting Notes
Band2	RAEDEVICEDATA.BAND2 <i>(LTE eNodeB, 5G gNodeB)</i>	
Beamwidth2	RAEDEVICEDATA.BEAM WIDTH2 <i>(LTE eNodeB, 5G gNodeB)</i>	
Gain2	RAEDEVICEDATA.GAIN2 <i>(LTE eNodeB, 5G gNodeB)</i>	
Band3	RAEDEVICEDATA.BAND3 <i>(LTE eNodeB, 5G gNodeB)</i>	
Beamwidth3	RAEDEVICEDATA.BEAM WIDTH3 <i>(LTE eNodeB, 5G gNodeB)</i>	
Gain3	RAEDEVICEDATA.GAIN3 <i>(LTE eNodeB, 5G gNodeB)</i>	
Band4	RAEDEVICEDATA.BAND4 <i>(LTE eNodeB, 5G gNodeB)</i>	
Beamwidth4	RAEDEVICEDATA.BEAM WIDTH4 <i>(LTE eNodeB, 5G gNodeB)</i>	
Gain4	RAEDEVICEDATA.GAIN4 <i>(LTE eNodeB, 5G gNodeB)</i>	
Installation Date	RAEDEVICEDATA.DATE <i>(LTE eNodeB, 5G gNodeB)</i>	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Installer's ID	RAEDEVICEDATA.INSTAL LERID <i>(LTE eNodeB, 5G gNodeB)</i>	
Base Station ID	RAEDEVICEDATA.BSID <i>(LTE eNodeB, 5G gNodeB)</i>	
AISG Sector ID	RAEDEVICEDATA.SECTO RID <i>(LTE eNodeB, 5G gNodeB)</i>	

Scenario 10: Connection to the RAE Through the RET Port (with an RET Antenna)

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **RETPORT** MO in this scenario, see [Table 4-3](#).

For key parameters that must be set in the **RAE**, **RAESUBUNIT**, and **RAEDEVICEDATA** MOs in this scenario, see [Table 4-24](#), [Table 4-25](#), and [Table 4-26](#).

Scenario 11: Connection to the RAE Through the Antenna Port

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-9](#).

For key parameters that must be set in the **RAE**, **RAESUBUNIT**, and **RAEDEVICEDATA** MOs in this scenario, see [Table 4-24](#), [Table 4-25](#), and [Table 4-26](#).

Scenario 12: Connection to the RAE Through the Antenna Port (with an RET Antenna)

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-9](#).

For key parameters that must be set in the **RAE**, **RAESUBUNIT**, and **RAEDEVICEDATA** MOs in this scenario, see [Table 4-24](#), [Table 4-25](#), and [Table 4-26](#).

Scenario 13: Connection to the RAE Through the Antenna Port (with a TMA)

For key parameters that must be set in the **TMA**, **TMASUBUNIT**, **RXBRANCH**, and **TMADEVICEDATA** MOs in this scenario, see [Table 4-11](#), [Table 4-12](#), [Table 4-13](#), and [Table 4-14](#).

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-9](#).

For key parameters that must be set in the **RAE**, **RAESUBUNIT**, and **RAEDEVICEDATA** MOs in this scenario, see [Table 4-24](#), [Table 4-25](#), and [Table 4-26](#).

Scenario 14: Connection to the RAE Through the Antenna Port (with a TMA and an RET Antenna)

For key parameters that must be set in the **TMA**, **TMASUBUNIT**, **RXBRANCH**, and **TMADEVICEDATA** MOs in this scenario, see [Table 4-11](#), [Table 4-12](#), [Table 4-13](#), and [Table 4-14](#).

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-9](#).

For key parameters that must be set in the **RAE**, **RAESUBUNIT**, and **RAEDEVICEDATA** MOs in this scenario, see [Table 4-24](#), [Table 4-25](#), and [Table 4-26](#).

Scenario 15: Connection to the RAS Unit Through the RET Port

In this scenario, the **RETPORT**, **RAS**, **RASSUBUNIT**, and **RASDEVICEDATA** MOs must be configured. The RAS azimuth is configured in the **RASSUBUNIT** MO. The **DIGIANTINFO** MO must be configured as well if digital antennas are involved.

[Table 4-3](#) describes the key parameters that must be set in the **RETPORT** MO.

[Table 4-27](#) describes the key parameters that must be set in the **RAS** MO.

[Table 4-30](#) describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-27 Key parameters that must be set in the **RAS** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RAS.DEVICENO	The device number of each ALD in a base station must be unique.
Device Name	RAS.DEVICENAME	This parameter specifies the device name ¹ .
Control Port Cabinet No.	RAS.CTRLN	This parameter specifies the cabinet number of the RRU or RFU that controls the ALD.
Control Port Subrack No.	RAS.CTRLNRN	This parameter specifies the subrack number of the RRU or RFU that controls the ALD.
Control Port Slot No.	RAS.CTRLN	This parameter specifies the slot number of the RRU or RFU that controls the ALD.
Number of RAS Subunits	RAS.SUBUNITNUM	This parameter specifies the number of configured RAS subunits.

Parameter Name	Parameter ID	Setting Notes
Antenna Scenario	RAS.SCENARIO	Set this parameter based on the actual hardware installation. The vendor code and serial number of the device must be correctly configured in a non-regular scenario. This parameter can be set to REGULAR or DAISY_CHAIN .
Vendor Code	RAS.VENDORCODE	This parameter specifies the vendor code.
Serial No.	RAS.SERIALNO	The vendor code and the serial number uniquely identify an ALD.

 NOTE

This parameter indicates the device name of the ALD, which identifies the ALD. It is recommended that you set the device name in the following format: **sector_device type_network type**. The meanings of the parts are as follows: Sector indicates the number of the sector. Device type indicates the type of currently connected device. It can be SINGLE_RET (Single-antenna Remote Electrical Tilt Unit), MULTI_RET (Multi-antenna Remote Electrical Tilt Unit), TMA (Tower-mounted Amplifier), SASU (Same-band Antenna Sharing Unit), RAE (Remote eAntenna Extension), RVD (Remote Vendor Defined Antenna Line Device), or RAS (Remote Azimuth Steering Unit). Network type indicates whether the current ALD is used for a 2G network, 3G network, or LTE network. The device name is optional, but the device name configured for each ALD must be unique.

Table 4-28 describes the parameters that must be configured in the **RASSUBUNIT MO**.

Table 4-28 Key parameters that must be set in the **RASSUBUNIT MO**

Parameter Name	Parameter ID	Setting Notes
Device No.	RASSUBUNIT.DEVICE NO	The device number of each ALD in a base station must be unique.
Subunit No.	RASSUBUNIT.SUBUNIT NO	This parameter specifies the number of the RAS subunit. The value starts from 1.
Subunit Name	RASSUBUNIT.SUBNAME	This parameter specifies the name of the RAS subunit. The RAS subunit name is optional, but the name configured for each subunit must be unique.
Connect Port 1 Cabinet No.	RASSUBUNIT.CONNC N1	These parameters specify the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 1, respectively.
Connect Port 1 Subrack No.	RASSUBUNIT.CONNS RN1	
Connect Port 1 Slot No.	RASSUBUNIT.CONNS N1	

Parameter Name	Parameter ID	Setting Notes
Connect Port 1 Port No.	RASSUBUNIT.CONNP N1	
Connect Port 2 Cabinet No.	RASSUBUNIT.CONNC N2	These parameters specify the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 2, respectively.
Connect Port 2 Subrack No.	RASSUBUNIT.CONNS RN2	
Connect Port 2 Slot No.	RASSUBUNIT.CONNS N2	
Connect Port 2 Port No.	RASSUBUNIT.CONNP N2	
Azimuth Offset	RASSUBUNIT.AZIMU TH	This parameter specifies the horizontal azimuth offset of the RAS subunit.
Alarm Error Range	RASSUBUNIT.AER	When the difference between the actual horizontal azimuth of an RET subunit and configured azimuth is larger than the value of this parameter, the RET Antenna Running Data and Configuration Mismatch alarm is reported.

Table 4-29 describes the key parameters that must be set in the **RASDEVICEDATA** MO.

Table 4-29 Key parameters that must be set in the **RASDEVICEDATA** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RASDEVICEDAT A.DEVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RAS device must differ from that of the RET antenna.
Subunit No.	RASDEVICEDAT A.SUBUNITNO	This parameter specifies the number of the RAS subunit. The value starts from 1.
Antenna Model Number	RASDEVICEDAT A.MODELNO	This parameter specifies the model of the RAS unit.
Antenna Serial No.	RASDEVICEDAT A.SERIALNO	This parameter specifies the serial number of the RAS unit.
Band1	RASDEVICEDAT A.BAND1	These parameters specify the bands supported by the antenna and the corresponding beamwidths.
Beamwidth1	RASDEVICEDAT A.BEAMWIDTH 1	

Parameter Name	Parameter ID	Setting Notes
Gain1	RASDEVICEDAT A.GAIN1	
Band2	RASDEVICEDAT A.BAND2	
Beamwidth2	RASDEVICEDAT A.BEAMWIDTH 2	
Gain2	RASDEVICEDAT A.GAIN2	
Band3	RASDEVICEDAT A.BAND3	
Beamwidth3	RASDEVICEDAT A.BEAMWIDTH 3	
Gain3	RASDEVICEDAT A.GAIN3	
Band4	RASDEVICEDAT A.BAND4	
Beamwidth4	RASDEVICEDAT A.BEAMWIDTH 4	
Gain4	RASDEVICEDAT A.GAIN4	
Installation Date	RASDEVICEDAT A.DATE	This parameter specifies the date when the antenna is installed.
Installer's ID	RASDEVICEDAT A.INSTALLERID	Indicates the ID of the installation technician who installed the antenna.
Base Station ID	RASDEVICEDAT A.BSID	According to AISG1.1, the length of the base station ID supported by the RAS device is limited to 12 characters. If the length of the base station ID entered by a user exceeds the limit, only the first 12 characters are stored on the RAS device.
AISG Sector ID	RASDEVICEDAT A.SECTORID	According to AISG1.1, the length of the sector ID supported by the RAS device is limited to 4 characters. If the length of the sector ID entered by a user exceeds the limit, only the first 4 characters are stored on the RAS device.
Antenna Bearing	RASDEVICEDAT A.ANTENNABE ARINGHP	If this parameter is set to 4294967295 , the parameter value is not delivered to the device. In this case, the parameter value in the LST RASDEVICEDATA command output is NULL .

Parameter Name	Parameter ID	Setting Notes
Installed Mechanical Tilt	RASDEVICEDAT A.TILT	If this parameter is set to 2147483647 , the parameter value is not delivered to the device. In this case, the parameter value in the LST RASDEVICEDATA command output is NULL .

Table 4-30 describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-30 Key parameters that must be set in the **DIGIANTINFO** MO

Parameter Name	Parameter ID	Setting Notes
Effective Range	DIGIANTINFO.EFFEC TIVERANGE (LTE eNodeB, 5G gNodeB)	This parameter specifies the effective range of the switch for a certain optional function related to digital antennas. The value SITE_LEVEL indicates that this optional function takes effect for all ALDs that support the function in the current base station. The value DEVICE_LEVEL indicates that this optional function takes effect for the specified ALD.
Subunit No.	DIGIANTINFO.SUBU NITNO (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the ALD subunit on which a certain optional function needs to take effect. The value 65535 indicates that this optional function takes effect for all subunits that support the function in the current ALD.
Optional Function	DIGIANTINFO.OPTIO NALFUNCTION (LTE eNodeB, 5G gNodeB)	This parameter specifies the switches for certain optional functions related to digital antennas. The INTRF_DIAGNOSIS , AZIMUTH_ADJ , and HBW_ADJ options control interference diagnosis, horizontal azimuth adjustment, and horizontal beamwidth adjustment, respectively.

Scenario 16: Connection to the RAS Unit Through the Antenna Port

For key parameters that must be set in the **ANTENNAPORT**, **RAS**, **RASSUBUNIT**, and **RASDEVICEDATA** MOs in this scenario, see **Table 4-9** in Scenario 1 and **Table 4-27**, **Table 4-28**, and **Table 4-29** in Scenario 15.

Table 4-9 describes key parameters that must be set in the **ANTENNAPORT** MO.

Scenario 17: Connection to the RVD Through the RET Port

In this scenario, the **RETPORT**, **RVD**, **RVDSUBUNIT**, and **RVDDEVICEDATA** MOs must be configured. The horizontal beamwidth of the RVD is configured in the **RVDSUBUNIT** MO. The **DIGIANTINFO** MO must be configured as well if digital antennas are involved.

Table 4-3 describes the key parameters that must be set in the **RETPORT** MO.

Table 4-31 describes the key parameters that must be set in the **RVD** MO.

Table 4-34 describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-31 Key parameters that must be set in the **RVD** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	RVD.DEVICENO (<i>LTE eNodeB, 5G gNodeB</i>)	The device number of the ALD must be unique.
Device Name	RVD.DEVICENAME (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the device name of the ALD, which identifies the ALD. The recommended format of the value is <i>sector_device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	RVD.CTRLN (<i>LTE eNodeB, 5G gNodeB</i>)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located.
Control Port Subrack No.	RVD.CTRLNRN (<i>LTE eNodeB, 5G gNodeB</i>)	
Control Port Slot No.	RVD.CTRLN (<i>LTE eNodeB, 5G gNodeB</i>)	
Number of RVD Subunits	RVD.SUBUNITNUM (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the number of configured RVD subunits.
Antenna Scenario	RVD.SCENARIO (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the actual hardware installation. The vendor code and serial number of the device must be correctly configured in a non-regular scenario. This parameter can be set to REGULAR or DAISY_CHAIN .
Vendor Code	RVD.VENDORCODE (<i>LTE eNodeB, 5G gNodeB</i>)	• If the RVD.SERIALNO (<i>LTE eNodeB, 5G gNodeB</i>) parameter is not set to REGULAR, the RVD.VENDORCODE (<i>LTE eNodeB, 5G gNodeB</i>) parameter must be set. • If the RVD.SERIALNO (<i>LTE eNodeB, 5G gNodeB</i>) parameter is set to REGULAR and the RVD.SERIALNO (<i>LTE eNodeB, 5G gNodeB</i>) parameter is set, the RVD.VENDORCODE (<i>LTE eNodeB, 5G gNodeB</i>) must be set.

Parameter Name	Parameter ID	Setting Notes
Serial No.	RVD.SERIALNO (LTE eNodeB, 5G gNodeB)	The vendor code and the serial number uniquely identify an ALD. - If the RVD.SERIALNO (LTE eNodeB, 5G gNodeB) parameter is not set to REGULAR, the RVD.VENDORCODE (LTE eNodeB, 5G gNodeB) parameter must be set. - If the RVD.SERIALNO (LTE eNodeB, 5G gNodeB) parameter is set to REGULAR and the RVD.VENDORCODE (LTE eNodeB, 5G gNodeB) parameter is set, the RVD.SERIALNO (LTE eNodeB, 5G gNodeB) must be set.

Table 4-32 describes the key parameters that must be set in the **RVDSUBUNIT MO**.

Table 4-32 Key parameters that must be set in the **RVDSUBUNIT MO**

Parameter Name	Parameter ID	Setting Notes
Device No.	RVDSUBUNIT.DEVICENO (LTE eNodeB, 5G gNodeB)	The device number of the ALD must be unique.
Subunit No.	RVDSUBUNIT.SUBUNITNO (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the RVD subunit. The value starts from 1.
Subunit Name	RVDSUBUNIT.SUBNAME (LTE eNodeB, 5G gNodeB)	This parameter indicates the name of the RVD subunit. The name of the RVD subunit is optional, but the name configured for each RVD subunit must be unique.
Horizontal Beamwidth	RVDSUBUNIT.HORIZONTALTALBW (LTE eNodeB, 5G gNodeB)	This parameter specifies the horizontal beamwidth of the RVD subunit.
Connect Port 1 Cabinet No.	RVDSUBUNIT.CONNCN1 (LTE eNodeB, 5G gNodeB)	These parameters indicate the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 1, respectively.
Connect Port 1 Subrack No.	RVDSUBUNIT.CONNSRN1 (LTE eNodeB, 5G gNodeB)	
Connect Port 1 Slot No.	RVDSUBUNIT.CONNSN1 (LTE eNodeB, 5G gNodeB)	

Parameter Name	Parameter ID	Setting Notes
Connect Port 1 Port No.	<i>RVDSUBUNIT.CONNPN1 (LTE eNodeB, 5G gNodeB)</i>	These parameters indicate the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 2, respectively. They are valid when a dual-polarized antenna is used.
Connect Port 2 Cabinet No.	<i>RVDSUBUNIT.CONNCN2 (LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Subrack No.	<i>RVDSUBUNIT.CONNSRN 2 (LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Slot No.	<i>RVDSUBUNIT.CONNSN2 (LTE eNodeB, 5G gNodeB)</i>	
Connect Port 2 Port No.	<i>RVDSUBUNIT.CONNPN2 (LTE eNodeB, 5G gNodeB)</i>	

Table 4-33 describes the key parameters that must be set in the RVDDEVICEDATA MO.

Table 4-33 Key parameters that must be set in the RVDDEVICEDATA MO

Parameter Name	Parameter ID	Setting Notes
Device No.	<i>RVDDEVICEDATA.DEVICENO (LTE eNodeB, 5G gNodeB)</i>	The device number of the ALD must be unique.
Subunit No.	<i>RVDDEVICEDATA.SUBUNITNO (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of the RVD subunit. The value starts from 1.
Antenna Model Number	<i>RVDDEVICEDATA.MODELNO (LTE eNodeB, 5G gNodeB)</i>	This parameter indicates the model of the RVD module.
Antenna Serial No.	<i>RVDDEVICEDATA.SERIALNO (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the serial number of the RVD module.

Parameter Name	Parameter ID	Setting Notes
Band1	RVDDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB)	These parameters specify the bands supported by the antenna and the corresponding beamwidths. - If the value of RVDDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB) is not UNUSED, the value of RVDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB) cannot be UNUSED. If the value of RVDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB) is not UNUSED, the value of RVDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB) cannot be UNUSED. If the value of RVDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB) is not UNUSED, the value of RVDDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB) cannot be UNUSED. - If the value of RVDDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB) is not UNUSED, it must be greater than the value of RVDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB). If the value of RVDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB) is not UNUSED, it must be greater than the value of RVDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB). If the value of RVDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB) is not UNUSED, it must be greater than the value of RVDDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB).
Beamwidth1	RVDDEVICEDATA.BEAMWIDTH1 (LTE eNodeB, 5G gNodeB)	
Gain1	RVDDEVICEDATA.GAIN1 (LTE eNodeB, 5G gNodeB)	
Band2	RVDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB)	
Beamwidth2	RVDDEVICEDATA.BEAMWIDTH2 (LTE eNodeB, 5G gNodeB)	
Gain2	RVDDEVICEDATA.GAIN2 (LTE eNodeB, 5G gNodeB)	
Band3	RVDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB)	
Beamwidth3	RVDDEVICEDATA.BEAMWIDTH3 (LTE eNodeB, 5G gNodeB)	
Gain3	RVDDEVICEDATA.GAIN3 (LTE eNodeB, 5G gNodeB)	
Band4	RVDDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB)	
Beamwidth4	RVDDEVICEDATA.BEAMWIDTH4 (LTE eNodeB, 5G gNodeB)	
Gain4	RVDDEVICEDATA.GAIN4 (LTE eNodeB, 5G gNodeB)	
Installation Date	RVDDEVICEDATA.DATE (LTE eNodeB, 5G gNodeB)	This parameter indicates the date when the antenna is installed.

Parameter Name	Parameter ID	Setting Notes
Installer's ID	RVDDEVICEDATA.INSTALLERID (LTE eNodeB, 5G gNodeB)	This parameter indicates the ID of the installation technician who installed the antenna.
Base Station ID	RVDDEVICEDATA.BSID (LTE eNodeB, 5G gNodeB)	This parameter indicates the ID of the base station served by the antenna.
AISG Sector ID	RVDDEVICEDATA.SECTORID (LTE eNodeB, 5G gNodeB)	This parameter indicates the AISG sector ID.
Antenna Bearing	RVDDEVICEDATA.BEARING (LTE eNodeB, 5G gNodeB)	This parameter indicates the antenna azimuth.
Installed Mechanical Tilt	RVDDEVICEDATA.TILT (LTE eNodeB, 5G gNodeB)	This parameter indicates the mechanical tilt of the installation.

Table 4-34 describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-34 Key parameters that must be set in the **DIGIANTINFO** MO

Parameter Name	Parameter ID	Setting Notes
Effective Range	DIGIANTINFO.EFFECTIVERANGE (LTE eNodeB, 5G gNodeB)	This parameter specifies the effective range of the switch for a certain optional function related to digital antennas. The value SITE_LEVEL indicates that this optional function takes effect for all ALDs that support the function in the current base station. The value DEVICE_LEVEL indicates that this optional function takes effect for the specified ALD.
Subunit No.	DIGIANTINFO.SUBUNITNO (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the ALD subunit on which a certain optional function needs to take effect. The value 65535 indicates that this optional function takes effect for all subunits that support the function in the current ALD.
Optional Function	DIGIANTINFO.OPTIONALFUNCTION (LTE eNodeB, 5G gNodeB)	This parameter specifies the switches for certain optional functions related to digital antennas. The INTRF_DIAGNOSIS , AZIMUTH_ADJ , and HBW_ADJ options control interference diagnosis, horizontal azimuth adjustment, and horizontal beamwidth adjustment, respectively.

Scenario 18: Connection to the RVD Through the Antenna Port

For key parameters that must be set in the **ANTENNA**PORT, **RVD**, **RVDSUBUNIT**, and **RVDDEVICEDATA** MOs in this scenario, see [Table 4-9](#) in scenario 1 and [Table 4-31](#), [Table 4-32](#), and [Table 4-33](#) in scenario 15.

[Table 4-9](#) describes key parameters that must be set in the **ANTENNA**PORT MO.

4.4.3.2 Data Preparation (New Model)

When the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**, the new model is used for data preparation.

NOTE

The new and old ALD models in this document are selected based on the value of the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter. The old ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**. The new ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**.

Scenario 1: Connection to the RET Antenna Through the RET Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU through the RET port. In this scenario, the AAS module functions as a conventional RET antenna. Therefore, the configuration can be performed in the same way as that for conventional RET antennas.

[Table 4-35](#) describes the key parameters that must be set to configure the control port (RET port).

Table 4-35 Key parameters that must be set to configure the control port (RET port)

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RETPORT.CN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located and the control port number. Set these parameters based on connections between the RET antenna and the RRU/RFU. Only one port on the RRU/RFU can be used as the control port for the RET antenna. In a daisy chain scenario, multiple RCUs share one control port.
Subrack No.	RETPORT.SRN (5G gNodeB, LTE eNodeB)	
Slot No.	RETPORT.SN (5G gNodeB, LTE eNodeB)	
Port No.	RETPORT.PN (5G gNodeB, LTE eNodeB)	
ALD Power Switch	RETPORT.PWRSWITCH (5G gNodeB, LTE eNodeB)	Set this parameter to ON when the RET function is enabled. The default value is OFF .
Current Alarm Threshold Type	RETPORT.THRESHOLDTYPE (5G gNodeB, LTE eNodeB)	When the RCU is connected through the RET port in a regular scenario, set this parameter to RET_ONLY_MULTICORE . In other scenarios, set this parameter to UER_SELF_DEFINE .

Parameter Name	Parameter ID	Setting Notes
Undercurrent Alarm Occur Threshold	RETPORT.UOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Undercurrent Alarm Clear Threshold	RETPORT.UCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Occur Threshold	RETPORT.OOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Clear Threshold	RETPORT.OCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, the **DEFCURRALMTHLD** MO can be used to modify the current alarm thresholds included in this type. The following table describes the key parameters involved.

Table 4-36 Key parameters in the default current alarm threshold MO **DEFCURRALMTHLD**

Parameter Name	Parameter ID	Setting Notes
Port Type	DEFCURRALMTHLD.PORTTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to RETPORT .
Threshold Type	DEFCURRALMTHLD.THRESHOLDTYPE (LTE eNodeB, 5G gNodeB)	This parameter specifies the to-be-adjusted type associated with a set of default current alarm thresholds.
Adjusted Undercurrent Alarm Occur Threshold	DEFCURRALMTHLD.UOOTH (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted undercurrent alarm occur threshold. For the specified type, the default undercurrent alarm occur threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.

Parameter Name	Parameter ID	Setting Notes
Adjusted Undercurrent Alarm Clear Threshold	DEFCURRALMTHLD.UCTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted undercurrent alarm clear threshold. For the specified type, the default undercurrent alarm clear threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Occur Threshold	DEFCURRALMTHLD.OOTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted overcurrent alarm occur threshold. For the specified type, the default overcurrent alarm occur threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Clear Threshold	DEFCURRALMTHLD.OCTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted overcurrent alarm clear threshold. For the specified type, the default overcurrent alarm clear threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.

Table 4-37 describes the key parameters that must be set to configure the RET device.

Table 4-37 Key parameters that must be set to configure the RET device

Parameter Name	Parameter ID	Setting Notes
Device No.	NGALD.DEVICENO (5G gNodeB, LTE eNodeB)	The device number of each ALD in a base station must be unique. Even ALDs of different types (such as an RET antenna and a TMA) cannot have the same device number.
Device Name	NGALD.DEVICENAME (5G gNodeB, LTE eNodeB)	This parameter identifies an RET antenna. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.

Parameter Name	Parameter ID	Setting Notes
Control Port Cabinet No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located. Set these parameters based on the control relationship between the RET antenna and the RRU/RFU.
Control Port Subrack No.	NGALD.CTRLRN (5G gNodeB, LTE eNodeB)	
Control Port Slot No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	
ALD Device type	NGALD.ALDDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter to SINGLE_RET or MULTI_RET based on the actual RET antenna. Set this parameter to SINGLE_RET if the base station uses the AAS module with passive antennas.
Number of ALD Subunits	NGALD.SUBUNITNUM (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual number of antenna subunits for a multi-subunit antenna.
Polar Type	NGALD.POLARTYPE (5G gNodeB, LTE eNodeB)	Set this parameter based on the RET antenna specifications. Set this parameter based on the AAS specifications when the AAS module with passive antennas is used.
Antenna Scenario	NGALD.SCENARIO (5G gNodeB, LTE eNodeB)	Antenna scenario ¹
Vendor Code	NGALD.VENDORCODE (5G gNodeB, LTE eNodeB)	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the manufacturer information, for example, KA for a Kathrein, AN for an Andrew, or HW for a Huawei Agissson. Set this parameter to HW for an AAS module with passive antennas.
Serial No.	NGALD.SERIALNO (5G gNodeB, LTE eNodeB)	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number. If an AAS module with passive antennas is used, run the SCN ALD command to obtain the serial number of the AAS module. Then set this parameter according to the mapping between the serial number and antenna based on the hardware description specific to the AAS module.

 NOTE

- When the RET antenna is connected to an AAS:
If an AAS module with passive antennas is used, set the antenna scenario parameter to **DAISY_CHAIN** even when you need to use only one set of antennas for the RET function. In this case, specify the **VENDORCODE** and **SERIALNO** parameters to determine the specific set of antennas whose RET function is used. Only the RET function of one set of antennas needs to be configured because the antennas of the AAS module are connected and controlled in daisy chain mode.
- When the RET is connected to an RRU/RFU:
 - Set the antenna scenario parameter to **REGULAR** if an RET antenna is directly connected to the RRU/RFU. In this scenario, the **VENDORCODE** and **SERIALNO** parameters do not need to be specified.
 - Set the antenna scenario parameter to **DAISY_CHAIN** when the RET antenna is cascaded with another RET antenna. In this scenario, the control port for the RET antenna must be the port on the RRU/RFU connected to the top-level RET antenna of the daisy chain. The **VENDORCODE** and **SERIALNO** parameters must be specified.

Table 4-38 describes the key parameters that must be set to configure the RET subunit.

Table 4-38 Key parameters that must be set to configure an RET subunit

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVICEN <i>O (5G gNodeB, LTE eNodeB)</i>	This parameter indicates the antenna device number provided during the RET configuration.
Subunit No.	ALDSUBUNIT.SUBUNIT <i>NO (5G gNodeB, LTE eNodeB)</i>	This parameter indicates the ID of the subunit, which starts from 1.
Tilt	ALDSUBUNIT.TILT (5G <i>gNodeB, LTE eNodeB)</i>	Set this parameter based on the engineering design.
RF Path ID	RFPATH.RFPATHID (5G <i>gNodeB, LTE eNodeB)</i>	This parameter indicates the ID of an RF path.
RF Path Alias	RFPATH.ALIAS (5G <i>gNodeB, LTE eNodeB)</i>	This parameter indicates the connection path between the RF module and the ALD.
Connect Port 1 Cabinet No.	RFPATH.CN (5G <i>gNodeB, LTE eNodeB)</i>	Set these parameters based on connections between the RET subunits and the RF ports on the RRU/RFU. If an AAS module with passive antennas is used, set these parameters based on the connections between the AAS module and RF ports on the RRU/RFU.
Connect Port 1 Subrack No.	RFPATH.SRN (5G <i>gNodeB, LTE eNodeB)</i>	
Connect Port 1 Slot No.	RFPATH.SN (5G <i>gNodeB, LTE eNodeB)</i>	
Connect Port 1 Port No.	RFPATH.PN (5G <i>gNodeB, LTE eNodeB)</i>	
Connect Port 2 Cabinet No.	RFPATH.CN (5G <i>gNodeB, LTE eNodeB)</i>	

Parameter Name	Parameter ID	Setting Notes
Connect Port 2 Subrack No.	RFPATH.SRN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Slot No.	RFPATH.SN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Port No.	RFPATH.PN (5G gNodeB, LTE eNodeB)	
Connection ID	ALDCONNINFO.CONNID (5G gNodeB, LTE eNodeB)	Set these parameters based on the connections between the ALD and the antenna ports on the RF module.
Connection Type	ALDCONNINFO.CONNTYPE (5G gNodeB, LTE eNodeB)	
Device No.	ALDCONNINFO.DEVICENO (5G gNodeB, LTE eNodeB)	
ALD Port No.	ALDCONNINFO.ALDPORTNO (5G gNodeB, LTE eNodeB)	
Subunit No.	ALDCONNINFO.SUBUNITNO (5G gNodeB, LTE eNodeB)	
RF Path ID	ALDCONNINFO.RFPATHID (5G gNodeB, LTE eNodeB)	

Table 4-39 describes the key parameter that must be set to configure the RET tilt.

Table 4-39 Key parameter that must be set to configure the RET tilt

Parameter Name	Parameter ID	Setting Notes
Tilt	ALDSUBUNIT.TILT (5G gNodeB, LTE eNodeB)	Set this parameter based on the engineering design.

Table 4-40 describes the key parameters that must be set to configure the RET device property group.

Table 4-40 Key parameters that must be set to configure the RET device property group

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDDEVICEDATA.DEVICEN O (5G gNodeB, LTE eNodeB)	This parameter indicates the antenna device number provided during the RET configuration.
Subunit No.	ALDDEVICEDATA.SUBUNIT NO (5G gNodeB, LTE eNodeB)	This parameter indicates the number of the RET subunit.
Antenna Model Number	ALDDEVICEDATA.MODEL O (5G gNodeB, LTE eNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see AISG3.0.
Antenna Serial No.	ALDDEVICEDATA.SERIAL O (5G gNodeB, LTE eNodeB)	
Band1	ALDDEVICEDATA.BAND1 (5G gNodeB, LTE eNodeB)	These parameters specify the frequency bands supported by an antenna and the corresponding beamwidths. These parameters are part of the device data defined by AISG protocols. For details, see AISG3.0.
Beamwidth1	ALDDEVICEDATA.BEAMWI DTH1 (5G gNodeB, LTE eNodeB)	
Gain1	ALDDEVICEDATA.GAIN1 (5G gNodeB, LTE eNodeB)	
Band2	ALDDEVICEDATA.BAND2 (5G gNodeB, LTE eNodeB)	
Beamwidth2	ALDDEVICEDATA.BEAMWI DTH2 (5G gNodeB, LTE eNodeB)	
Gain2	ALDDEVICEDATA.GAIN2 (5G gNodeB, LTE eNodeB)	
Band3	ALDDEVICEDATA.BAND3 (5G gNodeB, LTE eNodeB)	
Beamwidth3	ALDDEVICEDATA.BEAMWI DTH3 (5G gNodeB, LTE eNodeB)	
Gain3	ALDDEVICEDATA.GAIN3 (5G gNodeB, LTE eNodeB)	
Band4	ALDDEVICEDATA.BAND4 (5G gNodeB, LTE eNodeB)	
Beamwidth4	ALDDEVICEDATA.BEAMWI DTH4 (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
Gain4	ALDDEVICEDATA.GAIN4 <i>(5G gNodeB, LTE eNodeB)</i>	These parameters are part of the device data defined by AISG protocols. For details, see AISG3.0.
Installation Date	ALDDEVICEDATA.DATE <i>(5G gNodeB, LTE eNodeB)</i>	
Installer's ID	ALDDEVICEDATA.INSTALLERID <i>(5G gNodeB, LTE eNodeB)</i>	
Base Station ID	ALDDEVICEDATA.BSID <i>(5G gNodeB, LTE eNodeB)</i>	
AISG Sector ID	ALDDEVICEDATA.SECTORID <i>(5G gNodeB, LTE eNodeB)</i>	
Antenna Bearing	ALDDEVICEDATA.BEARING <i>(5G gNodeB, LTE eNodeB)</i>	
Installed Mechanical Tilt	ALDDEVICEDATA.TILT <i>(5G gNodeB, LTE eNodeB)</i>	

Scenario 2: Connection to the RET Antenna Through the Antenna Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU through the antenna port. In this scenario, the AAS module functions as a conventional RET antenna. Therefore, the configuration can be performed in the same way as that for conventional RET antennas.

For key parameters that must be set to configure the RET device, RET subunit, and RET device property group in this scenario, see [Table 4-37](#), [Table 4-38](#), and [Table 4-40](#) of scenario 1.

[Table 4-41](#) describes key parameters that must be set to configure the antenna port.

Table 4-41 Key parameters that must be set to configure the control port (antenna port)

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	ANTENNAPORT.CN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located and the control port number. Set these parameters based on connections between the RET antenna and the RRU/RFU. Only one port on the RRU/RFU can be used as the control port for the RET antenna. In a daisy chain scenario, multiple RCUs can share one control port.
Subrack No.	ANTENNAPORT.SRN (5G gNodeB, LTE eNodeB)	
Slot No.	ANTENNAPORT.SN (5G gNodeB, LTE eNodeB)	
Antenna Port No.	ANTENNAPORT.PN (5G gNodeB, LTE eNodeB)	
ALD Power Switch	ANTENNAPORT.PWRSWITCH (5G gNodeB, LTE eNodeB)	Set this parameter to ON when an ALD is used. The default value is OFF .
Feeder Length	ANTENNAPORT.FEEDERLENGTH (5G gNodeB, LTE eNodeB)	This parameter indicates the length of the feeder connected to the antenna port. Set this parameter to the actual feeder length.
Switch of Configuring Ultra Delay	ANTENNAPORT.ULTRADELAYSW (5G gNodeB, LTE eNodeB)	This parameter indicates the ultra-large antenna delay switch. This switch determines the configurable range of uplink delay and downlink delay. In normal sites, it is recommended that this parameter be set to OFF because the feeder delay does not exceed 4000 ns. In special sites such as optical fiber repeaters, this parameter needs to be set to ON when the ultra-large delay is required.
DL Time Delay	ANTENNAPORT.DLDELAY (5G gNodeB, LTE eNodeB)	Set this parameter based on the device specifications. Generally, the value is less than 30 ns.
UL Time Delay	ANTENNAPORT.ULDELAY (5G gNodeB, LTE eNodeB)	Set this parameter based on the device specifications. Generally, the value is less than 30 ns.
Current Alarm Threshold Type	ANTENNAPORT.THRESHOLDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter based on the device conditions and actual connection scenarios. For details, see Table 4-1 .
Undercurrent Alarm Occur Threshold	ANTENNAPORT.UOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

Parameter Name	Parameter ID	Setting Notes
Undercurrent Alarm Clear Threshold	ANTENNAPORT.UCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Occur Threshold	ANTENNAPORT.OOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Clear Threshold	ANTENNAPORT.OCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only when the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, the **DEFCURRALMTHLD** MO can be used to modify the current alarm thresholds included in this type. The following table describes the key parameters involved.

Table 4-42 Key parameters in the default current alarm threshold MO **DEFCURRALMTHLD**

Parameter Name	Parameter ID	Setting Notes
Port Type	DEFCURRALMTHLD.PORTTYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to ANTENNAPORT .
Threshold Type	DEFCURRALMTHLD.THRESHOLDTYPE (LTE eNodeB, 5G gNodeB)	This parameter specifies the to-be-adjusted type associated with a set of default current alarm thresholds.
Adjusted Undercurrent Alarm Occur Threshold	DEFCURRALMTHLD.UOTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted undercurrent alarm occur threshold. For the specified type, the default undercurrent alarm occur threshold defined in the ANTENNAPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Undercurrent Alarm Clear Threshold	DEFCURRALMTHLD.UCTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted undercurrent alarm clear threshold. For the specified type, the default undercurrent alarm clear threshold defined in the ANTENNAPORT MO no longer takes effect, and the adjusted threshold is used instead.

Parameter Name	Parameter ID	Setting Notes
Adjusted Overcurrent Alarm Occur Threshold	DEFCURRALMTHLD.OOTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted overcurrent alarm occur threshold. For the specified type, the default overcurrent alarm occur threshold defined in the ANTENNA MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Clear Threshold	DEFCURRALMTHLD.OCTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the adjusted overcurrent alarm clear threshold. For the specified type, the default overcurrent alarm clear threshold defined in the ANTENNA MO no longer takes effect, and the adjusted threshold is used instead.

Scenario 3: Connection to the RET Antenna Through the Antenna Port (with a TMA)

For key parameters that must be set to configure the RET device, RET subunit, and RET device property group in this scenario, see [Table 4-37](#), [Table 4-38](#), and [Table 4-40](#) of scenario 1.

For the key parameters that must be set to configure the antenna port in this scenario, see [Table 4-41](#) of scenario 2.

[Table 4-43](#) describes the key parameters that must be set to configure the TMA.

Table 4-43 Key parameters that must be set to configure the TMA

Parameter Name	Parameter ID	Setting Notes
Device No.	NGALD.DEVICENO (5G gNodeB, LTE eNodeB)	The device number of each ALD in a base station must be unique. Even ALDs of different types (such as an RET antenna and a TMA) cannot have the same device number.
Device Name	NGALD.DEVICENAME (5G gNodeB, LTE eNodeB)	This parameter identifies an RET antenna. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.

Parameter Name	Parameter ID	Setting Notes
Control Port Cabinet No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located. Set these parameters based on connections between the TMA and the RRU/RFU.
Control Port Subrack No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	
Control Port Slot No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	
ALD Device type	NGALD.ALDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter to TMA .
Number of ALD Subunits	NGALD.SUBUNITNUM (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual device conditions. Generally, a TMA has two subunits.
Vendor Code	NGALD.VENDORCODE (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual TMA manufacturer code. The manufacturer code and serial number are required in a non-regular scenario.
Serial No.	NGALD.SERIALNO (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual TMA serial number. The manufacturer code and serial number are required in a non-regular scenario.
TMA Type	NGALD.TMATYPE (5G gNodeB, LTE eNodeB)	This parameter specifies the TMA type. The value NORMAL_TMA indicates a normal TMA. The value VIRTUAL_TMA indicates a device other than a TMA which amplifies signals between an RF port and an antenna port.

[Table 4-44](#) describes the key parameters that must be set to configure a TMA subunit.

Table 4-44 Key parameters that must be set to configure a TMA subunit

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVICEN (5G gNodeB, LTE eNodeB)	This parameter indicates the device number of the antenna provided during the TMA configuration.
Subunit No.	ALDSUBUNIT.SUBUNITNO (5G gNodeB, LTE eNodeB)	This parameter indicates the number of a TMA subunit.

Parameter Name	Parameter ID	Setting Notes
Work Mode	ALDSUBUNIT.MODE (5G gNodeB, LTE eNodeB)	The TMA subunit supports two working modes, normal mode and bypass mode: - In normal mode, the TMA subunit functions and the TMA amplifies uplink signals. - In bypass mode, the TMA subunit works as a straight-through feeder. It does not amplify any uplink signals. The default value is NORMAL .
Gain	ALDSUBUNIT.GAIN (5G gNodeB, LTE eNodeB)	Set this parameter based on the engineering design. The gain value range supported by the TMA varies with the manufacturer and model. Run the DSP TMADEVICEDATA command (eGBTS/NodeB/eNodeB/gNodeB) or the DSP BTSTMADEVICEDATA command (GBTS) to query the gain value range supported by the TMA before setting the gain for the TMA subunit. If the gain is fixed, this parameter is optional, or you can set this parameter to its actual gain value.
RF Path ID	RFPATH.RFPATHID (5G gNodeB, LTE eNodeB)	This parameter indicates the ID of an RF path.
RF Path Alias	RFPATH.ALIAS (5G gNodeB, LTE eNodeB)	This parameter indicates the connection path between the RF module and the ALD.
Cabinet No.	RFPATH.CN (5G gNodeB, LTE eNodeB)	Set these parameters based on connections between the TMA and the RF port (control port) on the RRU/RFU.
Connect Port Subrack No.	RFPATH.SRN (5G gNodeB, LTE eNodeB)	
Connect Port Slot No.	RFPATH.SN (5G gNodeB, LTE eNodeB)	
Connect Port No.	RFPATH.PN (5G gNodeB, LTE eNodeB)	
Connection ID	ALDCONNINFO.CONNID (5G gNodeB, LTE eNodeB)	Set these parameters based on the connections between the ALD and the antenna ports on the RF module.
Connection Type	ALDCONNINFO.CONNTYPE (5G gNodeB, LTE eNodeB)	
Device No.	ALDCONNINFO.DEVICENO (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
ALD Port No.	ALDCONNINFO.ALDPORTNO (5G gNodeB, LTE eNodeB)	
Subunit No.	ALDCONNINFO.SUBUNITNO (5G gNodeB, LTE eNodeB)	
RF Path ID	ALDCONNINFO.RFPATHID (5G gNodeB, LTE eNodeB)	

Table 4-45 describes the key parameters that must be set to configure the RX channel attenuation.

Table 4-45 Key parameters that must be set to configure the RX channel attenuation

Parameter Name	Parameter ID	Setting Notes
RX Channel No.	RXBRANCH.RXNO (RFA LampSite, 5G gNodeB, LTE eNodeB)	This parameter specifies the number of the RX channel of the RRU/RFU.
Logical Switch of RX Channel	RXBRANCH.RXSW (RFA LampSite, 5G gNodeB, LTE eNodeB)	This parameter specifies the logical switch for the RX channel of the RRU/RFU. The default value is ON .
Attenuation	RXBRANCH.ATTEN (RFA LampSite, 5G gNodeB, LTE eNodeB)	If no TMA is used, set this parameter to 0 . If a 12 dB TMA is used, set this parameter to a value within the range from 4 dB to 11 dB. If a 24 dB TMA is used, set this parameter to a value within the range from 11 dB to 22 dB.

Table 4-46 describes the key parameters that must be set to configure TMA device properties.

Table 4-46 Key parameters that must be set to configure TMA device properties

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDDEVICEDATA.DEVICENO (5G gNodeB, LTE eNodeB)	This parameter indicates the device number of the antenna provided during the TMA configuration.
Subunit No.	ALDDEVICEDATA.SUBUNITNO (5G gNodeB, LTE eNodeB)	This parameter indicates the number of the TMA subunit.

Parameter Name	Parameter ID	Setting Notes
Antenna Model Number	ALDDEVICEDATA.MODE LNO (5G gNodeB, LTE eNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Antenna Serial No.	ALDDEVICEDATA.SERIAL NO (5G gNodeB, LTE eNodeB)	
Band1	ALDDEVICEDATA.BAND1 (5G gNodeB, LTE eNodeB)	These parameters specify the frequency bands supported by an antenna and the corresponding beamwidths. These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Beamwidth1	ALDDEVICEDATA.BEAM WIDTH1 (5G gNodeB, LTE eNodeB)	
Gain1	ALDDEVICEDATA.GAIN1 (5G gNodeB, LTE eNodeB)	
Band2	ALDDEVICEDATA.BAND2 (5G gNodeB, LTE eNodeB)	
Beamwidth2	ALDDEVICEDATA.BEAM WIDTH2 (5G gNodeB, LTE eNodeB)	
Gain2	ALDDEVICEDATA.GAIN2 (5G gNodeB, LTE eNodeB)	
Band3	ALDDEVICEDATA.BAND3 (5G gNodeB, LTE eNodeB)	
Beamwidth3	ALDDEVICEDATA.BEAM WIDTH3 (5G gNodeB, LTE eNodeB)	
Gain3	ALDDEVICEDATA.GAIN3 (5G gNodeB, LTE eNodeB)	
Band4	ALDDEVICEDATA.BAND4 (5G gNodeB, LTE eNodeB)	
Beamwidth4	ALDDEVICEDATA.BEAM WIDTH4 (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
Gain4	ALDDEVICEDATA.GAIN4 (5G gNodeB, LTE eNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Installation Date	ALDDEVICEDATA.DATE (5G gNodeB, LTE eNodeB)	
Installer's ID	ALDDEVICEDATA.INSTALLERID (5G gNodeB, LTE eNodeB)	
Base Station ID	ALDDEVICEDATA.BSID (5G gNodeB, LTE eNodeB)	
AISG Sector ID	ALDDEVICEDATA.SECTORID (5G gNodeB, LTE eNodeB)	
Antenna Bearing	ALDDEVICEDATA.BEARING (5G gNodeB, LTE eNodeB)	
Installed Mechanical Tilt	ALDDEVICEDATA.TILT (5G gNodeB, LTE eNodeB)	

Scenario 4: Connection to the RET Antenna Through the RET Port (with a TMA)

For key parameters that must be set to configure an RET device, RET subunit, and RET device properties in this scenario, see [Table 4-37](#), [Table 4-38](#), and [Table 4-40](#) of scenario 1.

For key parameters that must be set to configure an RET port (control port) in this scenario, see [Table 4-35](#) of scenario 1.

For key parameters that must be set to configure a TMA device, TMA subunit, RX channel attenuation, and TMA device properties in this scenario, see [Table 4-43](#), [Table 4-44](#), [Table 4-45](#), and [Table 4-46](#) of scenario 3.

Scenario 5: MU Controlling the RET Function of the AAS Module with Passive Antennas

For key parameters that must be set to configure the RET device properties for an AAS module with passive antennas, see [Table 4-40](#) of scenario 1.

[Table 4-47](#) describes the key parameters that must be set to configure the RET function for an AAS module with passive antennas.

Table 4-47 Key parameters that must be set for the MU to control the RET function for an AAS module with passive antennas

Parameter Name	Parameter ID	Setting Notes
Device No.	NGALD.DEVICENO (5G gNodeB, LTE eNodeB)	The device number of each ALD in a base station must be unique. Even ALDs of different types (such as an RET antenna and a TMA) cannot have the same device number.
Device Name	NGALD.DEVICENAME (5G gNodeB, LTE eNodeB)	This parameter identifies an RET antenna. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	These parameters specify the numbers of the cabinet, subrack, and slot where an AAS module is located.
Control Port Subrack No.	NGALD.CTRLRN (5G gNodeB, LTE eNodeB)	
Control Port Slot No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	
ALD Device type	NGALD.ALDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter to SINGLE_RET or MULTI_RET based on the actual RET antenna. Set this parameter to SINGLE_RET if the base station uses the AAS module with passive antennas.
Number of ALD Subunits	NGALD.SUBUNITNUM (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual number of antenna subunits for a multi-subunit antenna. Set this parameter to 1 for AAS modules with passive antennas.
Polar Type	NGALD.POLARTYPE (5G gNodeB, LTE eNodeB)	Set this parameter based on the AAS module specifications.
Antenna Scenario	NGALD.SCENARIO (5G gNodeB, LTE eNodeB)	If an AAS module with passive antennas is used, set the antenna scenario parameter to DAISY_CHAIN even when you need to use only one set of antennas for the RET function. In this case, specify the VENDORCODE and SERIALNO parameters to determine the specific set of antennas whose RET function is used. Only the RET function of one set of antennas needs to be configured because the antennas of the AAS module are connected and controlled in daisy chain mode.

Parameter Name	Parameter ID	Setting Notes
Vendor Code	NGALD.VENDORCODE <i>(5G gNodeB, LTE eNodeB)</i>	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the actual vendor code, for example, HW for Huawei Agissson.
Serial No.	NGALD.SERIALNO <i>(5G gNodeB, LTE eNodeB)</i>	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number.

Table 4-48 describes the key parameters that must be set to configure the RET subunit of an AAS module with passive antennas.

Table 4-48 Key parameters that must be set to configure the RET subunit of an AAS module with passive antennas

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVICENO <i>(5G gNodeB, LTE eNodeB)</i>	This parameter indicates the antenna device number provided during the RET configuration.
Subunit No.	ALDSUBUNIT.SUBUNITNO <i>(5G gNodeB, LTE eNodeB)</i>	This parameter indicates the ID of the subunit, which starts from 1.
Subunit Name	ALDSUBUNIT.SUBUNITNAME <i>(5G gNodeB, LTE eNodeB)</i>	The name of the RET subunit is optional, but the name configured for each RET subunit must be unique.
Tilt	ALDSUBUNIT.TILT <i>(5G gNodeB, LTE eNodeB)</i>	Set this parameter based on the engineering design.
RF Path ID	RFPATH.RFPATHID <i>(5G gNodeB, LTE eNodeB)</i>	This parameter indicates the ID of an RF path.
RF Path Alias	RFPATH.ALIAS <i>(5G gNodeB, LTE eNodeB)</i>	This parameter indicates the connection path between the RF module and the ALD.
Connect Port 1 Cabinet No.	RFPATH.CN <i>(5G gNodeB, LTE eNodeB)</i>	Set these parameters based on connections between the AAS module and the RF ports on the RRU/RFU.
Connect Port 1 Subrack No.	RFPATH.SRN <i>(5G gNodeB, LTE eNodeB)</i>	
Connect Port 1 Slot No.	RFPATH.SN <i>(5G gNodeB, LTE eNodeB)</i>	
Connect Port 1 Port No.	RFPATH.PN <i>(5G gNodeB, LTE eNodeB)</i>	

Parameter Name	Parameter ID	Setting Notes
Connect Port 2 Cabinet No.	RFPATH.CN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Subrack No.	RFPATH.SRN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Slot No.	RFPATH.SN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Port No.	RFPATH.PN (5G gNodeB, LTE eNodeB)	
Connection ID	ALDCONNINFO.CONN ID (5G gNodeB, LTE eNodeB)	Set these parameters based on the connections between the ALD and the antenna ports on the RF module.
Connection Type	ALDCONNINFO.CONN TYPE (5G gNodeB, LTE eNodeB)	
Device No.	ALDCONNINFO.DEVIC ENO (5G gNodeB, LTE eNodeB)	
ALD Port No.	ALDCONNINFO.ALDP ORTNO (5G gNodeB, LTE eNodeB)	
Subunit No.	ALDCONNINFO.SUBU NITNO (5G gNodeB, LTE eNodeB)	
RF Path ID	ALDCONNINFO.RFPAT HID (5G gNodeB, LTE eNodeB)	

Table 4-49 describes the key parameter that must be set to configure the RET tilt of an AAS module with passive antennas.

Table 4-49 Key parameter that must be set to configure the RET tilt of an AAS module with passive antennas

Parameter Name	Parameter ID	Setting Notes
Tilt	ALDSUBUNIT.TILT (5G gNodeB, LTE eNodeB)	Set this parameter based on the engineering design.

Scenario 6: AAS Active Module (RU) Controlling the RET Function of AAS Module with Passive Antennas

When the RU of the AAS module is used to control the RET function:

- For the AAU3910/AAU3911/AAU3961, the RET function is controlled by the RET port on the RU. [Table 4-35](#) in scenario 1 describes the key parameters that must be set to configure the RET port.
- For the AAU3920/AAU3940/AAU5240/AAU5243/AAU5940/AAU5942, the RET function is controlled by the antenna port on the RU. [Table 4-41](#) in scenario 2 describes the key parameters that must be set to configure the antenna port, and the **PN (5G gNodeB, LTE eNodeB)** parameter in this MO must be set to **ROA**.
- Configuring the RET part on AAS modules with passive antennas:
[Table 4-37](#) in scenario 1 describes the key parameters that must be set to configure the RET function of an AAS module with passive antennas.
[Table 4-38](#) in scenario 1 describes the key parameters that must be set to configure the RET subunit of an AAS module with passive antennas.
[Table 4-39](#) in scenario 1 describes the key parameters that must be set to configure the RET tilt of an AAS module with passive antennas.
[Table 4-40](#) in scenario 1 describes the key parameters that must be set to configure the RET device data of an AAS module with passive antennas.
- Configuring the RVD part on AAS modules with passive antennas:
[Table 4-50](#) describes the key parameters that must be set to configure the RVD of an AAS module with passive antennas.

Table 4-50 Key parameters that must be set to configure the RVD

Parameter Name	Parameter ID	Setting Notes
Device No.	NGALD.DEVICENO (5G gNodeB, LTE eNodeB)	The device number of the ALD must be unique.
Device Name	NGALD.DEVICENAME (5G gNodeB, LTE eNodeB)	This parameter specifies the device name of the ALD, which identifies the ALD. It is recommended that you set the device name in the following format: <i>sector_device type_network type</i> . For details, see the related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	NGALD.CTRL CN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located.
Control Port Subrack No.	NGALD.CTRL SRN (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
Control Port Slot No.	NGALD.CTRLN <i>(5G gNodeB, LTE eNodeB)</i>	
ALD Device type	NGALD.ALDTYPE <i>(5G gNodeB, LTE eNodeB)</i>	Set this parameter to RVD .
Number of ALD Subunits	NGALD.SUBUNITN <i>UM (5G gNodeB, LTE eNodeB)</i>	This parameter specifies the number of configured RVD subunits.
Antenna Scenario	NGALD.SCENARIO <i>(5G gNodeB, LTE eNodeB)</i>	Set this parameter based on the actual hardware installation. The vendor code and serial number of the device must be correctly configured in a non-regular scenario. This parameter can be set to REGULAR or DAISY_CHAIN .
Vendor Code	NGALD.VENDORCODE <i>(5G gNodeB, LTE eNodeB)</i>	- When SCENARIO in an RVD MO is not set to REGULAR, VENDORCODE must be set. - When SCENARIO in an RVD MO is set to REGULAR and SERIALNO is set, VENDORCODE must be set.
Serial No.	NGALD.SERIALNO <i>(5G gNodeB, LTE eNodeB)</i>	The vendor code and the serial number uniquely identify an ALD. - When SCENARIO in an RVD MO is not set to REGULAR, SERIALNO must be set. - When SCENARIO in an RVD MO is set to REGULAR and VENDORCODE is set, SERIALNO must be set.

Table 4-51 describes the key parameters that must be set to configure the RVD subunit of an AAS module with passive antennas.

Table 4-51 Key parameters that must be set to configure an RVD subunit

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVICENO <i>(5G gNodeB, LTE eNodeB)</i>	The device number of the ALD must be unique.
Subunit No.	ALDSUBUNIT.SUBUNITNO <i>(5G gNodeB, LTE eNodeB)</i>	This parameter indicates the number of the RVD subunit, which starts from 1.

Parameter Name	Parameter ID	Setting Notes
Subunit Name	ALDSUBUNIT.SUBNAME (5G gNodeB, LTE eNodeB)	This parameter indicates the name of the RVD subunit. The name of the RVD subunit is optional, but the name configured for each RVD subunit must be unique.
Vertical Beam Width	ALDSUBUNIT.VBEAMWIDTH (5G gNodeB, LTE eNodeB)	This parameter indicates the vertical beamwidth of an RVD subunit.
RF Path ID	RFPATH.RFPATHID (5G gNodeB, LTE eNodeB)	This parameter indicates the ID of an RF path.
RF Path Alias	RFPATH.ALIAS (5G gNodeB, LTE eNodeB)	This parameter indicates the connection path between the RF module and the ALD.
Connect Port 1 Cabinet No.	RFPATH.CN (5G gNodeB, LTE eNodeB)	These parameters indicate the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 1, respectively.
Connect Port 1 Subrack No.	RFPATH.SRN (5G gNodeB, LTE eNodeB)	
Connect Port 1 Slot No.	RFPATH.SN (5G gNodeB, LTE eNodeB)	
Connect Port 1 Port No.	RFPATH.PN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Cabinet No.	RFPATH.CN (5G gNodeB, LTE eNodeB)	These parameters indicate the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 2, respectively. They are valid when a dual-polarized antenna is used.
Connect Port 2 Subrack No.	RFPATH.SRN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Slot No.	RFPATH.SN (5G gNodeB, LTE eNodeB)	
Connect Port 2 Port No.	RFPATH.PN (5G gNodeB, LTE eNodeB)	
Connection ID	ALDCONNINFO.CONNID (5G gNodeB, LTE eNodeB)	Set these parameters based on the connections between the ALD and the antenna ports on the RF module.

Parameter Name	Parameter ID	Setting Notes
Connection Type	ALDCONNINFO.CO NNTYPE (5G gNodeB, LTE eNodeB)	
Device No.	ALDCONNINFO.DEV ICENO (5G gNodeB, LTE eNodeB)	
ALD Port No.	ALDCONNINFO.ALD PORTNO (5G gNodeB, LTE eNodeB)	
Subunit No.	ALDCONNINFO.SUB UNITNO (5G gNodeB, LTE eNodeB)	
RF Path ID	ALDCONNINFO.RFP ATHID (5G gNodeB, LTE eNodeB)	

Table 4-52 describes the key parameters that must be set to configure the vertical beamwidth or working mode of the programmable antenna for the RVD subunit of the AAS module with passive antennas.

Table 4-52 Key parameters that must be set in the **ALDSUBUNIT** MO for the RVD subunit of the AAS module with passive antennas

Parameter Name	Parameter ID	Setting Notes
Vertical Beam Width	ALDSUBUNIT.VBEAM WIDTH (5G gNodeB, LTE eNodeB)	This parameter indicates the vertical beamwidth of an RVD subunit.
Software-defined Antenna Mode	ALDSUBUNIT.SDAM ODE (5G gNodeB, LTE eNodeB)	This parameter indicates the working mode of the programmable antenna.

Table 4-53 describes the key parameters that must be set to configure the property group for the RVD of an AAS module with passive antennas.

Table 4-53 Key parameters that must be set to configure the RVD property group

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDDEVICEDATA.DEVICENO (5G gNodeB, LTE eNodeB)	The device number of the ALD must be unique.
Subunit No.	ALDDEVICEDATA.SUBUNITNO (5G gNodeB, LTE eNodeB)	This parameter indicates the number of the RVD subunit, which starts from 1.
Antenna Model Number	ALDDEVICEDATA.MODELNO (5G gNodeB, LTE eNodeB)	This parameter indicates the model of the RVD.
Antenna Serial No.	ALDDEVICEDATA.SERIALNO (5G gNodeB, LTE eNodeB)	This parameter specifies the serial number of the RVD.
Band1	ALDDEVICEDATA.BAND1 (5G gNodeB, LTE eNodeB)	These parameters specify the bands supported by the antenna and the corresponding beamwidths. - In an RVDDEVICEDATA MO, when BAND4 is not set to UNUSED, BAND3 cannot be set to UNUSED. When BAND3 is not set to UNUSED, BAND2 cannot be set to UNUSED. When BAND2 is not set to UNUSED, BAND1 cannot be set to UNUSED. - In an RVDDEVICEDATA MO, when BAND4 is not set to UNUSED, BAND4 must be greater than BAND3. When BAND3 is not set to UNUSED, BAND3 must be greater than BAND2. When BAND2 is not set to UNUSED, BAND2 must be greater than BAND1.
Beamwidth1	ALDDEVICEDATA.BEAMWIDTH1 (5G gNodeB, LTE eNodeB)	
Gain1	ALDDEVICEDATA.GAIN1 (5G gNodeB, LTE eNodeB)	
Band2	ALDDEVICEDATA.BAND2 (5G gNodeB, LTE eNodeB)	
Beamwidth2	ALDDEVICEDATA.BEAMWIDTH2 (5G gNodeB, LTE eNodeB)	
Gain2	ALDDEVICEDATA.GAIN2 (5G gNodeB, LTE eNodeB)	
Band3	ALDDEVICEDATA.BAND3 (5G gNodeB, LTE eNodeB)	
Beamwidth3	ALDDEVICEDATA.BEAMWIDTH3 (5G gNodeB, LTE eNodeB)	
Gain3	ALDDEVICEDATA.GAIN3 (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
Band4	ALDDEVICEDATA.BAND4 (5G gNodeB, LTE eNodeB)	
Beamwidth4	ALDDEVICEDATA.BEAMWIDTH4 (5G gNodeB, LTE eNodeB)	
Gain4	ALDDEVICEDATA.GAIN4 (5G gNodeB, LTE eNodeB)	
Installation Date	ALDDEVICEDATA.DATE (5G gNodeB, LTE eNodeB)	This parameter indicates the date when the antenna is installed.
Installer's ID	ALDDEVICEDATA.INSTALLERID (5G gNodeB, LTE eNodeB)	This parameter indicates the ID of the installation technician who installed the antenna.
Base Station ID	ALDDEVICEDATA.BSID	This parameter indicates the ID of the base station served by the antenna.
AISG Sector ID	ALDDEVICEDATA.SECTORID (5G gNodeB, LTE eNodeB)	This parameter indicates the AISG sector ID.
Antenna Bearing	ALDDEVICEDATA.BEARING (5G gNodeB, LTE eNodeB)	This parameter indicates the antenna azimuth.
Installed Mechanical Tilt	ALDDEVICEDATA.TILT (5G gNodeB, LTE eNodeB)	This parameter indicates the mechanical tilt of the installation.

Scenario 7: Connection to the RAE Unit Through the RET Port

For key parameters that must be set to configure an RET port (control port) in this scenario, see [Table 4-35](#) of scenario 1.

[Table 4-54](#) describes the key parameters that must be set to configure the RAE unit.

Table 4-54 Key parameters that must be set to configure the RAE unit

Parameter Name	Parameter ID	Setting Notes
Device No.	NGALD.DEVICENO (5G gNodeB, LTE eNodeB)	The device number of each ALD in a base station must be unique. Even ALDs of different types (such as an RET antenna and an RAE unit) cannot have the same device number.
Device Name	NGALD.DEVICENAME (5G gNodeB, LTE eNodeB)	This parameter identifies a device. The recommended format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RF module where the control port is located. Set these parameters based on connections between the RAE unit and the RF module.
Control Port Subrack No.	NGALD.CTRLRN (5G gNodeB, LTE eNodeB)	
Control Port Slot No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	
ALD Device type	NGALD.ALDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter to RAE .
Number of ALD Subunits	NGALD.SUBUNITNUM (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual device conditions. Generally, the value is 1 .
Antenna Scenario	NGALD.SCENARIO (5G gNodeB, LTE eNodeB)	This parameter indicates the scenario where an RAE unit is connected to the RF module. The antenna scenario must be set based on the hardware installation. Set this parameter to REGULAR if the RAE unit is directly connected to the RF module. In this scenario, the VENDORCODE and SERIALNO parameters do not need to be specified. Set this parameter to DAISY_CHAIN when the RAE unit is cascaded with another RAE unit. In this scenario, the control port for the RAE unit must be configured on the upper-level RF module of the daisy chain. The VENDORCODE and SERIALNO parameters must be specified.
Vendor Code	NGALD.VENDORCODE (5G gNodeB, LTE eNodeB)	Set this parameter based on the manufacturer information. This parameter is mandatory in daisy chain scenarios.
Serial No.	NGALD.SERIALNO (5G gNodeB, LTE eNodeB)	Set this parameter based on the actual antenna serial number. This parameter is mandatory in daisy chain scenarios.

Table 4-55 describes the key parameters that must be set to configure an RAE subunit.

Table 4-55 Key parameters that must be set to configure an RAE subunit

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVICENO (5G gNodeB, LTE eNodeB)	This parameter indicates the ALD number provided during the RAE configuration.
Subunit No.	ALDSUBUNIT.SUBUNITNO (5G gNodeB, LTE eNodeB)	This parameter indicates the number of the RAE subunit.
Subunit Name	ALDSUBUNIT.SUBNAME (5G gNodeB, LTE eNodeB)	This parameter indicates the name of the RAE subunit. The name of the RAE subunit is optional, but the name configured for each RAE subunit must be unique.
RF Path ID	RFPATH.RFPATHID (5G gNodeB, LTE eNodeB)	This parameter indicates the ID of an RF path.
RF Path Alias	RFPATH.ALIAS (5G gNodeB, LTE eNodeB)	This parameter indicates the connection path between the RF module and the ALD.
Cabinet No.	RFPATH.CN (5G gNodeB, LTE eNodeB)	Set these parameters based on connections between the RAE subunits and the antenna ports on the RF module.
Connect Port Subrack No.	RFPATH.SRN (5G gNodeB, LTE eNodeB)	
Connect Port Slot No.	RFPATH.SN (5G gNodeB, LTE eNodeB)	
Connection ID	ALDCONNINFO.CONNID (5G gNodeB, LTE eNodeB)	Set these parameters based on the connections between the ALD and the antenna ports on the RF module.
Connection Type	ALDCONNINFO.CONNTYPE (5G gNodeB, LTE eNodeB)	
Device No.	ALDCONNINFO.DEVICENO (5G gNodeB, LTE eNodeB)	
ALD Port No.	ALDCONNINFO.ALDPORTNO (5G gNodeB, LTE eNodeB)	
Subunit No.	ALDCONNINFO.SUBUNITNO (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
RF Path ID	ALDCONNINFO.RFPATHID (5G gNodeB, LTE eNodeB)	

Table 4-56 describes the key parameters that must be set to configure the RAE device property group.

Table 4-56 Key parameters that must be set to configure the RAE device property group

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDDEVICEDATA.DEVICENO (5G gNodeB, LTE eNodeB)	This parameter indicates the ALD number provided during the RAE configuration.
Subunit No.	ALDDEVICEDATA.SUBUNITNO (5G gNodeB, LTE eNodeB)	This parameter indicates the number of the RAE subunit.
Antenna Model Number	ALDDEVICEDATA.MODELNO (5G gNodeB, LTE eNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Antenna Serial No.	ALDDEVICEDATA.SERIALNO (5G gNodeB, LTE eNodeB)	
Band1	ALDDEVICEDATA.BAND1 (5G gNodeB, LTE eNodeB)	These parameters specify the frequency bands supported by an antenna and the corresponding beamwidths. These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Beamwidth1	ALDDEVICEDATA.BEAMWIDTH1 (5G gNodeB, LTE eNodeB)	
Gain1	ALDDEVICEDATA.GAIN1 (5G gNodeB, LTE eNodeB)	
Band2	ALDDEVICEDATA.BAND2 (5G gNodeB, LTE eNodeB)	
Beamwidth2	ALDDEVICEDATA.BEAMWIDTH2 (5G gNodeB, LTE eNodeB)	
Gain2	ALDDEVICEDATA.GAIN2 (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
Band3	ALDDEVICEDATA.BAND3 (5G gNodeB, LTE eNodeB)	
Beamwidth3	ALDDEVICEDATA.BEAMWIDTH3 (5G gNodeB, LTE eNodeB)	
Gain3	ALDDEVICEDATA.GAIN3 (5G gNodeB, LTE eNodeB)	
Band4	ALDDEVICEDATA.BAND4 (5G gNodeB, LTE eNodeB)	
Beamwidth4	ALDDEVICEDATA.BEAMWIDTH4 (5G gNodeB, LTE eNodeB)	
Gain4	ALDDEVICEDATA.GAIN4 (5G gNodeB, LTE eNodeB)	
Installation Date	ALDDEVICEDATA.INSTALLATIONDATE (5G gNodeB, LTE eNodeB)	These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Installer's ID	ALDDEVICEDATA.INSTALLERID (5G gNodeB, LTE eNodeB)	
Base Station ID	ALDDEVICEDATA.BSID (5G gNodeB, LTE eNodeB)	
AISG Sector ID	ALDDEVICEDATA.AISGSECTORID (5G gNodeB, LTE eNodeB)	

Scenario 8: Connection to the RET and RAE Devices Through the RET Port

For key parameters that must be set to configure the RET device, RET subunit, and RET device property group in this scenario, see [Table 4-37](#), [Table 4-38](#), and [Table 4-40](#) of scenario 1.

For key parameters that must be set to configure the RET port in this scenario, see [Table 4-35](#).

For the key parameters that must be set to configure the RAE unit, RAE subunit, and RAE device property group in this scenario, see [Table 4-54](#), [Table 4-55](#), and [Table 4-56](#) of scenario 9.

Scenario 9: Connection to the RAE Unit Through the Antenna Port

For the key parameters that must be set to configure the antenna port in this scenario, see [Table 4-41](#) of scenario 2.

For the key parameters that must be set to configure the RAE unit, RAE subunit, and RAE device property group in this scenario, see [Table 4-54](#), [Table 4-55](#), and [Table 4-56](#) of scenario 9.

Scenario 10: Connection to the RET and RAE Devices Through the Antenna Port

For key parameters that must be set to configure the RET device, RET subunit, and RET device property group in this scenario, see [Table 4-37](#), [Table 4-38](#), and [Table 4-40](#) of scenario 1.

For the key parameters that must be set to configure the antenna port in this scenario, see [Table 4-41](#) of scenario 2.

For the key parameters that must be set to configure the RAE unit, RAE subunit, and RAE device property group in this scenario, see [Table 4-54](#), [Table 4-55](#), and [Table 4-56](#) of scenario 9.

Scenario 11: Connection to the TMA and RAE Unit Through the Antenna Port

For the key parameters that must be set to configure the TMA, TMA subunit, RX channel attenuation, and TMA device properties in this scenario, see [Table 4-43](#), [Table 4-44](#), [Table 4-45](#), and [Table 4-46](#) of scenario 3.

For the key parameters that must be set to configure the antenna port in this scenario, see [Table 4-41](#) of scenario 2.

For the key parameters that must be set to configure the RAE unit, RAE subunit, and RAE device property group in this scenario, see [Table 4-54](#), [Table 4-55](#), and [Table 4-56](#) of scenario 9.

Scenario 12: Connection to the TMA, RET, and RAE Devices Through the Antenna Port

For the key parameters that must be set to configure the TMA, TMA subunit, RX channel attenuation, and TMA device properties in this scenario, see [Table 4-43](#), [Table 4-44](#), [Table 4-45](#), and [Table 4-46](#) of scenario 3.

For key parameters that must be set to configure the RET device, RET subunit, and RET device property group in this scenario, see [Table 4-37](#), [Table 4-38](#), and [Table 4-40](#) of scenario 1.

For the key parameters that must be set to configure the antenna port in this scenario, see [Table 4-41](#) of scenario 2.

For the key parameters that must be set to configure the RAE unit, RAE subunit, and RAE device property group in this scenario, see [Table 4-54](#), [Table 4-55](#), and [Table 4-56](#) of scenario 9.

Scenario 13: Connection to the SALD/MALD Through the RET Port

Table 4-57 describes the key parameters that must be set to configure the control port (RET port) in this scenario.

Table 4-57 Key parameters that must be set to configure the control port (RET port)

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RETPORT.CN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU where the control port is located and the control port number. Set these parameters based on connections between the RET antenna and the RRU. Only one port on the RRU can be used as the control port for the RET antenna. In a daisy chain scenario, multiple SALDs/MALDs can share one control port.
Subrack No.	RETPORT.SRN (5G gNodeB, LTE eNodeB)	
Slot No.	RETPORT.SN (5G gNodeB, LTE eNodeB)	
Port No.	RETPORT.PN (5G gNodeB, LTE eNodeB)	
ALD Power Switch	RETPORT.PWRSWITCH (5G gNodeB, LTE eNodeB)	Set this parameter to ON when an ALD is used. The default value is OFF .
Current Alarm Threshold Type	RETPORT.THRESHOLDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter based on the device conditions and actual connection scenarios. For details, see Table 4-1 .
Undercurrent Alarm Occur Threshold	RETPORT.UOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Undercurrent Alarm Clear Threshold	RETPORT.UCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Occur Threshold	RETPORT.OOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Clear Threshold	RETPORT.OCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, the **DEFCURRALMTHLD** MO can be used to modify the current alarm thresholds included in this type. The following table describes the key parameters involved.

Table 4-58 Key parameters in the default current alarm threshold MO
DEFCURRALMTHLD

Parameter Name	Parameter ID	Setting Notes
Port Type	DEFCURRALMTHLD.P ORTTYPE	Set this parameter to RETPORT .
Threshold Type	DEFCURRALMTHLD.T HRESHOLDTYPE	This parameter specifies the to-be-adjusted type associated with a set of default current alarm thresholds.
Adjusted Undercurrent Alarm Occur Threshold	DEFCURRALMTHLD.U OTH	This parameter specifies the adjusted undercurrent alarm occur threshold. For the specified type, the default undercurrent alarm occur threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Undercurrent Alarm Clear Threshold	DEFCURRALMTHLD.U CTH	This parameter specifies the adjusted undercurrent alarm clear threshold. For the specified type, the default undercurrent alarm clear threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Occur Threshold	DEFCURRALMTHLD. OOTH	This parameter specifies the adjusted overcurrent alarm occur threshold. For the specified type, the default overcurrent alarm occur threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Clear Threshold	DEFCURRALMTHLD. OCTH	This parameter specifies the adjusted overcurrent alarm clear threshold. For the specified type, the default overcurrent alarm clear threshold defined in the RETPORT MO no longer takes effect, and the adjusted threshold is used instead.

Table 4-59 describes the key parameters that must be set to configure the SALD/MALD device.

Table 4-59 Key parameters that must be set to configure the SALD/MALD device

Parameter Name	Parameter ID	Setting Notes
Device No.	NGALD.DEVICENO (5G gNodeB, LTE eNodeB)	The device number of each ALD in a base station must be unique. Even ALDs of different types (such as the SALD and MALD) cannot have the same device number.
Device Name	NGALD.DEVICENAME (5G gNodeB, LTE eNodeB)	This parameter identifies a SALD/MALD. The format of the value is site_sector+port+device_type_network type. For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RF module where the control port is located. Set these parameters based on connections between the SALD/MALD and the RF module.
Control Port Subrack No.	NGALD.CTRLRN (5G gNodeB, LTE eNodeB)	
Control Port Slot No.	NGALD.CTRLN (5G gNodeB, LTE eNodeB)	
ALD Device type	NGALD.ALDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter to SALD or MALD based on the actual device situation.
Antenna Scenario	NGALD.SCENARIO (5G gNodeB, LTE eNodeB)	This parameter specifies the scenario where a SALD/MALD is connected to the RF module. The antenna scenario must be set based on the hardware installation. Set this parameter to REGULAR if the SALD/MALD is directly connected to the RF module. If only one SALD is connected to the RRU, the VENDORCODE and SERIALNO parameters do not need to be specified. If only one MALD is connected to the RRU, the VENDORCODE and SERIALNO parameters need to be specified. Set this parameter to DAISY_CHAIN when two SALDs/MALDs are cascaded. In this scenario, the control port for the RET antenna must be configured on the upper-level RF module of the daisy chain. The VENDORCODE and SERIALNO parameters must be specified.
Vendor Code	NGALD.VENDORCODE (5G gNodeB, LTE eNodeB)	For the SALD, the vendor code must be configured in a daisy chain scenario. For the MALD, the vendor code must be configured in both a regular installation scenario and a daisy chain scenario.

Parameter Name	Parameter ID	Setting Notes
Serial No.	NGALD.SERIALNO (5G gNodeB, LTE eNodeB)	For the SALD, the serial number must be configured in a daisy chain scenario. For the MALD, the serial number must be configured in both a regular installation scenario and a daisy chain scenario. Set this parameter to the actual serial number.

Table 4-60 describes the key parameters that must be set to configure the SALD/MALD subunit.

Table 4-60 Key parameters that must be set to configure the SALD/MALD subunit

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVICENO (5G gNodeB, LTE eNodeB)	Device number of the ALD when configuring the SALD/MALD
Subunit No.	ALDSUBUNIT.SUBUNITNO (5G gNodeB, LTE eNodeB)	This parameter indicates the ID of the subunit, which starts from 1.
Subunit Name	ALDSUBUNIT.SUBNAME (5G gNodeB, LTE eNodeB)	The subunit name identifies a SALD/MALD subunit. It is recommended that this parameter be set to a proper value. When setting this parameter, ensure that the name of each ALD subunit is unique.
Subunit Type	ALDSUBUNIT.SUBTYPE (5G gNodeB, LTE eNodeB)	This parameter specifies the type of the SALD/MALD subunit. Set this parameter to RET , TMA , or ADB as required.
Tilt	ALDSUBUNIT.TILT (5G gNodeB, LTE eNodeB)	Set this parameter only when the SALD/MALD subunit type is RET. Set this parameter based on the antenna tilt in the engineering design.
Work Mode	ALDSUBUNIT.MODE (5G gNodeB, LTE eNodeB)	Set this parameter only when the SALD/MALD subunit type is TMA. The TMA subunit supports two working modes: normal mode and bypass mode. - In normal mode, the TMA subunit functions and the TMA amplifies uplink signals. - In bypass mode, the TMA subunit works as a straight-through feeder. It does not amplify any uplink signals. The default value is NORMAL .

Parameter Name	Parameter ID	Setting Notes
Gain	ALDSUBUNIT.GAIN (5G gNodeB, LTE eNodeB)	This parameter is required only when the SALD/MALD subunit type is TMA. Set this parameter based on the engineering design. The gain value range supported by the TMA varies with the manufacturer and model. Run the DSP ALDSUBUNIT (eGBTS/NodeB/eNodeB/gNodeB) command to query the value range before setting the gain. If the gain is fixed, this parameter is optional, or you can set this parameter to its actual gain value.
AISG Sector ID	ALDSUBUNIT.SECTORID (5G gNodeB, LTE eNodeB)	This parameter needs to be configured only when the SALD/MALD subunit type is ADB. Set this parameter based on site requirements.
Installation Position in the Sector	ALDSUBUNIT.POSITIONWITHINSECTOR (5G gNodeB, LTE eNodeB)	This parameter needs to be configured only when the SALD/MALD subunit type is ADB. Set this parameter based on site requirements.
Installation Mechanical Tilt	ALDSUBUNIT.MECHANICALTILT (5G gNodeB, LTE eNodeB)	This parameter needs to be configured only when the SALD/MALD subunit type is ADB. Set this parameter based on site requirements.
Installation Antenna Azimuth	ALDSUBUNIT.MECHANICALBEARING (5G gNodeB, LTE eNodeB)	This parameter needs to be configured only when the SALD/MALD subunit type is ADB. Set this parameter based on site requirements.
RF Path ID	RFPATH.RFPATHID (5G gNodeB, LTE eNodeB)	This parameter indicates the ID of an RF path.
RF Path Alias	RFPATH.ALIAS (5G gNodeB, LTE eNodeB)	This parameter indicates the connection path between the RF module and the ALD.
Cabinet No.	RFPATH.CN (5G gNodeB, LTE eNodeB)	Set these parameters based on connections between the SALD/MALD subunits and the RF ports on the RRU/RFU.
Connect Port Subrack No.	RFPATH.SRN (5G gNodeB, LTE eNodeB)	
Connect Port Slot No.	RFPATH.SN (5G gNodeB, LTE eNodeB)	
Connection ID	ALDCONNINFO.CONNECTIONID (5G gNodeB, LTE eNodeB)	Set these parameters based on the connections between the ALD and the antenna ports on the RF module.
Connection Type	ALDCONNINFO.CONNECTIONTYPE (5G gNodeB, LTE eNodeB)	
Device No.	ALDCONNINFO.DEVICENO (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
ALD Port No.	ALDCONNINFO.ALDPOR TNO (5G gNodeB, LTE eNodeB)	
Subunit No.	ALDCONNINFO.SUBUN ITNO (5G gNodeB, LTE eNodeB)	
RF Path ID	ALDCONNINFO.RFPAT HID (5G gNodeB, LTE eNodeB)	

Scenario 14: Connection to the SALD/MALD Through the Antenna Port

Table 4-61 describes the key parameters that must be set to configure the control port (antenna port) in this scenario.

Table 4-61 Key parameters that must be set to configure the control port (antenna port)

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	ANTENNAPORT.CN (5G gNodeB, LTE eNodeB)	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU where the control port is located and the control port number. Set these parameters based on connections between the SALD/MALD and the RRU. Only one port on the RRU can be used as the control port for the SALD/MALD. In a daisy chain scenario, multiple SALDs/MALDs can share one control port.
Subrack No.	ANTENNAPORT.SRN (5G gNodeB, LTE eNodeB)	
Slot No.	ANTENNAPORT.SN (5G gNodeB, LTE eNodeB)	
Port No.	ANTENNAPORT.PN (5G gNodeB, LTE eNodeB)	
ALD Power Switch	ANTENNAPORT.PWRSWIT CH (5G gNodeB, LTE eNodeB)	Set this parameter to ON when an ALD is used. The default value is OFF .
Feeder Length	ANTENNAPORT.FEEDERLE NGTH (5G gNodeB, LTE eNodeB)	This parameter indicates the length of the feeder connected to the antenna port. Set this parameter to the actual feeder length.

Parameter Name	Parameter ID	Setting Notes
Switch of Configuring Ultra Delay	ANTENNAPORT.ULTRADELAYSW (5G gNodeB, LTE eNodeB)	This parameter specifies the ultra-large antenna delay switch. This switch determines the configurable range of uplink delay and downlink delay. In normal sites, it is recommended that this parameter be set to OFF because the feeder delay does not exceed 4000 ns. In special sites such as optical fiber repeaters, this parameter needs to be set to ON when the ultra-large delay is required.
DL Time Delay	ANTENNAPORT.DLDELAY (5G gNodeB, LTE eNodeB)	Set this parameter based on the device specifications. Generally, the value is less than 30 ns.
UL Time Delay	ANTENNAPORT.ULDELAY (5G gNodeB, LTE eNodeB)	Set this parameter based on the device specifications. Generally, the value is less than 30 ns.
Current Alarm Threshold Type	ANTENNAPORT.THRESHOLDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter based on the device conditions and actual connection scenarios. For details, see Table 4-1 .
Undercurrent Alarm Occur Threshold	ANTENNAPORT.UOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Undercurrent Alarm Clear Threshold	ANTENNAPORT.UCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Occur Threshold	ANTENNAPORT.OOTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .
Overcurrent Alarm Clear Threshold	ANTENNAPORT.OCTHD (5G gNodeB, LTE eNodeB)	Set this parameter only if the THRESHOLDTYPE parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see 4.4.2 Precautions .

If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, the **DEFCURRALMTHLD** MO can be used to modify the current alarm thresholds included in this type. The following table describes the key parameters involved.

Table 4-62 Key parameters in the default current alarm threshold MO
DEFCURRALMTHLD

Parameter Name	Parameter ID	Setting Notes
Port Type	DEFCURRALMTHLD.P ORTTYPE	Set this parameter to ANTENNA PORT .
Threshold Type	DEFCURRALMTHLD.T HRESHOLDTYPE	This parameter specifies the to-be-adjusted type associated with a set of default current alarm thresholds.
Adjusted Undercurrent Alarm Occur Threshold	DEFCURRALMTHLD.U OTH	This parameter specifies the adjusted undercurrent alarm occur threshold. For the specified type, the default undercurrent alarm occur threshold defined in the ANTENNA PORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Undercurrent Alarm Clear Threshold	DEFCURRALMTHLD.U CTH	This parameter specifies the adjusted undercurrent alarm clear threshold. For the specified type, the default undercurrent alarm clear threshold defined in the ANTENNA PORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Occur Threshold	DEFCURRALMTHLD. OOTH	This parameter specifies the adjusted overcurrent alarm occur threshold. For the specified type, the default overcurrent alarm occur threshold defined in the ANTENNA PORT MO no longer takes effect, and the adjusted threshold is used instead.
Adjusted Overcurrent Alarm Clear Threshold	DEFCURRALMTHLD. OCTH	This parameter specifies the adjusted overcurrent alarm clear threshold. For the specified type, the default overcurrent alarm clear threshold defined in the ANTENNA PORT MO no longer takes effect, and the adjusted threshold is used instead.

Scenario 15: Connection to the RAS Unit Through the RET Port

In this scenario, the control port (RET port), RAS device, RAS subunit, and RAS device property group must be configured. The RAS azimuth is configured in the **ALDSUBUNIT** MO. The **DIGIANTINFO** MO must be configured as well if digital antennas are involved.

Table 4-35 describes the key parameters that must be set to configure the RET port.

Table 4-63 describes the key parameters that must be set to configure the RAS device.

Table 4-66 describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-63 Key parameters that must be set to configure the RAS device

Parameter Name	Parameter ID (New Model)	Setting Notes
Device No.	NGALD.DEVICENO	The device number of each ALD in a base station must be unique.
Device Name	NGALD.DEVICENAME	This parameter specifies the device name ¹ .
Control Port Cabinet No.	NGALD.CTRLN	This parameter specifies the cabinet number of the RRU or RFU that controls the ALD.
Control Port Subrack No.	NGALD.CTRLN	This parameter specifies the subrack number of the RRU or RFU that controls the ALD.
Control Port Slot No.	NGALD.CTRLN	This parameter specifies the slot number of the RRU or RFU that controls the ALD.
ALD Device type	NGALD.ALDTYPE (5G gNodeB, LTE eNodeB)	Set this parameter to RAS .
Number of ALD Subunits	NGALD.SUBUNITN UM	This parameter specifies the number of configured RAS subunits.
Antenna Scenario	NGALD.SCENARIO	Set this parameter based on the actual hardware installation. The vendor code and serial number of the device must be correctly configured in a non-regular scenario. This parameter can be set to REGULAR or DAISY_CHAIN .
Vendor Code	NGALD.VENDORCODE	This parameter specifies the vendor code.
Serial No.	NGALD.SERIALNO	The vendor code and the serial number uniquely identify an ALD.

 NOTE

This parameter indicates the device name of the ALD, which identifies the ALD. It is recommended that you set the device name in the following format: **sector_device type_network type**. The meanings of the parts are as follows: Sector: indicates the number of the sector. Device type: indicates the type of currently connected device. It can be SINGLE_RET (Single-antenna Remote Electrical Tilt Unit), MULTI_RET (Multi-antenna Remote Electrical Tilt Unit), TMA (Tower-mounted Amplifier), SASU (Same-band Antenna Sharing Unit), RAE (Remote eAntenna Extension), RVD (Remote Vendor Defined Antenna Line Device), or RAS (Remote Azimuth Steering Unit). Network type: indicates whether the current ALD is used for a 2G network, 3G network, or LTE network. The device name is optional, but the device name configured for each ALD must be unique.

Table 4-64 describes the key parameters that must be set to configure an RAS subunit.

Table 4-64 Key parameters that must be set to configure an RAS subunit

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVICE NO	The device number of each ALD in a base station must be unique.
Subunit No.	ALDSUBUNIT.SUBUNIT NO	This parameter specifies the number of the RAS subunit. The value starts from 1.
Subunit Name	ALDSUBUNIT.SUBNAME	This parameter specifies the name of the RAS subunit. The RAS subunit name is optional, but the name configured for each subunit must be unique.
Connect Port 1 Cabinet No.	RFPATH.CN	These parameters indicate the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 1, respectively.
Connect Port 1 Subrack No.	RFPATH.SRN	
Connect Port 1 Slot No.	RFPATH.SN	
Connect Port 1 Port No.	RFPATH.PN	
Connect Port 2 Cabinet No.	RFPATH.CN	These parameters indicate the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to antenna port 2, respectively.
Connect Port 2 Subrack No.	RFPATH.SRN	
Connect Port 2 Slot No.	RFPATH.SN	
Connect Port 2 Port No.	RFPATH.PN	
Azimuth	ALDSUBUNIT.AZIMUTH	This parameter specifies the horizontal azimuth of the RAS subunit.

Parameter Name	Parameter ID	Setting Notes
Alarm Error Range	ALDSUBUNIT.AER	When the difference between the actual horizontal azimuth of an RET subunit and configured azimuth is larger than the value of this parameter, the RET Antenna Running Data and Configuration Mismatch alarm is reported.

Table 4-65 describes the key parameters that must be set to configure the RAS device property group.

Table 4-65 Key parameters that must be set to configure the RAS device property group

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDDEVICEDATA.D EVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RAS device must differ from that of the RET.
Subunit No.	ALDDEVICEDATA.S UBUNITNO	This parameter specifies the number of the RAS subunit. The value starts from 1.
Antenna Model Number	ALDDEVICEDATA.M ODELNO	This parameter specifies the model of the RAS unit.
Antenna Serial No.	ALDDEVICEDATA.S ERIALNO	This parameter specifies the serial number of the RAS unit.
Band1	ALDDEVICEDATA.B AND1	These parameters specify the bands supported by the antenna and the corresponding beamwidths.
Beamwidth1	ALDDEVICEDATA.B EAMWIDTH1	
Gain1	ALDDEVICEDATA.G AIN1	
Band2	ALDDEVICEDATA.B AND2	
Beamwidth2	ALDDEVICEDATA.B EAMWIDTH2	
Gain2	ALDDEVICEDATA.G AIN2	
Band3	ALDDEVICEDATA.B AND3	
Beamwidth3	ALDDEVICEDATA.B EAMWIDTH3	

Parameter Name	Parameter ID	Setting Notes
Gain3	ALDDEVICEDATA.GAIN3	
Band4	ALDDEVICEDATA.BAND4	
Beamwidth4	ALDDEVICEDATA.BEAMWIDTH4	
Gain4	ALDDEVICEDATA.GAIN4	
Installation Date	ALDDEVICEDATA.DATE	This parameter specifies the date when the antenna is installed.
Installer's ID	ALDDEVICEDATA.INSTALLERID	This parameter specifies the ID of the installation technician who installed the antenna.
Base Station ID	ALDDEVICEDATA.BSID	According to AISG1.1, the length of the base station ID supported by the RAS device is limited to 12 characters. If the length of the base station ID entered by a user exceeds the limit, only the first 12 characters are stored on the RAS device.
AISG Sector ID	ALDDEVICEDATA.SECTORID	According to AISG1.1, the length of the sector ID supported by the RAS device is limited to 4 characters. If the length of the sector ID entered by a user exceeds the limit, only the first 4 characters are stored on the RAS device.
Antenna Bearing	ALDDEVICEDATA.BEARING	If this parameter is set to 4294967295 , the parameter value is not delivered to the device. In this case, the parameter value in the LST RASDEVICEDATA command output is NULL .
Installed Mechanical Tilt	ALDDEVICEDATA.TILT	If this parameter is set to 2147483647 , the parameter value is not delivered to the device. In this case, the parameter value in the LST RASDEVICEDATA command output is NULL .

[Table 4-66](#) describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-66 Key parameters that must be set in the **DIGIANTINFO** MO

Parameter Name	Parameter ID	Setting Notes
Effective Range	DIGIANTINFO.EFFEC TIVERANGE (LTE eNodeB, 5G gNodeB)	This parameter specifies the effective range of the switch for a certain optional function related to digital antennas. The value SITE_LEVEL indicates that this optional function takes effect for all ALDs that support the function in the current base station. The value DEVICE_LEVEL indicates that this optional function takes effect for the specified ALD.
Subunit No.	DIGIANTINFO.SUBU NITNO (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the ALD subunit on which a certain optional function needs to take effect. The value 65535 indicates that this optional function takes effect for all subunits that support the function in the current ALD.
Optional Function	DIGIANTINFO.OPTIO NALFUNCTION (LTE eNodeB, 5G gNodeB)	This parameter specifies the switches for certain optional functions related to digital antennas. The INTRF_DIAGNOSIS , AZIMUTH_ADJ , and HBW_ADJ options control interference diagnosis, horizontal azimuth adjustment, and horizontal beamwidth adjustment, respectively.

Scenario 16: Connection to the RAS Unit Through the Antenna Port

In this scenario, the control port (antenna port), RAS device, RAS subunit, and RAS device property group must be configured. The RAS azimuth is configured in the **ALDSUBUNIT** MO.

[Table 4-41](#) describes key parameters that must be set in the **ANTENNAPORT** MO. For key parameters that must be set for configuring the RAS device, RAS subunit, and RAS device property group, see [Table 4-63](#), [Table 4-64](#), and [Table 4-65](#) in scenario 15, respectively.

Scenario 17: Connection to the RVD Through the RET Port

In this scenario, the control port (RET port), RVD, RVD subunit, and RVD property group must be configured. The horizontal beamwidth of the RVD is configured in the **ALDSUBUNIT** MO.

[Table 4-35](#) describes the key parameters that must be set to configure the RET port. The **DIGIANTINFO** MO must be configured as well if digital antennas are involved.

[Table 4-67](#) describes the key parameters that must be set to configure the RVD.

[Table 4-70](#) describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-67 Key parameters that must be set to configure the RVD

Parameter Name	Parameter ID	Setting Notes
Device No.	NGALD.DEVICENO <i>(LTE eNodeB, 5G gNodeB)</i>	The device number of the ALD must be unique.
Device Name	ALD.DEVICENAME <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the device name of the ALD, which identifies the ALD. It is recommended that you set the device name in the following format: <i>sector_device type_network type</i> . For details, see the related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	ALD.CTRLN <i>(LTE eNodeB, 5G gNodeB)</i>	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located.
Control Port Subrack No.	ALD.CTRLN <i>(LTE eNodeB, 5G gNodeB)</i>	
Control Port Slot No.	ALD.CTRLN <i>(LTE eNodeB, 5G gNodeB)</i>	
Number of ALD Subunits	NGALD.SUBUNITN <i>UM (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of configured RVD subunits.
Antenna Scenario	NGALD.SCENARIO <i>(LTE eNodeB, 5G gNodeB)</i>	Set this parameter based on the actual hardware installation. The vendor code and serial number of the device must be correctly configured in a non-regular scenario.
Vendor Code	ALD.VENDORCODE <i>(LTE eNodeB, 5G gNodeB)</i>	- If the NGALD.SCENARIO <i>(LTE eNodeB, 5G gNodeB)</i> parameter is not set to REGULAR, the ALD.VENDORCODE <i>(LTE eNodeB, 5G gNodeB)</i> parameter must be set. - If the NGALD.SCENARIO <i>(LTE eNodeB, 5G gNodeB)</i> parameter is set to REGULAR and the ALD.SERIALNO <i>(LTE eNodeB, 5G gNodeB)</i> parameter is set, the ALD.VENDORCODE <i>(LTE eNodeB, 5G gNodeB)</i> parameter must be set.

Parameter Name	Parameter ID	Setting Notes
Serial No.	ALD.SERIALNO <i>(LTE eNodeB, 5G gNodeB)</i>	The vendor code and the serial number uniquely identify an ALD. • If the NGALD.SCENARIO <i>(LTE eNodeB, 5G gNodeB)</i> parameter is not set to REGULAR, the ALD.SERIALNO <i>(LTE eNodeB, 5G gNodeB)</i> parameter must be set. • If the NGALD.SCENARIO <i>(LTE eNodeB, 5G gNodeB)</i> parameter is set to REGULAR and the ALD.VENDORCODE <i>(LTE eNodeB, 5G gNodeB)</i> parameter is set, the ALD.SERIALNO <i>(LTE eNodeB, 5G gNodeB)</i> parameter must be set.

Table 4-68 describes the key parameters that must be set to configure an RVD subunit.

Table 4-68 Key parameters that must be set to configure an RVD subunit

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDSUBUNIT.DEVIC ENO <i>(LTE eNodeB, 5G gNodeB)</i>	The device number of the ALD must be unique.
Subunit No.	ALDSUBUNIT.SUBU NITNO <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of the RVD subunit. The value starts from 1.
Subunit Name	ALDSUBUNIT.SUBN AME <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the name of the RVD subunit. The name of the RVD subunit is optional, but the name configured for each RVD subunit must be unique.
Horizontal Beamwidth	ALDSUBUNIT.HORIZ ONTALBW <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the horizontal beamwidth of the RVD subunit.
RF Path ID	RFPATH.RFPATHID <i>(LTE eNodeB, 5G gNodeB)</i>	These parameters specify the connection path between the RF module and the ALD.
RF Path Alias	RFPATH.ALIAS <i>(LTE eNodeB, 5G gNodeB)</i>	
Cabinet No.	RFPATH.CN <i>(LTE eNodeB, 5G gNodeB)</i>	These parameters specify the cabinet number, subrack number, slot number, and port number of the RRU/RFU that is connected to a port on the antenna, respectively.

Parameter Name	Parameter ID	Setting Notes
Subrack No.	RFPATH.SRN (<i>LTE eNodeB, 5G gNodeB</i>)	
Slot No.	RFPATH.SN (<i>LTE eNodeB, 5G gNodeB</i>)	
Port No.	RFPATH.PN (<i>LTE eNodeB, 5G gNodeB</i>)	
Connection ID	ALDCONNINFO.CO NNID (<i>LTE eNodeB, 5G gNodeB</i>)	Set these parameters based on the connections between the ALD and the antenna ports on the RF module.
Connection Type	ALDCONNINFO.CO NNTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	

[Table 4-69](#) describes the key parameters that must be set to configure the RVD property group.

Table 4-69 Key parameters that must be set to configure the RVD property group

Parameter Name	Parameter ID	Setting Notes
Device No.	ALDDEVICEDATA.DEVI CENO (<i>LTE eNodeB, 5G gNodeB</i>)	The device number of the ALD must be unique.
Subunit No.	ALDDEVICEDATA.SUB UNITNO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the number of the RVD subunit. The value starts from 1.
Antenna Model Number	ALDDEVICEDATA.MO DELNO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the model of the RVD.
Antenna Serial No.	ALDDEVICEDATA.SERI ALNO (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the serial number of the RVD.

Parameter Name	Parameter ID	Setting Notes
Band1	ALDDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB)	These parameters specify the bands supported by the antenna and the corresponding beamwidths. • If the value of ALDDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB) is not UNUSED, the value of ALDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB) cannot be UNUSED. If the value of ALDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB) is not UNUSED, the value of ALDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB) cannot be UNUSED. If the value of ALDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB) is not UNUSED, the value of ALDDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB) cannot be UNUSED. • If the value of ALDDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB) is not UNUSED, it must be greater than the value of ALDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB). If the value of ALDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB) is not UNUSED, it must be greater than the value of ALDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB). If the value of ALDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB) is not UNUSED, it must be greater than the value of ALDDEVICEDATA.BAND1 (LTE eNodeB, 5G gNodeB).
Beamwidth1	ALDDEVICEDATA.BEAMWIDTH1 (LTE eNodeB, 5G gNodeB)	
Gain1	ALDDEVICEDATA.GAIN1 (LTE eNodeB, 5G gNodeB)	
Band2	ALDDEVICEDATA.BAND2 (LTE eNodeB, 5G gNodeB)	
Beamwidth2	ALDDEVICEDATA.BEAMWIDTH2 (LTE eNodeB, 5G gNodeB)	
Gain2	ALDDEVICEDATA.GAIN2 (LTE eNodeB, 5G gNodeB)	
Band3	ALDDEVICEDATA.BAND3 (LTE eNodeB, 5G gNodeB)	
Beamwidth3	ALDDEVICEDATA.BEAMWIDTH3 (LTE eNodeB, 5G gNodeB)	
Gain3	ALDDEVICEDATA.GAIN3 (LTE eNodeB, 5G gNodeB)	
Band4	ALDDEVICEDATA.BAND4 (LTE eNodeB, 5G gNodeB)	
Beamwidth4	ALDDEVICEDATA.BEAMWIDTH4 (LTE eNodeB, 5G gNodeB)	
Gain4	ALDDEVICEDATA.GAIN4 (LTE eNodeB, 5G gNodeB)	
Installation Date	ALDDEVICEDATA.DATE (LTE eNodeB, 5G gNodeB)	This parameter specifies the date when the antenna is installed.

Parameter Name	Parameter ID	Setting Notes
Installer's ID	ALDDEVICEDATA.INSTALLERID (LTE eNodeB, 5G gNodeB)	This parameter specifies the ID of the installation technician who installed the antenna.
Base Station ID	ALDDEVICEDATA.BSID (LTE eNodeB, 5G gNodeB)	This parameter specifies the ID of the base station served by the antenna.
AISG Sector ID	ALDDEVICEDATA.SECTORID (LTE eNodeB, 5G gNodeB)	This parameter specifies the AISG sector ID.
Antenna Bearing	ALDDEVICEDATA.BEARING (LTE eNodeB, 5G gNodeB)	This parameter specifies the antenna azimuth.
Installed Mechanical Tilt	ALDDEVICEDATA.TILT (LTE eNodeB, 5G gNodeB)	This parameter specifies the mechanical tilt of the installation.

Table 4-70 describes the key parameters that must be set in the **DIGIANTINFO** MO.

Table 4-70 Key parameters that must be set in the **DIGIANTINFO** MO

Parameter Name	Parameter ID	Setting Notes
Effective Range	DIGIANTINFO.EFFECTIVERANGE (LTE eNodeB, 5G gNodeB)	This parameter specifies the effective range of the switch for a certain optional function related to digital antennas. The value SITE_LEVEL indicates that this optional function takes effect for all ALDs that support the function in the current base station. The value DEVICE_LEVEL indicates that this optional function takes effect for the specified ALD.
Subunit No.	DIGIANTINFO.SUBUNITNO (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the ALD subunit on which a certain optional function needs to take effect. The value 65535 indicates that this optional function takes effect for all subunits that support the function in the current ALD.
Optional Function	DIGIANTINFO.OPTIONALFUNCTION (LTE eNodeB, 5G gNodeB)	This parameter specifies the switches for certain optional functions related to digital antennas. The INTRF_DIAGNOSIS , AZIMUTH_ADJ , and HBW_ADJ options control interference diagnosis, horizontal azimuth adjustment, and horizontal beamwidth adjustment, respectively.

Scenario 18: Connection to the RVD Through the Antenna Port

In this scenario, the control port (antenna port), RVD, RVD subunit, and RVD property group must be configured. The horizontal beamwidth of the RVD is configured in the **ALDSUBUNIT** MO.

Table 4-41 describes key parameters that must be set in the **ANTENNAPORT** MO. For key parameters that must be set for configuring the RVD, RVD subunit, and RVD property group, see **Table 4-67**, **Table 4-68**, and **Table 4-69** in scenario 15, respectively.

4.4.3.3 Initial Configuration

Configuring a Single Base Station

- eGBTS
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based) > Creating a Single Co-MPT Base Station > Configuring Device Data About the Co-MPT Base Station > Configuring ALDs > Procedure**. Configure the initial data based on the parameters and configuration scenarios in data preparation descriptions.
- NodeB
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **NodeB Initial Configuration Guide (MAE-Deployment-based) > Creating a Single NodeB > Configuring NodeB Device Data > Configuring ALDs > Procedure**. Configure the initial data based on the parameters and configuration scenarios in data preparation descriptions.
- eNodeB
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **eNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating a Single eNodeB > Configuring eNodeB Device Data > Configuring ALDs > Procedure**. Configure the initial data based on the parameters and configuration scenarios in data preparation descriptions.
- gNodeB
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **gNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating a Single gNodeB > Configuring gNodeB Device Data > Configuring ALDs > Procedure**. Configure the initial data based on the parameters and configuration scenarios in data preparation descriptions.

Configuring Base Stations in Batches

Customize a template based on a base station where ALDs have been configured, and save this template. Prepare a summary data file by referencing the user-defined template. Configure base stations in batches based on the summary data file.

- eGBTs
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based) > Creating Co-MPT Base Stations in Batches (MOC Export)/Creating Co-MPT Base Stations in Batches (Customization Tool)**.
- NodeBs
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **NodeB Initial Configuration Guide (MAE-Deployment-based) > Creating NodeBs in Batches (MOC Export)/Creating NodeBs in Batches (Customization Tool)**.
- eNodeBs
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **eNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating eNodeBs in Batches**.
- gNodeBs
For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **gNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating gNodeBs in Batches**.

4.4.3.4 Using MML Commands (Old Model)

Before using MML commands, refer to [4.2.2 Impacts](#) and [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

When the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**, the old model is used for data preparation.

NOTE

The new and old ALD models in this document are selected based on the value of the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter. The old ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**. The new ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**.

Scenario 1: Connection to the RET Antenna Through the RET Port

The following operations also apply to the scenario in which the RET function on an AAS module is controlled by the RRU through the RET port.

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-3](#).
- Step 2** (Optional) If the current alarm threshold included in the current alarm threshold type does not match the actual ALD power consumption, run the **ADD DEFCURRALMTHLD** command to modify the current alarm threshold included in this type. For details, see [Table 4-4](#).
- Step 3** Run the **SCN ALD** command to scan ALDs.

Step 4 Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).

Step 5 Run the **CLB RET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 6 Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).

Step 7 (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 8 (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data, as listed in [Table 4-8](#).

----End

Scenario 2: Connection to the RET Antenna Through the Antenna Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU through the antenna port.

Step 1 Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).

 NOTE

The RET antenna is also supplied with power and controlled through the antenna port when it is connected through the CAL port. In this case, the **Antenna Port No.** is set to **R0A**.

Step 2 (Optional) If the current alarm threshold included in the current alarm threshold type does not match the actual ALD power consumption, run the **ADD DEFCURRALMTHLD** command to modify the current alarm threshold included in this type. For details, see [Table 4-10](#).

Step 3 Run the **SCN ALD** command to scan ALDs.

Step 4 Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).

Step 5 Run the **CLB RET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 6 Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).

Step 7 (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt, as listed in [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 8 (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

----End

Scenario 3: Connection to the RET Antenna Through the Antenna Port (with a TMA)

Step 1 Run the **MOD ANTENNA** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).

Step 2 Run the **SCN ALD** command to scan ALDs.

Step 3 Run the **ADD TMA** command to add a TMA and set related parameters, as listed in [Table 4-11](#).

Step 4 Run the **MOD TMA** command to set parameters related to a TMA subunit, as listed in [Table 4-12](#).

 NOTE

The value range of TMA gain varies with the manufacturer and model. Run the **DSP TMADEVICEDATA** command to query the value range before setting the gain.

Step 5 Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).

Step 6 Run the **CLB RET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 7 Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).

Step 8 (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- Step 9** (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).
- Step 10** (Optional) Run the **MOD RXBRANCH** command to configure RX channel attenuation. For details, see [Table 4-13](#).
- Step 11** (Optional) Run the **MOD TMADEVICEDATA** command to set the TMA device data. For details, see [Table 4-14](#).
- End

Scenario 4: Connection to the RET Antenna Through the RET Port (with a TMA)

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-3](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD TMA** command to add a TMA and set related parameters, as listed in [Table 4-11](#).
- Step 4** Run the **MOD TMASUBUNIT** command to set parameters related to a TMA subunit, as listed in [Table 4-12](#).

NOTE

The value range of TMA gain varies with the manufacturer and model. Run the **DSP TMADEVICEDATA** command to query the value range before setting the gain.

- Step 5** Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).
- Step 6** Run the **CLB RET** command to calibrate the RET antenna.

NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 7** Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).
- Step 8** (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- Step 9** (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).
- Step 10** (Optional) Run the **MOD RXBRANCH** command to configure RX channel attenuation. For details, see [Table 4-13](#).

- Step 11** (Optional) Run the **MOD TMADEVICEDATA** command to set the TMA device data. For details, see [Table 4-14](#).

----End

Scenario 5: Connection to the RET Antenna Through the Antenna Port (with an SASU, Supported only by NodeBs)

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD SASU** command to add an SASU and set related parameters. For details, see [Table 4-15](#).
- Step 4** Run the **MOD SASUSUBUNIT** command to set parameters related to an SASU subunit, as listed in [Table 4-16](#).
- Step 5** Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).
- Step 6** Run the **CLB RET** command to calibrate the RET antenna.

NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 7** Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).
- Step 8** (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- Step 9** (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

----End

Scenario 6: Connection to the RET Antenna Through the Antenna Port (with an SASU and a TMA, Supported only by NodeBs)

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).
- Step 2** Run the **SCN ALD** command to scan ALDs and obtain SASU information.
- Step 3** Run the **ADD SASU** command to add an SASU and set related parameters. For details, see [Table 4-15](#).

- Step 4** Run the **MOD SASUSUBUNIT** command to set parameters related to an SASU subunit, as listed in [Table 4-16](#).
- Step 5** Run the **SCN ALD** command to scan ALDs and obtain information about the TMA and RET antenna. Skip this step if you have obtained the information about the TMA and RET antenna in Step 2.
- Step 6** Run the **ADD TMA** command to add a TMA and set related parameters, as listed in [Table 4-11](#).
- Step 7** Run the **MOD TMASUBUNIT** command to set parameters related to a TMA subunit, as listed in [Table 4-12](#).

 NOTE

The value range of TMA gain varies with the manufacturer and model. Run the **DSP TMADEVICEDATA** command to query the value range before setting the gain.

- Step 8** Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).
- Step 9** Run the **CLB RET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 10** Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).
- Step 11** (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- Step 12** (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).
- Step 13** (Optional) Run the **MOD RXBRANCH** command to configure RX channel attenuation. For details, see [Table 4-13](#).
- Step 14** (Optional) Run the **MOD TMADEVICEDATA** command to set the TMA device data. For details, see [Table 4-14](#).

----End

Scenario 7: Using the AAS MU to Control the RET Function for AAS Passive Antennas

- Step 1** Run the **SCN ALD** command to scan ALDs.
- Step 2** Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-17](#).

Step 3 Run the **CLB RET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 4 Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-18](#).

Step 5 (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-19](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 6 (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

----End

Scenario 8: Using the AAS RU to Control the RET Function for AAS Passive Antennas

Operation procedures for the RET part on the AAS modules:

Step 1 For the AAU3910/AAU3911/AAU3961/AAU5246, run the **MOD RETPORT** command to configure parameters related to the power switch and current alarm threshold of the **RETPORT** MO. For details, see [Table 4-3](#).

For the AAU3920/AAU3940/AAU5240/AAU5243/AAU5940/AAU5942, run the **MOD ANTENNAPORT** command to configure parameters related to the power switch and current alarm threshold on the antenna port. For details, see [Table 4-9](#).

Step 2 Run the **SCN ALD** command to scan ALDs.

 NOTE

If multiple RF modules share the AAS module, running the **SCN ALD** command may fail to scan all the RET antennas of all RF modules at a time. In this scenario, you can run this command separately to scan the RF module of which the RET antenna cannot be found.

Step 3 Run the **ADD RET** command to add an RET antenna and set related parameters. For details, see [Table 4-5](#).

 NOTE

In this example, an RU in an AAU3911 controls all the RET antennas and the device serial numbers queried by ALD scanning are AAU3911xxxxxxxxHL, AAU3911xxxxxxxxHR, and AAU3911xxxxxxxxLR.

```
ADD RET: DEVICENO=0, CTRLCN=0, CTRLSRN=63, CTRLSN=0, RETTYPE=SINGLE_RET,  
SCENARIO=DAISY_CHAIN, VENDORCODE="HW", SERIALNO="AAU3911xxxxxxxxHL";  
ADD RET: DEVICENO=1, CTRLCN=0, CTRLSRN=63, CTRLSN=0, RETTYPE=SINGLE_RET,  
SCENARIO=DAISY_CHAIN, VENDORCODE="HW", SERIALNO="AAU3911xxxxxxxxHR";  
ADD RET: DEVICENO=2, CTRLCN=0, CTRLSRN=63, CTRLSN=0, RETTYPE=SINGLE_RET,  
SCENARIO=DAISY_CHAIN, VENDORCODE="HW", SERIALNO="AAU3911xxxxxxxxLR";
```

In the preceding information:

DEVICENO is configured according to the data plan and is unique.

CTRLCN, **CTRLSRN**, and **CTRLSN** are the cabinet number, subrack number, and slot number of the control unit. They must be configured based on site conditions. **RETTYE** must be set to **SINGLE_RET**.

SCENARIO must be set to **DAISY_CHAIN**.

VENDORCODE must be set to **HW**.

Step 4 Run the **CLB RET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 5 Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).

Step 6 (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- When an AAU5940/AAU5942 is horizontally installed, adjusting its electrical downtilt is not recommended.
- For an AAU5940/AAU5942 with the vertical beamwidth of 68°, the electrical downtilt is fixed to 0°. For an AAU5940/AAU5942 with the vertical beamwidth of 12°, the electrical downtilt can be adjusted within a range of -3° to +12°.

Step 7 (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

----End

Some AAS modules support Huawei-defined functions. For example, the vertical beamwidth adjustment using the RVD is provided by the AAU5940/AAU5942/AAU5246. The following are specific procedures:

Step 1 Run the **SCN ALD** command to scan ALDs.

 NOTE

This step is the same as scanning RETs by running the **SCN ALD** command. If the information of the RVD is ready, skip this step.

- Step 2** Run the **ADD RVD** command to add an RVD and set related parameters described in [Table 4-20](#).

 NOTE

In this example, an RU in an AAU5940/AAU5942 controls all the RET antennas and the device serial number queried by ALD scanning is HWVxxxxxxxx.

ADD RVD: DEVICENO=0, DEVICENAME="XXX", CTRLCN=0, CTRLSRN=63, CTRLSN=0, SUBUNITNUM=1, SCENARIO=DAISY_CHAIN, VENDORCODE="HW", SERIALNO="HWVxxxxxxxx";

In the preceding information:

DEVICENO is configured according to the data plan and is unique.

CTRLCN, **CTRLSRN**, and **CTRLSN** are the cabinet number, subrack number, and slot number of the control unit. They must be configured based on site conditions.

SCENARIO must be set to **DAISY_CHAIN**.

DEVICENAME must be set according to the configuration planning.

VENDORCODE must be set to **HW**.

- Step 3** Run the **CLB ALD** command to calibrate the RVD.

- Step 4** Run the **MOD RVDSUBUNIT** command to set parameters related to an RVD subunit, as listed in [Table 4-21](#).

- Step 5** (Optional) Run the **MOD RVDINFO** command to configure the vertical beamwidth for the RVD. For details, see [Table 4-22](#).

 NOTE

The RVDs of different models may support different adjustable ranges of vertical beamwidth. Run the **DSP RVDSUBUNIT** command to query the supported adjustable ranges of vertical beamwidth before setting the vertical beamwidth.

- Step 6** (Optional) Run the **MOD RVDDEVICEDATA** command to set the RVD device data. For details, see [Table 4-23](#).

----End

Scenario 9: Connection to the RAE Through the RET Port

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-3](#).

- Step 2** Run the **SCN ALD** command to scan ALDs.

- Step 3** Run the **ADD RAE** command to add an RAE and set related parameters, as listed in [Table 4-24](#).

- Step 4** Run the **MOD RAESUBUNIT** command to set parameters related to an RAE subunit, as listed in [Table 4-25](#).

- Step 5** (Optional) Run the **MOD RAEDEVICEDATA** command to set the RAE device data. For details, see [Table 4-26](#).

----End

Scenario 10: Connection to the RAE Through the RET Port (with an RET Antenna)

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-3](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD RAE** command to add an RAE and set related parameters, as listed in [Table 4-24](#).
- Step 4** Run the **MOD RAESUBUNIT** command to set parameters related to an RAE subunit, as listed in [Table 4-25](#).
- Step 5** Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).
- Step 6** Run the **CLB RET** command to calibrate the RET antenna.

NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 7** Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).
- Step 8** (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- Step 9** (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).
- Step 10** (Optional) Run the **MOD RAEDEVICEDATA** command to set the RAE device data. For details, see [Table 4-26](#).

----End

Scenario 11: Connection to the RAE Through the Antenna Port

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD RAE** command to add an RAE and set related parameters, as listed in [Table 4-24](#).
- Step 4** Run the **MOD RAESUBUNIT** command to set parameters related to an RAE subunit, as listed in [Table 4-25](#).

- Step 5** (Optional) Run the **MOD RAEDEVICEDATA** command to set the RAE device data. For details, see [Table 4-26](#).

----End

Scenario 12: Connection to the RAE Through the Antenna Port (with an RET Antenna)

- Step 1** Run the **MOD ANTENNA** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD RAE** command to add an RAE and set related parameters, as listed in [Table 4-24](#).
- Step 4** Run the **MOD RAESUBUNIT** command to set parameters related to an RAE subunit, as listed in [Table 4-25](#).
- Step 5** Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).
- Step 6** Run the **CLB RET** command to calibrate the RET antenna.

NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 7** Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit. For details, see [Table 4-6](#).
- Step 8** (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- Step 9** (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).
- Step 10** (Optional) Run the **MOD RAEDEVICEDATA** command to set the RAE device data. For details, see [Table 4-26](#).

----End

Scenario 13: Connection to the RAE Through the Antenna Port (with a TMA)

- Step 1** Run the **MOD ANTENNA** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.

Step 3 Run the **ADD TMA** command to add a TMA and set related parameters. For details, see [Table 4-11](#).

Step 4 Run the **MOD TMASUBUNIT** command to set parameters related to a TMA subunit, as listed in [Table 4-12](#).

 NOTE

The value range of TMA gain varies with the manufacturer and model. Run the **DSP TMADEVICEDATA** command to query the value range before setting the gain.

Step 5 Run the **ADD RAE** command to add an RAE and set related parameters, as listed in [Table 4-24](#).

Step 6 Run the **MOD RAESUBUNIT** command to set parameters related to an RAE subunit. For details, see [Table 4-25](#).

Step 7 (Optional) Run the **MOD RXBRANCH** command to configure RX channel attenuation. For details, see [Table 4-13](#).

Step 8 (Optional) Run the **MOD TMADEVICEDATA** command to set the TMA device data. For details, see [Table 4-14](#).

Step 9 (Optional) Run the **MOD RAEDEVICEDATA** command to set the RAE device data. For details, see [Table 4-26](#).

----End

Scenario 14: Connection to the RAE Through the Antenna Port (with an RET Antenna and a TMA)

Step 1 Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).

Step 2 Run the **SCN ALD** command to scan ALDs.

Step 3 Run the **ADD TMA** command to add a TMA and set related parameters, as listed in [Table 4-11](#).

Step 4 Run the **MOD TMASUBUNIT** command to set parameters related to a TMA subunit, as listed in [Table 4-12](#).

 NOTE

The value range of TMA gain varies with the manufacturer and model. Run the **DSP TMADEVICEDATA** command to query the value range before setting the gain.

Step 5 Run the **ADD RAE** command to add an RAE and set related parameters, as listed in [Table 4-24](#).

Step 6 Run the **MOD RAESUBUNIT** command to set parameters related to an RAE subunit, as listed in [Table 4-25](#).

Step 7 Run the **ADD RET** command to add an RET antenna and set related parameters, as listed in [Table 4-5](#).

Step 8 Run the **CLB RET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 9 Run the **MOD RETSUBUNIT** command to set parameters related to an RET subunit, as listed in [Table 4-6](#).

Step 10 (Optional) Run the **MOD RETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP RETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 11 (Optional) Run the **MOD RETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

Step 12 (Optional) Run the **MOD RXBRANCH** command to configure RX channel attenuation. For details, see [Table 4-13](#).

Step 13 (Optional) Run the **MOD TMADEVICEDATA** command to set the TMA device data. For details, see [Table 4-14](#).

Step 14 (Optional) Run the **MOD RAEDEVICEDATA** command to set the RAE device data. For details, see [Table 4-26](#).

----End

Scenario 15: Connection to the RAS Unit Through the RET Port

The configuration procedure is as follows:

Step 1 Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-3](#).

Step 2 Run the **SCN ALD** command to scan ALDs.

Step 3 Run the **ADD RAS** command to add an RAS unit and set related parameters, as listed in [Table 4-27](#).

Step 4 Run the **CLB ALD** command to calibrate the RAS unit.

Step 5 Run the **DSP ALDFUNCTION** command to check whether the ALD is a digital antenna. The ALD is a digital antenna if the command output shows that at least one non-standard protocol function is supported by the ALD. If the ALD is confirmed as a digital antenna, run the **ADD DIGIANTINFO** command to configure the corresponding license.

Step 6 Run the **MOD RASSUBUNIT** command to set parameters related to an RAS subunit, as listed in [Table 4-28](#).

Step 7 (Optional) Run the **MOD RASDEVICEDATA** command to set the RAS device data. For details, see [Table 4-29](#).

----End

Scenario 16: Connection to the RAS Unit Through the Antenna Port

The configuration procedure is as follows:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-9](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD RAS** command to add an RAS unit and set related parameters, as listed in [Table 4-27](#).
- Step 4** Run the **CLB ALD** command to calibrate the RAS unit.
- Step 5** Run the **DSP ALDFUNCTION** command to check whether the ALD is a digital antenna. The ALD is a digital antenna if the command output shows that at least one non-standard protocol function is supported by the ALD. If the ALD is confirmed as a digital antenna, run the **ADD DIGIANTINFO** command to configure the corresponding license.
- Step 6** Run the **MOD RASSUBUNIT** command to set parameters related to an RAS subunit. For details, see [Table 4-28](#).

(Optional) Run the **MOD RASDEVICEDATA** command to set the RAS device data. For details, see [Table 4-29](#).

----End

Scenario 17: Connection to the RVD Through the RET Port

When the old ALD model is used:

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-3](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD RVD** command to add an RVD and set related parameters described in [Table 4-31](#).
- Step 4** Run the **CLB ALD** command to calibrate the RVD.
- Step 5** Run the **MOD RVDSUBUNIT** command to set parameters related to an RVD subunit. For details, see [Table 4-32](#).
- Step 6** (Optional) Run the **MOD RVDINFO** command to configure the horizontal beamwidth for the RVD.

NOTE

The horizontal beamwidth adjustment range may vary with the module model. Run the **DSP RVDSUBUNIT** command to query the horizontal beamwidth adjustment range supported by the RVD before setting.

- Step 7** (Optional) Run the **MOD RVDDEVICEDATA** command to set the RVD device data. For details, see [Table 4-32](#).

----End

Scenario 18: Connection to the RVD Through the Antenna Port

When the old ALD model is used:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-3](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD RVD** command to add an RVD and set related parameters described in [Table 4-31](#).
- Step 4** Run the **CLB ALD** command to calibrate the RVD.
- Step 5** Run the **MOD RVDSUBUNIT** command to set parameters related to an RVD subunit. For details, see [Table 4-32](#).
- Step 6** (Optional) Run the **MOD RVDINFO** command to configure the horizontal beamwidth for the RVD.

NOTE

The horizontal beamwidth adjustment range may vary with the module model. Run the **DSP RVDSUBUNIT** (recommended)/**DSP ALDSUBUNIT** command to query the horizontal beamwidth adjustment range supported by the RVD before setting.

- Step 7** (Optional) Run the **MOD RVDDEVICE DATA** command to set the RVD device data. For details, see [Table 4-32](#).

----End

4.4.3.5 Using MML Commands (New Model)

Before using MML commands, refer to [4.2.2 Impacts](#) and [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

When the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**, the new model is used for data preparation.

NOTE

The new and old ALD models in this document are selected based on the value of the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter. The old ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**. The new ALD model is used when the **EQUIPMENT.ALDCFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**.

Scenario 1: Connection to the RET Device Through the RET Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU through the RET port.

When the new ALD model is used:

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-35](#).

Step 2 (Optional) If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, run the **ADD DEFCURRALMTHLD** command to modify the current alarm thresholds included in this type. For details, see [Table 4-36](#).

Step 3 Run the **SCN ALD** command to scan ALDs.

Step 4 Run the **ADD ALD** command to add an RET device and set related parameters, as listed in [Table 4-37](#).

Step 5 Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 6 Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-38](#).

Step 7 Run the **ADD RFPATH** command to set parameters related to the RF path between the RET device and RF module, as listed in [Table 4-38](#).

Step 8 Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-38](#).

Step 9 (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-38](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

Step 10 (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RET device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-40](#).

----End

Scenario 2: Connection to the RET Device Through the Antenna Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU through the antenna port.

When the new ALD model is used:

Step 1 Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-41](#).

 NOTE

The RET antenna is also supplied with power and controlled through the antenna port when it is connected through the CAL port. In this case, the **Antenna Port No.** is set to **R0A**.

Step 2 (Optional) If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, run the **ADD**

DEFCURRALMTHLD command to modify the current alarm thresholds included in this type. For details, see [Table 4-42](#).

Step 3 Run the **SCN ALD** command to scan ALDs.

Step 4 Run the **ADD ALD** command to add an RET device and set related parameters, as listed in [Table 4-37](#).

Step 5 Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 6 Run the **MOD ALDSUBUNIT** command to set the parameters related to the ALD subunit (RET in this case), and run the **ADD RFPATH** command to set the parameters related to the path between the RET antenna and the RF module. For details, see [Table 4-38](#).

Step 7 Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-38](#).

Step 8 (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-39](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

Step 9 (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RET device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-40](#).

----End

Scenario 3: Connection to TMA and RET Devices Through the Antenna Port

When the new ALD model is used:

Step 1 Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-41](#).

Step 2 Run the **SCN ALD** command to scan ALDs.

Step 3 Run the **ADD ALD** command to add a TMA and set related parameters, as listed in [Table 4-43](#).

Step 4 Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (TMA in this case), as listed in [Table 4-44](#).

 NOTE

The range of TMA gains varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the range of TMA gains before setting the gain.

Step 5 Run the **ADD ALD** command to add an RET device and set related parameters, as listed in [Table 4-37](#).

Step 6 Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 7 Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-38](#).

Step 8 Run the **ADD RFPATH** command to set parameters related to the RF path between the RET device and RF module, as listed in [Table 4-38](#).

Step 9 Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-38](#).

Step 10 (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-39](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

Step 11 (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RET device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-40](#).

Step 12 (Optional) Run the **MOD RXBRANCH** command to configure the RX channel attenuation, as listed in [Table 4-45](#).

Step 13 (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the TMA device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-46](#).

----End

Scenario 4: Connection to TMA and RET Devices Through the RET Port

When the new ALD model is used:

Step 1 Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-35](#).

Step 2 Run the **SCN ALD** command to scan ALDs.

Step 3 Run the **ADD ALD** command to add a TMA and set related parameters, as listed in [Table 4-43](#).

Step 4 Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (TMA in this case), as listed in [Table 4-44](#).

 NOTE

The range of TMA gains varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the range of TMA gains before setting the gain.

Step 5 Run the **ADD RFPATH** command to set parameters related to the RF path between the TMA and RF module, as listed in [Table 4-44](#).

- Step 6** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the TMA and RF module, as listed in [Table 4-44](#).
- Step 7** Run the **ADD ALD** command to add an RET device and set related parameters, as listed in [Table 4-37](#).
- Step 8** Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 9** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-38](#).
- Step 10** (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-39](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

- Step 11** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RET device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-40](#).
- Step 12** (Optional) Run the **MOD RXBRANCH** command to configure the RX channel attenuation, as listed in [Table 4-45](#).
- Step 13** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the TMA device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-46](#).

----End

Scenario 5: MU Controlling the RET Function of the AAS Module with Passive Antennas

When the new ALD model is used:

- Step 1** Run the **SCN ALD** command to scan ALDs.
- Step 2** Run the **ADD ALD** command to add an RET device and set related parameters, as listed in [Table 4-47](#).
- Step 3** Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 4** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-48](#).
- Step 5** Run the **ADD RFPATH** command to set parameters related to the RF path between the RET device and RF module, as listed in [Table 4-48](#).

Step 6 Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-48](#).

Step 7 (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-48](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

Step 8 (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RET subunit, as listed in [Table 4-40](#).

----End

Scenario 6: AAS RU Controlling the RET Function of the AAS Module with Passive Antennas

When the new ALD model is used, perform the following procedure to enable the RET function of an AAS module:

Step 1 For the AAU3910/AAU3911/AAU3961, run the **MOD RETPORT** command to configure parameters related to the power switch and current alarm threshold of the RET port, as listed in [Table 4-35](#).

For the AAU3920/AAU3940/AAU5240/AAU5243/AAU5940/AAU5942, run the **MOD ANTENNA** command to configure parameters related to the power switch and current alarm threshold of the antenna port, as listed in [Table 4-41](#).

Step 2 Run the **SCN ALD** command to scan ALDs.

 NOTE

If multiple RF modules share the AAS module, the **SCN ALD** command execution may fail to identify the RET antennas for all RF modules at a time. In this scenario, you can run this command separately for the RF module whose RET antenna has not been identified.

Step 3 Run the **ADD ALD** command to add an RET device and set related parameters, as listed in [Table 4-37](#).

Step 4 Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 5 Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-38](#).

Step 6 Run the **ADD RFPATH** command to set parameters related to the RF path between the RET device and RF module, as listed in [Table 4-38](#).

Step 7 Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-38](#).

Step 8 (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-38](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

- When an AAU5940/AAU5942 is horizontally installed, adjusting its electrical tilt is not recommended.
- For an AAU5940/AAU5942 with the vertical beamwidth of 68°, the downtilt is fixed to 0°. For an AAU5940/AAU5942 with the vertical beamwidth of 12°, the downtilt can be adjusted within a range of -3° to +12°.

Step 9 (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RET device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-40](#).

----End

Some AAS modules support Huawei-defined functions. For example, the AAU5940/AAU5942 provides the vertical beamwidth adjustment using the RVD. When the new ALD model is used, the configuration procedure is as follows:

Step 1 Run the **SCN ALD** command to scan ALDs.

 NOTE

This step is the same as scanning the AAS RET part by running the **SCN ALD** command. Skip this step if the information of the AAS RVD is obtained.

Step 2 Run the **ADD ALD** command to add an RVD and set related parameters, as listed in [Table 4-50](#).

Step 3 Run the **CLB ALD** command to calibrate an RVD.

Step 4 Run the **MOD ALDSUBUNIT** command to set parameters related to the RVD subunit (**ALDSUBUNIT MO**), as listed in [Table 4-51](#).

Step 5 Run the **ADD RFPATH** command to set parameters related to the RF path between the RVD and RF module, as listed in [Table 4-51](#).

Step 6 Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RVD and RF module, as listed in [Table 4-51](#).

Step 7 (Optional) Run the **MOD ALDSUBUNIT** command to configure the vertical beamwidth for the RVD, as listed in [Table 4-52](#).

 NOTE

The adjustable vertical beamwidth range may vary with the module model. Run the **DSP ALDSUBUNIT** command to query the adjustable vertical beamwidth range supported by the RVD before setting the vertical beamwidth.

Step 8 (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RVD property group (**ALDDEVICEDATA MO**), as listed in [Table 4-53](#).

----End

Scenario 7: Connection to the RAE Unit Through the RET Port

When the new ALD model is used:

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-35](#).
 - Step 2** Run the **SCN ALD** command to scan ALDs.
 - Step 3** Run the **ADD ALD** command to add an RAE unit and set related parameters, as listed in [Table 4-54](#).
 - Step 4** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RAE in this case), as listed in [Table 4-55](#).
 - Step 5** Run the **ADD RFPATH** command to set parameters related to the RF path between the RAE unit and RF module, as listed in [Table 4-55](#).
 - Step 6** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RAE unit and RF module, as listed in [Table 4-55](#).
 - Step 7** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RAE device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-56](#).
- End

Scenario 8: Connection to RAE and RET Devices Through the RET Port

When the new ALD model is used:

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-35](#).
- Step 2** (Optional) If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, run the **ADD DEFCURRALMTHLD** command to modify the current alarm thresholds included in this type. For details, see [Table 4-36](#).
- Step 3** Run the **SCN ALD** command to scan ALDs.
- Step 4** Run the **ADD ALD** command to add an RAE unit and set related parameters, as listed in [Table 4-54](#).
- Step 5** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RAE in this case), as listed in [Table 4-55](#).
- Step 6** Run the **ADD ALD** command to add an RET antenna and set related parameters, as listed in [Table 4-37](#).
- Step 7** Run the **CLB ALD** command to calibrate an RET device.

NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 8** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-38](#).
- Step 9** Run the **ADD RFPATH** command to set parameters related to the RF path between the RET device and RF module, as listed in [Table 4-38](#).

- Step 10** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-38](#).
- Step 11** (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-38](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

- Step 12** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RET subunit, as listed in [Table 4-40](#).
- Step 13** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RAE subunit, as listed in [Table 4-56](#).

----End

Scenario 9: Connection to the RAE Unit Through the Antenna Port

When the new ALD model is used:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-41](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD ALD** command to add an RAE unit and set related parameters, as listed in [Table 4-54](#).
- Step 4** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RAE in this case), as listed in [Table 4-55](#).
- Step 5** Run the **ADD RFPATH** command to set parameters related to the RF path between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 6** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 7** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RAE device property group (**ALDDEVICEDATA** MO), as listed in [Table 4-55](#).

----End

Scenario 10: Connection to RAE and RET Devices Through the Antenna Port

When the new ALD model is used:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-41](#).
- Step 2** (Optional) If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, run the **ADD DEFcurrALMTHLD** command to modify the current alarm thresholds included in this type. For details, see [Table 4-36](#).

- Step 3** Run the **SCN ALD** command to scan ALDs.
- Step 4** Run the **ADD ALD** command to add an RAE unit and set related parameters, as listed in [Table 4-54](#).
- Step 5** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RAE in this case), as listed in [Table 4-55](#).
- Step 6** Run the **ADD RFPATH** command to set parameters related to the RF path between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 7** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 8** Run the **ADD ALD** command to add an RET antenna and set related parameters, as listed in [Table 4-37](#).
- Step 9** Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 10** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-38](#).
- Step 11** Run the **ADD RFPATH** command to set parameters related to the RF path between the RET device and RF module, as listed in [Table 4-38](#).
- Step 12** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-38](#).
- Step 13** (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-38](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

- Step 14** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RET subunit, as listed in [Table 4-40](#).
- Step 15** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RAE subunit, as listed in [Table 4-56](#).

----End

Scenario 11: Connection to TMA and RAE Devices Through the Antenna Port

When the new ALD model is used:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-41](#).

- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD ALD** command to add a TMA and set related parameters, as listed in [Table 4-43](#).
- Step 4** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (TMA in this case), as listed in [Table 4-44](#).

 NOTE

The range of TMA gains varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the range of TMA gains before setting the gain.

- Step 5** Run the **ADD RFPATH** command to set parameters related to the RF path between the RAE unit and RF module, as listed in [Table 4-44](#).
- Step 6** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RAE unit and RF module, as listed in [Table 4-44](#).
- Step 7** Run the **ADD ALD** command to add an RAE unit and set related parameters, as listed in [Table 4-54](#).
- Step 8** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RAE in this case), as listed in [Table 4-55](#).
- Step 9** Run the **ADD RFPATH** command to set parameters related to the RF path between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 10** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 11** (Optional) Run the **MOD RXBRANCH** command to configure the RX channel attenuation, as listed in [Table 4-45](#).
- Step 12** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the TMA subunit, as listed in [Table 4-46](#).
- Step 13** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RAE subunit, as listed in [Table 4-56](#).

----End

Scenario 12: Connection to TMA, RAE, and RET Devices Through the Antenna Port

When the new ALD model is used:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-41](#).
- Step 2** (Optional) If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, run the **ADD DEFCCURRALMTHLD** command to modify the current alarm thresholds included in this type. For details, see [Table 4-36](#).
- Step 3** Run the **SCN ALD** command to scan ALDs.
- Step 4** Run the **ADD ALD** command to add a TMA and set related parameters, as listed in [Table 4-43](#).

- Step 5** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (TMA in this case), as listed in [Table 4-44](#).

 NOTE

The range of TMA gains varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the range of TMA gains before setting the gain.

- Step 6** Run the **ADD RFPATH** command to set parameters related to the RF path between the TMA and RF module, as listed in [Table 4-44](#).
- Step 7** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the TMA and RF module, as listed in [Table 4-44](#).
- Step 8** Run the **ADD ALD** command to add an RAE unit and set related parameters, as listed in [Table 4-54](#).
- Step 9** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RAE in this case), as listed in [Table 4-55](#).
- Step 10** Run the **ADD RFPATH** command to set parameters related to the RF path between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 11** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RAE unit and RF module, as listed in [Table 4-55](#).
- Step 12** Run the **ADD ALD** command to add an RET antenna and set related parameters, as listed in [Table 4-37](#).
- Step 13** Run the **CLB ALD** command to calibrate an RET device.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 14** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (RET in this case), as listed in [Table 4-38](#).
- Step 15** Run the **ADD RFPATH** command to set parameters related to the RF path between the RET device and RF module, as listed in [Table 4-38](#).
- Step 16** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RET device and RF module, as listed in [Table 4-38](#).
- Step 17** (Optional) Run the **MOD RETTILT** command to set the RET tilt, as listed in [Table 4-38](#).

 NOTE

The tilt range supported by the RET antenna varies with the manufacturer and model. Run the **DSP ALDDEVICEDATA** command to query the supported tilt range before setting the tilt.

- Step 18** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RET subunit, as listed in [Table 4-40](#).
- Step 19** (Optional) Run the **MOD RXBRANCH** command to configure the RX channel attenuation, as listed in [Table 4-45](#).

- Step 20** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the TMA subunit, as listed in [Table 4-46](#).
- Step 21** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RAE subunit, as listed in [Table 4-56](#).
- End

Scenario 13: Connection to the SALD/MALD Through the RET Port

When the new ALD model is used:

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-57](#).
- Step 2** (Optional) If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, run the **ADD DEFCURRALMTHLD** command to modify the current alarm thresholds included in this type. For details, see [Table 4-36](#).
- Step 3** Run the **SCN ALD** command to scan ALDs. The protocol version used in the scanning must include AISG3.0.
- Step 4** Run the **ADD ALD** command to add a SALD/MALD and set related parameters, as listed in [Table 4-59](#).
- Step 5** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (SALD/MALD in this case), as listed in [Table 4-60](#).
- Step 6** Run the **ADD RFPATH** command to add path information about the RF module connected to the RET/TMA subunit in the SALD/MALD, as listed in [Table 4-60](#).
- Step 7** Run the **ADD ALDCONNINFO** command to configure connection parameters between the RET/TMA subunit in the SALD/MALD and the RF module, as listed in [Table 4-60](#).
- Step 8** (Optional) Run the **CLB ALD** command to calibrate the RET subunits in the SALD/MALD.

NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

----End

Scenario 14: Connection to the SALD/MALD Through the Antenna Port

When the new ALD model is used:

- Step 1** Run the **MOD ANTENNAPOORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-61](#).
- Step 2** (Optional) If the current alarm thresholds included in a current alarm threshold type do not match the actual ALD power consumption, run the **ADD**

DEFCURRALMTHLD command to modify the current alarm thresholds included in this type. For details, see [Table 4-62](#).

- Step 3** Run the **SCN ALD** command to scan ALDs. The protocol version used in the scanning must include AISG3.0.
- Step 4** Run the **ADD ALD** command to add a SALD/MALD and set related parameters, as listed in [Table 4-59](#).
- Step 5** Run the **MOD ALDSUBUNIT** command to set parameters related to the ALD subunit (SALD/MALD in this case), as listed in [Table 4-60](#).
- Step 6** Run the **ADD RFPATH** command to add path information about the RF module connected to the RET/TMA subunit in the SALD/MALD, as listed in [Table 4-60](#).
- Step 7** Run the **ADD ALDCONNINFO** command to configure connection parameters between the RET/TMA subunit in the SALD/MALD and the RF module, as listed in [Table 4-60](#).
- Step 8** (Optional) Run the **CLB ALD** command to calibrate the RET subunits in the SALD/MALD.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

----End

Scenario 15: Connection to the RAS Unit Through the RET Port

When the new ALD model is used:

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-35](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD ALD** command to add an RAS unit and set related parameters, as listed in [Table 4-63](#).
- Step 4** Run the **CLB ALD** command to calibrate the RAS unit.
- Step 5** Run the **DSP ALDFUNCTION** command to check whether the antenna in use is a digital antenna. The antenna is a digital antenna if the command output shows that it supports at least one non-standard protocol function. If the antenna is confirmed as a digital antenna, run the **ADD DIGIANTINFO** command to configure the corresponding licensed functions.
- Step 6** Run the **MOD ALDSUBUNIT** command to set parameters related to the RAS subunit, as listed in [Table 4-64](#).
- Step 7** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters in the **ALDDEVICEDATA** MO related to the RAS subunit, as listed in [Table 4-65](#).

----End

Scenario 16: Connection to the RAS Unit Through the Antenna Port

When the new ALD model is used:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-41](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD ALD** command to add an RAS unit and set related parameters, as listed in [Table 4-63](#).
- Step 4** Run the **CLB ALD** command to calibrate the RAS unit.
- Step 5** Run the **DSP ALDFUNCTION** command to check whether the antenna in use is a digital antenna. The antenna is a digital antenna if the command output shows that it supports at least one non-standard protocol function. If the antenna is confirmed as a digital antenna, run the **ADD DIGIANTINFO** command to configure the corresponding licensed functions.
- Step 6** Run the **MOD ALDSUBUNIT** command to set parameters related to the RAS subunit, as listed in [Table 4-64](#).
- Step 7** (Optional) Run the **MOD RASDEVICEDATA** command to set parameters related to the RAS device property group (**ALDDEVICEDATA MO**), as listed in [Table 4-65](#).

----End

Scenario 15: Connection to the RVD Through the RET Port

When the new ALD model is used:

- Step 1** Run the **MOD RETPORT** command to set parameters related to the RET port, including the power switch and current alarm thresholds, as listed in [Table 4-35](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD ALD** command to add an RVD and set related parameters, as listed in [Table 4-67](#).
- Step 4** Run the **CLB ALD** command to calibrate the RVD.
- Step 5** Run the **MOD ALDSUBUNIT** command to set parameters related to the RVD subunit (**ALDSUBUNIT MO**), as listed in [Table 4-68](#).
- Step 6** Run the **ADD RFPATH** command to set parameters related to the RF path between the RVD and RF module, as listed in [Table 4-68](#).
- Step 7** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RVD and RF module, as listed in [Table 4-68](#).
- Step 8** (Optional) Run the **MOD ALDSUBUNIT** command to configure the horizontal beamwidth for the RVD, as listed in [Table 4-68](#).

NOTE

The horizontal beamwidth adjustment range may vary with the module model. Run the **DSP ALDSUBUNIT** command to query the horizontal beamwidth adjustment range supported by the RVD before setting.

- Step 9** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RVD property group (**ALDDEVICEDATA MO**), as listed in [Table 4-69](#).

----End

Scenario 16: Connection to the RVD Through the Antenna Port

When the new ALD model is used:

- Step 1** Run the **MOD ANTENNAPORT** command to set parameters related to the antenna port, including the power switch and current alarm thresholds, as listed in [Table 4-35](#).
- Step 2** Run the **SCN ALD** command to scan ALDs.
- Step 3** Run the **ADD ALD** command to add an RVD and set related parameters, as listed in [Table 4-67](#).
- Step 4** Run the **CLB ALD** command to calibrate the RVD.
- Step 5** Run the **MOD ALDSUBUNIT** command to set parameters related to the RVD subunit (**ALDSUBUNIT MO**), as listed in [Table 4-68](#).
- Step 6** Run the **ADD RFPATH** command to set parameters related to the RF path between the RVD and RF module, as listed in [Table 4-68](#).
- Step 7** Run the **ADD ALDCONNINFO** command to set parameters related to the connection between the RVD and RF module, as listed in [Table 4-68](#).
- Step 8** (Optional) Run the **MOD ALDSUBUNIT** command to configure the horizontal beamwidth for the RVD, as listed in [Table 4-68](#).

NOTE

The horizontal beamwidth adjustment range may vary with the module model. Run the **DSP ALDSUBUNIT** command to query the horizontal beamwidth adjustment range supported by the RVD before setting.

- Step 9** (Optional) Run the **MOD ALDDEVICEDATA** command to set parameters related to the RVD property group (**ALDDEVICEDATA MO**), as listed in [Table 4-69](#).

----End

4.4.4 Commissioning

- Step 1** Run the **DSP ALDVER** command to query the software version of an ALD. If the software version of the ALD needs to be updated, download the required software as follows:
- If the ALD is an RET antenna, see "RCU Software Downloading" in [4.1.2 Operations on RET Antennas](#).
 - If the ALD is a TMA, see "TMA software downloading" in [5.1.2 Operations on the TMA](#).
 - If the ALD is an SASU, see "SASU software downloading" in [6.1.2 Operations on the SASU](#).
 - If the ALD is an RAE, see "RAE software downloading" in [7.1.2 Operations on the RAE](#).

- If the ALD is an AAS, see "RCU software downloading" in [8.1.2 Operations on AAS Modules](#).
- If the ALD is an RAS unit, see "RAS software downloading" in [10.1.2 Operations on RAS Units](#).

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*.

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

Step 2 If the RET or AAS has no configuration file, load its configuration file as follows:

- If an RET is used, see "Configuration File Loading" in [4.1.2 Operations on RET Antennas](#).
- If an AAS is used, see "Configuration file loading" in [8.1.2 Operations on AAS Modules](#).

----End

4.4.5 Activation Verification

RET/TMA/SASU/RAE/RAS/RVD

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU that is connected to this AAS module or by the active module on the AAS module.

Step 1 Run the **DSP RETSUBUNIT** (old model) (recommended) or **DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the working status and tilt of each RET subunit. When an RET subunit is working properly, **Online Status** is **AVAILABLE** in the command output.

Step 2 Run the **DSP RET** (old model)/**DSP ALD** (new model) command to query the dynamic information about the RET antenna.

 NOTE

If only one antenna port on the RF module supports RET, the RF module is unable to report the **Control Port No.** parameter setting. The value of this parameter is displayed as **NULL**.

- Step 3** If a TMA is used, run the **DSP TMA** (old model)/**DSP ALD** (new model) command to query the dynamic information about the TMA.
- Step 4** If a TMA is used, run the **DSP TMASUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the dynamic information about TMA subunits.
- Step 5** If an SASU is used, run the **DSP SASU** command to query the SASU dynamic information (supported only on the NodeB).
- Step 6** If an SASU is used, run the **DSP SASUSUBUNIT** command to query the dynamic information about SASU subunits (supported only on the NodeB).
- Step 7** If an RAE unit is used, run the **DSP RAE** (old model)/**DSP ALD** (new model) command to query the dynamic information about the RAE unit.
- Step 8** If an RAE unit is used, run the **DSP RAESUBUNIT** (old model) (recommended) or **DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the dynamic information about RAE subunits.
- Step 9** If an RAE unit is used, run the **DSP RAEDEVICEDATA** (old model)/**DSP ALDDEVICEDATA** (new model) command to query the dynamic information about the RAE device property group.
- Step 10** If an RAS unit is used, run the **DSP RAS** (old model)/**DSP ALD** (new model) command to query the dynamic information about the RAS unit.
- Step 11** If an RAS unit is used, run the **DSP RASSUBUNIT** (old model) (recommended) or **DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the dynamic information about RAS subunits.
- Step 12** If an RAS unit is used, run the **DSP RASDEVICEDATA** (old model)/**DSP ALDDEVICEDATA** (new model) command to query the horizontal azimuth convergence status of RAS subunits.
- Step 13** If an RAS unit is used, run the **DSP DIGIANTINFO** command to check whether the RAS unit supports horizontal azimuth adjustment.
- Step 14** If an RVD is used and the antenna is a digital antenna, run the **DSP DIGIANTINFO** command to check whether the RVD supports horizontal azimuth adjustment.
- Step 15** If an RVD is used, run the **DSP RVD** (old model)/**DSP ALD** (new model) command to query the dynamic information about the RVD.
- Step 16** If an RVD is used, run the **DSP RVDSUBUNIT** (old model) (recommended) or **DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the dynamic information about RVD subunits.
- Step 17** If an RVD is used, run the **SCN ANTINFO** command with **SCANTYPE** set to **RF_CHANNEL_INFO** to query the mapping between the cells and RVD.
- Step 18** If an RVD is used, run the **DSP RVDSUBUNIT** (old model) (recommended) or **DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the association between RVD subunits and the antenna.
- Step 19** If an RVD is used, run the **DSP ALDFUNCTION** command to query the digital antenna capability of the RET antenna.
- Step 20** If an RVD is used and the antenna is a digital antenna, run the **DSP DIGIANTINFO** command to check whether the RVD supports horizontal azimuth adjustment.

- Step 21** If an RVD is used, run the **DSP ALDADJUSTINFO** command to query the current adjustment times and maximum adjustment times of the RVD.

 NOTE

Only digital antennas support the query using commands in [Step 15](#) to [Step 21](#).

----End

SALD/MALD

- Step 1** If an SALD/MALD is used, run the **DSP ALD** command to query the dynamic information about the SALD/MALD.

- Step 2** If an SALD/MALD is used, run the **DSP ALDSUBUNIT** command to query dynamic information about RET/TMA/ADB subunits.

----End

AAS Modules with Passive Antennas (RET Function Controlled by MU)

- Step 1** Run the **DSP RETSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the working status and tilt of each RET subunit. When an RET subunit is working properly, **Online Status** is **AVAILABLE** in the command output.

- Step 2** Run the **DSP RET** (old model)/**DSP ALD** (new model) command to query the dynamic information about the RET antenna.

 NOTE

The MU is unable to report the **Control Port No.** parameter setting. The value of this parameter is displayed as **NULL**.

----End

4.4.6 Deactivation

RET/TMA/SASU/RAE/RAS/SALD/MALD/RVD

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU that is connected to this AAS module or by the active module on the AAS module.

If an ALD is no longer used, remove it by running the following commands. Then, turn off the power switch and set the attenuation to its default value.

- Run the **RMV RET** (old model)/**RMV ALD** (new model) command to remove an RET antenna, its subunits, and device properties.
- Run the **RMV TMA** (old model)/**RMV ALD** (new model) command to remove a TMA, its subunits, and device properties.
- Run the **RMV SASU** command to remove an SASU, its subunits, and device properties (supported only by NodeBs).
- Run the **RMV RAE** (old model)/**RMV ALD** (new model) command to remove an RAE unit, its subunits, and device properties.
- Run the **RMV RAS** (old model)/**RMV ALD** (new model) command to remove an RAS unit, its subunits, and device properties.

- Before removing an SALD or MALD, you need to remove its subunits by running the **RMV ALDSUBUNIT** command. Then run the **RMV ALD** command to remove the SALD/MALD.
- Run the **RMV RVD** (old model)/**RMV ALD** (new model) command to remove an RVD, its subunits, and device properties.

AAS Modules with Passive Antennas (RET Function Controlled by MU)

If an AAS module with passive antennas is no longer used, run the **RMV RET** (old model)/**RMV ALD** (new model) command to remove an RET antenna, its subunits, and device properties.

4.4.7 Reconfiguration

RET/TMA/SASU/RAE/RAS/SALD/MALD/RVD

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU that is connected to this AAS module or by the active module on the AAS module.

When ALD data needs to be reconfigured in the old model, collect information about the parameters to be modified based on connections between the RRU/RFU and the RET antenna. For details, see [4.4.3.1 Data Preparation \(Old Model\)](#).

- Run the **MOD RETPORT** command to modify parameter settings related to an RET port.
- Run the **MOD RET** command to modify parameter settings related to an RET antenna.
- Run the **MOD RETSUBUNIT** command to modify the RET subunit parameter settings.
- Run the **MOD RETTILT** command to adjust an RET tilt.
- Run the **MOD RETDEVICEDATA** command to modify parameter settings in the **RETDEVICEDATA** MO.
- Run the **MOD ANTENNAPORT** command to modify parameter settings related to an antenna port.
- Run the **MOD TMA** command to modify parameter settings related to a TMA.
- Run the **MOD TMASUBUNIT** command to modify parameter settings related to a TMA subunit.
- Run the **MOD RXBRANCH** command to adjust the RX channel attenuation.
- Run the **MOD TMADEVICEDATA** command to modify parameter settings in the **TMADEVICEDATA** MO.
- Run the **MOD SASU** command to modify parameter settings related to an SASU. This step applies to NodeBs only.
- Run the **MOD SASUSUBUNIT** command to modify parameter settings related to an SASU subunit. This step applies to NodeBs only.
- Run the **MOD RAE** command to modify parameter settings related to an RAE unit.
- Run the **MOD RAESUBUNIT** command to modify parameter settings related to an RAE subunit.

- Run the **MOD RAEDEVICEDATA** command to modify parameter settings in the **RAEDEVICEDATA** MO.
- Run the **MOD RAS** command to modify parameter settings related to an RAS unit.
- Run the **MOD RASSUBUNIT** command to modify parameter settings related to an RAS subunit.
- Run the **MOD RASDEVICEDATA** command to modify parameter settings in the **RASDEVICEDATA** MO.
- Run the **MOD RVD** command to modify parameter settings related to an RVD.
- Run the **MOD RVDSUBUNIT** command to modify parameter settings related to an RVD subunit.
- Run the **MOD RVDINFO** command to modify RVD device information.
- Run the **MOD RVDDEVICEDATA** command to modify parameter settings in the **RVDDEVICEDATA** MO.

When ALD data needs to be reconfigured in the new model, collect information about the parameters to be modified based on connections between the RRU/RFU and the RET antenna. For details, see [4.4.3.2 Data Preparation \(New Model\)](#).

- Run the **MOD RETPORT** command to modify parameter settings related to an RET port.
- Run the **MOD ALD** command to modify parameter settings related to an RET antenna.
- Run the **MOD ALDSUBUNIT** command to modify parameter settings related to an RET subunit.
- Run the **MOD RETTILT** command to adjust an RET tilt.
- Run the **MOD ALDDEVICEDATA** command to modify parameter settings in the **ALDDEVICEDATA** MO related to an RET antenna.
- Run the **MOD ANTENNAPORT** command to modify parameter settings related to an antenna port.
- Run the **MOD ALD** command to modify parameter settings related to a TMA.
- Run the **MOD ALDSUBUNIT** command to modify parameter settings related to a TMA subunit.
- Run the **MOD RXBRANCH** command to adjust the RX channel attenuation.
- Run the **MOD ALDDEVICEDATA** command to modify parameter settings in the **ALDDEVICEDATA** MO related to a TMA.
- Run the **MOD ALD** command to modify parameter settings related to an RAE unit.
- Run the **MOD ALDSUBUNIT** command to modify parameter settings related to an RAE subunit.
- Run the **MOD ALDDEVICEDATA** command to modify parameter settings in the **ALDDEVICEDATA** MO related to an RAE unit.
- Run the **MOD ALD** command to modify parameter settings related to an RAS unit.
- Run the **MOD ALDSUBUNIT** command to modify parameter settings related to an RAS subunit.

- Run the **MOD ALDDEVICEDATA** command to modify parameter settings in the **ALDDEVICEDATA** MO related to an RAS unit.
- Run the **MOD ALD** command to modify parameter settings related to an SALD/MALD.
- Run the **MOD ALDSUBUNIT** command to modify parameter settings related to ADB/RET/TMA subunits in an SALD/MALD.
- Run the **MOD ALD** command to modify parameter settings related to an RVD.
- Run the **MOD ALDSUBUNIT** command to modify parameter settings related to an RVD subunit.
- Run the **MOD ALDDEVICEDATA** command to modify parameter settings in the **ALDDEVICEDATA** MO related to the RVD.

 NOTE

Before changing the power port from an RET port to an antenna port or from an antenna port to an RET port, turn off the power switch if it has been turned on for the reconfiguration. This is necessary because the power switches of the antenna port and RET port on one RRU cannot be turned on simultaneously.

AAS Modules with Passive Antennas (RET Function Controlled by MU)

When the old ALD model is used:

- Run the **MOD RET** command to modify parameter settings related to an RET antenna.
- Run the **MOD RETSUBUNIT** command to modify the RET subunit parameter settings.
- Run the **MOD RETTILT** command to adjust an RET tilt.
- Run the **MOD RETDEVICEDATA** command to modify parameter settings in the **RETDEVICEDATA** MO related to the RET device property group.

When the new ALD model is used:

- Run the **MOD ALD** command to modify parameter settings related to an RET antenna.
- Run the **MOD ALDSUBUNIT** command to modify parameter settings related to an RET subunit.
- Run the **MOD RETTILT** command to adjust an RET tilt.
- Run the **MOD ALDDEVICEDATA** command to modify parameter settings in the **ALDDEVICEDATA** MO related to the RET device property group.

4.5 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

4.5.1 When to Use

For details, see [4.4.1 When to Use](#).

4.5.2 Precautions

Determine the location of the ALD control port on the RRU/RFU. The ALD control port provides power and OOK signals for the ALD. You can locate the port based on the site's cable connections.

Device data in a co-MPT multimode base station needs to be configured only once. For a co-MPT multimode base station, you only need to determine which port on the RRU/RFU is the control port for the ALD device and configure the power switch, current thresholds and ALD data once.

For a separate-MPT base station, the RRU/RFU can be managed by one RAT or multiple RATs. Determine the RAT that manages the RRU/RFU and configure all data for the ALD powered by this RRU/RFU in this RAT. This procedure involves the following two scenarios:

Scenario 1: RRU/RFU Is Managed by One RAT

The RRU/RFU, which might support multiple RATs, is managed by one RAT, and provides power and OOK signals for the ALD. Determine the RAT that manages the RRU/RFU and configure all ALD data in this RAT.

Scenario 2: RRU/RFU Is Managed by Multiple RATs

The RRU/RFU is managed by multiple RATs, and provides power and OOK signals for the ALD.

Parameters related to the ALD control port in this scenario are RF module common parameters, as listed in [Table 4-71](#) and [Table 4-72](#). During the data preparation, initial configuration, feature reconfiguration, the common parameters must be set to the consistent values in all RATs served by the multimode RRU/RFU.

Table 4-71 Common ALD parameters for RF modules in each RAT (1)

Object	GBTS Parameter Name	eGBTS/NodeB/ eNodeB/ gNodeB/Co- MPT Base Station Parameter Name	Recommended Setting
Control port (RET port as a control port)	RET ALD Power Switch	ALD Power Switch	For a separate-MPT base station, set this parameter to ON in all RATs. The power switches for the RET port and RF port on one RRU cannot be turned on simultaneously.

Object	GBTS Parameter Name	eGBTS/NodeB/ eNodeB/ gNodeB/Co- MPT Base Station Parameter Name	Recommended Setting
	RET ALD Current Alarm Threshold Type	Current Alarm Threshold Type	For a separate-MPT base station, each of these parameters must be set to the same value in all RATs. For recommended values, see Reference current alarm thresholds for different current alarm threshold types (RET port) .
	RET ALD Under Current Occur Threshold(mA)	Undercurrent Alarm Occur Threshold	
	RET ALD Under Current Clear Threshold(mA)	Undercurrent Alarm Clear Threshold	
	RET ALD Over Current Occur Threshold(mA)	Overcurrent Alarm Occur Threshold	
	RET ALD Over Current Clear Threshold(mA)	Overcurrent Alarm Clear Threshold	

Table 4-72 Common ALD parameters for RF modules in each RAT (2)

Object	GBTS Parameter Name	eGBTS/NodeB/ eNodeB/ gNodeB/Co- MPT Base Station Parameter Name	Recommended Setting
Control port (RF port as a control port and ANT_A as an example)	ANT_A ALD Power Switch	ALD Power Switch	For a separate-MPT base station, set this parameter to ON in all RATs. The power switches for the RET port and RF port on one RRU cannot be turned on simultaneously.
	ANT_A ALD Current Alarm Threshold Type	Current Alarm Threshold Type	For a separate-MPT base station, each of these parameters must be set to the same value in all RATs. For recommended values, see Reference current alarm thresholds for different current alarm threshold types (antenna port) .
	ANT_A ALD Over Current Occur Threshold(mA)	Undercurrent Alarm Occur Threshold	
	ANT_A ALD Over Current Clear Threshold(mA)	Undercurrent Alarm Clear Threshold	

Object	GBTS Parameter Name	eGBTS/NodeB/ eNodeB/ gNodeB/Co- MPT Base Station Parameter Name	Recommended Setting
	ANT_A ALD Under Current Occur Threshold(mA)	Overcurrent Alarm Occur Threshold	
	ANT_A ALD Under Current Clear Threshold(mA)	Overcurrent Alarm Clear Threshold	
RRU/RFU RX channel attenuati on	Antenna Tributary 1 Factor	Attenuation	If no TMA is used, set this parameter to 0. If a 12 dB TMA is used, set this parameter to a value within the range from 4 dB to 11 dB. If a 24 dB TMA is used, set this parameter to a value within the range from 11 dB to 22 dB. For a separate-MPT base station, each of these parameters must be set to the same value in all RATs.
	Antenna Tributary 2 Factor	Attenuation	

NOTE

When an RF port except ANT_A is used as a control port, common ALD parameters are named in the following ways: For a GBTS, common ALD parameters are named by analogy based on [Table 4-71](#). For an eGBTS, NodeB, eNodeB, gNodeB, or co-MPT base station, common ALD parameters are named in the same way as those listed in [Table 4-72](#).

Other ALD-related parameters, except the RF module common parameters, must be set only in one RAT of the separate-MPT base station. Choose one from the RATs that manage the RRU/RFU providing power and OOK signals for the ALD. Then, configure all data for the ALD only in this RAT.

NOTE

The RAT in which the other ALD-related parameters are set must be the same in the data preparation, initial configuration, activation verification, and reconfiguration.

Only one maintenance link can be established between the RRU/RFU and the ALDs. Therefore, configuring ALD data in two RATs is prohibited and must be avoided in a multimode base station. If ALD data is configured in two RATs in this multimode base station, unexpected faults may occur. For example, ALDs cannot be detected, or the ALM-26541 ALD Maintenance Link Failure is reported. If such faults occur, remove the ALD data configuration in both RATs, and scan and configure the ALDs in the desired RAT. If ALDs cannot be detected after the ALD data configuration is removed from both RATs, turn off the power switches in both RATs, and set the power switches and current alarm thresholds in both RATs again.

4.5.3 Data Configuration

4.5.3.1 Data Preparation

For a co-MPT base station, data preparation is the same as that for an eGBTS/NodeB/eNodeB/gNodeB. For details, see [4.4.3.1 Data Preparation \(Old Model\)](#).

For a separate-MPT base station, data preparations differ as follows:

Scenario 1: RRU/RFU Is Managed by One RAT

This scenario does not involve RF module common parameters. Determine the RAT that manages the RRU/RFU according to [4.5.2 Precautions](#) and then prepare all ALD data in the RAT. For details, see the following data preparation sections:

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.3.1 Data Preparation \(Old Model\)](#)
- GBTS: [4.6.3.1 Data Preparation](#)

Scenario 2: RRU/RFU Is Managed by Multiple RATs

To prepare data for RF module common parameters in this scenario, see [Table 4-71](#) and [Table 4-72](#).

Select one of the RATs that manage the RF modules according to [4.5.2 Precautions](#) and then prepare other ALD parameters except the RF module common parameters in this RAT. For details, see the following data preparation sections:

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.3.1 Data Preparation \(Old Model\)](#)
- GBTS: [4.6.3.1 Data Preparation](#)

Scenario 3: AAS Modules Are Used

Determine the RAT in which the RET function is set for the AAS module according to [4.5.2 Precautions](#). The data preparation is the same as that for a single-mode base station in this RAT.

If an AAS module with passive antennas is used, see scenario 1, 2, 7, or 8 in [4.4.3.1 Data Preparation \(Old Model\)](#).

4.5.3.2 Initial Configuration

For a co-MPT base station, initial configuration is the same as that for an eGBTS/NodeB/eNodeB/gNodeB. For details, see [Initial Configuration](#) and [4.4.3.4 Using MML Commands \(Old Model\)](#).

For a separate-MPT base station, initial configurations differ as follows:

Scenario 1: RRU/RFU Is Managed by One RAT

The RRU/RFU, which might support multiple RATs, is managed by one RAT, and provides power and OOK signals for the ALD. Determine the RAT that manages the RRU/RFU and configure all ALD data in this RAT. The configuration procedure is the same as that for a single-mode base station of the RAT. For details, see the following initial configuration sections:

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.3.3 Initial Configuration](#) and [4.4.3.4 Using MML Commands \(Old Model\)](#)

- GBTS: [4.6.3.2 Initial Configuration](#) and [4.6.3.3 Using MML Commands](#)

Scenario 2: RRU/RFU Is Managed by Multiple RATs

The RRU/RFU is managed by multiple RATs, and provides power and OOK signals for the ALD.

Before setting other ALD-related parameters, set the RF module common parameters to consistent values for all RATs that manage the RRU/RFU, as shown in [Table 4-71](#) and [Table 4-72](#). If ALD power switches are turned on for all RATs managing the RRU/RFU but current alarm threshold configurations are inconsistent for these RATs, the ALM-26272 Inter-System RF Unit Parameter Settings Conflict will be reported. Run the **SET BTSRXUBP** (GBTS) or **MOD RETPORT/MOD ANTENNA** (eGBTS/NodeB/eNodeB/gNodeB) command to set RF module common parameters.

NOTE

The RRU/RFU managed by multiple RATs can supply power to ALDs only after the ALD power switches are turned on for all involved RATs. The ALD can be detected only after the power supply is normal.

The power switches for the RET port and antenna port on one RRU cannot be turned on simultaneously.

Then, select one of the RATs that manage the RRU/RFU and set the other ALD-related parameters in this RAT. For example, if the RRU/RFU is managed by the GBTS and NodeB simultaneously, set the other ALD-related parameters on the GBTS or NodeB. The configuration procedure is the same as that for a single-mode base station of the RAT.

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.3.3 Initial Configuration](#) and [4.4.3.4 Using MML Commands \(Old Model\)](#)
- GBTS: [4.6.3.2 Initial Configuration](#) and [4.6.3.3 Using MML Commands](#)

Scenario 3: AAS Modules Are Used

Determine the RAT in which the RET function is configured for the AAS module according to [4.5.2 Precautions](#). The configuration is the same as that for a single-mode base station of the RAT.

- To use a graphical user interface (GUI) to configure the module, see [4.6.3.2 Initial Configuration](#).
- To use MML commands to configure an AAS module with passive antennas, see scenario 1, 2, or 7 in [4.6.3.2 Initial Configuration](#).

4.5.3.3 Using MML Commands

Before using MML commands, refer to [4.2.2 Impacts](#) and [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

For details, see [4.4.3.4 Using MML Commands \(Old Model\)](#).

4.5.4 Commissioning

For a co-MPT base station, the commissioning procedure is the same as that for an eGBTS/NodeB/eNodeB/gNodeB. For details, see [4.4.4 Commissioning](#).

For a separate-MPT base station, this procedure does not involve the RF module command parameters. Determine the RAT for configuring the ALD according to [4.5.2 Precautions](#) and perform this procedure accordingly. For details, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.4 Commissioning](#)
- GBTS: [4.4.4 Commissioning](#)

4.5.5 Activation Verification

For a co-MPT base station, the activation verification procedure is the same as that for an eGBTS/NodeB/eNodeB/gNodeB. For details, see [4.4.5 Activation Verification](#).

For a separate-MPT base station, this procedure does not involve the RF module command parameters. Determine the RAT for configuring the ALD according to [4.5.2 Precautions](#) and perform this procedure accordingly. For details, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.5 Activation Verification](#)
- GBTS: [4.6.5 Activation Verification](#)

4.5.6 Deactivation

For a co-MPT base station, the deactivation procedure is the same as that for an eGBTS/NodeB/eNodeB/gNodeB. For details, see [4.4.6 Deactivation](#).

For a separate-MPT base station, this procedure does not involve the RF module command parameters. Determine the RAT for configuring the ALD according to [4.5.2 Precautions](#) and perform this procedure accordingly. For details, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.6 Deactivation](#)
- GBTS: [4.6.6 Deactivation](#)

4.5.7 Reconfiguration

For a co-MPT base station, the reconfiguration procedure is the same as that for an eGBTS/NodeB/eNodeB/gNodeB. For details, see [4.6.6 Deactivation](#).

For a separate-MPT base station, reconfigurations differ as follows:

Scenario 1: RRU/RFU Is Managed by One RAT

This scenario does not involve RF module common parameters. Determine the RAT that manages the RRU/RFU based on [4.5.2 Precautions](#) and reconfigure ALD data in this RAT. The configuration procedure is the same as that for a single-mode base station of the RAT. For details, see the following reconfiguration sections:

- eGBTS/NodeB/eNodeB/gNodeB/co-MPT base station: [4.6.6 Deactivation](#)
- GBTS: [4.6.7 Reconfiguration](#)

Scenario 2: RRU/RFU Is Managed by Multiple RATs

If the RF module common parameters listed in [Table 4-71](#) need to be reconfigured, reconfigure the parameters consistently for all RATs that manage the RRU/RFU. Run the **SET BTSRXUBP** (GBTS) or **MOD RETPORT/MOD ANTENNAPORT** (eGBTS/NodeB/eNodeB/gNodeB) command to set RF module common parameters.

If other ALD-related parameters except the RF module common parameters need to be reconfigured, determine the RAT for configuring the ALD based on [4.5.2 Precautions](#) and perform this procedure accordingly.

- eGBTS/NodeB/eNodeB/gNodeB: [4.4.7 Reconfiguration](#)
- GBTS: [4.6.7 Reconfiguration](#)

Scenario 3: AAS Modules Are Used

Determine the RAT in which the RET function is configured for the AAS module according to [4.5.2 Precautions](#). The data preparation is the same as that for a single-mode base station of the RAT. For details, see [4.4.7 Reconfiguration](#).

4.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

4.6.1 When to Use

For details, see [4.4.1 When to Use](#).

4.6.2 Precautions

[4.4.2 Precautions](#) describes the common precautions for the GBTS and eGBTS/NodeB/eNodeB/gNodeB.

The common TMA does not support the AISG protocol. To configure a common TMA, you only need to run the **SET BTSRXUBP** command to turn on the power switch, set current alarm thresholds, and configure the RX channel attenuation based on the network plan. For data preparation details for turning on the power switch and setting the current alarm thresholds, see [Table 4-79](#). For data preparation details for configuring the RX channel attenuation, see [Table 4-80](#).

After subunits are added to an AISG1.1-based TMA, all TMA subunits start to work only after you run the **STR BTSALDSCAN** command.

[Table 4-1](#) and [Table 4-2](#) list the current alarm threshold types for the control port and the reference current alarm thresholds.

Table 4-73 Reference current alarm thresholds for different current alarm threshold types (antenna port)

Reference Value	Description	Undercurrent Alarm Occur Threshold (mA)	Undercurrent Alarm Clear Threshold (mA)	Overcurrent Alarm Occur Threshold (mA)	Overcurrent Alarm Clear Threshold (mA)
TMA12DB_ONLY_NON_AISG	For 12 dB TMA only	30	40	170	150
TMA24DB_ONLY_NON_AISG	For 24 dB TMA only	40	60	310	280
RET_ONLY_COAXIAL	For RET antenna only (coaxial cable)	25	33	150	120
TMA12DB_AISG	For 12 dB TMA +RET antenna or 12 dB TMA only (AISG)	30	40	450	400
TMA24DB_AISG	For 24 dB TMA +RET antenna or 24 dB TMA only (AISG)	40	60	850	750
UER_SELF_DEFINE1	User-defined	For details, see the description below.			
UER_SELF_DEFINE2	User-defined	For details, see the description below.			
UER_SELF_DEFINE3	User-defined	For details, see the description below.			

Table 4-74 Reference current alarm thresholds for different current alarm threshold types (RET port)

Reference Value	Description	Undercurrent Alarm Occur Threshold (mA)	Undercurrent Alarm Clear Threshold (mA)	Overcurrent Alarm Occur Threshold (mA)	Overcurrent Alarm Clear Threshold (mA)
RET_ONLY_MULTICORE	For RET antenna only (multi-wire cable)	10	15	150	120
UER_SELF_DEFINE1	User-defined	For details, see the description below.			

Reference Value	Description	Undercurrent Alarm Occur Threshold (mA)	Undercurrent Alarm Clear Threshold (mA)	Overcurrent Alarm Occur Threshold (mA)	Overcurrent Alarm Clear Threshold (mA)
UER_SELF_DEFINE2	User-defined	For details, see the description below.			
UER_SELF_DEFINE3	User-defined	For details, see the description below.			

When the current alarm threshold type is user-defined, set current alarm thresholds based on the actual ALD type. Pay attention to the following restrictions:

- User-defined current alarm thresholds must meet the requirements: Under Current Occur Threshold < Under Current Clear Threshold < Over Current Clear Threshold < Over Current Occur Threshold.
- Generally, set the Under Current Occur Threshold to 20% to 30% of the device rated operating current, and set the Under Current Clear Threshold to about 20 mA greater than the Under Current Occur Threshold. Set the Over Current Occur Threshold to 150% to 200% of the device rated operating current, and set the Over Current Clear Threshold to about 50 mA less than the Over Current Occur Threshold.
- If RET antennas are connected in a non-regular scenario, set the Under Current Occur Threshold to 20% to 30% of the total rated current of all ALDs controlled by the RRU. Set the Over Current Occur Threshold to a value that is 150% to 200% of the total rated current of all ALDs controlled by the RRU.
- If the ALD model in use is not recommended by Huawei, set the Under Current Occur Threshold to 20% to 30% of the total rated current of all ALDs controlled by the RRU. Set the Over Current Occur Threshold to 150% to 200% of the total rated current of all ALDs controlled by the RRU.
- There are three user-define types for GBTs: UER_SELF_DEFINE1, UER_SELF_DEFINE2, and UER_SELF_DEFINE3. Generally, the value is UER_SELF_DEFINE1.

4.6.3 Data Configuration

4.6.3.1 Data Preparation

Overview

This section includes only key parameters, not parameters in all scenarios.

ALD data configuration varies with scenarios. Scenarios are categorized by the type of the RRU/RFU port through which control signals are sent to an RET antenna, and whether the TMA is used. The scenarios are as follows:

- Scenario 1: connection to the RET antenna through the RET port
- Scenario 2: connection to the RET antenna through the antenna port
- Scenario 3: connection to the RET antenna through the antenna port (with a TMA)
- Scenario 4: connection to the RET antenna through the RET port (with a TMA)
- Scenario 5: connection to the RET antenna through the GATM

Generic Data

For details, see [Generic Data](#) in [4.4.3.1 Data Preparation \(Old Model\)](#).

Scenario 1: Connection to the RET Antenna Through the RET Port

The following operations also apply to the scenario where the RET function on an AAS module with passive antennas is controlled by the RRU/RFU through its RET port. In this scenario, the AAS module functions as the conventional RET antennas. Therefore, the configuration can be performed in the same way as that for conventional RET antennas.

[Table 4-75](#) describes the key parameters that must be set in the **RETPORT** MO.

Table 4-75 Key parameters that must be set in the **RETPORT** MO

Parameter Name	Parameter ID	Setting Notes
RET ALD Power Switch	BTSRXUBP.PwrSwitchRET	Set this parameter to ON when an RET antenna is used. The default value is OFF .
RET ALD Current Alarm Threshold Type	BTSRXUBP.THRESHOLDTYPERET	Set this parameter as required.
RET ALD Under Current Occur Threshold(mA)	BTSRXUBP.UnderCurAlmThdRET	Set this parameter only if the THRESHOLDTYPERET parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.4.2 Precautions .
RET ALD Under Current Clear Threshold(mA)	BTSRXUBP.UnderCurClrThdRET	Set this parameter only if the THRESHOLDTYPERET parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.6.2 Precautions .
RET ALD Over Current Occur Threshold(mA)	BTSRXUBP.OverCurAlmThdRET	Set this parameter only if the THRESHOLDTYPERET parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.6.2 Precautions .
RET ALD Over Current Clear Threshold(mA)	BTSRXUBP.OverCurClrThdRET	Set this parameter only if the THRESHOLDTYPERET parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.6.2 Precautions .

[Table 4-76](#) describes the key parameters that must be set in the **RET** MO.

Table 4-76 Key parameters that must be set in the **RET MO**

Parameter Name	Parameter ID	Setting Notes
Device No.	BTSRET.DEVICE NO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.
Device Name	BTSRET.DEVICE NAME	This parameter identifies an RET antenna. The format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
Control Port Cabinet No.	BTSRET.CTRLPO RTCN	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located. Set these parameters based on the control relationship between the RET antenna and the RRU/RFU.
Control Port Subrack No.	BTSRET.CTRLPO RTSRN	
Control Port Slot No.	BTSRET.CTRLPO RTSN	
Control Port No.	BTSRET.CTRLPO RTNO	This parameter specifies the control port number. The value ranges from 0 to 2. Control ports 0, 1, and 2 correspond to the ports ANT_A, ANT_B, and RET, respectively. Only one port on the RRU can be used as the control port for the RET antenna. In a daisy chain scenario, multiple RCUs share one control port.
RET Type	BTSRET.RETTYP E	Set this parameter to SINGLE_RET for the RET antenna with a single RET subunit. Set this parameter to MULTI_RET for the RET antenna with multiple RET subunits. Set this parameter to SINGLE_RET for the AAS module with passive antennas.
Number of RET Subunits	BTSRET.SUBUNI TNUM	This parameter specifies the number of RET subunits used by a base station. Set this parameter only when the RETTYPE parameter is set to MULTI_RET . Set this parameter to 1 for an AAS module with passive antennas.
Polar Type	BTSRET.POLAR TYPE	Set this parameter based on the RET antenna specifications. Set this parameter based on the AAS specifications only when the base station uses the AAS module with passive antennas.
Antenna Scenario	BTSRET.SCENA RIO	Antenna scenario ¹

Parameter Name	Parameter ID	Setting Notes
Vendor Code	BTSRET.VENDORCODE	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the manufacturer information, for example, KA for a Kathrein RET antenna, AN for an Andrew RET antenna, or HW for a Huawei Agissson RET antenna. Set this parameter to HW for an AAS module with passive antennas.
Serial No.	BTSRET.SERIALNO	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number. If an AAS module with passive antennas is used, run the STR BTSALDSCAN command to obtain the serial number of the AAS module. Then set this parameter according to the mapping between the serial number and antenna based on the hardware description specific to the AAS module.

 NOTE

This parameter specifies how the RET antenna is connected to an RRU/RFU.

- Set this parameter to **REGULAR** if the RET antenna is directly connected to the RRU/RFU. In this scenario, **VENDORCODE** and **SERIALNO** parameters do not need to be specified.
- Set this parameter to **DAISY_CHAIN** when two RET antennas are cascaded. In this scenario, the control port for RET antennas must be configured on the upper-level RRU/RFU of the daisy chain. The **VENDORCODE** and **SERIALNO** parameters must be specified.

If an AAS module with passive antennas is used, set this parameter to **DAISY_CHAIN** even when you need to use only one set of antennas for the RET function. In this case, specify the **VENDORCODE** and **SERIALNO** parameters because the antennas in the AAS module are working in daisy chain mode.

For key parameters that must be set in the **BTSRETSUBUNIT** MO, see [Table 4-77](#).

Table 4-77 Key parameters related to the **BTSRETSUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	BTSRETSUBUNIT.DEVICENO	This parameter specifies the antenna device number of the RET.
SubUnit No.	BTSRETSUBUNIT.SUBUNITNO	This parameter specifies the ID of the subunit, which starts from 1.
Tilt(0.1degree)	BTSRETSUBUNIT.TILT	Set this parameter based on the engineering design.

Parameter Name	Parameter ID	Setting Notes
Connect Port 1 Cabinet No.	BTSRETSUBUNIT.CON NCN1	Set these parameters based on connections between the RET subunits and the RF ports on the RRU/RFU. If an AAS module with passive antennas is used, set these parameters based on the connections between the AAS module and RF ports on the RRU/RFU.
Connect Port 1 Subrack No.	BTSRETSUBUNIT.CON NSRN1	
Connect Port 1 Slot No.	BTSRETSUBUNIT.CON NSN1	
Connect Port 1 Port No.	BTSRETSUBUNIT.CON NPN1	
Connect Port 2 Cabinet No.	BTSRETSUBUNIT.CON NCN2	
Connect Port 2 Subrack No.	BTSRETSUBUNIT.CON NSRN2	
Connect Port 2 Slot No.	BTSRETSUBUNIT.CON NSN2	
Connect Port 2 Port No.	BTSRETSUBUNIT.CON NPN2	

For key parameters that must be set to configure the RET antenna downtilt, see [Table 4-7](#).

For key parameters that must be set to configure the RET device data, see [Table 4-8](#).

Scenario 2: Connection to the RET Antenna Through the Antenna Port

The following operations also apply to the scenario where the RET function of an AAS module with passive antennas is controlled by the RRU/RFU through its antenna port. In this scenario, the AAS module functions as the conventional RET antennas. Therefore, the configuration can be performed in the same way as that for conventional RET antennas.

For key parameters that must be set in the **RET**, **BTSRETSUBUNIT**, and **RETDEVICEDATA** MOs and key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-76](#), [Table 4-77](#), [Table 4-7](#), and [Table 4-8](#).

[Table 4-78](#) describes key parameters that must be set in the **ANTENNAPORT** MO. This table assumes that ANT_A is a control port. When any other antenna port is the control port, the key parameters can be similarly configured.

Table 4-78 Key parameters that must be set in the **ANTENNAPORT** MO

Parameter Name	Parameter ID	Setting Notes
ANT_A ALD Power Switch	BTSRXUBP.PwrSwitchA	Set this parameter to ON when an ALD is used. The default value is OFF . If this parameter is set to ON , current alarm thresholds for this port must be specified.
ANT_A ALD Current Alarm Threshold Type	BTSRXUBP.ChkModA	Set this parameter as required. For details, see Table 4-73 .
ANT_A ALD Over Current Occur Threshold(mA)	BTSRXUBP.OverCurAlmThdA	Set this parameter only if the ChkModA parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT_A ALD Over Current Clear Threshold(mA)	BTSRXUBP.OverCurClrThdA	Set this parameter only if the ChkModA parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT_A ALD Under Current Occur Threshold(mA)	BTSRXUBP.UnderCurAlmThdA	Set this parameter only if the ChkModA parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT_A ALD Under Current Clear Threshold(mA)	BTSRXUBP.UnderCurClrThdA	Set this parameter only if the ChkModA parameter is set to UER_SELF_DEFINE . Set this parameter as required. For details, see section 4.6.2 Precautions .

Scenario 3: Connection to the RET Antenna Through the Antenna Port (with a TMA)

For key parameters that must be set in the **RET**, **RETSUBUNIT**, and **RETDEVICEDATA** MOs and key parameters that must be set to configure the RET antenna downtilt in this scenario, see [Table 4-76](#), [Table 4-6](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set in the **ANTENNAPORT** MO in this scenario, see [Table 4-78](#).

[Table 4-79](#) describes the key parameters that must be set in the **TMA** MO.

Table 4-79 Key parameters that must be set in the **TMA** MO

Parameter Name	Parameter ID	Setting Notes
Device No.	BTSTMA.DEVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.
Device Name	BTSTMA.DEVICENAME	This parameter specifies the device name of the ALD, which identifies the ALD. The format of the value is <i>site_sector+port+device type_network type</i> . For details, see the device name-related parameter descriptions. This parameter is optional. If this parameter is specified, the device name of each ALD must be unique.
TMA Power Supply Type	BTSTMA.PWRSUPPLYTYPE	This parameter specifies the power supply type of a TMA. Set this parameter based on the specifications provided by the TMA manufacturer.
Control Port Cabinet No.	BTSTMA.CTRLPORTCN	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU/RFU where the control port is located. Set these parameters based on the control relationship between the TMA and the RRU/RFU.
Control Port Subrack No.	BTSTMA.CTRLPORTSRN	
Control Port Slot No.	BTSTMA.CTRLPORTSN	
Number of TMA Subunits	BTSTMA.SUBUNITNUM	Set this parameter as required. Generally, there are two TMA subunits.
Vendor Code	BTSTMA.VENDORCODE	Set this parameter to the actual TMA manufacturer code.
Serial No.	BTSTMA.SERIALNO	Set this parameter to the actual TMA serial number.

[Table 4-12](#) describes the key parameters that must be set in the **TMASUBUNIT** MO.

[Table 4-80](#) describes the key parameters that must be set in the **BTSRXUBP** MO.

Table 4-80 Key parameters that must be set in the **BTSRXUBP** MO

Parameter Name	Parameter ID	Setting Notes
Antenna Tributary 1 Flag	BTSRXUBP.HAVETT1	This parameter specifies whether a TMA is connected to RF port ANT_A. If a TMA is connected, set this parameter to YES .

Parameter Name	Parameter ID	Setting Notes
Antenna Tributary 1 Factor	BTSRXUBP.ATT ENFACTOR1	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is set to DRRU or DRFU .
Antenna Tributary 1 Factor	BTSRXUBP.MRR UATTENFACTO R1	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is not set to DRRU or DRFU .
Antenna Tributary 2 Flag	BTSRXUBP.HAV ETT2	This parameter specifies whether a TMA is connected to RF port ANT_B. If a TMA is connected, set this parameter to YES .
Antenna Tributary 2 Factor	BTSRXUBP.ATT ENFACTOR2	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is set to DRRU or DRFU .
Antenna Tributary 2 Factor	BTSRXUBP.MRR UATTENFACTO R2	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is not set to DRRU or DRFU .
Antenna Tributary 3 Flag	BTSRXUBP.HAV ETT3	This parameter specifies whether a TMA is connected to RF port ANT_C. If a TMA is connected, set this parameter to YES .
Antenna Tributary 3 Factor	BTSRXUBP.MRR UATTENFACTO R3	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is set to MRRU or GRRU .
Antenna Tributary 4 Flag	BTSRXUBP.HAV ETT4	This parameter specifies whether a TMA is connected to RF port ANT_D. If a TMA is connected, set this parameter to YES .
Antenna Tributary 4 Factor	BTSRXUBP.MRR UATTENFACTO R4	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is set to MRRU or GRRU .

[Table 4-81](#) describes the parameters that must be set to configure the TMA device data.

Table 4-81 Key parameters related to the TMA device data

Parameter Name	Parameter ID	Setting Notes
Device No.	BTSTMADEVICEDATA.DE VICENO	This parameter specifies the device number of the ALD. Set this parameter to the device number in the TMA MO.
Subunit No.	BTSTMADEVICEDATA.SU BUNITNO	This parameter specifies the number of an RET subunit. Set this parameter when the TMASUBUNIT MO has been configured.

Parameter Name	Parameter ID	Setting Notes
Antenna Bearing	BTSTMADEVICEDATA.BEARING	This parameter specifies the azimuth of an antenna. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Antenna Model Number	BTSTMADEVICEDATA.MODELNO	This parameter specifies the antenna model. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Base Station ID	BTSTMADEVICEDATA.BSID	This parameter specifies the ID of a base station served by an antenna. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Band1	BTSTMADEVICEDATA.BAND1	These parameters specify the frequency bands supported by an antenna and the corresponding beamwidths. These parameters are part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Beamwidth1	BTSTMADEVICEDATA.BEAMWIDTH1	
Gain1	BTSTMADEVICEDATA.GAIN1	
Band2	BTSTMADEVICEDATA.BAND2	
Beamwidth2	BTSTMADEVICEDATA.BEAMWIDTH2	
Gain2	BTSTMADEVICEDATA.GAIN2	
Band3	BTSTMADEVICEDATA.BAND3	
Beamwidth3	BTSTMADEVICEDATA.BEAMWIDTH3	
Gain3	BTSTMADEVICEDATA.GAIN3	
Band4	BTSTMADEVICEDATA.BAND4	
Beamwidth4	BTSTMADEVICEDATA.BEAMWIDTH4	
Gain4	BTSTMADEVICEDATA.GAIN4	

Parameter Name	Parameter ID	Setting Notes
Installation Date	BTSTMADEVICEDATA.D ATE	This parameter specifies the date on which an antenna is to be installed. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Installed Mechanical Tilt	BTSTMADEVICEDATA.T/ LT	This parameter specifies the mechanical tilt of an antenna. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Installer's ID	BTSTMADEVICEDATA.I/ STALLERID	This parameter specifies the ID of the person who installs an antenna. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
AISG Sector ID	BTSTMADEVICEDATA.S/ CTORID	This parameter specifies the ID of a sector served by an antenna. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Serial No.	BTSTMADEVICEDATA.S/ RIALNO	This parameter specifies the equipment serial number of an antenna. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Gain Resolution	BTSTMADEVICEDATA.G AINRESOLUTION	These parameters are usually set when a TMA is delivered. If these parameters have not been set upon a TMA delivery, you can set them according to manuals delivered with the TMA. This parameter is part of the device data defined by AISG protocols. For details, see 3GPP TS 25.466 or AISG2.0.
Subunit Type	BTSTMADEVICEDATA.S/ BUNITTYPE	
Received Max Frequency	BTSTMADEVICEDATA.RX MAXFQ	
Received Min Frequency	BTSTMADEVICEDATA.RX MINFQ	
Transmit Max Frequency	BTSTMADEVICEDATA.TX MAXFQ	
Transmit Min Frequency	BTSTMADEVICEDATA.TX MINFQ	

Scenario 4: Connection to the RET Antenna Through the RET Port (with a TMA)

For key parameters that must be set to configure an RET port, RET antenna, RET subunit, RET tilt, and RET device data in this scenario, see [Table 4-75](#), [Table 4-76](#), [Table 4-77](#), [Table 4-7](#), and [Table 4-8](#).

For key parameters that must be set to configure a TMA, TMA subunit, RX channel attenuation, and TMA device data in this scenario, see [Table 4-79](#), [Table 4-12](#), [Table 4-80](#), and [Table 4-81](#).

Scenario 5: Connection to the RET Antenna Through the GATM

[Table 4-82](#) describes the key parameters that must be set to configure the GATM.

Table 4-82 Key parameters that must be set to configure the GATM

Parameter Name	Parameter ID	Setting Notes
ANT0 ALD Power Switch	BTSDATUBP.AM PC0	Set this parameter to ON when this port is connected to the RET antenna. (The default value is OFF .)
ANT0 Alarm Mode	BTSDATUBP.M ODE0	Set this parameter as required. MODE1 (Mode 1) is recommended.
ANT0 ALD Over Critical Alarm Threshold(mA)	BTSDATUBP.MA JORALMU0	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT0 ALD Over Warning Alarm Threshold(mA)	BTSDATUBP.MI NORALMU0	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT0 ALD Low Current Alarm Threshold(mA)	BTSDATUBP.AL MD0	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT1 ALD Power Switch	BTSDATUBP.AM PC1	Set this parameter to ON when this port is connected to the RET antenna. (The default value is OFF .)
ANT1 Alarm Mode	BTSDATUBP.M ODE1	Set this parameter as required. MODE1 (Mode 1) is recommended.
ANT1 ALD Over Critical Alarm Threshold(mA)	BTSDATUBP.MA JORALMU1	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT1 ALD Over Warning Alarm Threshold(mA)	BTSDATUBP.MI NORALMU1	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT1 ALD Low Current Alarm Threshold(mA)	BTSDATUBP.AL MD1	Set this parameter as required. For details, see section 4.6.2 Precautions .

Parameter Name	Parameter ID	Setting Notes
ANT2 ALD Power Switch	BTSDATUBP.AM PC2	Set this parameter to ON when this port is connected to the RET antenna. (The default value is OFF .)
ANT2 Alarm Mode	BTSDATUBP.M ODE2	Set this parameter as required. MODE1(Mode 1) is recommended.
ANT2 ALD Over Critical Alarm Threshold(mA)	BTSDATUBP.MA JORALMUP2	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT2 ALD Over Warning Alarm Threshold(mA)	BTSDATUBP.MI NORALMUP2	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT2 ALD Low Current Alarm Threshold(mA)	BTSDATUBP.AL MD2	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT3 ALD Power Switch	BTSDATUBP.AM PC3	Set this parameter to ON when this port is connected to the RET antenna. (The default value is OFF .)
ANT3 Alarm Mode	BTSDATUBP.M ODE3	Set this parameter as required. MODE1(Mode 1) is recommended.
ANT3 ALD Over Critical Alarm Threshold(mA)	BTSDATUBP.MA JORALMUP3	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT3 ALD Over Warning Alarm Threshold(mA)	BTSDATUBP.MI NORALMUP3	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT3 ALD Low Current Alarm Threshold(mA)	BTSDATUBP.AL MD3	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT4 ALD Power Switch	BTSDATUBP.AM PC4	Set this parameter to ON when this port is connected to the RET antenna. (The default value is OFF .)
ANT4 Alarm Mode	BTSDATUBP.M ODE4	Set this parameter as required. MODE1(Mode 1) is recommended.
ANT4 ALD Over Critical Alarm Threshold(mA)	BTSDATUBP.MA JORALMUP4	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT4 ALD Over Warning Alarm Threshold(mA)	BTSDATUBP.MI NORALMUP4	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT4 ALD Low Current Alarm Threshold(mA)	BTSDATUBP.AL MD4	Set this parameter as required. For details, see section 4.6.2 Precautions .

Parameter Name	Parameter ID	Setting Notes
ANT5 ALD Power Switch	BTSDATUBP.AM PC5	Set this parameter to ON when this port is connected to the RET antenna. (The default value is OFF .)
ANT5 Alarm Mode	BTSDATUBP.M ODE5	Set this parameter as required. MODE1 (Mode 1) is recommended.
ANT5 ALD Over Critical Alarm Threshold(mA)	BTSDATUBP.MA JORALMUP5	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT5 ALD Over Warning Alarm Threshold(mA)	BTSDATUBP.MI NORALMUP5	Set this parameter as required. For details, see section 4.6.2 Precautions .
ANT5 ALD Low Current Alarm Threshold(mA)	BTSDATUBP.AL MD5	Set this parameter as required. For details, see section 4.6.2 Precautions .
Manager Cabinet No.	BTSDATUBP.MC N	This parameter specifies the cabinet number of the managing board. The managing board is the GTMU or CCU board that directly communicates with the board.
Manager Subrack No.	BTSDATUBP.MS RN	This parameter specifies the subrack number of the managing board. The managing board is the GTMU or CCU board that directly communicates with the board.
Manager Port No.	BTSDATUBP.MP N	This parameter specifies the number of the monitoring port on the manager to which the board is connected.

[Table 4-83](#) describes the key parameter that must be set to enable the RET antenna.

Table 4-83 Key parameters that must be set to enable the RET antenna

Parameter Name	Parameter ID	Setting Notes
Device No.	BTSRET.DEVICE NO	The device number of each ALD in a base station must be unique.

For key parameters that must be set in the **RET** and **RETDEVICEDATA** MOs and key parameters that must be set to configure the RET antenna downtilt, see [Table 4-76](#), [Table 4-7](#), and [Table 4-8](#).

4.6.3.2 Initial Configuration

Configuring a Single Base Station

For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **Initially Configuring GBTs Using the MAE-Deployment > Creating a Single GBT > Configuring GBT Device Data > Configuring ALDs > Procedure**. Configure the initial data based on the parameters and configuration scenarios in data preparation descriptions.

NOTE

When you navigate in the document, locate the correct node based on the model of the base station controller connected to the base station.

Configuring Base Stations in Batches

Customize a template based on a base station where ALDs have been configured, and save this template. Prepare a summary data file by referencing the user-defined template. Configure base stations in batches based on the summary data file.

For details, see *3900 & 5900 Series Base Station Initial Configuration Guide* and navigate in the following sequence: **Initially Configuring GBTs Using the MAE-Deployment > Creating GBTs in Batches**.

NOTE

When you navigate in the document, locate the correct node based on the model of the base station controller connected to the base station.

4.6.3.3 Using MML Commands

Before using MML commands, refer to [4.2.2 Impacts](#) and [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Scenario 1: Connection to the RET Antenna Through the RET Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU through the RET port.

- Step 1** Run the **SET BTSRXUBP** command to set parameters related to the RET port, including the power switch and current alarm thresholds. For details, see [Table 4-78](#).
- Step 2** Run the **STR BTSALDSCAN** command to scan ALDs.
- Step 3** Run the **ADD BTSRET** command to add an RET antenna and set related parameters. For details, see [Table 4-76](#).
- Step 4** Run the **CLB BTSRET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 5 Run the **MOD BTSRETSUBUNIT** command to set parameters related to an RET subunit. For details, see [Table 4-77](#).

Step 6 (Optional) Run the **MOD BTSRETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP BTSRETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 7 (Optional) Run the **MOD BTSRETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

----End

Scenario 2: Connection to the RET Antenna Through the Antenna Port

The following operations also apply to the scenario where the RET function of an AAS module is controlled by the RRU/RFU through the antenna port.

Step 1 Run the **SET BTSRXUBP** command to set parameters related to the antenna port, including the power switch and current alarm thresholds. For details, see [Table 4-9](#).

Step 2 Run the **STR BTSALDSCAN** command to scan ALDs.

Step 3 Run the **ADD BTSRET** command to add an RET antenna and set related parameters. For details, see [Table 4-76](#).

Step 4 Run the **CLB BTSRET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 5 Run the **MOD BTSRETSUBUNIT** command to set parameters related to an RET subunit. For details, see [Table 4-77](#).

Step 6 (Optional) Run the **MOD BTSRETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP BTSRETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 7 (Optional) Run the **MOD BTSRETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

----End

Scenario 3: Connection to the RET Antenna Through the Antenna Port (with a TMA)

Step 1 Run the **SET BTSRXUBP** command to set parameters related to the antenna port, including the power switch and current alarm thresholds. For details, see [Table 4-9](#).

Step 2 Run the **STR BTSALDSCAN** command to scan ALDs.

Step 3 Run the **ADD BTSTMA** command to add a TMA and set related parameters. For details, see [Table 4-79](#).

Step 4 Run the **MOD BTSTMASUBUNIT** command to set parameters related to a TMA subunit. For details, see [Table 4-12](#).

NOTE

The value range of TMA gain varies with the manufacturer and model. Run the **DSP BTSTMADEVICEDATA** command to query the value range before setting the gain.

Step 5 Run the **ADD BTSRET** command to add an RET antenna and set related parameters. For details, see [Table 4-76](#).

Step 6 Run the **CLB BTSRET** command to calibrate the RET antenna.

NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 7 Run the **MOD BTSRETSSUBUNIT** command to set parameters related to an RET subunit. For details, see [Table 4-77](#).

Step 8 (Optional) Run the **MOD BTSRETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP BTSRETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 9 (Optional) Run the **MOD BTSRETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

Step 10 (Optional) Run the **SET BTSRXUBP** command to configure RX channel attenuation. For details, see [Table 4-80](#).

Step 11 (Optional) Run the **MOD BTSTMADEVICEDATA** command to set the TMA device data. For details, see [Table 4-81](#).

----End

Scenario 4: Connection to the RET Antenna Through the RET Port (with a TMA)

- Step 1** Run the **SET BTSRXUBP** command to set parameters related to the RET port, including the power switch and current alarm thresholds. For details, see [Table 4-75](#).
- Step 2** Run the **STR BTSALDSCAN** command to scan ALDs.
- Step 3** Run the **ADD BTSTMA** command to add a TMA and set related parameters. For details, see [Table 4-79](#).
- Step 4** Run the **MOD BTSTMASUBUNIT** command to set parameters related to a TMA subunit. For details, see [Table 4-12](#).

 NOTE

The value range of TMA gain varies with the manufacturer and model. Run the **DSP BTSTMADEVICEDATA** command to query the value range before setting the gain.

- Step 5** Run the **ADD BTSRET** command to add an RET antenna and set related parameters. For details, see [Table 4-76](#).
- Step 6** Run the **CLB BTSRET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

- Step 7** Run the **MOD BTSRETSUBUNIT** command to set parameters related to an RET subunit. For details, see [Table 4-77](#).
- Step 8** (Optional) Run the **MOD BTSRETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP BTSRETDEVICEDATA** command to query the supported tilt before setting the downtilt.

- Step 9** (Optional) Run the **MOD BTSRETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).
- Step 10** (Optional) Run the **SET BTSRXUBP** command to configure RX channel attenuation. For details, see [Table 4-80](#).
- Step 11** (Optional) Run the **MOD BTSTMADEVICEDATA** command to set the TMA device data. For details, see [Table 4-81](#).

----End

Scenario 5: Connection to the RET Antenna Through the GATM

- Step 1** Run the **SET BTSDATUBP** command to set parameters related to a GATM, including the power switch and current alarm thresholds. For details, see [Table 4-82](#).

Step 2 Run the **SET BTSRETANTENB** command to enable the RET function. For details, see [Table 4-83](#).

Step 3 Run the **STR BTSALDSCAN** command to scan ALDs.

Step 4 Run the **ADD BTSRET** command to add an RET antenna and set related parameters. For details, see [Table 4-76](#).

Step 5 Run the **CLB BTSRET** command to calibrate the RET antenna.

 NOTE

Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Step 6 (Optional) Run the **MOD BTSRETTILT** command to set the RET antenna downtilt. For details, see [Table 4-7](#).

 NOTE

The value range of the tilt supported by the RET antenna varies with the manufacturer and model. Run the **DSP BTSRETDEVICEDATA** command to query the supported tilt before setting the downtilt.

Step 7 (Optional) Run the **MOD BTSRETDEVICEDATA** command to set the RET device data. For details, see [Table 4-8](#).

----End

4.6.4 Commissioning

Step 1 Run the **DSP BTSALDVER** command to query the software version of an ALD. If the ALD needs to be upgraded, download the required software as follows:

- If the ALD is an RET, see "RCU Software Downloading" in [4.1.2 Operations on RET Antennas](#).
- If the ALD is a TMA, see "TMA software downloading" in [5.1.2 Operations on the TMA](#).

Step 2 If the RET has no configuration file, load its configuration file. For details, see "Configuration file loading" in section [4.1.2 Operations on RET Antennas](#).

----End

4.6.5 Activation Verification

Step 1 Run the **DSP BTSRETSUBUNIT** command to query the working status and tilt of each RET subunit. When an RET subunit is working properly, **Online Status** is **AVAILABLE** in the command output.

Step 2 Run the **DSP BTSRET** command to query dynamic information about the RET antenna.

Step 3 If a TMA is used, run the **DSP BTSTMA** command to query dynamic information about the TMA.

Step 4 If a TMA is used, run the **DSP BTSTMASUBUNIT** command to query dynamic information about TMA subunits.

----End

4.6.6 Deactivation

If an ALD is no longer used, remove it by running the following commands. Then, turn off the power switch and set the attenuation factor to its default value.

- Run the **RMV BTSRET** command to remove an RET antenna, its subunits, and device data.
- Run the **RMV BTSTMA** command to remove a TMA, its subunits, and device data.

4.6.7 Reconfiguration

When ALD data needs to be reconfigured, collect information about the parameters to be modified based on connections between the RRU/RFU and the RET antenna. For details, see [4.5.2 Precautions](#).

- Run the **SET BSRXUBP** command to modify parameters related to an RET port.
- Run the **MOD BTSRET** command to modify parameters related to an RET antenna.
- Run the **MOD BTSRETSUBUNIT** command to modify parameters related to an RET subunit.
- Run the **MOD BTSRETTILT** command to adjust an RET antenna downtilt.
- Run the **MOD BTSRETDEVICEDATA** command to modify parameters related to the RET device data.
- Run the **SET BSRXUBP** command to modify parameters related to an antenna port.
- Run the **MOD BTSTMA** command to modify parameters related to a TMA.
- Run the **MOD BTSTMASUBUNIT** command to modify parameters related to a TMA subunit.
- Run the **SET BSRXUBP** command to adjust RX channel attenuation.
- Run the **MOD BTSTMADEVICEDATA** command to modify parameters related to the TMA device data.

NOTE

Before changing the power port from an RET port to an antenna port or from an antenna port to an RET port, turn off the power switch if it has been turned on for the reconfiguration. This is necessary because the power switches of the antenna port and RET port on one RRU cannot be turned on simultaneously.

For the scenarios of connection to the RET antenna through the GATM:

- Run the **SET BTSDATUBP** command to modify parameters related to the port where the GATM connects to the RET antenna.
- Run the **MOD BTSRET** command to modify parameters related to an RET antenna.

- Run the **MOD BTSRETTILT** command to adjust an RET antenna downtilt.
- Run the **MOD BTSRETDEVICEDATA** command to modify parameters related to the RET device data.

4.7 Operation and Maintenance (Automatic ALD Deployment)

4.7.1 When to Use

It is recommended that this feature be enabled when ALDs complying with AISG protocols have been installed. AISG protocols include AISG1.1, AISG2.0, and AISG3.0. SRAN9.0 and later versions support AISG1.1 and AISG2.0.

Automatic ALD deployment applies to RETs and TMAs but not SASUs or AAS modules. In addition, automatic ALD deployment is not supported in GATM scenarios.

NOTE

SRAN17.1 supports only the old ALD model complying with AISG1.1 and AISG2.0. The new ALD model complying with AISG3.0 is not supported.

4.7.2 Data Configuration

4.7.2.1 Data Preparation

Overview

In automatic deployment of ALDs, the system automatically performs initial configuration for most ALD data. Only a small amount of ALD data needs to be manually modified or added.

- For details of the automatic configuration, see [4.7.8 Appendix: Automatic ALD Configuration Process](#).
- The ALD data that needs manual operation varies with the number of RET subunits, whether RET antennas are cascaded, number of TMA subunits, and connections between the TMA and RRU/RFU. The related scenarios are as follows:
 - Scenario 1: Single-subunit RET not in daisy chain mode (see [4.1.1 Connections Between RET Antennas and RRUs/RFUs](#))
 - Scenario 2: AISG2.0-based TMA connected to the RRU/RFU with two RF ports (one TMA with two subunits) (see [5.1.1 Connections Between the TMA, RRU/RFU, and RET Antenna](#))
 - Scenario 3: Single-subunit RET in daisy chain mode (see [4.1.1 Connections Between RET Antennas and RRUs/RFUs](#))
 - Scenario 4: Multi-subunit RET (see [4.1.1 Connections Between RET Antennas and RRUs/RFUs](#))
 - Scenario 5: AISG1.1-based TMA (two TMAs, each with one TMA subunit)

- Scenario 6: TMA connected to two cascaded RRUs or RFUs (see [5.1.1 Connections Between the TMA, RRU/RFU, and RET Antenna](#))
- Scenario 7: TMA connected to the RRU with four RF ports (see [5.1.1 Connections Between the TMA, RRU/RFU, and RET Antenna](#))

NOTE

In scenario 7, if the RRU with four RF ports is connected to two RET antennas, the TMA and RET antenna on the RF port ANT_A connecting to the control port can be automatically deployed, but the TMA and RET antenna on the RF port ANT_B connecting to the control port for the RET antenna cannot be automatically deployed. For details about the data configurations, see scenario 3 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#) or scenario 3 in [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

Obtain an RET antenna configuration file from the RET antenna manufacturer in advance because the configuration file may be required for commissioning an ALD after initial configuration is complete.

Manually Configured Data

After automatic ALD configuration is complete, manually modify and add required ALD data, which must be obtained locally on the base station and recorded.

1. Common data that needs to be modified

[Table 4-84](#) or [Table 4-85](#) describes the parameters that need to be manually modified after automatic configuration is complete.

Table 4-84 Common parameters that need to be manually modified (eGBTS/NodeB/eNodeB/gNodeB)

Parameter Name	Parameter ID (Old Model)	Parameter ID (New Model)	Setting Notes
Tilt	RETSUBUNIT.TILT (5G gNodeB, LTE eNodeB)	ALDSUBUNIT.TILT (5G gNodeB, LTE eNodeB)	Check whether manual modification is required. If yes, obtain the method of modifying this parameter.
Work Mode	TMASUBUNIT.MODE (5G gNodeB, LTE eNodeB)	ALDSUBUNIT.MODE (5G gNodeB, LTE eNodeB)	Check whether manual modification is required. If yes, obtain the method of modifying these parameters.
Gain	TMASUBUNIT.GAIN (5G gNodeB, LTE eNodeB)	ALDSUBUNIT.GAIN (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID (Old Model)	Parameter ID (New Model)	Setting Notes
Attenuation	RXBRANCH.ATTEN (5G gNodeB, LTE eNodeB)	RXBRANCH.ATTEN (5G gNodeB, LTE eNodeB)	After manually modifying the values of the TMASUBUNIT.MODE and TMASUBUNIT.GAIN parameters, and obtaining the accurate antenna loss, calculate the modification value for the RX channel attenuation using the following formula: RXBRANCH.ATTEN = TMASUBUNIT.GAIN – Antenna loss.

Table 4-85 Common parameters that need to be manually modified (GBTS)

Parameter Name	Parameter ID	Setting Notes
Tilt	BTSRETSUBUNIT.TILT	Check whether manual modification is required. If yes, obtain the method of modifying this parameter.
Work Mode	BTSTMASUBUNIT.MODE	Check whether manual modification is required. If yes, obtain the method of modifying these parameters.
Gain(0.25db)	BTSTMASUBUNIT.GAIN	
Antenna Tributary 1 Factor	BTSRETSUBUNIT.ATTENFACTOR1	After manually modifying the values of the BTSTMASUBUNIT.MODE and BTSTMASUBUNIT.GAIN parameters, and obtaining the accurate antenna loss, calculate the modification value for the RX channel attenuation using the following formula: BTSRXUBP.ATTENFACTOR = BTSTMASUBUNIT.GAIN – Antenna loss.
Antenna Tributary 1 Factor	BTSRETSUBUNIT.MRUATTENFACTOR1	
Antenna Tributary 2 Factor	BTSRETSUBUNIT.ATTENFACTOR2	
Antenna Tributary 2 Factor	BTSRETSUBUNIT.MRUATTENFACTOR2	

2. Scenario-specific parameters that need to be manually added

Only parameters listed in [Table 4-84](#) or [Table 4-85](#) need to be manually configured in scenarios 1 and 2. However, apart from those parameters in [Table 4-84](#) or [Table 4-85](#), parameters in [Table 4-86](#) also need to be manually added in scenarios 3 to 7.

Table 4-86 Scenario-specific parameters that need to be manually added

Scenario Item	Scenario Description	Configuration Data	Setting Notes
Scenario 3	Single-subunit RET in a daisy chain mode	RET subunit connection port	Add data of the connection port for the RETSUBUNIT MO based on connections between the RET antenna and RF module. The vendor code and serial number of the device identify an RET antenna. The cabinet number, subrack number, and slot number identify the RF module where an RF port connecting to the RET antenna is located and the port number identifies the RF port.
Scenario 4	Multi-subunit RET	RET subunit connection port	Add data of the connection port for the RETSUBUNIT MO based on connections between the RET antenna and RF module. The ALD device number and subunit number identify an RET subunit. The cabinet number, subrack number, and slot number identify the RF module where an RF port connecting to the RET antenna is located and the port number identifies the RF port.
Scenario 5	AISG1.1-based TMA	TMA subunit connection port	Add data of the connection port for the TMASUBUNIT MO based on connections between the TMA and RF module. The vendor code and serial number of the device identify a TMA. The cabinet number, subrack number, and slot number identify the RF module where an RF port connecting to the TMA is located and the port number identifies the RF port.
Scenario 6	TMA connected to two cascaded RRUs or RFUs	TMA subunit connection port	Manually modify the configuration of the connection port for the TMASUBUNIT MO after automatic configuration is complete. This is because automatic configuration cannot identify RF ports ANT_A (R0A) on the two cascaded RF modules as TMA subunit connection ports.
		RX channel attenuation	Manually modify attenuation on the four RX channels of two RF modules.
Scenario 7	TMA connected to the RRU with four RF ports	TMA subunit connection port	Manually modify the configuration of the connection port for the TMASUBUNIT MO after automatic configuration is complete. This is because automatic configuration cannot identify any of the four RF ports on RF modules as the TMA subunit connection port.

Scenario Item	Scenario Description	Configuration Data	Setting Notes
		RX channel attenuation	Manually modify attenuation on the RX channels of the RF modules.

RET Antenna Configuration File

- Check whether the RET antenna configuration file needs to be downloaded.
Download the RET antenna configuration file in scenarios where an RCU and antenna are delivered separately and installed onsite. There is no need to download the configuration file in scenarios where an RCU and antenna are combined or delivered together or where an existing RET antenna is reused.
- Obtain the RET antenna configuration file from the RET antenna manufacturer.
After obtaining the configuration file, record RCU and antenna models of all sectors in a base station in the site survey report and determine which configuration file to download for a specific RET antenna based on the model information.

4.7.2.2 Creating an Automatic ALD Deployment Task

Automatic ALD deployment must be performed on the MAE-Access Antenna Management System (AMS) client. This section describes how to start the MAE-Access AMS client and create an automatic ALD deployment task.

Starting the MAE-Access AMS

You can start the MAE-Access AMS client by entering the IP address in the Internet Explorer (IE) address bar or by using the MAE-Access client.

Method 1: Entering the IP address in the address bar of the IE

In the address bar of the IE, enter **https://MAE-Access IP address:31040/ams** (for example, **https://10.141.143.253:31040/ams**) and press **Enter**. In the displayed login window of the MAE-Access AMS client, input **User Name**, **Password**, and **Verification Code**.

Note that the user name and password for the MAE-Access AMS client are the same as those for the MAE-Access client.

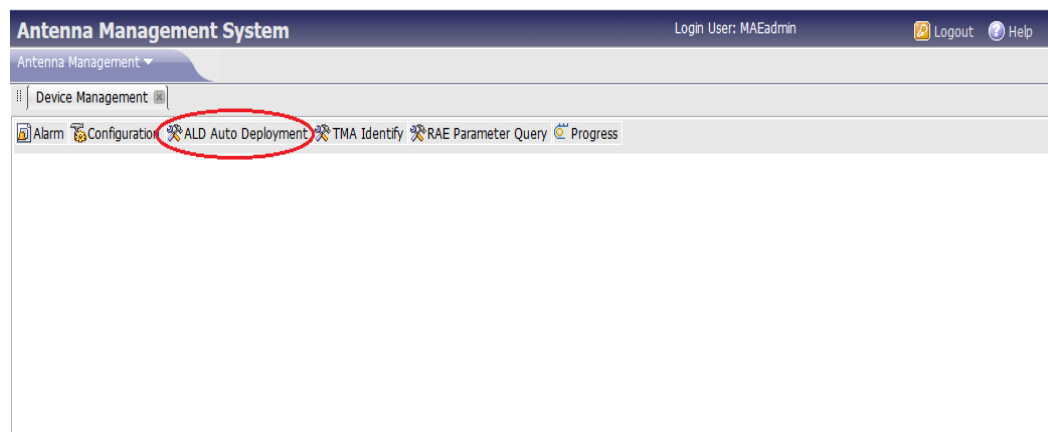
Method 2: Using the MAE-Access client

On the MAE-Access client, choose **Maintenance > Antenna Management > Device Management**. The MAE-Access AMS client is started.

Creating an Automatic ALD Deployment Task

After the MAE-Access AMS client is started, click the **Device Management** tab. In the displayed **Device Management** tab page, click the **ALD Auto Deployment** tab.

Figure 4-11 Opening the ALD Auto Deployment tab page



Operators can manage and monitor automatic ALD deployment on a single NE after an automatic ALD deployment task has been created on the MAE-Access AMS client.

To create an automatic ALD deployment task for an NE, you can manually select the NE or import a deployment list that contains the NE information.

NOTE

The preceding deployment list is exported through the MAE-Deployment. For details, see *3900 & 5900 Series Base Station Initial Configuration*.

Method 1: Manually selecting an NE

Step 1 On the **ALD Auto Deployment** tab page, click **Create**.

Step 2 Select target NEs and click **OK**.

You can click  to move one or multiple NEs from the **Available NEs** list to the **Select NEs** list.

You can click  to move all NEs from the **Available NEs** list to the **Select NEs** list.

Step 3 The value of **Status** is **Wait to be started**, indicating that an automatic ALD deployment task has been created on the MAE-Access for the selected NEs.

----End

Method 2: Importing a deployment list

Step 1 On the **ALD Auto Deployment** tab page, click **Import NE List**.

Step 2 In the displayed **Import** dialog box, click **Browse** to select a deployment list and click **OK**.

Step 3 The value of **Status** is **Wait to be started**, indicating that an automatic ALD deployment task has been created on the MAE-Access for the imported NEs.

----End

4.7.2.3 Initial Configuration

Initial configuration for automatic ALD deployment includes automatic configuration and manual configuration. Manual configuration is performed after automatic configuration.

Automatic Configuration

NOTICE

Do not stop an ongoing automatic ALD configuration task. Perform other operations only after the automatic configuration is complete.

- Step 1** On the list of automatic ALD deployment tasks, select a target NE and click **Start**.
- Step 2** Wait until automatic configuration is complete. During the automatic configuration, **Status** is **Running**.
- Step 3** When automatic configuration is complete, the **Progress** becomes **100%** and the **Status** changes to **Wait to be acknowledged**.
- Step 4** Click **Export** in the **Report** column to download an automatic ALD configuration report.
- Step 5** Check the configuration process and results in the automatic ALD configuration report.

----End

4.7.2.4 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Run MML commands to manually configure ALD data for a GBTS.

[Table 4-87](#) describes the MML commands used to manually modify common ALD data.

Table 4-87 MML commands used to manually modify common ALD data

MO	Parameter Name	Parameter ID	MML Command
BTSRETSUBUNIT	Tilt	<i>TILT</i>	MOD BTSRETSUBUNIT
BTSTMASUBUNIT	Work Mode	<i>MODE</i>	MOD BTSTMASUBUNIT
BTSTMASUBUNIT	Gain(0.25db)	<i>GAIN</i>	MOD BTSTMASUBUNIT

MO	Parameter Name	Parameter ID	MML Command
BTSRXUBP	Antenna Tributary 1 Factor	<i>ATTENFACTOR1</i>	SET BTSRXUBP
	Antenna Tributary 1 Factor	<i>MRRUATTENFACTOR1</i>	
	Antenna Tributary 2 Factor	<i>ATTENFACTOR2</i>	
	Antenna Tributary 2 Factor	<i>MRRUATTENFACTOR2</i>	

Table 4-88 describes the MML commands used to manually modify scenario-specific ALD data.

Table 4-88 MML commands used to manually modify scenario-specific ALD data

Scenario Item	Scenario Description	Configuration Data	MML Command
Scenario 3	Single-subunit RET in a daisy chain mode	Parameters related to the "Connect Port" in the BTSRETSUBUNIT MO	MOD BTSRETSUBUNIT
Scenario 4	Multi-subunit RET	Parameters related to the "Connect Port" in the BTSRETSUBUNIT MO	MOD BTSRETSUBUNIT
Scenario 6	TMA connected to two cascaded RRUs or RFUs	Parameters related to the "Connect Port" in the BTSTMASUBUNIT MO	MOD BTSTMASUBUNIT
		RX channel attenuation	SET BTSRXUBP
Scenario 7	TMA connected to the RRU/RFU with four RF ports	Parameters related to the "Connect Port" in the BTSTMASUBUNIT MO	MOD BTSTMASUBUNIT
		RX channel attenuation	SET BTSRXUBP

4.7.2.5 Using the MAE-Deployment

Using the MAE-Deployment to Perform Single Configuration

For detailed operations, see Feature Configuration Using the MAE-Deployment.

Using the MAE-Deployment to Perform Batch Configuration

You are advised to use the batch configuration function on the MAE-Deployment to manually configure ALD data for base stations of the same type (eGBTS/NodeB/eNodeB/gNodeB) to improve configuration efficiency.

For detailed operations, see Feature Configuration Using the MAE-Deployment.

1. Export ALD data of multiple base stations using the MAE-Deployment.
The ALD-related MOs that need to be exported include **ANTENNAPORT**, **RETPORT**, **RXBRANCH**, **RET**, **RETSUBUNIT**, **TMA**, and **TMASUBUNIT**.
2. Modify the exported ALD data.
Export ALD data of multiple base stations to data tables using the MAE-Deployment. Modify the data for a specific scenario as described in [4.7.2.1 Data Preparation](#).
3. Import ALD data of multiple base stations using the MAE-Deployment.

4.7.3 Commissioning

Downloading RET Antenna Configuration Files in Batches

To download RET antenna configuration files in batches, perform the following steps on the MAE-Access AMS client:

- Step 1** On the MAE-Access AMS client, click the **Configuration** tab.
- Step 2** Select an NE and then an RET subunit on the displayed **RET Subunit** tab page. Multiple NEs of the same version can be selected and multiple RET subunits can be selected for the same NE.
- Step 3** Choose **Transfer GFG File > From OSS Client to OSS Server** to upload the target RET antenna configuration files from local client to the MAE-Access server.
- Step 4** On the **RET Subunit** tab page, click **Export Configuration Template** to export the template for downloading RET antenna configuration files in batches.
- Step 5** Input the configuration file name and tilt for each RET subunit in the exported template. If the tilt is not specified for an RET subunit, the original tilt remains unchanged.
- Step 6** On the **RET Subunit** tab page, click **Import Configuration Template** to import the template for downloading configuration files for the RET antenna in batches. The system automatically downloads configuration files for the RET antenna, calibrates the RET antenna, and sets the downtilt.
- Step 7** On the **RET Subunit** tab page, click **Export Configuration Report** to export the report of downloading configuration files for the RET antenna in batches.
- Step 8** Check the process and results in the report of downloading configuration files for the RET antenna in batches.

----End

4.7.4 Activation Verification

eGBTS/NodeB/eNodeB/gNodeB

- Step 1** Run the **DSP RETSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the working status and tilt of each RET subunit. If **Online Status** is **AVAILABLE** in the command output, the RET subunit is working properly.
- Step 2** Run the **DSP RET** (old model)/**DSP ALD** (new model) command to query the dynamic information about an RET antenna. If the values of the actual vendor code and serial number for the device, and actual number of RET subunits are valid, the RET antenna has started properly.

NOTE

If only one antenna port on the RF module supports RET, the RF module is unable to report the **Control Port No.** parameter setting. The value of this parameter is displayed as **NULL**.

- Step 3** If a TMA is used, run the **DSP TMA** (old model)/**DSP ALD** (new model) command to query the dynamic information about the TMA. If the values of the actual vendor code and serial number for the device, and actual number of TMA subunits are valid, the TMA has started properly.
- Step 4** If a TMA is used, run the **DSP TMASUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the working status of TMA subunits. If **Online Status** is **AVAILABLE** in the command output, the TMA subunit is working properly.

----End

GBTS

- Step 1** Run the **DSP BTSRETSUBUNIT** command to query the working status and tilt of each RET subunit. If **Online Status** is **AVAILABLE** in the command output, the RET subunit is working properly.
- Step 2** Run the **DSP BTSRET** command to query dynamic information about the RET antenna. If the values of the actual vendor code and serial number for the device, and actual number of RET subunits are valid, the RET antenna has started properly.
- Step 3** If a TMA is used, run the **DSP BTSTMA** command to query dynamic information about the TMA. If the values of the actual vendor code and serial number for the device, and actual number of TMA subunits are valid, the TMA has started properly.
- Step 4** If a TMA is used, run the **DSP BTSTMASUBUNIT** command to query the working status of TMA subunits. If **Online Status** is **AVAILABLE** in the command output, the TMA subunit is working properly.

----End

4.7.5 Checking that Automatic ALD Deployment Is Complete

After an automatic ALD deployment task is completed for an NE, you need to confirm that the automatic ALD deployment is completed on the **ALD Auto Deployment** tab page of the MAE-Access AMS.

Step 1 In the list of automatic ALD deployment tasks, select NEs whose **Status** is **Wait to be acknowledged** and click **Acknowledge**.

Step 2 The values of **Status** for the NEs change to **Completed**.

----End

4.7.6 Deactivation

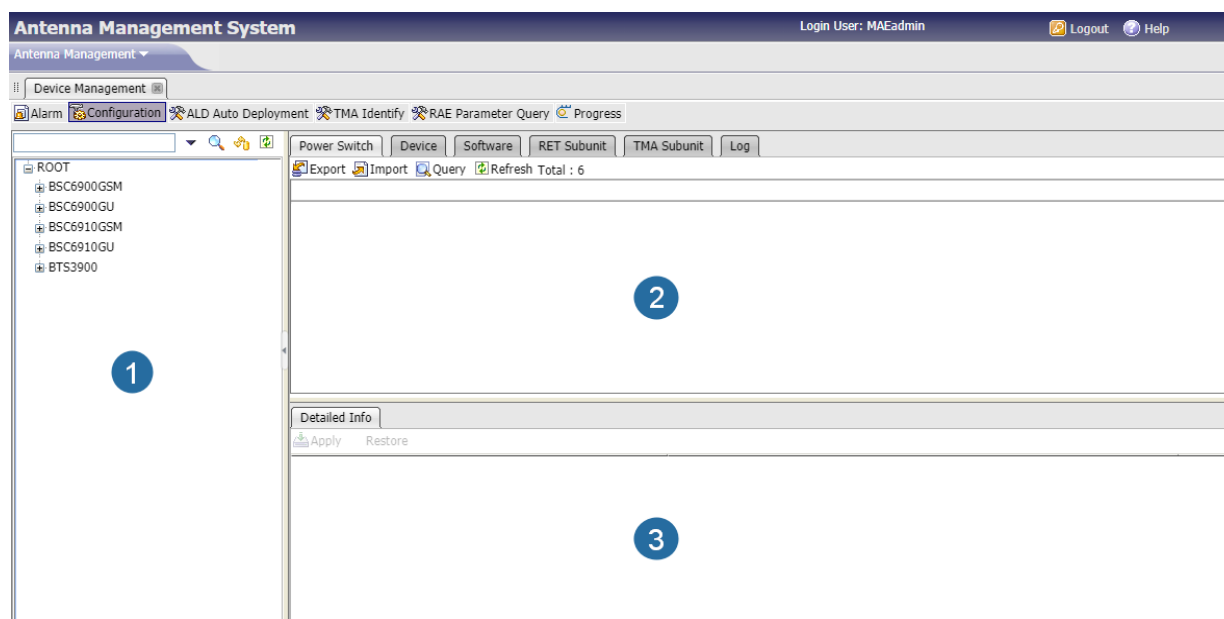
If an ALD is no longer used, remove it by running the following commands. Then, turn off the power switch and set the attenuation to its default value.

eGBTS/NodeB/eNodeB/gNodeB

Step 1 Remove the ALD data.

- Using the MAE-Access AMS
 - a. In the MAE-Access AMS, click **Configuration** on the **Device Management** tab page, as shown in the following figure.

Figure 4-12 ALD configuration tab page



- b. On the displayed **Configuration** tab page, choose the target NE from which the ALD is to be removed in area 1.
 - c. In area 2, click the **Device** tab, choose the ALD to be removed, and click **Remove**. Data of the ALD and its subunits are removed.
- Using MML commands
 - Run the **RMV RET** command to remove an RET antenna and its subunits.
 - Run the **RMV TMA** command to remove a TMA and its subunits.

Step 2 Turn off the power switch.

- Using the MAE-Access AMS

- a. In area 2 on the **Configuration** tab page, click the **Power Switch** tab, choose the port where the power switch for the ALD removed in step 1 is located.
- b. In area 3, set the power switch for the port to **OFF**.
- Using MML commands
 - If all ALDs that are connected to an RET port are removed, run the **MOD RETPORT** command to turn off the power switch for the port.
 - If all ALDs that are connected to an antenna port are removed, run the **MOD ANTENNAPORT** command to turn off the power switch for the port.

Step 3 Set the corresponding attenuation parameters to the default value.

Run the **MOD RXBRANCH** command to set the corresponding RX channel attenuation parameters to the default value.

-----End

GBTS

Step 1 Remove the ALD data.

- Using the MAE-Access AMS
 - a. In the MAE-Access AMS, click **Configuration** on the **Device Management** tab page, as shown in [Figure 4-12](#).
 - b. On the displayed **Configuration** tab page, choose the target NE from which the ALD is to be removed in area 1.
 - c. In area 2, click the **Device** tab, choose the ALD to be removed, and click **Remove**. Data of the ALD and its subunits are removed.
- Using MML commands
 - Run the **RMV BTSRET** command to remove an RET antenna and its subunits.
 - Run the **RMV BTSTMA** command to remove a TMA and its subunits.

Step 2 Turn off the power switch.

- Using the MAE-Access AMS
 - a. In area 2 on the **Configuration** tab page, click the **Power Switch** tab, choose the port where the power switch for the ALD removed in step 1 is located.
 - b. In area 3, set the power switch for the port to **OFF**. You can set the power switch only on one port at a time.
- Using MML commands
 - If all ALDs that are connected to an RET port are removed, run the **SET BTSRXUBP** command to turn off the power switch for the port.
 - If all ALDs that are connected to an antenna port are removed, run the **SET BTSRXUBP** command to turn off the power switch for the port.

Step 3 Set the corresponding attenuation parameters to the default value.

Run the **SET BTSRXUBP** command to set the corresponding RX channel attenuation parameters to the default value.

----End

4.7.7 Reconfiguration

If ALD data needs to be reconfigured after initial configuration is complete, you can use either of the following ways:

- Remove ALDs that require data reconfiguration and then use the automatic ALD deployment to restart initial configuration.
- Run MML commands to reconfigure ALD data.

Removing ALDs and Using Automatic ALD Deployment (eGBTS/NodeB/eNodeB/gNodeB)

1. Remove ALDs that require data reconfiguration:
For details, see [eGBTS/NodeB/eNodeB/gNodeB](#).
2. Use automatic ALD deployment to restart initial configuration.
For details, see [4.7.2.3 Initial Configuration](#).

Removing ALDs and Using Automatic ALD Deployment (GBTS)

1. Remove ALDs that require data reconfiguration:
For details, see [GBTS](#).
2. Use automatic ALD deployment to restart initial configuration.
For details, see [4.7.2.3 Initial Configuration](#).

Running MML Commands (eGBTS/NodeB/eNodeB/gNodeB)

For details, see [4.4.7 Reconfiguration](#).

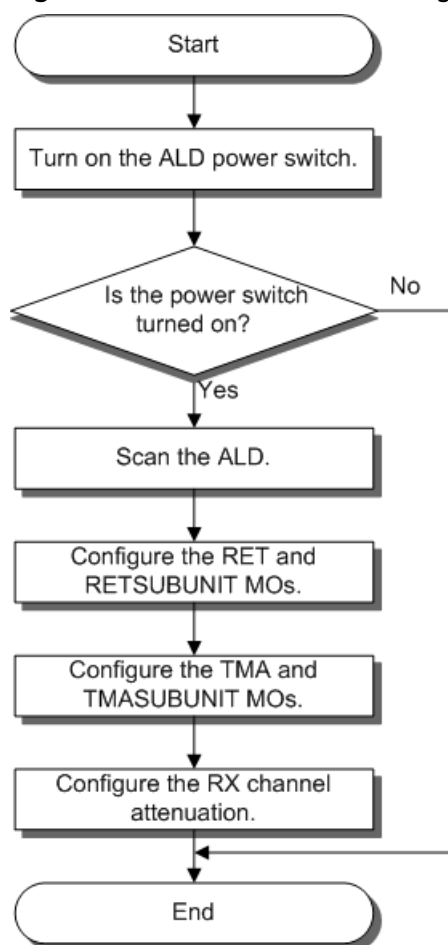
Running MML Commands (GBTS)

For details, see [4.4.7 Reconfiguration](#).

4.7.8 Appendix: Automatic ALD Configuration Process

[Figure 4-13](#) describes the process of automatic ALD configuration.

Figure 4-13 Automatic ALD configuration process



Turning on the ALD Power Switch

The system attempts to turn on the ALD power switch in the **RETPORT** or **ANTENNAPORT** MO.

- If an ALD is connected to the RET port or antenna port, the ALD power switch is turned on successfully. The system starts scanning the ALD and configuration continues.
- If no ALD is connected to the RET port or antenna port, the ALD power switch fails to be turned on. The system turns off the ALD power switch and automatic ALD configuration ends.

Table 4-89 and **Table 4-90** describe the key parameters that must be set in the **RETPORT** and **ANTENNAPORT** MOs for an eGBTS/NodeB/eNodeB/gNodeB, respectively.

Table 4-89 Key parameters that must be set in the **RETPORT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Cabinet No.	CN	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU where the control port is located and the control port number.	These parameters are set to the cabinet number, subrack number, and slot number of the module where the RET port is located and the port number of the RET port. The system attempts to turn on the power switch in the RETPORT MO.
Subrack No.	SRN		
Slot No.	SN		
Port No.	PN		
ALD Power Switch	PWRSWITCH	Set this parameter to ON when an ALD is used. The default value is OFF .	This parameter is set to ON when the power switch is turned on successfully or to OFF when the power switch cannot be turned on.
Current Alarm Threshold Type	THRESHOLDTYPE	-	This parameter is set to the default value RET_ONLY_MULTICORE .

Table 4-90 Key parameters that must be set in the **ANTENNAPORT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Cabinet No.	CN	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located and the control port number.	These parameters are set to the cabinet number, subrack number, and slot number of the module where the antenna port is located and the port number of the antenna port. The system attempts to turn on the ALD power switch in the ANTENNAPORT MO.
Subrack No.	SRN		
Slot No.	SN		
Port No.	PN		
ALD Power Switch	PWRSWITCH	Set this parameter to ON when an ALD is used. The default value is OFF .	This parameter is set to ON when the power switch is turned on successfully or to OFF when the power switch cannot be turned on.

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Current Alarm Threshold Type	<i>THRESHOLDTYPE</i>	N/A	This parameter is set to the default value TMA24DB_AISG .

Table 4-91 and **Table 4-92** describe the key parameters that must be set in the **RETPORT** and **ANTENNAPORT** MOs for a GBTS, respectively.

Table 4-91 Key parameters that must be set in the **RETPORT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
RET ALD Power Switch	<i>PwrSwitchRET</i>	Set this parameter to ON when an ALD is used. The default value is OFF .	This parameter is set to ON when the power switch is turned on successfully or to OFF when the power switch cannot be turned on.
RET ALD Current Alarm Threshold Type	<i>THRESHOLDTYPERET</i>	N/A	This parameter is set to the default value RET_ONLY_MULTICORE .

Table 4-92 Key parameters that must be set in the **ANTENNAPORT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
ANT_A ALD Power Switch	<i>PwrSwitchA</i>	Set this parameter to ON when an ALD is used. The default value is OFF .	This parameter is set to ON when the power switch is turned on successfully or to OFF when the power switch cannot be turned on.
ANT_A ALD Current Alarm Threshold Type	<i>ChkModA</i>	N/A	This parameter is set to the default value TMA24DB_AISG .

Scanning ALDs

The system starts scanning ALDs after the ALD power switch is turned on. The system will detect a connected ALD and existing data of the ALD. The ALD data is used for later automatic configuration.

Configuring the RET and RETSUBUNIT MOs

After an RET antenna is detected, the system automatically configures the **RET** and **RETSUBUNIT** MOs.

[Table 4-93](#) and [Table 4-94](#) describe the key parameters that must be set in the **RET** and **RETSUBUNIT** MOs for an eGBTS/NodeB/eNodeB/gNodeB, respectively.

Table 4-93 Key parameters that must be set in the **RET** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.	The system automatically allocates the numbers with no duplicates.
Device Name	DEVICENAME	This parameter identifies an RET antenna.	The format of the device name is as follows: <i>device type_cabinet number of the control port_subrack number of the control port_slot number of the control port_random number</i> .
Control Port Cabinet No.	CTRLCN	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located.	These parameters are set based on the ALD scanning results.
Control Port Subrack No.	CTRLSRN		
Control Port Slot No.	CTRLSN		
RET Type	RETTYPE	Set this parameter to SINGLE_RET for the RET antenna with a single RET subunit. Set this parameter to MULTI_RET for the RET antenna with multiple RET subunits as required.	This parameter is set based on the ALD scanning results.
Number of RET Subunits	SUBUNITNUM	Set this parameter only when the RETTYPE parameter is set to MULTI_RET .	This parameter is set based on the ALD scanning results.
Polar Type	POLARTYPE	N/A	This parameter is set to DUAL .

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Antenna Scenario	SCENARIO	This parameter specifies how the RET antenna is connected to an RRU or RFU. ¹	If only one RET antenna is detected on a control port, this parameter is set to REGULAR . If multiple RET antennas are detected on a control port, this parameter is set to DAISY_CHAIN .
Vendor Code	VENDORCODE	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the manufacturer information, for example, KA for a Kathrein RET antenna, AN for an Andrew RET antenna, or HW for a Huawei Agissson RET antenna.	This parameter is set based on the ALD scanning results.
Serial No.	SERIALNO	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number.	This parameter is set based on the ALD scanning results.

 NOTE

- Set this parameter to **REGULAR** if the RET antenna is directly connected to the RRU or RFU. In this scenario, the **VENDORCODE** and **SERIALNO** parameters do not need to be specified.
- Set this parameter to **DAISY_CHAIN** when two RET antennas are cascaded. In this scenario, the control port for RET antennas must be configured on the upper-level RRU or RFU of the daisy chain. The **VENDORCODE** and **SERIALNO** parameters must be specified.

Table 4-94 Key parameters that must be set in the **RETSUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	This parameter specifies the device number of the ALD. Set this parameter to the device number in the RET MO.	Device number of the ALD configured with an RET unit
Subunit No.	SUBUNITNO	This parameter specifies the RET subunit number, which starts from 1.	The system automatically allocates the numbers with no duplicates.

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Subunit Name	<i>SUBNAME</i>	This parameter specifies the name of the RET subunit. The name of the RET subunit is optional, but the name configured for each RET subunit must be unique.	This parameter must be manually configured.
Connect Port 1 Cabinet No.	<i>CONN CN1</i>	-	If single-subunit RET antennas are used and the RET antennas are not in daisy chain mode, connection port 1 is set to ROA on the RRU where the control port is located, and connection port 2 is set to ROB on the RRU where the control port is located. If a multi-subunit RET antenna is used or if single-subunit RET antennas work in daisy chain mode, neither connection port 1 nor connection port 2 is set.
Connect Port 1 Subrack No.	<i>CONNSRN1</i>		
Connect Port 1 Slot No.	<i>CONNSN1</i>		
Connect Port 1 Port No.	<i>CONNPN1</i>		
Connect Port 2 Cabinet No.	<i>CONN CN2</i>		
Connect Port 2 Subrack No.	<i>CONNSRN2</i>		
Connect Port 2 Slot No.	<i>CONNSN2</i>		
Connect Port 2 Port No.	<i>CONNPN2</i>		
Tilt	<i>TILT</i>	N/A	This parameter is set to the actual RET antenna downtilt obtained from the output of the DSP RETSUBUNIT command, which is automatically executed by the system.

[Table 4-95](#) and [Table 4-96](#) describe the key parameters that must be set in the **RET** and **RETSUBUNIT** MOs for a GBTS, respectively.

Table 4-95 Key parameters that must be set in the **RET MO**

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.	The system automatically allocates the numbers with no duplicates.
Device Name	DEVICENAME	This parameter identifies an RET antenna.	The format of the device name is as follows: <i>device type_cabinet number of the control port_subrack number of the control port_slot number of the control port_random number</i> .
Control Port Cabinet No.	CTRLPORTC N	These parameters specify location information about the control port for an RET antenna, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located.	These parameters are set based on the ALD scanning results.
Control Port Subrack No.	CTRLPORTS RN		
Control Port Slot No.	CTRLPORTS N		
Control Port No.	CTRLPORTN O	This parameter specifies the control port number. The value ranges from 0 to 2. Control ports 0, 1, and 2 correspond to the ports ANT_A, ANT_B, and RET, respectively. Only one port on the RRU can be used as the control port for the RET antenna. In a daisy chain scenario, multiple RCUs share one control port.	This parameter is set based on the ALD scanning results.
RET Type	RETTYPE	Set this parameter to SINGLE_RET for the RET antenna with a single RET subunit. Set this parameter to MULTI_RET for the RET antenna with multiple RET subunits as required.	This parameter is set based on the ALD scanning results.
Number of RET Subunits	SUBUNITN UM	This parameter specifies the number of RET subunits used by a base station. Set this parameter only when the RETTYPE parameter is set to MULTI_RET .	This parameter is set based on the ALD scanning results.
Polar Type	POLARTYPE	N/A	This parameter is set to DUAL .

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Antenna Scenario	SCENARIO	This parameter specifies how the RET antenna is connected to an RRU or RFU. ¹	If only one RET antenna is detected on a control port, this parameter is set to REGULAR . If multiple RET antennas are detected on a control port, this parameter is set to DAISY_CHAIN .
Vendor Code	VENDORCODE	This parameter is mandatory in daisy chain scenarios. Set this parameter based on the manufacturer information, for example, KA for a Kathrein RET antenna, AN for an Andrew RET antenna, or HW for a Huawei Agissson RET antenna.	This parameter is set based on the ALD scanning results.
Serial No.	SERIALNO	This parameter is mandatory in daisy chain scenarios. Set this parameter according to the antenna serial number.	This parameter is set based on the ALD scanning results.

 NOTE

- Set this parameter to **REGULAR** if the RET antenna is directly connected to the RRU or RFU. In this scenario, the **VENDORCODE** and **SERIALNO** parameters do not need to be specified.
- Set this parameter to **DAISY_CHAIN** when two RET antennas are cascaded. In this scenario, the control port for RET antennas must be configured on the upper-level RRU or RFU of the daisy chain. The **VENDORCODE** and **SERIALNO** parameters must be specified.

Table 4-96 Key parameters that must be set in the **RETSUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	This parameter specifies the device number of the ALD. Set this parameter to the device number in the RET MO.	Device number when an RET unit is configured
Subunit No.	SUBUNITNO	This parameter specifies the RET subunit number, which starts from 1.	The system automatically allocates the numbers with no duplicates.

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Connect Port 1 Cabinet No.	CONN1CN1	-	If single-subunit RET antennas are used and the RET antennas are not in daisy chain mode, connection port 1 is set to 0 (ANT_A) on the RRU where the control port is located, and connection port 2 is set to 1 (ANT_B) on the RRU where the control port is located. If a multi-subunit RET antenna is used or if single-subunit RET antennas work in daisy chain mode, neither connection port 1 nor connection port 2 is set.
Connect Port 1 Subrack No.	CONNSRN1		
Connect Port 1 Slot No.	CONNSN1		
Connect Port 1 Port No.	CONNPN1		
Connect Port 2 Cabinet No.	CONN1CN2		
Connect Port 2 Subrack No.	CONNSRN2		
Connect Port 2 Slot No.	CONNSN2		
Connect Port 2 Port No.	CONNPN2		
Tilt (0.1degree)	TILT	N/A	This parameter is set to the actual RET antenna downtilt obtained from the output of the DSP BTSRETSUBUNIT command, which is automatically executed by the system.

Configuring the TMA and TMASUBUNIT MOs

After a TMA is detected, the system automatically configures the **TMA** and **TMASUBUNIT** MOs.

[Table 4-97](#) and [Table 4-98](#) describe the key parameters that must be set in the **TMA** and **TMASUBUNIT** MOs for an eGBTS/NodeB/eNodeB/gNodeB, respectively.

Table 4-97 Key parameters that must be set in the **TMA** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.	The system automatically allocates the numbers with no duplicates.
Device Name	DEVICENAME	This parameter identifies an RET antenna.	The format of the device name is as follows: <i>device type_cabinet number of the control port_subrack number of the control port_slot number of the control port_random number</i> .
Control Port Cabinet No.	CTRLCN	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located. Set these parameters based on connections between the TMA and the RRU or RFU.	These parameters are set based on the ALD scanning results.
Control Port Subrack No.	CTRLSRN		
Control Port Slot No.	CTRLSN		
Number of TMA Subunits	SUBUNITNUM	Set this parameter as required. Generally, there are two TMA subunits.	This parameter is set based on the ALD scanning results.
Vendor Code	VENDORCODE	This parameter is required in a non-regular scenario. Set this parameter to the actual TMA manufacturer code.	This parameter is set based on the ALD scanning results.
Serial No.	SERIALNO	This parameter is required in a non-regular scenario. Set this parameter to the actual TMA serial number.	This parameter is set based on the ALD scanning results.
TMA Type	TMA TYPE	This parameter specifies the TMA type. The value NORMAL_TMA indicates a normal TMA. The value VIRTUAL_TMA indicates a device other than a TMA which amplifies signals between an RF port and an antenna port.	This parameter is set based on the ALD scanning results.

Table 4-98 Key parameters that must be set in the **TMASUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	This parameter specifies the device number of the ALD. Set this parameter to the device number in the TMA MO.	Device number when an RET unit is configured
Subunit No.	SUBUNITNO	Number of a TMA subunit	The system automatically allocates the numbers with no duplicates.
Connect Port Cabinet No.	CONN CN	-	If the number of TMA subunits is 2, connection port 1 is set to ROA on the RRU where the control port is located, and connection port 2 is set to ROB on the RRU where the control port is located. If the number of TMA subunits is not 2, connection ports are not set.
Connect Port Subrack No.	CONN SRN		
Connect Port Slot No.	CONN SN		
Connect Port No.	CONN PN		
Work Mode	MODE	The TMA subunit supports two working modes, normal mode and bypass mode: • In normal mode, the TMA subunit functions and the TMA amplifies uplink signals. • In bypass mode, the TMA subunit works as a straight-through feeder. It does not amplify any uplink signals. The default value is NORMAL .	This parameter is set to the actual working mode of the TMA subunit obtained from the output of the DSP TMASUBUNIT command, which is automatically executed by the system.
Gain	GAIN	Set this parameter based on the engineering design. The gain value range supported by the TMA varies with the manufacturer and model. Run the DSP TMADEVICEDATA command to query the value range before setting the gain. If the gain is fixed, this parameter is optional, or you can set this parameter to its actual gain value.	This parameter is set to the actual gain of the TMA subunit obtained from the output of the DSP TMASUBUNIT command, which is automatically executed by the system.

Table 4-99 and **Table 4-100** describe the key parameters that must be set in the **TMA** and **TMASUBUNIT** MOs for a GBTS, respectively.

Table 4-99 Key parameters that must be set in the **TMA** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	The device number of each ALD in a base station must be unique. Note that the DEVICENO parameter value of the RET antenna must differ from that of the TMA.	The system automatically allocates the numbers with no duplicates.
Device Name	DEVICENAME	This parameter identifies an RET antenna.	The format of the device name is as follows: <i>device type_cabinet number of the control port_subrack number of the control port_slot number of the control port_random number.</i>
TMA Power Supply Type	PWRSUPPLY TYPE	N/A	This parameter is set to the default value SINGLE_PORT_POWER .
Control Port Cabinet No.	CTRLPORTC N	These parameters specify location information about the control port, including the cabinet number, subrack number, and slot number of the RRU or RFU where the control port is located. Set these parameters based on the control relationship between the TMA and the RRU or RFU.	These parameters are set based on the ALD scanning results.
Control Port Subrack No.	CTRLPORTS RN		
Control Port Slot No.	CTRLPORTS N		
Number of TMA Subunits	SUBUNITN UM	Set this parameter as required. Generally, there are two TMA subunits.	This parameter is set based on the ALD scanning results.
Vendor Code	VENDORCODE	N/A	This parameter is set based on the ALD scanning results.
Serial No.	SERIALNO	N/A	This parameter is set based on the ALD scanning results.

Table 4-100 Key parameters that must be set in the **TMASUBUNIT** MO

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Device No.	DEVICENO	This parameter specifies the device number of the ALD. Set this parameter to the device number in the TMA MO.	Device number of the ALD configured with an RET unit

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Subunit No.	<i>SUBUNITN</i> <i>O</i>	This parameter specifies the number of a TMA subunit.	The system automatically allocates the numbers with no duplicates.
Connect Port Cabinet No.	<i>CONN CN</i>	-	If the number of TMA subunits is 2, connection port 1 is set to 0 (ANT_A) on the RRU where the control port is located, and connection port 2 is set to 1 (ANT_B) on the RRU where the control port is located. If the number of TMA subunits is not 2, connection ports are not set.
Connect Port Subrack No.	<i>CONN SRN</i>		
Connect Port Slot No.	<i>CONN SN</i>		
Connect Port No.	<i>CONN PN</i>		
Work Mode	<i>MODE</i>	The TMA subunit supports two working modes, normal mode and bypass mode: • In normal mode, the TMA subunit functions and the TMA amplifies uplink signals. • In bypass mode, the TMA subunit works as a straight-through feeder. It does not amplify any uplink signals. The default value is NORMAL.	This parameter is set to the actual working mode of the TMA subunit obtained from the output of the DSP BTSTMASUBUNIT command, which is automatically executed by the system.
Gain(0.25db)	<i>GAIN</i>	Set this parameter based on the engineering design. The gain value range supported by the TMA varies with the manufacturer and model. Run the DSP BTSTMADEVICE DATA command to query the value range before setting the gain. If the gain is fixed, this parameter is optional, or you can set this parameter to its actual gain value.	This parameter is set to the actual working mode of the TMA subunit obtained from the output of the DSP BTSTMASUBUNIT command, which is automatically executed by the system.

Configuring the RX Channel Attenuation

You need to configure the corresponding RX channel attenuation after ***MODE*** and ***GAIN*** have been configured in the **TMASUBUNIT MO**.

Table 4-101 describes the key parameters that must be set to configure the RX channel attenuation for an eGBTS/NodeB/eNodeB/gNodeB.

Table 4-101 Key parameters that must be set to configure the RX channel attenuation

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
RX Channel No.	<i>RXNO</i>	This parameter specifies the number of the RX channel of the RRU/RFU.	RX channel number of port 0 is set to that of RF port R0A and RX channel number of port 1 is set to that of RF port R0B.
Attenuation	<i>ATTEN</i>	If no TMA is used, set this parameter to 0 . If a 12 dB TMA is used, set this parameter to a value within the range from 4 dB to 11 dB. If a 24 dB TMA is used, set this parameter to a value within the range from 11 dB to 22 dB.	If the number of TMA subunits is 2, this parameter is set to TMASUBUNIT.GAIN minus 4 dB for TMA subunits working in normal mode or to the default value 0 for TMA subunits working in bypass mode. If the number of TMA subunits is not 2, the RX channel attenuation is not set.

Table 4-102 describes the key parameters that must be set to configure the RX channel attenuation for a GBTS.

Table 4-102 Key parameters that must be set to configure the RX channel attenuation

Parameter Name	Parameter ID	Setting Notes	Automatic Configuration
Antenna Tributary 1 Flag	HAVETT1	This parameter specifies whether a TMA is connected to RF port ANT_A. If a TMA is connected, set this parameter to YES .	Antenna tributary 1 is the RX channel on port 0 (ANT_A). Antenna tributary 2 is the RX channel on port 1 (ANT_B). If the number of TMA subunits is 2, the RX channel attenuation is set to a corresponding value. When the TMA subunits are working in normal mode, set the parameters specifying TMA configuration for corresponding antenna tributaries to YES , and set the parameters specifying the RX channel attenuation to the value of TMASUBUNIT.GAIN minus 4 dB. When the TMA subunits are working in bypass mode, set the parameters specifying TMA configuration for corresponding antenna tributaries to YES , and set the parameters specifying the RX channel attenuation to 0 (default value). If the number of TMA subunits is not 2, the RX channel attenuation is not set.
Antenna Tributary 1 Factor	ATTENFACTOR1	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is set to DRRU or DRFU .	
Antenna Tributary 1 Factor	MRRUATTENFACTOR1	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is not set to DRRU or DRFU .	
Antenna Tributary 2 Flag	HAVETT2	This parameter specifies whether a TMA is connected to RF port ANT_B. If a TMA is connected, set this parameter to YES .	
Antenna Tributary 2 Factor	ATTENFACTOR2	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is set to DRRU or DRFU .	
Antenna Tributary 2 Factor	MRRUATTENFACTOR2	Set this parameter based on the site conditions after the TMA is installed. This parameter can be set only if the RXUTYPE parameter is not set to DRRU or DRFU .	

Multimode Base Station Automatic ALD Configuration

If an RRU or RFU connected to ALDs works in multiple RATs, the system uses the following rules when implementing automatic ALD deployment:

1. Parameters related to the ALD power switch and RX channel attenuation of the RRU or RFU must be set to the same values in the RATs that manage the RRU or RFU.
2. ALD parameters other than the preceding ones must be configured only in one of all RATs that manage the RRU or RFU.

The RAT can be selected on the MAE-Access by choosing **Configuration > MBTS Management > MBTS Priority Settings**. An RAT of a higher priority is preferred. Therefore, you need to set RAT priorities before starting automatic ALD deployment.

5 TMA

5.1 Principles

A TMA is a low noise amplifier (LNA) installed next to the antenna. It improves the signal-to-noise ratio (SNR), sensitivity, and uplink coverage of a base station.

A TMA is one of two types: common TMA or smart TMA.

- Common TMAs do not support the Antenna Interface Standards Group (AISG) protocol.
- Smart TMAs support the AISG protocol.

A smart TMA has a built-in smart bias-tee (SBT) and performs the following functions:

- Converts RS485 signals received from the RCU to on-off-keying (OOK) signals and sends the OOK signals to the RRU/RFU through the feeder
- Converts OOK signals received from the RRU/RFU to RS485 signals and sends the RS485 signals to the RCU
- Feeds DC power from the RRU/RFU to the RCU

Unless otherwise stated in this document, "TMA" refers to a smart TMA.

NOTE

- The SBT provides DC power supply and control commands for the RCU through the feeder. It is applied on the RET antenna side.
- When an RRU/RFU connects to a TMA, at least one RF port supporting the TMA must connect to this TMA so that the RRU/RFU can transmit control signals to the TMA. For details about capabilities of RF ports on the RRU/RFU to support the TMA, see [3 General Principles](#).

The TMA provides the following functions:

- Amplifies uplink signals to compensate for attenuation from an antenna to an RRU/RFU.
- Balances signal amplification between the uplink and downlink.

A TMA has one or multiple subunits to support amplification of uplink RF signals on one or multiple paths.

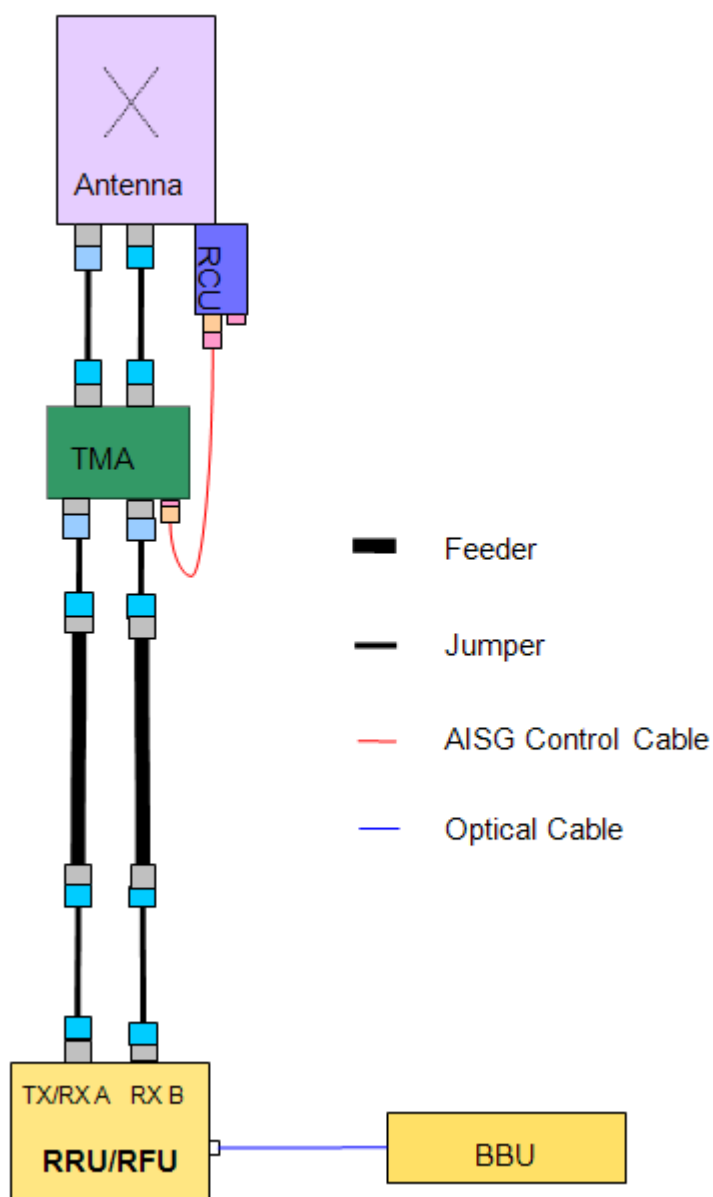
5.1.1 Connections Between the TMA, RRU/RFU, and RET Antenna

If an RRU/RFU is to be connected to an RET antenna through a TMA, the RRU/RFU control port must be connected to the TMA control port.

Connection to the RET Antenna Through the Antenna Port (with a TMA)

A TMA is connected to the RRU/RFU and RET antenna, and is powered by the RRU/RFU. [Figure 5-1](#) shows how an RRU or RFU with two RF ports is connected to the TMA and RET antenna. With the integrated SBT, the TMA splits combined signals from the RRUs/RFUs into two paths. On one path, RF signals are sent to the RET antenna. On the other path, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RCU through an AISG multi-wire cable.

Figure 5-1 Connection to the RET antenna through the antenna port (with a TMA)



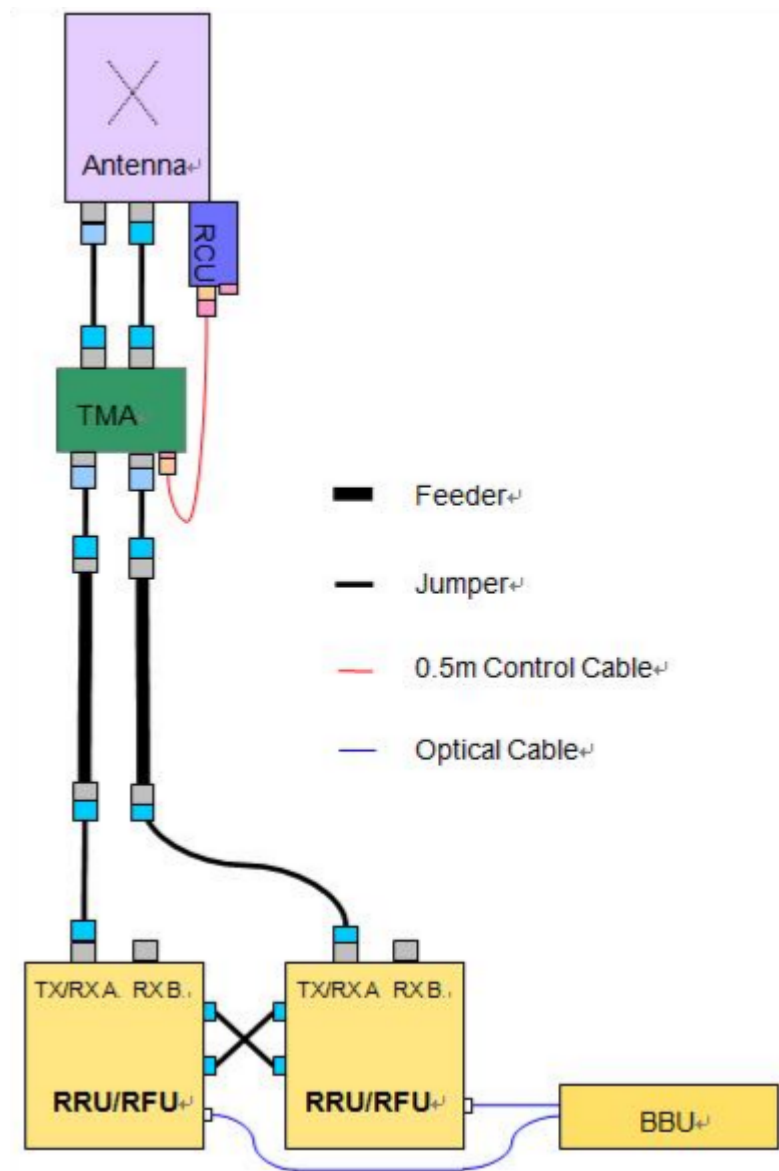
For the data configurations in this connection scheme, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 3 in [5.5 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).
- GBTS: scenario 3 in [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).
- Base station using automatic ALD deployment: scenario 2 or 5 in [5.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

Figure 5-2 shows how two RF modules are cascaded and then connected to the TMA and RET antenna. In this scheme, RF ports ANT_A on the two RF modules are

connected to the TMA and RET antenna, and either RF port ANT_A can be the control port for the RET antenna connected to the two RF modules.

Figure 5-2 Two cascaded RF modules connected to the RET antenna through the TMA



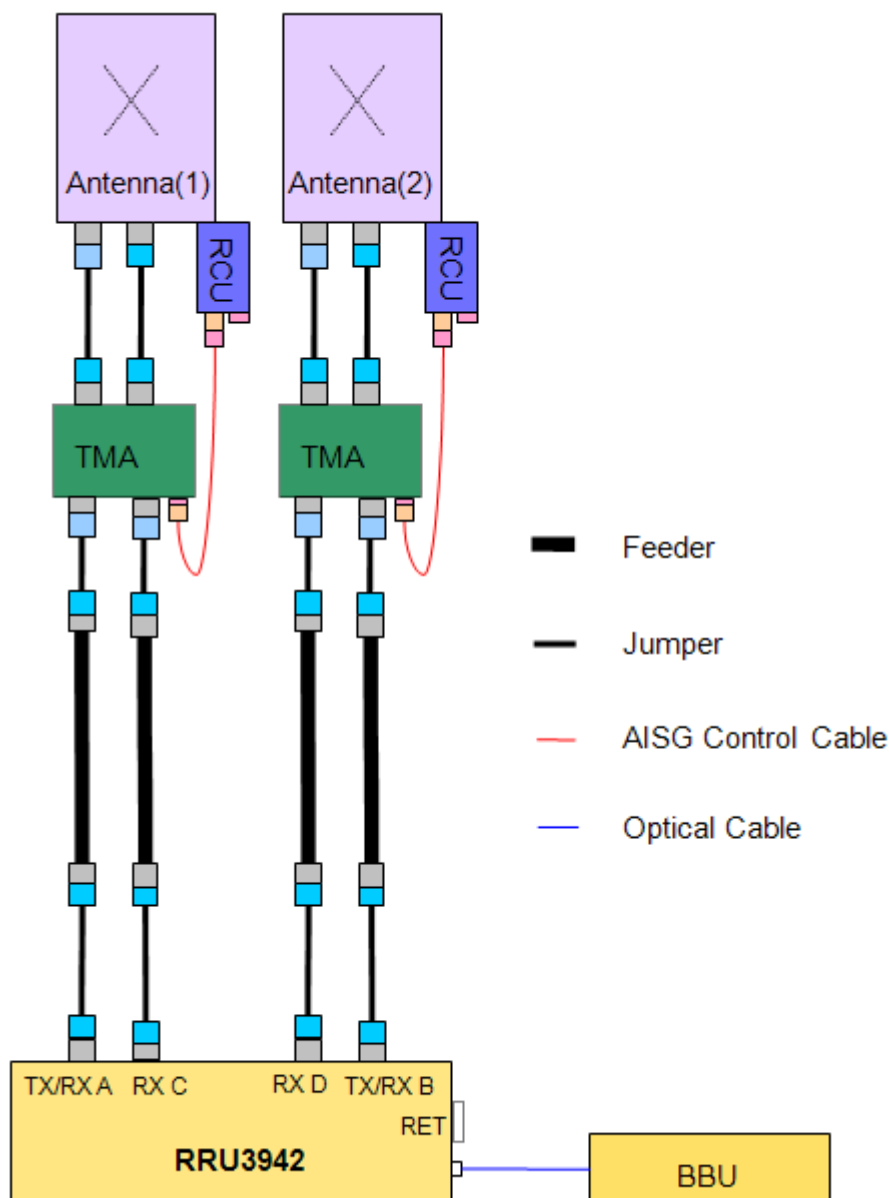
For the data configurations in this connection scheme, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 3 in [5.5 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).
- GBTS: scenario 3 in [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).
- Base station using automatic ALD deployment: scenario 6 in [5.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

When RRUs with four RF ports are connected to two antennas, two RF ports on these RRUs can be paired and connected to one antenna. The following

description assumes that the RRU3942 uses a 2T4R configuration. RF ports ANT_A and ANT_C on the RRU3942 are paired and connected to one antenna, and RF ports ANT_B and ANT_D are paired and connected to the other antenna. [Figure 5-3](#) shows how the RRU3942 is connected to RET antenna through the TMA.

Figure 5-3 RRU3942 (2T4R) connected to the RET antenna through the TMA



For the data configurations in [Figure 5-3](#), see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 3 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).
- GBTS: scenario 3 in [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

- Base station using automatic ALD deployment: scenario 7 in [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

In [Figure 5-3](#), RF ports ANT_A and ANT_B are the control ports for Antenna(1) and Antenna(2), respectively.

Connection to the RET Antenna Through the RET Port (with a TMA)

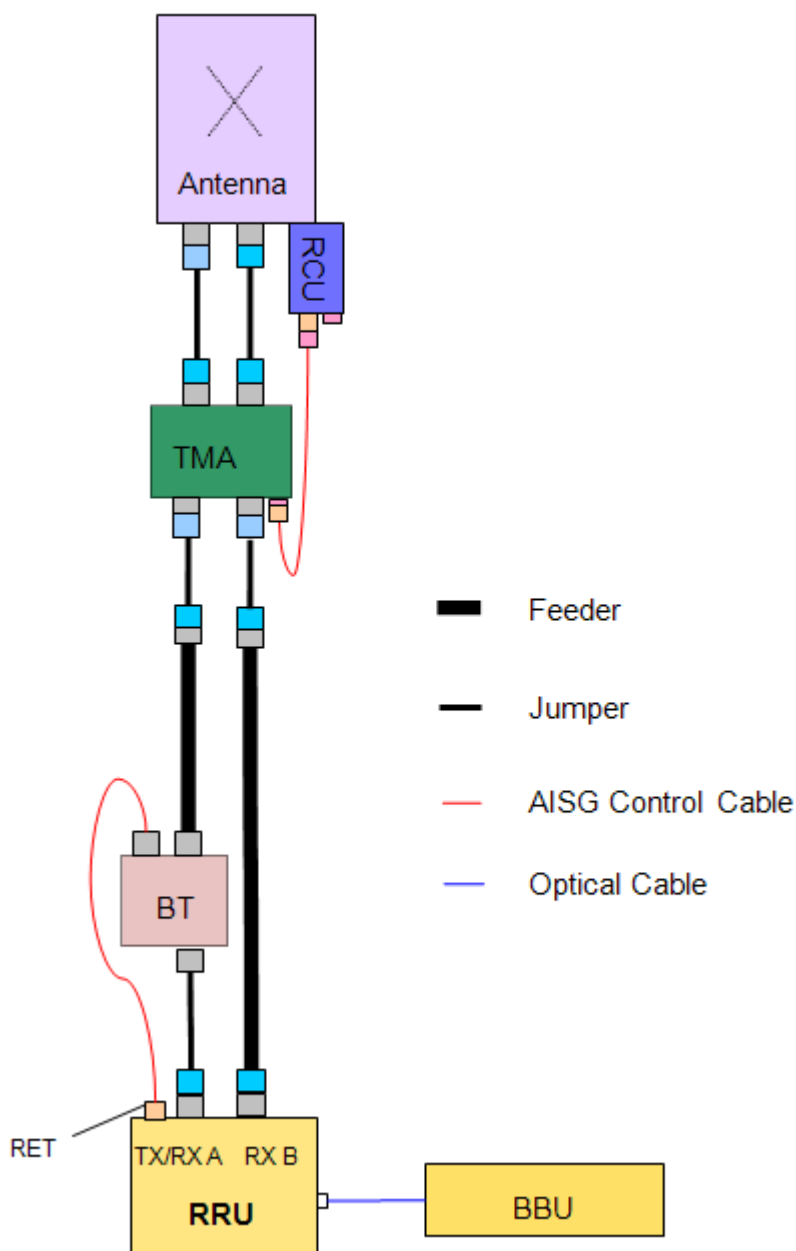
For RRUs without a BT such as the RRU3004 for GSM, RRU3801C (20 W) for UMTS, and RRU3220 for LTE, RF ports cannot provide 12 V DC power for antennas. In this case, the RET port on an RRU connects to an external BT to provide power for antennas.

[Figure 5-4](#) shows the connections between the RRU, BT, TMA, and RET antenna. The signal transmission process is as follows:

1. The RRU supplies 12 V DC power and sends RS485 control signals to the external BT through an AISG multi-wire cable.
2. The BT converts RS485 control signals to OOK signals, combines the OOK signals and RF signals, transparently forwards 12 V DC power, and sends the combined signals to the TMA.
3. The TMA splits the combined signals into two paths. On one path, RF signals are sent to the RET antenna. On the other path, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RCU through an AISG multi-wire cable.

For the data configurations in this connection scheme, see the following sections:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 4 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).
- GBTS: scenario 4 in [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

Figure 5-4 Connection to the RET antenna through the RET port (with a TMA)

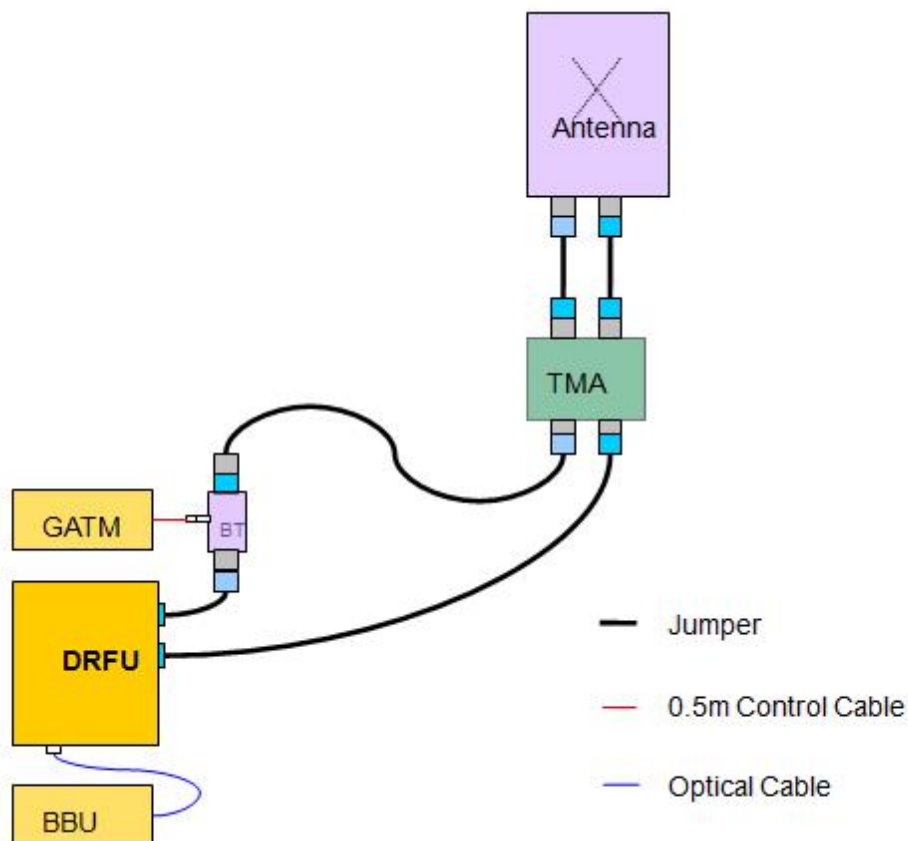
GATM Scenario

The TMA needs to be connected to the GATM only for a GBTS. In this scenario, a DRFU is connected to a TMA through a GATM and BT, as shown in [Figure 5-5](#). The GATM feeds 12 V DC power to the BT. Then, the BT combines the OOK signals, 12 V DC power, and RF signals and sends the combined signals to the TMA.

The GATM can supply power only to the connected TMA. The TMA provides the default gains, which are not configurable. You need to first configure parameters related to the power switch and current alarm threshold for the port on the GATM connecting to the TMA. For details about data preparation, see [Table 4-82](#). Then,

configure parameters related to the RX channel attenuation. For details about data preparation, see [Table 4-80](#). RET data is not involved in this connection scheme.

Figure 5-5 Connection to the TMA through the GATM



5.1.2 Operations on the TMA

TMA gains and working modes are configurable.

- **Setting TMA working mode**
By default, the working mode of the TMA is set to **NORMAL** to ensure normal operation. If you run the **MOD BTSTMASUBUNIT** (GBTS)/**MOD TMASUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) or **MOD ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command with the **MODE** parameter set to **BYPASS**, the TMA serves as a straight-through feeder and does not amplify uplink signals.
- **Setting TMA gain**
If a TMA is used, run the **MOD BTSTMASUBUNIT** (GBTS)/**MOD TMASUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) or **MOD ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command to adjust the gain of the TMA subunit. To query the value range of the gain, run the **DSP BTSTMADEVICEDATA** (GBTS)/**DSP TMADEVICEDATA** (eGBTS/NodeB/eNodeB/gNodeB) command. The gain of the TMA with fixed gain is not configurable.

The base station can perform software downloading and TMA resetting on the TMA separately.

 NOTE

After the software is downloaded, the ALD automatically resets. The new software takes effect after the reset of the ALD. After the ALD is reset, run the **DSP ALDVER** command to query the software version of the target ALD to check whether the ALD has been successfully upgraded.

- TMA software downloading

The TMA manufacturers provide TMA software. For details, see the documents provided by the manufacturers.

Run the **DLD BTSALDFILE** command for a GBTS to download the TMA software from the file server to the BSC OMU. Run the **LOD BTSALDSW** command to download the TMA software.

Perform the following operations for an eGBTS/NodeB/eNodeB/gNodeB to download the TMA software.

1. Run the **DLD ALDSW** command and set **Device Type** to **TMA**.
2. Run the **DLD ALDFILE** command. Set **Device Type** to **TMA** and **File Type** to **FIRMWARE**.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*.

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

-
- TMA resetting

Run the **RST BTSALD** (GBTS) or **RST ALD** (eGBTS/NodeB/eNodeB/gNodeB) command to reset the TMA. Resetting the TMA does not change the TMA gain and working mode.

5.2 Network Analysis

5.2.1 Benefits

None

5.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

For details, see [4.3.1 Licenses](#).

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

For details, see [3 General Principles](#).

5.3.4 Others

None

5.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

For details, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

5.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

For details, see [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

5.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

For details, see [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

5.7 Operation and Maintenance (Automatic ALD Deployment)

For details, see [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

6 SASU

6.1 Principles

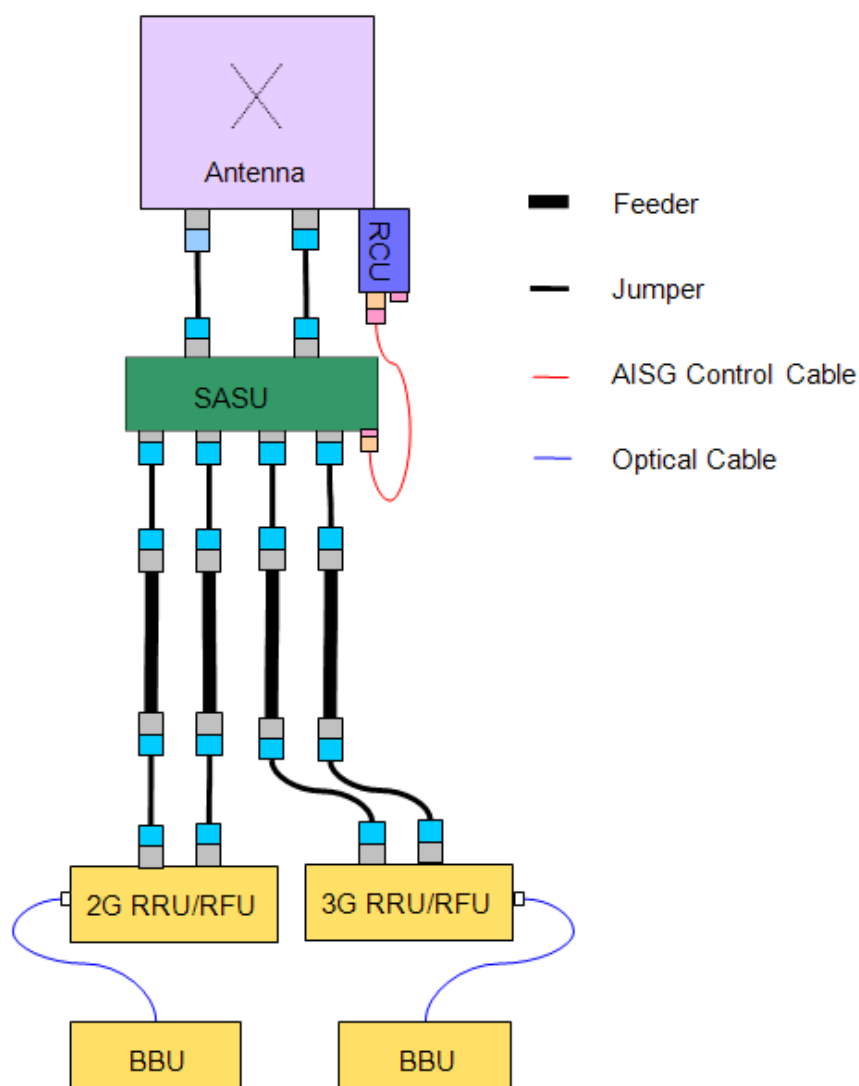
An SASU is a Huawei customized device for antenna sharing between intra-band GSM and UMTS modes at a multimode site. Antenna sharing helps operators reduce their capital expenditure (CAPEX). Currently, the SASU supports only the 900 MHz and 2100 MHz frequency bands.

An SASU comprises two subunits and each subunit amplifies an uplink RF signal for both GSM and UMTS.

6.1.1 Connections Between the SASU, RRU/RFU, and RET Antenna

Figure 6-1 shows how the SASU, RRUs/RFUs, and RET antenna are connected when RRUs/RFUs for GSM (referred to as 2G) and UMTS (referred to as 3G) share one RET antenna and the SASU is installed less than 20 m away from the RET antenna.

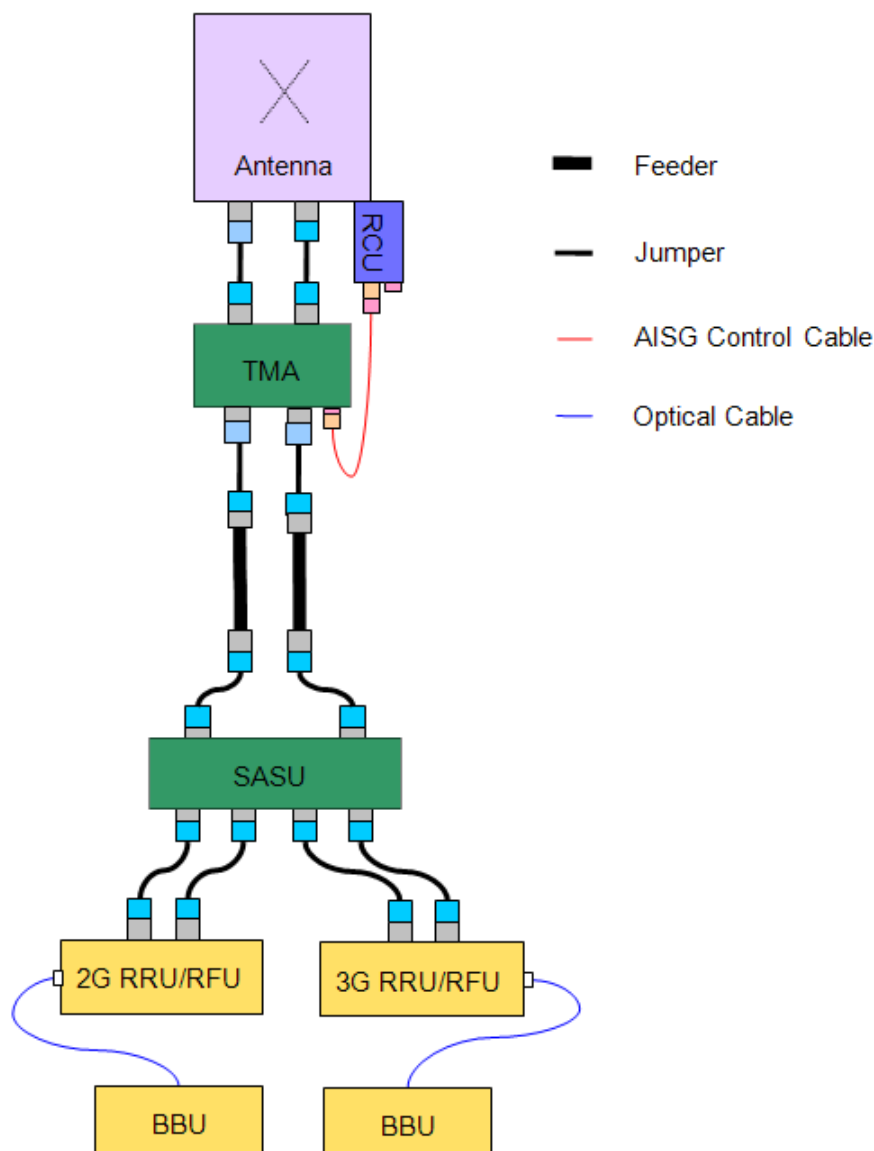
With the integrated SBT function, the SASU splits combined 3G signals from the RRUs/RFUs into two links. RF signals and 2G signals are combined and sent to the RET antenna on one link. On the other link, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RCU through an AISG multi-wire cable.

Figure 6-1 SASU directly connected to the RET antenna

SASU data can be configured only on the NodeB side. Because the SASU is an active device, you need to turn on the power switch for the 3G RRU control port on the NodeB side, and specify current alarm thresholds. For connection scheme in [Figure 6-1](#), set the **DCSWITCH** parameter for the SASU to **OFF**. Otherwise, the RET antenna will short-circuit. For the data configuration for this connection scheme, see scenario 5 in section [5.5 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

[Figure 6-2](#) shows how the SASU, RRUs/RFUs, TMA, and RET antenna are connected when the SASU is installed greater than 20 m away from the RET antenna. The SASU combines 2G and 3G signals and sends the combined signals to the TMA. The TMA splits the combined signals into two links. RF signals are sent to the RET antenna on one link. On the other link, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RCU through an AISG multi-wire cable.

Figure 6-2 SASU connected to the TMA and RET antenna



In this connection scheme, the **DCSWITCH** parameter for the SASU cannot be set to **OFF** because the TMA is an active device. The value of this parameter specifies whether the TMA is powered by the GBTS or NodeB. If the **DCSWITCH** parameter is set to **UMTS**, specify the **DCLOAD** parameter for SASU subunits so that the TMA connected to the SASU can be acknowledged by the base transceiver station (BTS). For the data configuration for this connection scheme, see scenario 6 in section [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

6.1.2 Operations on the SASU

The SASU DC power switch, gain, and working mode are configurable only on the NodeB side.

- Setting the DC power switch

When the SASU is connected to the RET antenna through a TMA, run the **MOD SASU** command with the **DCSWITCH** parameter set to any value except **OFF**. When the SASU is directly connected to the RET antenna, set the **DCSWITCH** parameter to **OFF**.

- Setting SASU working mode

By default, the SASU is set to **NORMAL** working mode to ensure normal operation. If you run the **MOD SASUSUBUNIT** command with **MODE** set to **BYPASS**, the SASU serves as a straight-through feeder and does not amplify signals.

- Setting SASU gain

Run the **MOD SASUSUBUNIT** command to modify the SASU subunit gain. You can run the **DSP SASUDEVICEDATA** command to query the value range of the SASU gain.

The base station can perform software downloading and SASU resetting on the SASU separately.

NOTE

After the software is downloaded, the ALD automatically resets. The new software takes effect after the reset of the ALD. After the ALD is reset, run the **DSP ALDVER** command to query the software version of the target ALD to check whether the ALD has been successfully upgraded.

- SASU software downloading

Run the **DLD ALDSW** command to download the SASU software. For details about the SASU software, see the documents provided by the SASU manufacturer.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*.

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

-
- SASU resetting

Run the **RST ALD** command to reset the SASU. Resetting the SASU does not change the SASU gain and working mode.

6.2 Network Analysis

6.2.1 Benefits

None

6.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The function has no impact on the network.

6.3 Requirements

6.3.1 Licenses

No license is required for GSM, UMTS, LTE, and NR.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

For details, see [3 General Principles](#).

6.3.4 Others

None

6.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

For details, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

6.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

For details, see [5.5 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

6.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

For details, see [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

6.7 Operation and Maintenance (Automatic ALD Deployment)

For details, see [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

7 RAE

7.1 Principles

The RAE includes the antenna information sensor unit (AISU).

The AISU is a device for measuring the following antenna engineering parameters: azimuth, mechanical tilt, longitude, latitude, and height. Installed on the top of an antenna, the unit performs GNSS-based direction finding, and works with the base station and operations support system (OSS) to implement the antenna auto-sensing solution. That is, the engineering parameters described above can be queried remotely in real time on the OSS. In this way, site visits can be reduced, saving network optimization costs.

NOTE

For details about Huawei AISU hardware information and installation methods, see *AISU User Manual* in *Antenna&ALD Product Documentation*.

7.1.1 Connections Between the RAE and RRU/RFU/AAU

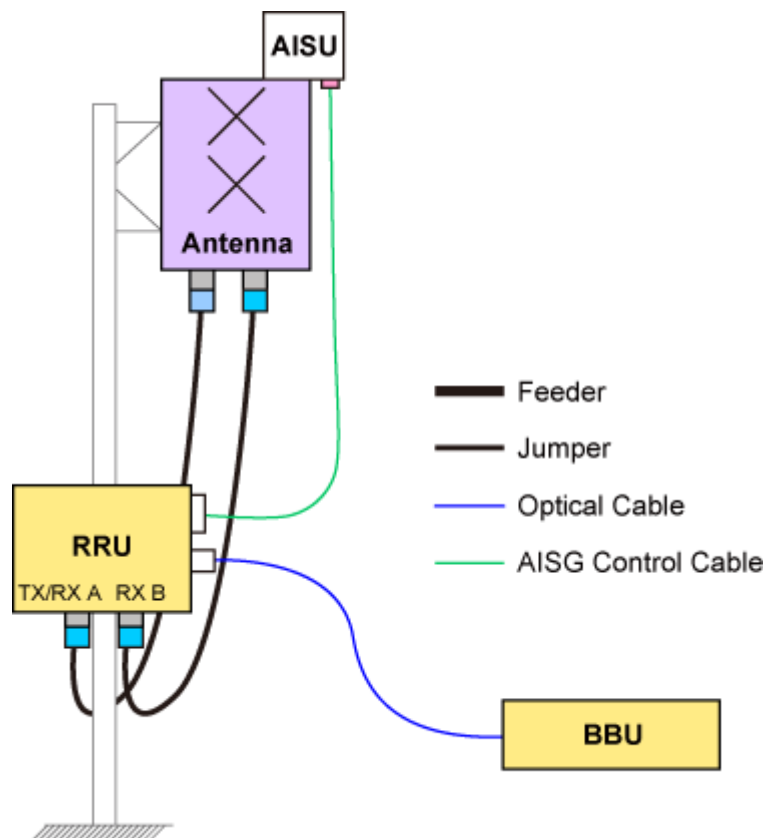
When an RAE connects to an RRU/RFU/AAU, only one AISU is installed for each sector, and other connected RET devices are not equipped with the RAE. One RRU can control only one AISU. The AISU must be installed on the antenna connected to the RRU, forming a control link.

- When an RAE connects to an RRU or RFU, see [Scenario 1: Connection to the RAE Through the RET Port](#) to [Scenario 6: Connection to the RAE Through the Antenna Port \(with a TMA and an RET Antenna\)](#).
- When an RAE connects to an AAU, see [Scenario 7: Connection Between the AAU and RAE Through the AUX Port](#).

Scenario 1: Connection to the RAE Through the RET Port

[Figure 7-1](#) shows a connection to the RAE through the RET port.

Figure 7-1 Connection to the RAE through the RET port

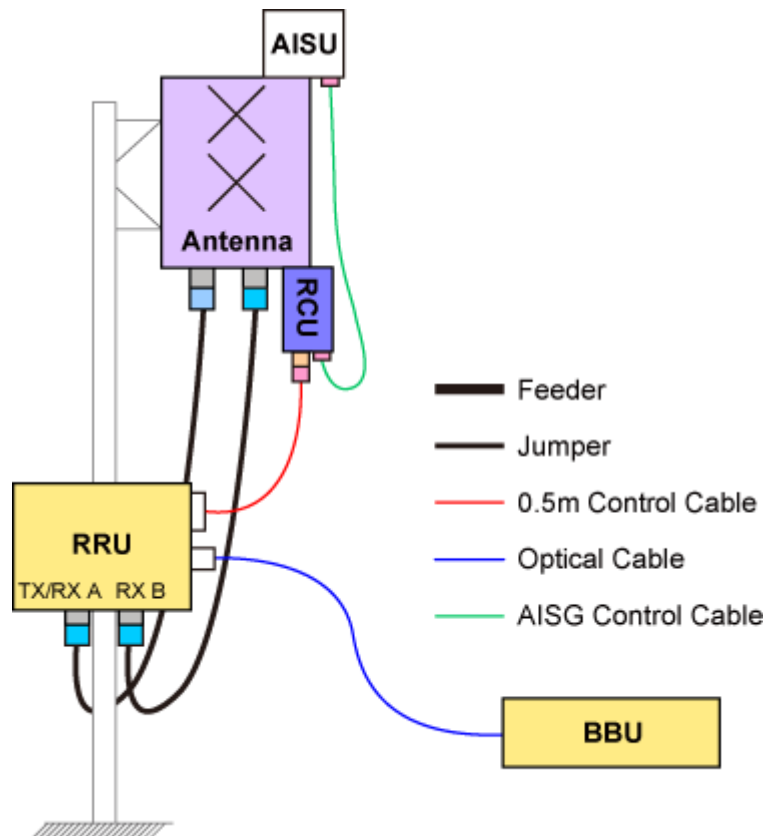


For details of data configuration in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 9 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

Scenario 2: Connection to the RAE Through the RET Port (with an RET Antenna)

[Figure 7-2](#) shows a connection to the RAE through the RET port (with an RET antenna).

Figure 7-2 Connection to the RAE through the RET port (with an RET antenna)

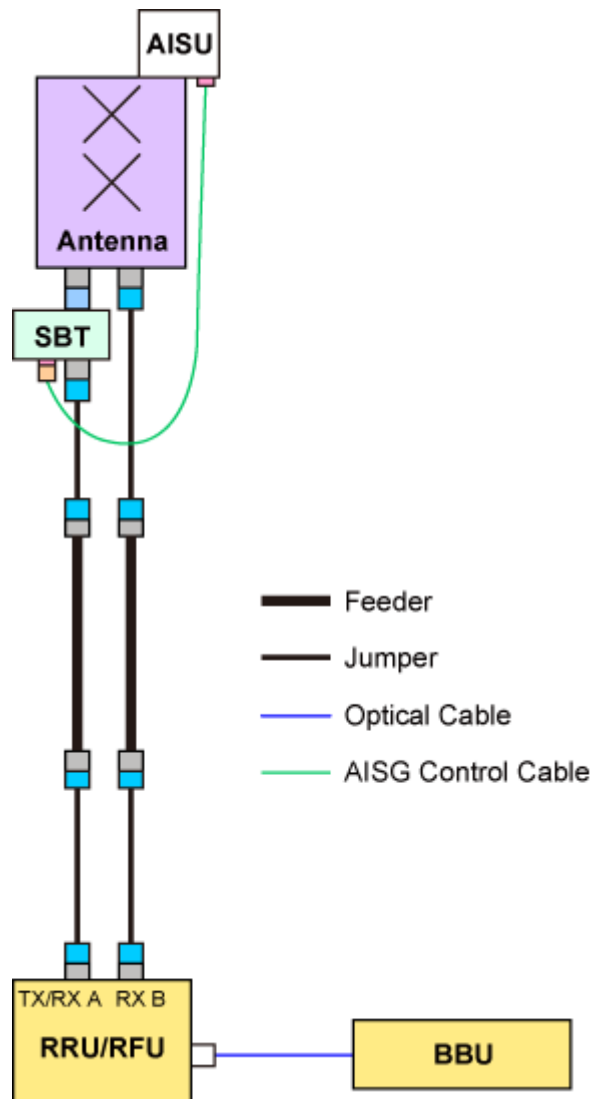


For details of data configuration in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 10 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

Scenario 3: Connection to the RAE Through the Antenna Port

Figure 7-3 shows a connection to the RAE through the antenna port.

Figure 7-3 Connection to the RAE through the antenna port

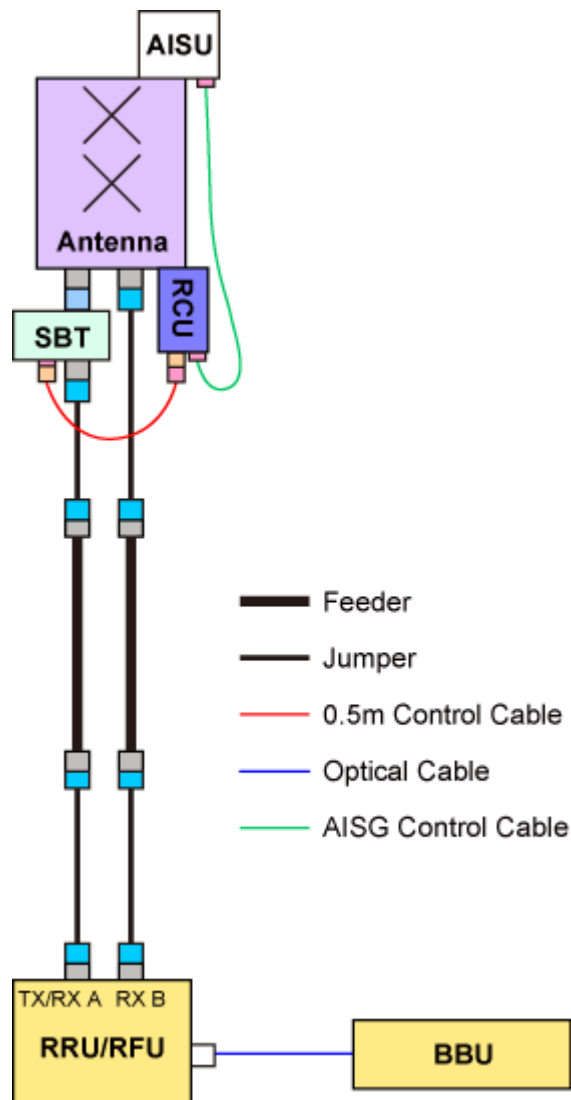


For details of data configuration in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 11 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

Scenario 4: Connection to the RAE Through the Antenna Port (with an RET Antenna)

Figure 7-4 shows a connection to the RAE through the Antenna port (with an RET antenna).

Figure 7-4 Connection to the RAE through the antenna port (with an RET antenna)

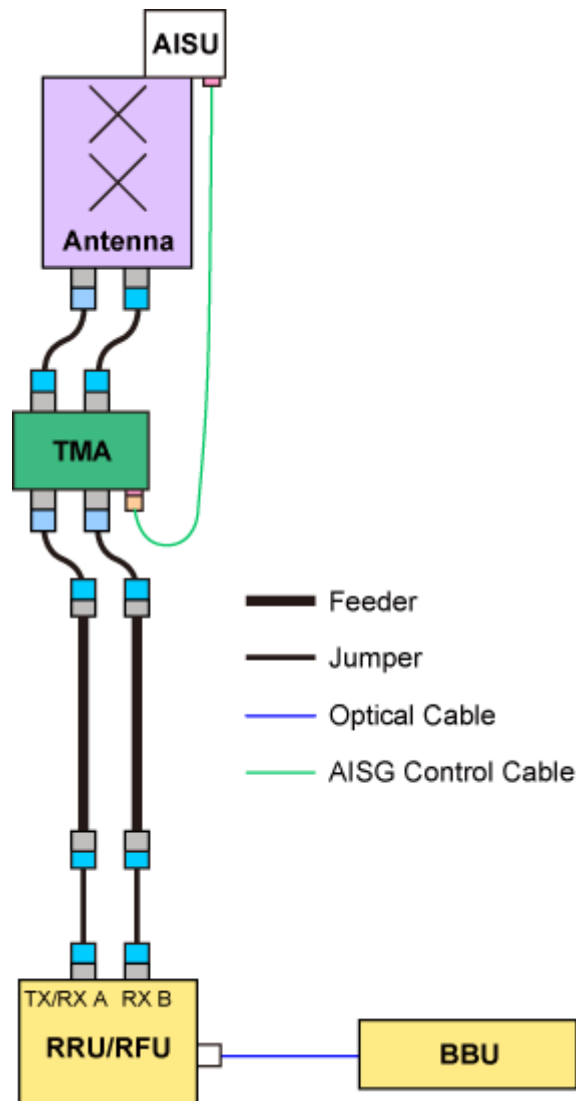


For details of data configuration in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 12 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

Scenario 5: Connection to the RAE Through the Antenna Port (with a TMA)

Figure 7-5 shows a connection to the RAE through the antenna port (with a TMA).

Figure 7-5 Connection to the RAE through the antenna port (with a TMA)

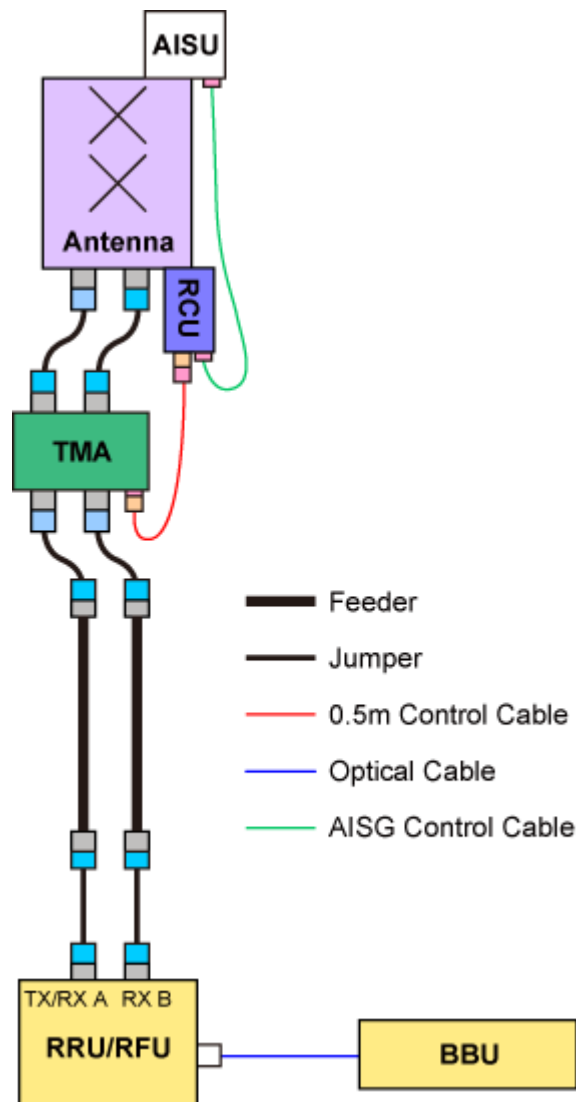


For details of data configuration in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 13 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

Scenario 6: Connection to the RAE Through the Antenna Port (with a TMA and an RET Antenna)

Figure 7-6 shows a connection to the RAE through the antenna port (with a TMA and an RET antenna).

Figure 7-6 Connection to the RAE through the antenna port (with a TMA and an RET antenna)

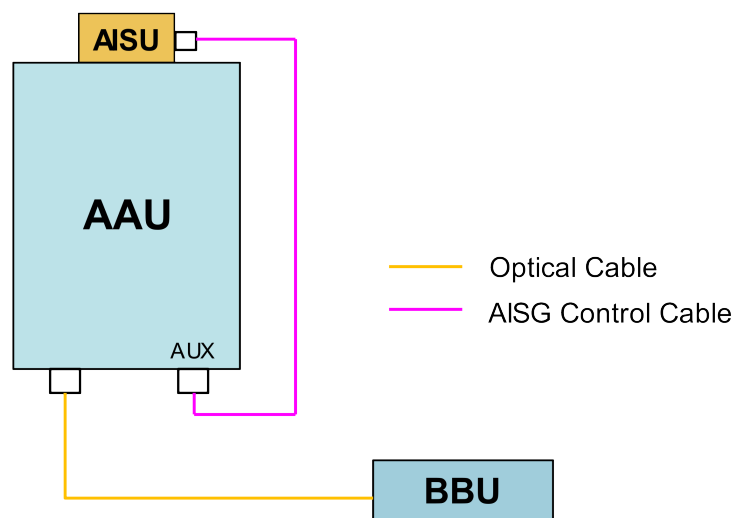


For details of data configuration in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 14 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

Scenario 7: Connection Between the AAU and RAE Through the AUX Port

[Figure 7-7](#) shows the connection between an AAU and RAE.

Figure 7-7 Connection between an AAU and RAE through the AUX port



For details of data configuration in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 9 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

NOTE

The current AUX ports on an AAU include DB9 and DB15. (For details, see the corresponding AAU hardware description.) The AISU used must match the port type.

7.1.2 Operations on the RAE

Antenna engineering parameters can be remotely queried.

- Querying the RAE function list
Run the **DSP RAEFUNCTION** command to query the functions supported by the RAE.
- Querying antenna engineering parameters
Run the **DSP RAEDEVICEDATA** command to query the antenna engineering parameters, including antenna azimuth, mechanical tilt, height, latitude, and longitude.

The system can perform software downloading and RAE resetting on the RAE separately.

NOTE

After the software is downloaded, the ALD automatically resets. The new software takes effect after the reset of the ALD. After the ALD is reset, run the **DSP ALDVER** command to query the software version of the target ALD to check whether the ALD has been successfully upgraded.

- Setting antenna attitude detection
If the antenna attitude changes and services are affected, the base station cannot detect this issue. It depends on manual inspections, which slows down problem-solving. Against this backdrop, the antenna attitude detection function is introduced for digital antennas. You can enable or disable this

function using the **ANT_ATT_DETECT_SW** option of the **Equipment.EQMOPTFEATSW** parameter.

Before enabling this function, you need to run the **DSP ALDFUNCTION** or **DSP RAEFUNCTION** command to check whether the digital antenna supports antenna attitude detection. You can enable this function only if the command output confirms support (**RAE Non-Standard Protocol Function List = Antenna Attitude Detection Function:YES**).

After this function is enabled, manually adjust the antenna attitude to the optimal state, then you can run the **UPD ANTATTDETECTREF** command to update the reference values for antenna attitude. The antenna device will record the detected azimuth, mechanical tilt, and roll angle as reference values.

The base station reports EVT-26764 Antenna Attitude Detection Event if the antenna azimuth, mechanical tilt, or roll angle of an antenna goes out of range or returns to normal. The antenna attitude adjustment cannot be detected during power-off. The base station will report this event after a reset to promptly notify any antenna attitude changes.

- RAE software downloading

The RAE manufacturers provide RAE software. For details, see the documents provided by the manufacturers. Run the following commands to download the RAE software.

- a. Run the **DLD ALDSW** command and set **Device Type** to **RAE**.
- b. Run the **DLD ALDFILE** command. Set **Device Type** to **RAE** and **File Type** to **FIRMWARE**.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*:

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

- RAE resetting

Run the **RST ALD** command to reset the RAE.

7.2 Network Analysis

7.2.1 Benefits

None

7.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The function has no impact on the network.

7.3 Requirements

7.3.1 Licenses

No license is required for GSM, UMTS, LTE, and NR.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

None

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

None

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

For details, see [3 General Principles](#).

7.3.4 Others

None

7.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

For details, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

7.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

For details, see [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

7.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

For details, see [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

7.7 Operation and Maintenance (Automatic ALD Deployment)

For details, see [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

7.8 Operation and Maintenance

This section covers only the antenna attitude detection function. For details about the operation and maintenance when the RAE unit is connected through an RET port or antenna port, see [7.5 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

7.8.1 Data Configuration

7.8.1.1 Data Preparation

Table 7-1 describes the parameters used for activating the antenna attitude detection function.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Equipment Optional Feature Switch	Equipment.EQMOPTFEATSW	ANT_ATT_DETECT_SW	Select this option if the antenna attitude detection function needs to be enabled for compatible digital antennas. To verify compatibility, use either the DSP ALDFUNCTION or DSP RAEFUNCTION command.

7.8.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Turning on the antenna attitude detection switch  
SET EQUIPMENT: EQMOPTFEATSW=ANT_ATT_DETECT_SW-1;
```

Deactivation Command Examples

```
//Turning off the antenna attitude detection switch  
SET EQUIPMENT: EQMOPTFEATSW=ANT_ATT_DETECT_SW-0;
```

7.8.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.8.2 Activation Verification

After the **UPD ANTATTDETECTREF** command is executed to update the antenna attitude reference value, wait a certain period of time and then run the **DSP**

RAESUBUNIT or **DSP ALDSUBUNIT** command. If the value of **Ref Ant Attitude Gen Status** in the command output is **Complete**, the antenna attitude detection function has been enabled.

 NOTE

The time the azimuth reference value is updated during antenna attitude detection depends on how long it takes for the azimuth to converge. For details about the maximum time, see the section describing AISU in the *Antenna&ALD Product Documentation*.

7.8.3 Network Monitoring

This basic function ensures basic service performance. Separate monitoring is not required.

8 AAS

8.1 Principles

Overview

An AAS module incorporates the functions of RF modules and antennas. It consists of the following function units:

- Antenna unit (AU): antenna unit.
- Radio unit (RU): RF unit.
- Management Unit (MU): control unit. Currently, only the AAU3902 has this function unit.

The passive antennas and RCUs in the AAS provide the antenna and RET functions, respectively, for RRUs/RFUs that are connected to the AAS module.

For details about an AAS product, see the hardware description of the AAS product.

Different types of AAS modules have different sets of integrated dual-polarized antennas. For details about the mapping between integrated antennas and either of the following: inserted modules, RET functions, and external ports, see the hardware description of the AAS product.

AAS Passive Antennas

The passive antenna and RCU built in an AAS module provide the RET function for RRUs/RFUs in the same way as a conventional RET antenna does. The AAS module with the passive antenna complies with the AISG2.0 protocol. Each RET subunit of the passive antennas in the AAS module functions as a single antenna working in daisy chain mode.

The AAS module supports the RET function in the following modes:

- By using the MU of the AAS
- By using the RETPORT on the RU of the AAS
- By using the RRU/RFU connected to the AAS

Table 8-1 lists the capabilities of AAS modules to support the RET function.

Table 8-1 Capabilities of AAS modules to support the RET function

AAS Module	Using the AAS MU (Antenna Cascading Mode)	Using the AAS RU	Using the RRU/RFU
AAU3902	Supported	Not supported	Supported
AAU3910	Not supported	Supported	Supported
AAU3911	Not supported	Supported	Supported
AAU3920	Not supported	Supported	Supported
AAU3940	Not supported	Supported	Not supported
AAU3961	Not supported	Supported	Supported
AAU5271	Not supported	Not supported	Not supported
AAU5281	Not supported	Not supported	Not supported
AAU5243	Not supported	Supported	Not supported
AAU5246	Not supported	Supported	Not supported
AAU5310	Not supported	Not supported	Not supported
AAU5313	Not supported	Not supported	Not supported
AAU5336	Not supported	Not supported	Not supported
AAU5336w	Not supported	Not supported	Not supported
AAU5339	Not supported	Not supported	Not supported
AAU5339w	Not supported	Not supported	Not supported
AAU5349	Not supported	Not supported	Not supported
AAU5356	Not supported	Not supported	Not supported
AAU5366	Not supported	Not supported	Not supported
AAU5368	Not supported	Not supported	Not supported
AAU5368md	Not supported	Not supported	Not supported
AAU5613	Not supported	Not supported	Not supported
AAU5614	Not supported	Not supported	Not supported
AAU5631	Not supported	Not supported	Not supported
AAU5636	Not supported	Not supported	Not supported
AAU5636m	Not supported	Not supported	Not supported
AAU5636w	Not supported	Not supported	Not supported

AAS Module	Using the AAS MU (Antenna Cascading Mode)	Using the AAS RU	Using the RRU/RFU
AAU5639	Not supported	Not supported	Not supported
AAU5639m	Not supported	Not supported	Not supported
AAU5639w	Not supported	Not supported	Not supported
AAU5646	Not supported	Not supported	Not supported
AAU5649	Not supported	Not supported	Not supported
AAU5656	Not supported	Not supported	Not supported
AAU5656m	Not supported	Not supported	Not supported
AAU5665md	Not supported	Not supported	Not supported
AAU5668	Not supported	Not supported	Not supported
AAU5668m	Not supported	Not supported	Not supported
AAU5685md	Not supported	Not supported	Not supported
AAU5711	Not supported	Not supported	Not supported
AAU5711a	Not supported	Not supported	Not supported
AAU5726	Not supported	Not supported	Not supported
AAU5726e	Not supported	Not supported	Not supported
AAU5733	Not supported	Not supported	Not supported
AAU5736	Not supported	Not supported	Not supported
AAU5765	Not supported	Not supported	Not supported
AAU5767t	Not supported	Not supported	Not supported
AAU5768	Not supported	Not supported	Not supported
AAU5785t	Not supported	Not supported	Not supported
AAU5811	Not supported	Not supported	Supported
AAU5821g	Not supported	Not supported	Supported
AAU5822a	Not supported	Not supported	Supported
AAU5825a	Not supported	Not supported	Supported
AAU5825g	Not supported	Not supported	Supported
AAU5826g	Not supported	Not supported	Supported
AAU5831	Not supported	Not supported	Supported
AAU5832a	Not supported	Not supported	Supported

AAS Module	Using the AAS MU (Antenna Cascading Mode)	Using the AAS RU	Using the RRU/RFU
AAU5832f	Not supported	Not supported	Supported
AAU5833f	Not supported	Not supported	Supported
AAU5835f	Not supported	Not supported	Supported
AAU5835h	Not supported	Not supported	Supported
AAU5836f	Not supported	Not supported	Not supported
AAU5940	Not supported	Supported	Not supported
AAU5942	Not supported	Supported	Not supported
AAU5972	Not supported	Supported	Not supported
HAAU5213	Not supported	Not supported	Not supported
HAAU5222	Not supported	Not supported	Not supported
HAAU5233w	Not supported	Not supported	Not supported
HAAU5235w	Not supported	Not supported	Not supported
HAAU5323	Not supported	Not supported	Not supported

When site cable connections support all the three methods, you can choose only one method. When the AAS module supports the RET function by using the AAS RU, you are advised to use only one RU in the AAS to control all the RET antennas.

8.1.1 Application of AAS Passive Antennas

An AAS module has two types of external ports: an RET port and an antenna port. The antenna port is at the bottom of the AU. Some AAS modules have AISG ports, which are connected to the RRU/RFU for the RET function.

When the RET function is required, **RETTYPE** must be set to **SINGLE_RET** and **SCENARIO** must be set to **DAISY_CHAIN**. You can determine which antenna to use for the RET function according to the serial number of the antenna. **SCENARIO** must be set to **DAISY_CHAIN** even if you need to configure the RET function for only one antenna. In addition, the **VENDORCODE** and **SERIALNO** parameters must be specified to indicate the antenna to be configured for the RET function, as the antennas in the AAS module are working in daisy chain mode. For details about the serial numbers of the integrated antennas in the AAS module, see *AAU Hardware Description* of the corresponding AAS product in *3900 & 5900 Series Base Station Product Documentation*.

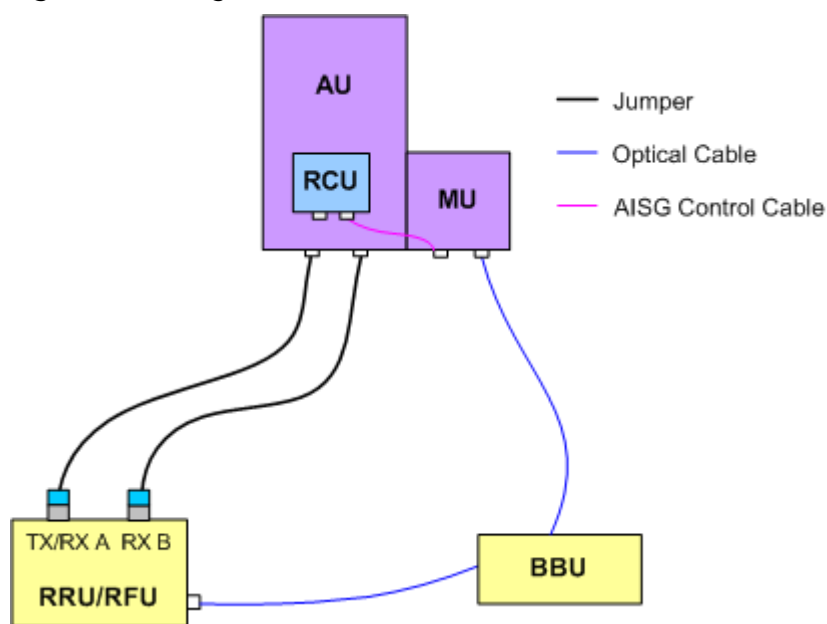
Scenario 1: Using the MU to Provide the RET Function on the AAS Module

As shown in this scenario, RRU/RFUs connect to an AAS module with passive antennas through external ports on the AAS module. Choose an appropriate

external port to connect the RRU/RFU to the AAS module based on the supported frequency band, as shown in [Figure 8-1](#). In this scenario, the RET function for the AAS module with passive antennas must be set through the RAT that manages the AAS module. For example, to enable GSM RRU/RFUs to use the RET function for the AAS module with passive antennas when the AAS module is managed by UMTS, the RET function must be set in UMTS.

Before setting the RET function, you must configure the AAS module. Compared with the RET function on conventional antennas, the RET function on the AAS module with passive antennas does not require you to configure power supply switches or current alarm thresholds. For the data configuration for this connection scheme, see scenario 7 in [Operation and Maintenance](#).

Figure 8-1 Using the MU to control the RET function



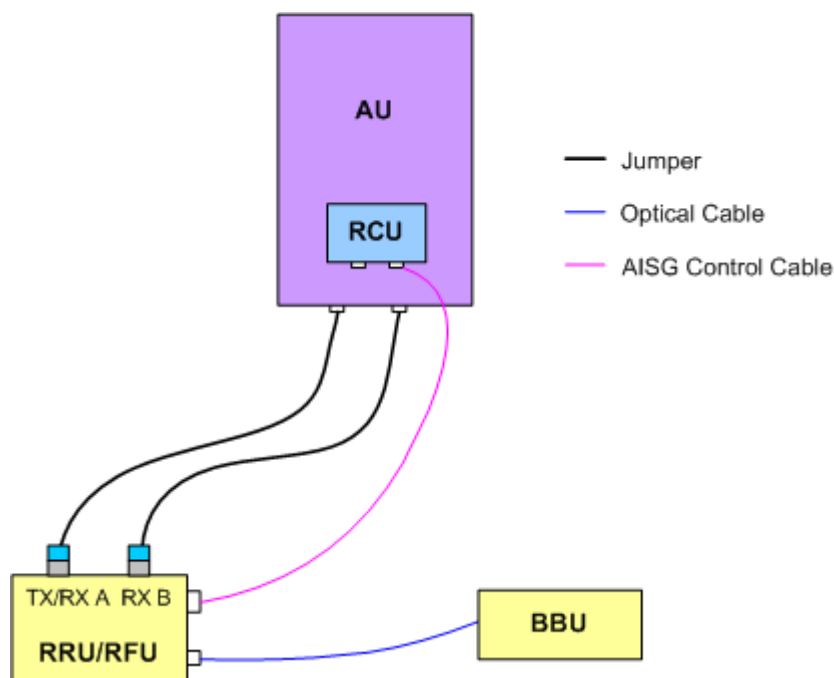
Scenario 2: Using the RRU/RFU to Provide the RET Function on the AAS Module

When the RET function for an AAS module with passive antennas is provided by an RRU/RFU connecting to the AAS module, the connections between the RRU/RFU and the AAS module are similar to the connections between the RRU/RFU and the conventional RET antennas except for the AISG port configuration, as shown in [Figure 8-2](#). In this scenario, the AAS module functions as the conventional RET antennas. Therefore, the configuration can be performed in the same way as that for conventional RET antennas.

- In scenarios in which the RET function is controlled by the **RETPORT** MO of an external RRU connected to an AAS module with passive antennas, the eGBTS/NodeB/eNodeB/gNodeB corresponds to scenario 1 in [Operation and Maintenance](#), and the GBTS corresponds to scenario 1 in [Operation and Maintenance](#).
- In scenarios in which the RET function is controlled by the **ANTENNAPORT** MO of an external RRU/RFU connected to an AAS module with passive antennas, the eGBTS/NodeB/eNodeB/gNodeB corresponds to scenario 2 in

Operation and Maintenance and the GBTS corresponds to scenario 2 in **Operation and Maintenance**.

Figure 8-2 Using the RRU/RFU to provide the RET function on the AAS module



Scenario 3: Using the AAS RU to Provide the RET Function

When the RET function for AAS passive antennas is controlled by the AAS RU:

- For the AAU3910/AAU3911/AAU3920/AAU3961/AAU5246, the function is controlled by the RET port, as shown in **Figure 8-3**.
- For the AAU3940/AAU5940/AAU5942, the function is controlled by the antenna port and the port must be R0A, as shown in **Figure 8-4**.

In this scenario, the AAS module functions as the conventional RET antennas. Therefore, the configuration can be performed in the same way as that for conventional RET antennas. In addition, some AAS modules provide not only the RET function but also some Huawei-defined functions. For example, the AAU5940/AAU5942/AAU5246 provides the vertical beamwidth adjustment feature which is configured and managed by the RVD.

For details of data configurations in this connection scheme for the eGBTS/NodeB/eNodeB/gNodeB, see scenario 8 in **Operation and Maintenance**.

Figure 8-3 RET function controlled by the RU of an AAS module (AAU3910/AAU3911/AAU3920/AAU3961/AAU5246)

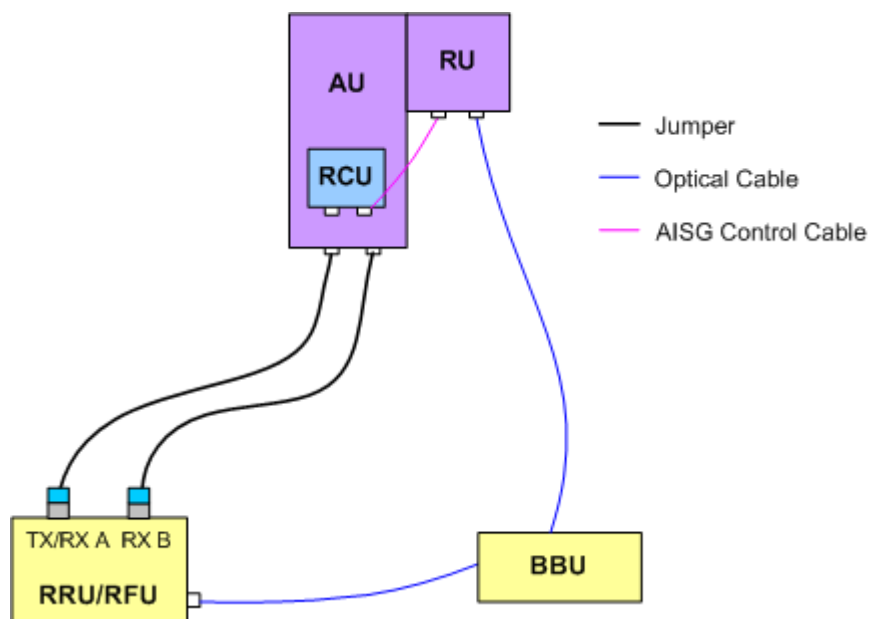
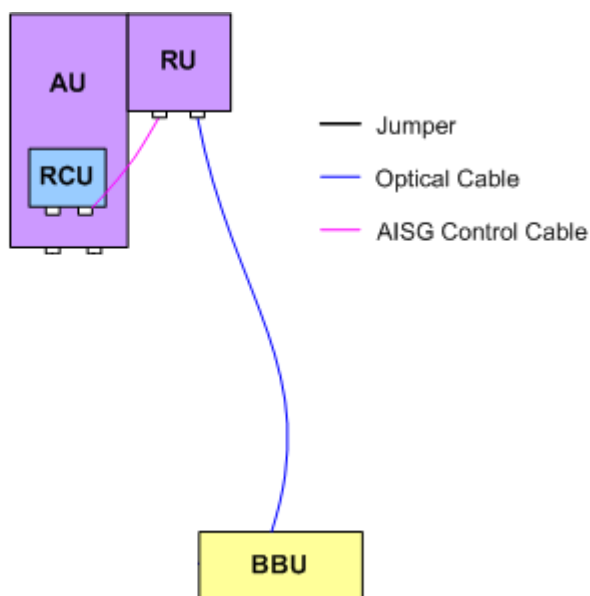


Figure 8-4 RET function controlled by the RU of an AAS module (AAU3940/AAU5940/AAU5942)



8.1.2 Operations on AAS Modules

Operations on the RET of AAS Modules with Passive Antennas

Each RET subunit of the passive antennas in the AAS module functions as a single antenna working in daisy chain mode.

- Scanning

You can scan all the AAS modules connected to an RF module. When the RET function for a specific AAS module is controlled by an RF module, the RET function for the module cannot be scanned or controlled by other RF modules that are also connected to the AAS module.

If multiple RF modules share the AAS module, running the **SCN ALD** command may fail to scan all the RET antennas of all RF modules at a time. In this scenario, you can run this command separately to scan the RF module of which the RET antenna cannot be found.

- Configuration file loading

Loading configuration files is not required because AAS modules with passive antennas have built-in configuration files. If ALM-26754 RET Antenna Data Loss is reported, the configuration files have been lost. In this case, you can run the **DLD RETCFGDATA** command (eGBTS/NodeB/eNodeB/gNodeB) to reload configuration files for the AAS module with passive antennas.

 NOTE

- An AAS does not need to be calibrated again across a reset or power-off.
 - Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and the ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.
- Calibration
After an AAS module is installed, run the **CLB RET** command (eGBTS/NodeB/eNodeB/gNodeB) to calibrate the RET antennas. During the calibration, RCUs adjust the RET antennas within the tilt range until the RET antennas operate properly. If the calibration fails, ALM-26753 RET Antenna Not Calibrated is generated.

 NOTE

The tilt ranges supported by different RET subunits vary according to the AAS model.

- Tilt setting

After the calibration is completed, run the **DSP RETDEVICEDATA** command (eGBTS/NodeB/eNodeB/gNodeB) to query the supported tilt range. Then, run the **MOD RETSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) (recommended)/**DSP ALDSUBUNIT** command to set the tilts for RET subunits. Setting the tilt of an RET subunit affects the coverage of the RET antenna. Therefore, set the tilt according to the engineering design.

 NOTE

After the software is loaded, the ALD automatically resets. The new software takes effect after the reset of the ALD. After the ALD is reset, run the **DSP ALDVER** command to query the software version of the target ALD to check whether the ALD has been successfully upgraded.

The built-in RCU of the AAS can be operated separately, including RCU software download and RCU reset.

- RCU software downloading

The AAS manufacturers provide software for the RCU. For details, see the AAS-related documents. Run the following commands for an eGBTS/NodeB/eNodeB/gNodeB to download the RCU software.

- a. Run the **DLD ALDSW** command and set **Device Type** to **RET**.
- b. Run the **DLD ALDFILE** command. Set **Device Type** to **RET** and **File Type** to **FIRMWARE**.
- c. After the software is loaded, the ALD automatically resets. The new software takes effect after the reset of the ALD. After the ALD is reset, run the **DSP ALDVER** command to query the software version of the target ALD to check whether the ALD has been successfully upgraded.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*.

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

-
- RCU resetting
Run the **RST ALD** command (eGBTS/NodeB/eNodeB/gNodeB) to reset the RCU. Resetting the RCU does not change the RET tilt.

Operations on the RVD of AAS Modules with Passive Antennas

The RVD is a configuration model for managing Huawei-defined devices. Some AAS modules support certain Huawei-defined functions. For example, the AAU5940/AAU5942/AAU5246 provides the vertical beamwidth adjustment feature, which is controlled by this model.

The operation processes of the RVD are described as follows:

- Scanning
Run the **SCN ALD** command to scan the RET antennas controlled by RF modules. An RVD can be scanned when it exists.
- Configuration file loading
Run the following commands for an eGBTS/NodeB/eNodeB/gNodeB to reload the configuration data file for the RVD:
 - a. Run the **DLD ALDCFGDATA** command and set **Device Type** to **RVD**.
 - b. Run the **DLD ALDFILE** command. Set **Device Type** to **RVD** and **File Type** to **CFGFILE**.

- Calibration

Run the **CLB ALD** command (eGBTS/NodeB/eNodeB/gNodeB) to calibrate an RVD. If the calibration fails, ALM-26753 RET Antenna Not Calibrated is generated. Set the **Device Type** parameter in the command to **RVD**.

Vertical beamwidth adjustment method (taking the AAU5940/AAU5942/AAU5246 as the example):

- Vertical beamwidth configuration

- Before the configuration, run the **DSP RVDSUBUNIT** command (eGBTS/NodeB/eNodeB/gNodeB) to query the tilt range supported by the RVD subunit.
- Run the **MOD RVDSUBUNIT** command (eGBTS/NodeB/eNodeB/gNodeB) to configure the vertical beamwidth.

 NOTE

Because setting vertical beamwidth using the RVD affects the coverage of the related antennas, the related parameters must be specified based on the engineering design.

The base station can download software and reset the RVD separately.

- Software downloading

The AAS manufacturers provide software for the RVD. For details, see the AAS-related documents. Run the following commands to download the RVD software.

- Run the **DLD ALDSW** command for an eGBTS/NodeB/eNodeB/gNodeB with **Device Type** set to **RVD**.
- Run the **DLD ALDFILE** command for an eGBTS/NodeB/eNodeB/gNodeB. Set **Device Type** to **RVD** and **File Type** to **FIRMWARE**.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*:

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

- RVD resetting

Run the **RST ALD** command (eGBTS/NodeB/eNodeB/gNodeB) to reset an RVD. When an RVD is reset, the new parameter configurations take effect.

8.2 Network Analysis

8.2.1 Benefits

None

8.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The function has no impact on the network.

8.3 Requirements

8.3.1 Licenses

No license is required for GSM, UMTS, LTE, and NR.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

None

8.3.2.3 Function Impacts

None

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

For details, see [8.1 Principles](#).

8.3.4 Others

None

8.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

For details, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

8.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

For details, see [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

8.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

For details, see [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

8.7 Operation and Maintenance (Automatic ALD Deployment)

For details, see [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

9 SALD/MALD

9.1 Principles

SALD and MALD are AISG3.0-compliant ALDs. For ALDs of the SALD/MALD type, one physical ALD corresponds to one ALD serial number. A SALD/MALD generally consists of multiple types of ALD subunits. AISG3.0 defines three types of ALD subunits: ADB, RET, and TMA. Each SALD/MALD contains at least one ADB subunit.

A SALD can communicate with only one RRU/AAU at a time, whereas an MALD can communicate with multiple RRUs/AAUs at the same time.

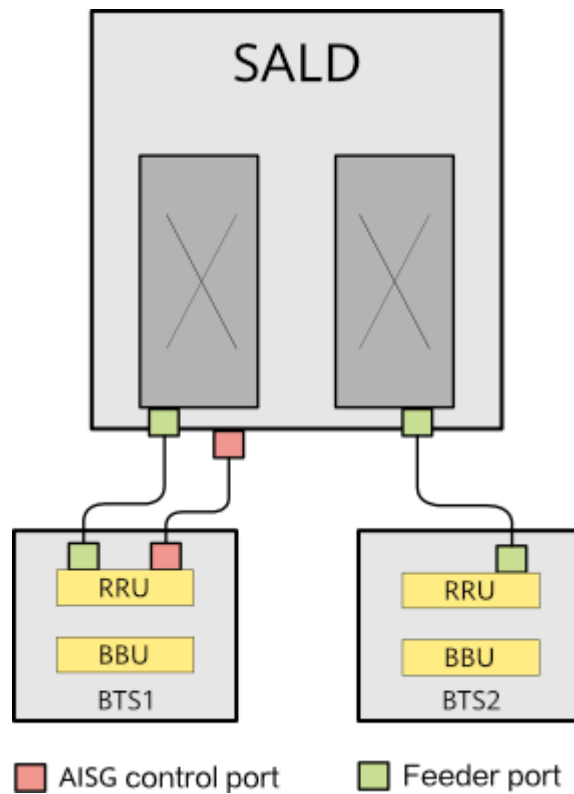
9.1.1 Connection Between the SALD/MALD and the RRU/AAU

An RF module can be connected to a SALD/MALD through an RET port (**RETPORT**) or an RF port (**ANTENNAPORT**).

Scenario 1: Connection to the SALD Through the RETPORT or ANTENNAPORT

This scenario applies only to RRUs/AAUs. The RRUs/AAUs are directly connected to the SALD through the RET ports or RF modules, as shown in [Figure 9-1](#).

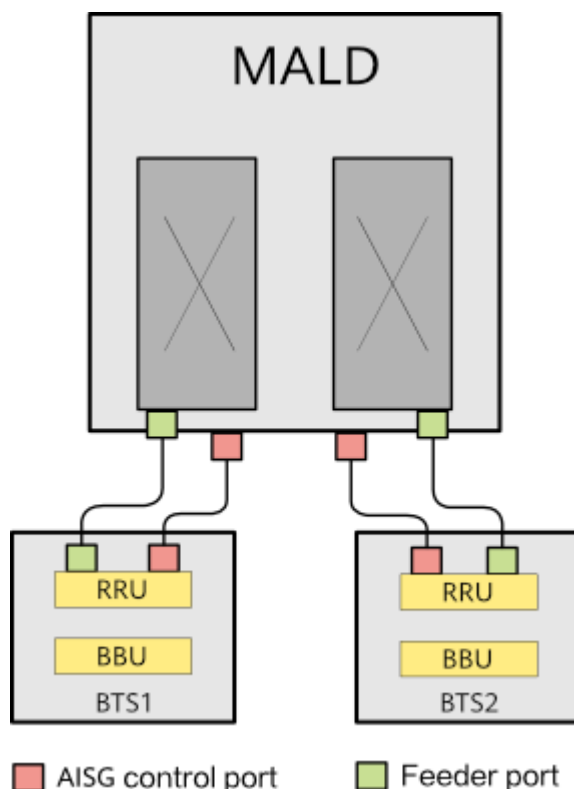
Figure 9-1 Connecting to the SALD through the RETPORT or ANTENNAPORT



Scenario 2: Connection to the MALD Through the RETPORT or ANTENNAPORT

This scenario applies only to RRUs/AAUs. RRUs/AAUs in different base stations are directly connected to the MALD through the RET ports or RF modules, as shown in [Figure 9-2](#).

Figure 9-2 Connecting to the MALD through the RETPORT or ANTENNAPORT



9.1.2 SALD/MALD Operations

The SALD/MALD supports three types of ALD subunits: RET, TMA, and ADB. The RET subunit supports configuration file loading, calibration, and tilt setting. The TMA subunit supports gain and working mode setting. The ADB subunit supports antenna installation information setting.

Configuration File Loading

A configuration file, which is provided by the SALD/MALD manufacturer, describes the relationship between the RCU and the ALD subunit tilt. Some SALDs/MALDs have been loaded with default configuration files before delivery. If no configuration file is loaded for a SALD/MALD before delivery, download the correct configuration file for the ALD subunit.

For an eGBTS/NodeB/eNodeB/gNodeB, run the **DLD ALDCFGDATA** command to load the RET configuration file.

If an incorrect RET configuration file is loaded, the RET antenna will experience unexpected errors. In this case, you can run the **DSP ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command to query the name of the last loaded RET configuration file and the time when the file is loaded. During the implementation, you are advised to select one or two base stations, and check whether the actual RET downtilts onsite are the same as the configured tilts to determine whether the loaded RET configuration file is correct. Ensure that a correct configuration file is loaded to the RET subunit before the calibration.

NOTE

- Run the **DSP ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command to query dynamic information about SALD/MALD subunits. If any information is incorrect in the command output, for example, the values of **Max tilt** and **Min tilt** are **NULL**, no RET configuration file is loaded or the configuration file is lost.
- If ALM-26754 RET Antenna Data Loss is not reported, the RET device has been loaded with a configuration file. File reloading is not required unless the RCU is changed.
- If the antennas with built-in RCUs have loaded configuration files before delivery, reloading is not required onsite. The antennas with external RCUs may need reloading configuration files according to the site conditions.

Calibration

After an RET antenna is installed, run the **CLB ALD** (eGBTS/NodeB/eNodeB/gNodeB) command to calibrate the RET antenna. During the calibration, RCUs adjust the RET antennas within the tilt range until the antennas operate properly. If the calibration fails, ALM-26753 RET Antenna Not Calibrated is generated.

After the preceding calibration command is executed:

- If the RET antenna tilt has been configured on the base station, it will be restored to the configured value.
- If the RET antenna tilt has not been configured on the base station, it will be determined by the actual RET antenna implementation.

NOTE

- An RET antenna does not need to be calibrated again after it is reset or powered off.
- Skip this step if the RET antenna has a built-in RCU (that has been calibrated before delivery) and ALM-26753 RET Antenna Not Calibrated is not reported. Perform this step if the RET antenna has an external RCU or ALM-26753 RET Antenna Not Calibrated is reported.

Downtilt Setting

After the RET antenna is calibrated, run the **MOD ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command to set the RET subunit tilt.

Before the configuration, run the **DSP ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command to query the supported tilt range. Setting the tilt of an RET subunit affects the coverage of the RET antenna. Therefore, set the tilt according to the engineering design.

NOTE

Each RET subunit supports a unique tilt range.

Gain and Working Mode Setting

TMA gains and working modes are configurable.

- Set the working mode.
By default, the working mode of the TMA is set to **NORMAL** to ensure normal operation. If you run the **MOD ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command with **ALDSUBUNIT.MODE (LTE eNodeB, 5G gNodeB)** set

to **BYPASS**, the TMA serves as a straight-through feeder and does not amplify uplink signals.

- Set the gain.

If a TMA is used, run the **MOD ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command to adjust the gain of the TMA subunit. To query the value range of the gain, run the **DSP ALDSUBUNIT** (eGBTS/NodeB/eNodeB/gNodeB) command. The gain of the TMA with fixed gain is not configurable.

SALD/MALD Antenna Installation Information Setting

The ADB supports the settings of the AISG sector ID, installation position in the sector, mechanical tilt, and antenna azimuth. The settings of these parameters are used only for recording the installation information and do not affect the operation of the SALD/MALD antenna.

- Set the AISG sector ID.

Run the **ADD/MOD ALDSUBUNIT** command with **Subunit Type** set to **ADB** to set the AISG sector ID.

- Set the installation position in the sector.

Run the **ADD/MOD ALDSUBUNIT** command with **Subunit Type** set to **ADB** to set the installation position in a sector.

- Set the mechanical tilt.

Run the **ADD/MOD ALDSUBUNIT** command with **Subunit Type** set to **ADB** to set the mechanical tilt.

- Set the antenna azimuth.

Run the **ADD/MOD ALDSUBUNIT** command with **Subunit Type** set to **ADB** to set the antenna azimuth.

RCU Software Downloading

- Traditional RCU:

The SALD/MALD manufacturers provide RCU software. For details, see the documents provided by the manufacturers.

For an eGBTS/NodeB/eNodeB/gNodeB, run the following command to download RCU software:

Run the **DLD ALDFILE** command with **Device Type** set to **SALD/MALD** and **File Type** set to **FIRMWARE**.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*:

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

RCU Resetting

Run the **RST ALD** (eGBTS/NodeB/eNodeB/gNodeB) command to reset the RCU antenna. Resetting the RCU does not affect the SALD/MALD configuration that has taken effect.

9.2 Network Analysis

9.2.1 Benefits

None

9.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The function has no impact on the network.

9.3 Requirements

9.3.1 Licenses

No license is required for GSM, UMTS, LTE, and NR.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

None

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

None

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

UMPT series main control boards

RF Modules

RRU5000/AAU5000 series RF modules

9.3.4 Others

None

9.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

For details, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

9.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

For details, see [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

9.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

For details, see [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

9.7 Operation and Maintenance (Automatic ALD Deployment)

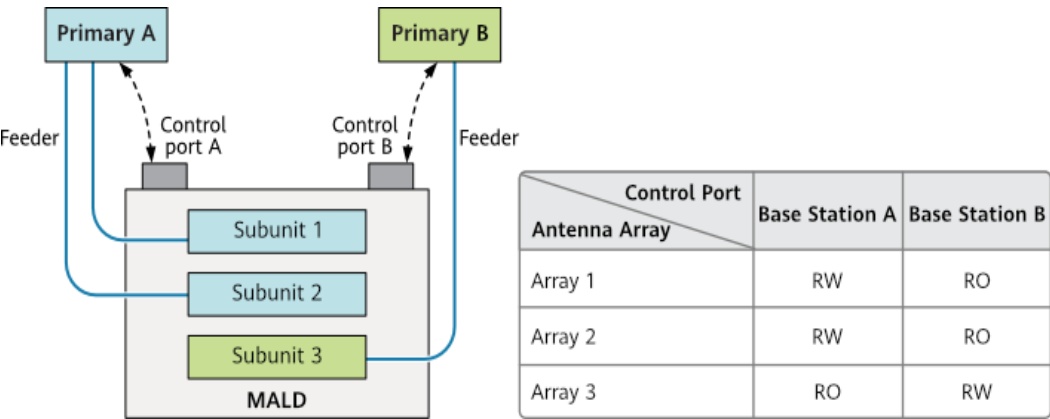
For details, see [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

9.8 Operation and Maintenance (MALD Subunit Access Permission Setting)

The MALD supports allocation of subunit access permissions among all base stations that manage the MALD. The AISG3.0 protocol provides instructions to allocate permissions to each subunit on each control port of the MALD. According to the AISG3.0 protocol, the subunit access permission has three values: RW, RO, and NA. RW means that the target subunit is readable and writable to the base station, and is used when the base station needs to modify the configurations of the subunit. RO means that the target subunit is read-only to the base station, and is used when the base station needs only to query information about the subunit. NA indicates that the target subunit is invisible to the base station.

As shown in [Figure 9-3](#), subunit 1 is readable and writable to base station A and is read-only to base station B. That is, only base station A can modify the configurations (such as the electrical downtilt) of subunit 1. Base station B can read information about subunit 1, but cannot modify its configurations.

Figure 9-3 Example of MALD subunit access permission setting



10_{RAS}

10.1 Principles

Remote azimuth steering (RAS) units are used to adjust the horizontal azimuth of antennas.

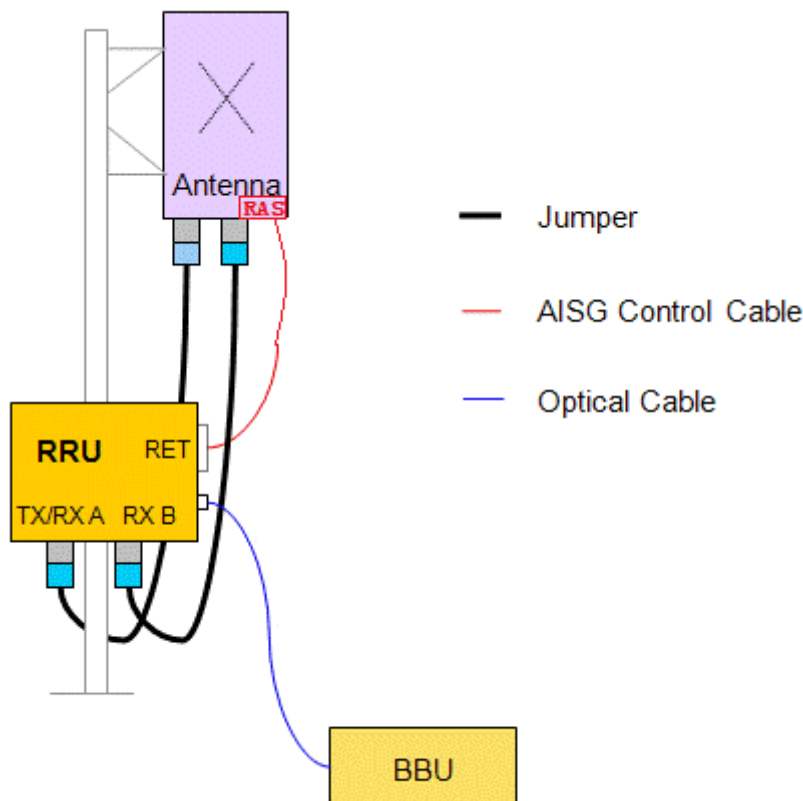
The RAS unit can be remotely set to adjust the horizontal azimuth of the antenna and change the signal coverage.

10.1.1 Connections Between RAS Units and RRUs/RFUs

An RRU can be connected to an RAS unit through the RET port (RETPORT) or RF port (ANTENNAPORT). The RFU does not have any RET port and can be connected to the RAS only through the antenna port.

- Connection to the RAS unit through the RET port
This scenario applies only to RRUs. An AISG multi-wire cable connects the RAS port, as shown in [Figure 10-1](#). With the integrated Bias Tee (BT), the RRU can send RS485 control signals and feed 12 V DC power to the RAS unit through the RET port. Therefore, no Smart Bias-Tee (SBT) is required. This connection scheme is recommended when the distance between the RRU and the RAS unit is shorter than 20 m.

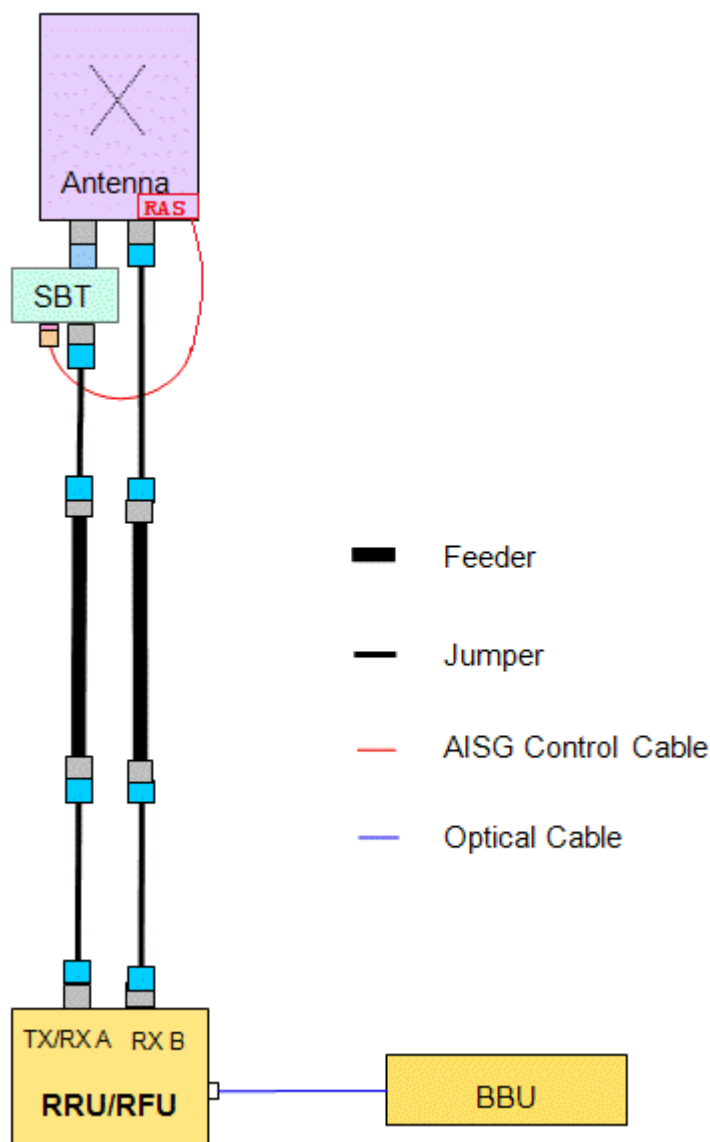
Figure 10-1 Connection to the RAS unit through the RET port



For the data configurations in this connection scheme, see the following section:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 15 in **Operation and Maintenance**
- Connection to the RET antenna through the antenna port
An SBT is required in this connection scheme. Feeders and jumpers connect the RRU/RFU and SBT antenna, and an AISG multi-wire cable connects the SBT to the RAS unit, as shown in **Figure 10-2**.
 - a. The RRU/RFU combines OOK signals, 12 V DC power, and RF signals. Then, it sends the combined signals to the SBT through feeders.
 - b. The SBT splits the combined signals into two paths. On one path, the RF signals are sent to the antenna. On the other path, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RAS unit through an AISG multi-wire cable.

Figure 10-2 Connection to the RAS unit through the antenna port



For the data configurations in this connection scheme, see the following section:

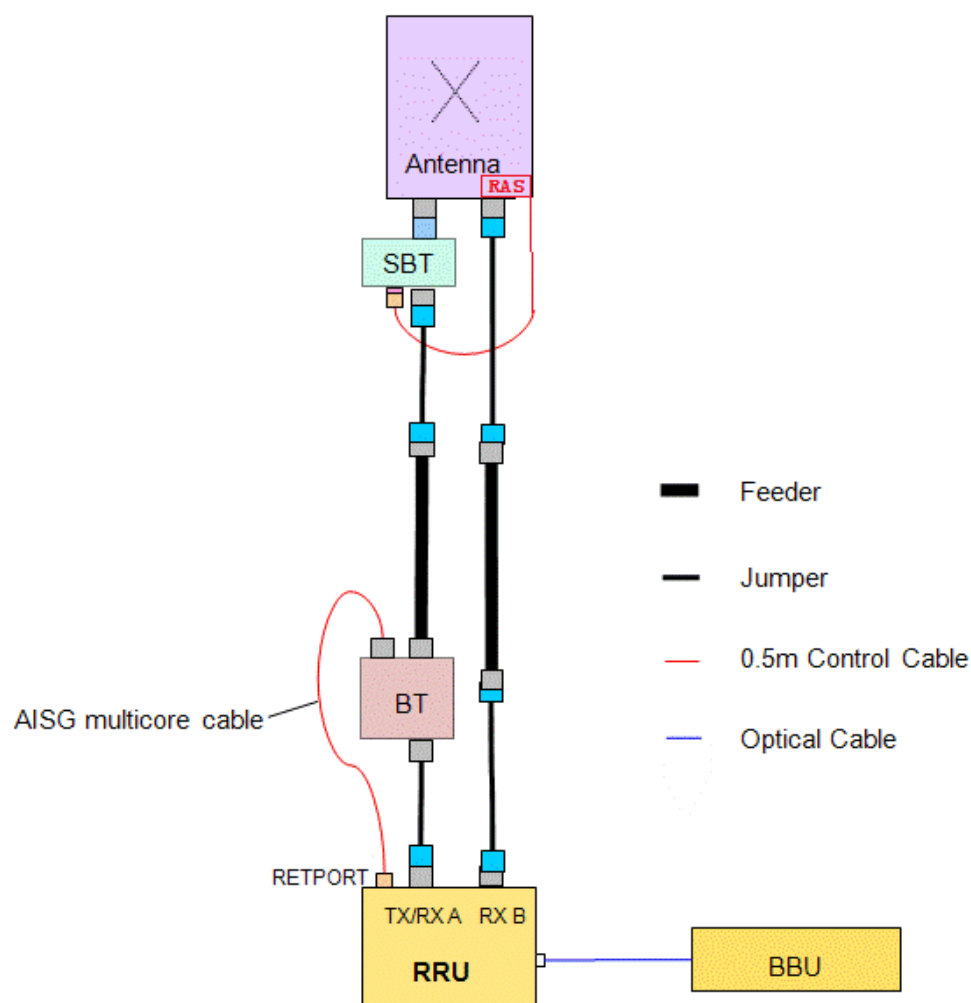
- eGBTS/NodeB/eNodeB/gNodeB: scenario 16 in [Operation and Maintenance](#)

The connections between the RAS unit and the RRU/RFU in multimode base stations are the same as those shown in [Figure 10-1](#) and [Figure 10-2](#). The RRU/RFU in the figures must support multiple modes. For details about data configuration, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

- Connection to the RET antenna through the RET port (with an external BT)
When an RRU without a BT such as the RRU3801C (20 W) for UMTS or RRU3220 for LTE is installed 20 m or further away from the RET antenna, RF ports on the RRU cannot provide 12 V DC power for the RET antenna. In this case, the RET port on the RRU must connect to an external BT, which connects

to the RET antenna through an external SBT to provide power for the antenna. The connections between the RRU, external BT, external SBT, and RAS unit are shown in [Figure 10-3](#). The BT is a passive component used on the base station side. It couples the RF signals and the OOK signals to the feeder.

Figure 10-3 Connection to the RAS unit through the RET port (with an external BT)



For the data configurations in this connection scheme, see the following section:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 15 in [Operation and Maintenance](#)

NOTE

If the association between RAS subunits and antennas is NULL in the **DSP RASSUBUNIT** (old model) (recommended)/**DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command output for an eGBTS/NodeB/eNodeB/gNodeB, run the **DSP ALDFUNCTION** command. If **RAS Non-Standard Protocol Function List = Reporting Device Association Relationships:YES** is displayed in the command output, the RAS association relationship can be reported.

10.1.2 Operations on RAS Units

The base station allows users to perform operations on each antenna subunit separately, including configuration file loading, calibration, azimuth offset setting, software downloading, and RAS unit reset.

- Loading a configuration file
A configuration file, which is provided by the RAS unit manufacturer, describes the relationships between the RAS azimuth and other antenna subunit azimuths.
 - For an eGBTS/NodeB/eNodeB/gNodeB, run the **DLD ALDCFGDATA** command to load the configuration file for the RAS unit.
If an incorrect configuration file is loaded, the RAS unit will experience unexpected errors. In this case, you can run the **DSP RASSUBUNIT** (old model) (recommended)/**DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command for an eGBTS/NodeB/eNodeB/gNodeB to query the name of the last loaded configuration file and the time when the file was loaded. During the implementation, you are advised to select one or two base stations, and check whether the actual RAS azimuth offsets are the same as the configured ones onsite to determine whether the loaded RAS configuration file is correct. Before the calibration, ensure that a correct configuration file has been loaded to the antenna subunit.

NOTE

- Run the **DSP RASSUBUNIT** (old model) (recommended)/**DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command for an eGBTS/NodeB/eNodeB/gNodeB to query dynamic information about RAS unit parameters. If any information is incorrect in the command output, for example, if the values of **Max Azimuth Offset** and **Min Azimuth Offset** are **NULL**, no configuration file is loaded or the configuration file is lost.
- If ALM-26754 RET Antenna Data Loss is not reported, the RAS device has been loaded with a configuration file. File reloading is not required unless the device is replaced.
- If a configuration file has been loaded to the RAS unit before delivery, reloading is not required onsite.
- Calibrating the RAS unit
Run the **CLB ALD** command for an eGBTS/NodeB/eNodeB/gNodeB to calibrate the RAS unit after it is installed. During the calibration, the RAS unit is adjusted within the azimuth offset range until it operates properly. If the calibration fails, ALM-26753 RET Antenna Not Calibrated will be reported.
- Setting the azimuth offset
After the RAS unit is calibrated, run the **MOD RASSUBUNIT** (old ALD model)/**MOD ALDSUBUNIT** (new ALD model) command for an eGBTS/NodeB/eNodeB/gNodeB to set the azimuth offset of each RAS subunit.
Before setting the azimuth offset, run the **DSP RASSUBUNIT** (old model) (recommended) or **DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command for an eGBTS/NodeB/eNodeB/gNodeB to query the azimuth offset range supported by the RAS unit. Setting the azimuth offset of an RAS subunit affects the coverage of the antenna. Therefore, set the azimuth offset according to the engineering design.
For digital antennas, before setting the azimuth offset, run the **DSP RASSUBUNIT** (old model) (recommended)/**DSP ALDSUBUNIT** (old model)/

DSP ALDSUBUNIT (new model) command to query the azimuth adjustable status of the RAS. The azimuth offset of the RAS subunit can be set only when the status is adjustable. If the RAS azimuth cannot be adjusted and the **MOD RASSUBUNIT** (old model)/**MOD ALDSUBUNIT** (new model) is executed to set the azimuth offset for the RAS subunit, the base station returns an error message, and periodically queries the azimuth adjustable status. When the azimuth is adjustable, the base station sets the azimuth offset for the RAS subunit.

NOTE

Before setting the azimuth offset for a digital antenna, ensure that the license of the corresponding type has been configured (using the **DIGIANTINFO** MO). You can run the **DSP ALDFUNCTION** command to check whether the antenna is a digital antenna. The ALD is a digital antenna if the command output shows that at least one non-standard protocol function is supported by the ALD.

The azimuth offset range of an RAS unit varies with the manufacturer and model.

- Downloading RAS software

RAS manufacturers provide RAS software. For details, see the documents provided by the manufacturers.

For an eGBTS/NodeB/eNodeB/gNodeB, run the **DLD ALDSW** command to download the RAS software.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*:

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

-
- Resetting the RAS unit

Run the **RST ALD** command for an eGBTS/NodeB/eNodeB/gNodeB to reset an RAS unit. RAS unit reset does not change its azimuth offsets.

10.2 Network Analysis

10.2.1 Benefits

None

10.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

The function has no impact on the network.

10.3 Requirements

10.3.1 Licenses

No license is required for GSM, UMTS, LTE, and NR.

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

None

10.3.2.2 Mutually Exclusive Functions

None

10.3.2.3 Function Impacts

None

10.3.3 Hardware

Base Station Models

3900 and 5900 series base stations

Boards

N/A

RF Modules

N/A

10.3.4 Others

None

10.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

For details, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

10.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

For details, see [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

10.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

For details, see [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

10.7 Operation and Maintenance (Automatic ALD Deployment)

For details, see [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

11 RVD

11.1 Principles

The RVD is a Huawei-defined device that supports functions such as horizontal beamwidth adjustment.

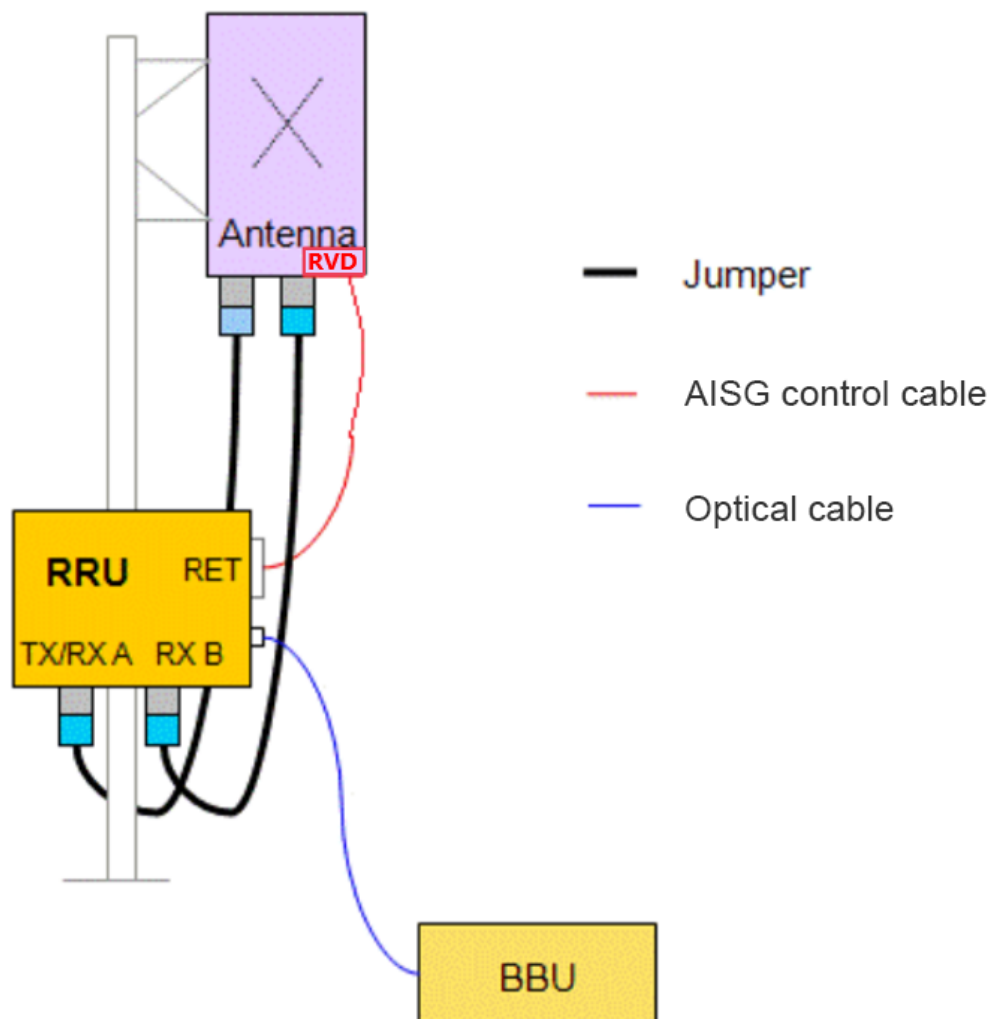
The horizontal beamwidth of the RVD can be remotely adjusted to change the signal coverage.

11.1.1 Connections Between the RVD and RRU/RFU

The RRU can be connected to the RVD through the RET port (**RET**PORT MO) or RF port (**ANTENNA**PORT MO). The RFU does not have the RET port and can be connected to the RVD only through the RF port.

- Connection to the RVD through the RET port
This connection scheme only applies to RRUs. An AISG multi-wire cable connects the RVD port, as shown in [Figure 11-1](#). With the integrated BT, the RRU can send RS485 control signals and feed 12 V DC power to the RVD through the RET port. Therefore, no SBT is required. This connection scheme is recommended when the distance between the RRU and the RVD is shorter than 20 m.

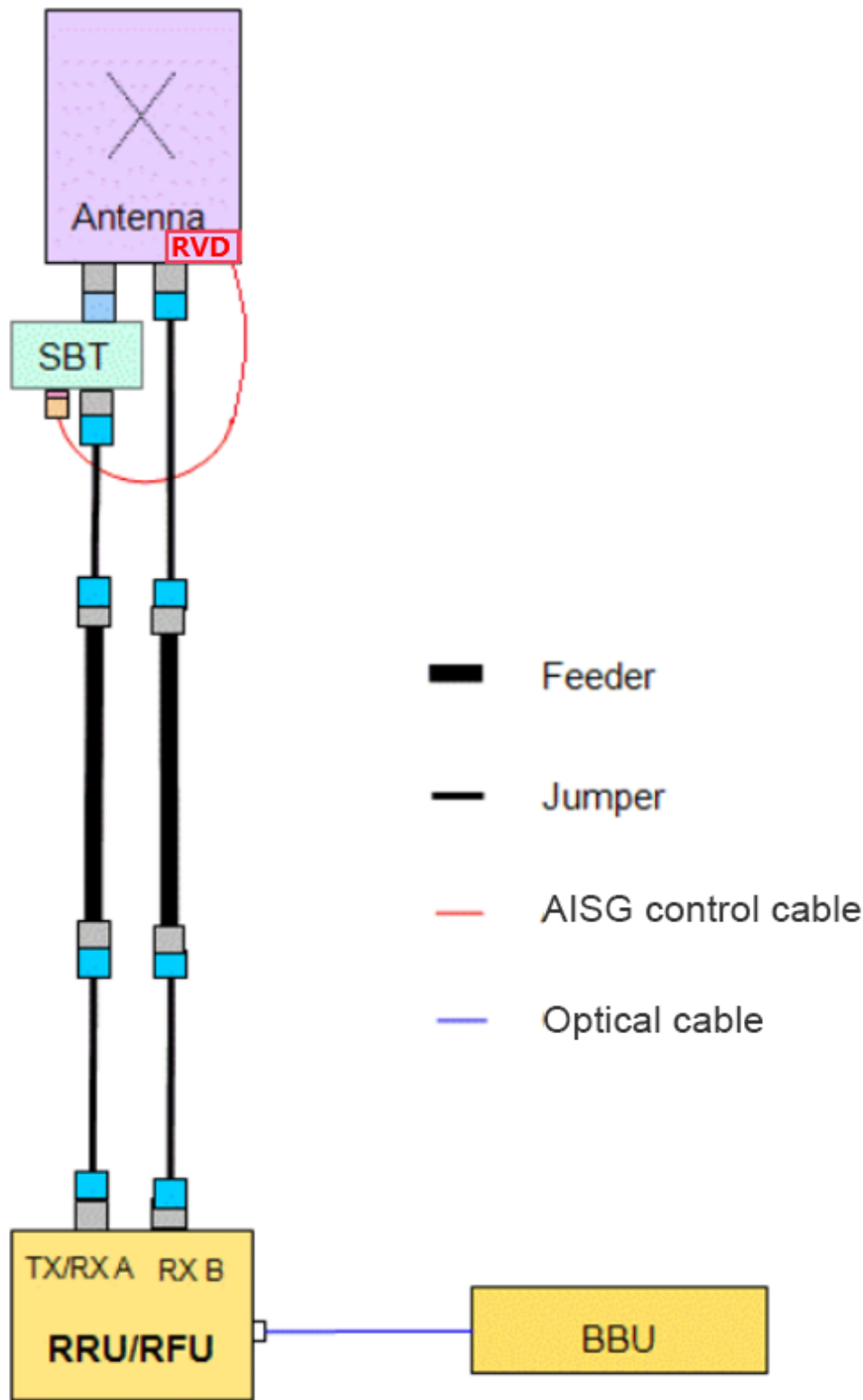
Figure 11-1 Connection to the RVD through the RET port



For the data configurations in this connection scheme, see the following section:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 17 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#)
- Connection to the RVD through the antenna port
An SBT is required in this connection scheme. Feeders and jumpers connect the RRU/RFU and SBT, and an AISG multi-wire cable connects the SBT to the RVD, as shown in [Figure 11-2](#).
 - a. The RRU/RFU combines OOK signals, 12 V DC power, and RF signals. Then, it sends the combined signals to the SBT through feeders and jumpers.
 - b. The SBT splits the combined signals into two paths. On one path, the RF signals are sent to the antenna. On the other path, OOK signals are converted to RS485 control signals and combined with 12 V DC power, both of which are forwarded to the RVD through the AISG multi-wire cable.

Figure 11-2 Connection to the RVD through the antenna port



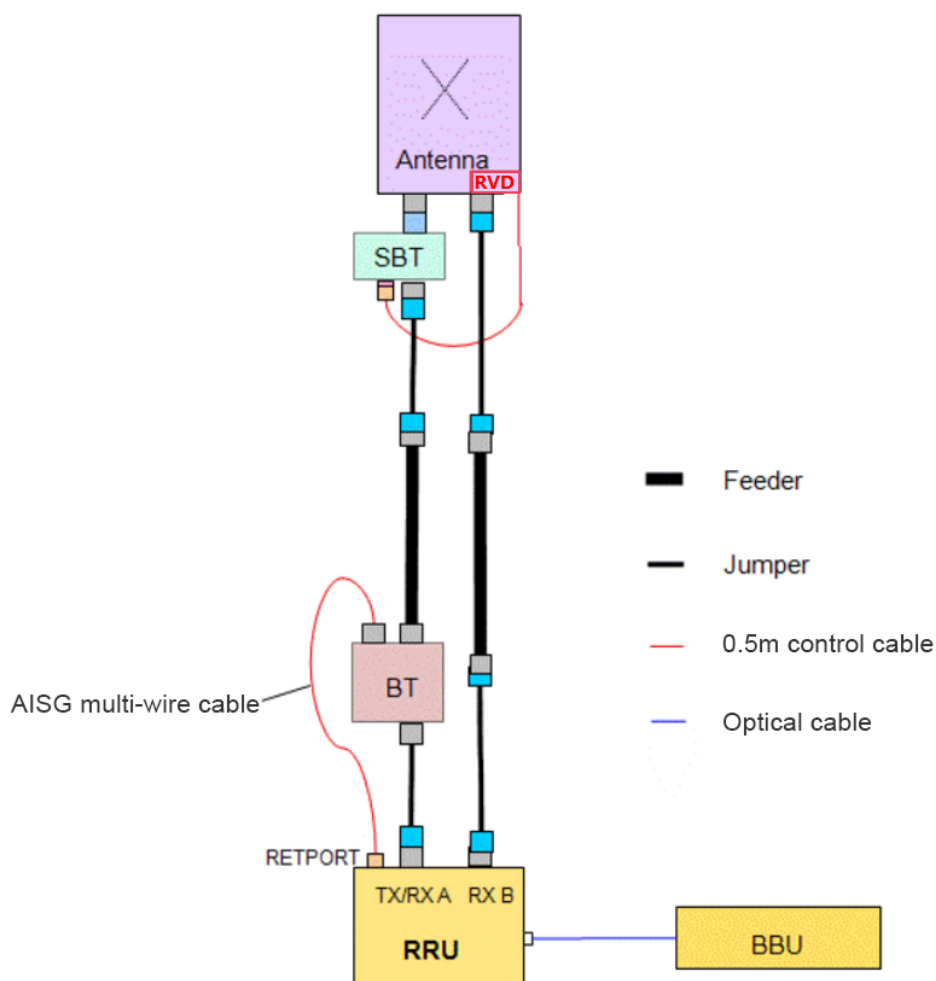
For the data configurations in this connection scheme, see the following section:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 18 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#)

The connections between the RVD and the RRU/RFU in multimode base stations are the same as those shown in [Figure 11-3](#) and the RRU/RFU in the figure must support multiple modes. For details about data configuration, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

- Connection to the RVD through the RET port (with an external BT)
When an RRU without a BT, for example, the RRU3801C (20 W) for UMTS or RRU3220 for LTE, is installed over 20 m away from the RVD, RF ports on the RRU cannot provide 12 V DC power for the RVD. In this case, the RET port on the RRU is used to supply power to the RVD and the RRU must connect to an external BT. [Figure 11-3](#) shows the connections between the RRU, BT, SBT, and RVD. The BT is a passive component used on the base station side. It couples the RF signals and the OOK signals to the feeder.

Figure 11-3 Connection to the RVD through the RET port (with an external BT)



For the data configurations in this connection scheme, see the following section:

- eGBTS/NodeB/eNodeB/gNodeB: scenario 17 in [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#)

11.1.2 Operations on the RVD

The base station allows users to perform operations on each antenna subunit separately, including configuration file loading, calibration, horizontal beamwidth setting, software downloading, and RVD resetting.

- Loading a configuration file
A configuration file, which is provided by the RVD manufacturer, describes the relationships between the RVD's beamwidths and other antenna subunits' beamwidths.
 - For an eGBTS/NodeB/eNodeB, run the **DLD ALDCFGDATA** command to load the configuration file for the RVD.
If an incorrect configuration file is loaded, the RVD may experience unexpected errors. In this case, you can run the **DSP RVDSUBUNIT** (old model) (recommended)/**DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the name of the most recently loaded configuration file and the time when the file was loaded. Before the calibration, ensure that a correct configuration file has been loaded to the antenna subunit.

NOTE

- Run the **DSP RVDSUBUNIT** (old model) (recommended)/**DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query dynamic information about RVD additional parameters. If any information is incorrect in the command output, for example, if the value of the field indicating the horizontal beamwidth is **NULL**, no configuration file is loaded to the RVD or the configuration file is lost.
- If ALM-26754 RET Antenna Data Loss is not reported, the RET device has been loaded with a configuration file. File reloading is not required unless the device is changed.
- If a configuration file has been loaded to the RVD before delivery, reloading is not required onsite.
- Calibration
Run the **CLB ALD** (eGBTS/NodeB/eNodeB) command to calibrate the RVD after it is installed. This calibration process enables the RVD to operate with the supported horizontal beamwidth once, guaranteeing proper operation. If the calibration fails, ALM-26753 RET Antenna Not Calibrated will be reported.
- Setting the horizontal beamwidth
After the RVD is calibrated, run the **MOD RVDSUBUNIT** (eGBTS/NodeB/eNodeB) command to set the horizontal beamwidth.
Before setting the horizontal beamwidth, you can run the **DSP RVDSUBUNIT** (old model) (recommended)/**DSP ALDSUBUNIT** (old model)/**DSP ALDSUBUNIT** (new model) command to query the horizontal beamwidth range supported by the RVD. Setting the horizontal beamwidth affects the antenna coverage. Therefore, the configured horizontal beamwidth must be consistent with the engineering design.
- Downloading RVD software
RVD manufacturers provide RVD software. For details, see the documents provided by the manufacturers.

For an eGBTS/NodeB/eNodeB, run the **DLD ALDSW** command to download the RVD software.

NOTICE

To prevent a software package from being maliciously tampered with during transmission or storage, download the corresponding digital signature file for integrity verification when downloading the software package.

After the software package is downloaded, verify its PGP digital signature according to the *OpenPGP Signature Verification Guide*. If the software package fails the verification, do not use the software package, and contact Huawei technical support engineers.

Before a software package is used in installation or upgrade, its digital signature also needs to be verified according to the *OpenPGP Signature Verification Guide* to ensure that the software package is not tampered with. Visit either of the following websites to obtain the *OpenPGP Signature Verification Guide*:

- Carrier customers: <https://support.huawei.com/carrier/digitalSignatureAction>
- Enterprise customers: <https://support.huawei.com/enterprise/en/tool/pgp-verify-TL1000000054>

- Resetting the RVD

Run the **RST ALD** (eGBTS/NodeB/eNodeB) command to reset the RVD. Resetting the RVD does not affect its horizontal beamwidth.

11.2 Network Analysis

11.2.1 Benefits

None

11.2.2 Impacts

The impacts between this function and other functions are described in [11.3.2.3 Function Impacts](#).

This function has no impact on the network.

11.3 Requirements

11.3.1 Licenses

No license is required for GSM, UMTS, LTE, and NR.

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

None

11.3.2.2 Mutually Exclusive Functions

None

11.3.2.3 Function Impacts

None

11.3.3 Hardware

Base Station Models

3900 and 5900 series base stations

Boards

N/A

RF Modules

N/A

11.3.4 Others

None

11.4 Operation and Maintenance (Manual ALD Deployment on the Multimode Base Station)

For details, see [4.5 Operation and Maintenance \(Manual ALD Deployment on the Multimode Base Station\)](#).

11.5 Operation and Maintenance (Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB)

For details, see [4.4 Operation and Maintenance \(Manual ALD Deployment on the eGBTS/NodeB/eNodeB/gNodeB\)](#).

11.6 Operation and Maintenance (Manual ALD Deployment on the GBTS)

For details, see [4.6 Operation and Maintenance \(Manual ALD Deployment on the GBTS\)](#).

11.7 Operation and Maintenance (Automatic ALD Deployment)

For details, see [4.7 Operation and Maintenance \(Automatic ALD Deployment\)](#).

12 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

13 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

14 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

15 Reference Documents

1. 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
2. Documents in *3900 & 5900 Series Base Station Product Documentation*:
 - *3900 & 5900 Series Base Station Initial Configuration Guide*
 - *RRU Hardware Description*
 - *AAU Hardware Description*
3. *AISU User Manual* in *Antenna&ALD Product Documentation*
4. *OpenPGP Signature Verification Guide*